

# AS320 Series Elevator-Used Inverter

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## Foreword

AS320 series elevator-used inverter is a new inverter developed according to elevator carrying characteristics. It adopts 32-bit motor-specific microprocessor, programmable extra-large scale logic device CPLD and state-of-the-art power module, as well as international leading closed-loop vector VC control technology, while supporting voltage vector V/F, non-speed sensor vector SVC and torque control mode, it makes elevator running stably, comfortably and efficiently in combination with characteristics of potential energy load.

### General

This operating instruction gives a comprehensive and systematic description of installation, operation, setting of functional parameters, maintenance and troubleshooting for AS320 series elevator-used inverter. This manual can also be used as reference for designing elevator control system with specific frequency converter for AS320 series elevator, as well as materials on system installation, commissioning and maintenance.

To ensure correct installation, please read this operating instruction carefully before use of the frequency converter.

### Audience

User

Lift control designer

Engineering maintenance staff

Technical support staff of user side

## **Innovative characteristics**

- a) New no-load sensor starting compensation technology, making elevator starting comfortable without installing weighting device.
- b) Realizing synchronous motor control with incremental ABZ encoder, and achieve superior startup comfort by no-load sensor starting compensation technology.
- c) New PWM dead-time compensation technology, reducing motor noise and loss effectively.
- d) Dynamic PWM carrier modulation technology, reducing motor noise effectively.
- e) Synchronous motor can adjust phase angle by itself without encoder.
- f) If motor parameters are set accurately, asynchronous motor can adjust by itself without encoder either. If it can not learn accurate motor on site, it is also possible to make frequency converter get accurate motor parameter automatically with simple self-learn mode of static motor instead of complex working such as car lifting.
- g) Hardware uses the 6<sup>th</sup> generation of new module, which could withstand the high temperature of 175°C, and with low switch loss.

## **Simple and quick elevator commissioning way**

On completion of inverter wiring, convenience of elevator commissioning is critical. To complete inverter commissioning, operator always takes a lot of time and efforts to set up a large number of parameters and perform complex operation. Because the inverter is used specially for elevator, it is vary simple and quick to drive elevator commissioning, only 3 steps are required.

### **1. Parameter setting**

- a) Reset all parameter to ex-factory setting with operating device;
- b) Set parameters of motor and encoder according to motor name plate.

### **2. Adjustment of operating direction**

It is possible to check encoder wiring and motor operating direction by operating device, if any error is found, parameters can be adjusted simply.

### **3 Comfort adjustment**

- a) Even ex-factory setting can achieve considerable comfort without any parameter change;
- b) PID adjuster tuning will gain best comfort.

## **Content description**

Supplement and modification may be made to the content of this operating instruction, please visit our company website regularly to update. Website: [www.stepelectric.com](http://www.stepelectric.com).

## Signs and notices related to safety

This operating instruction uses following signs to hint safety-related content. Description and content noted with safety sign is very important, please observe them strictly.



In case of use by mistake, caused hazard may lead to human injury or even death.



In case of use by mistake, caused hazard may lead to minor or major human injury and equipment damage.



**Important:** user must observe important notices.

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## Chapter 1 Notice for inverter use

User familiar with this inverter can read Appendix C “Quick Operating Guide of Elevator Control” directly.

This chapter mainly introduces general information, including voltage level of inverter, adaptive motor capacity, and how to carry out Ooba etc. In addition, it also details notices during inverter installation, wiring, operation, maintenance and reject, facilitating safe inverter operation and extending inverter service life. Please read this chapter carefully.

### 1.1 Voltage level and adaptive motor capacity

Voltage level of AS320 series inverter is 200V and 400V, supporting the asynchronous and synchronous motors. Currently the adaptive motor capacity is 1.1~75kW. For configuration beyond this range, please contact our engineering center.

### 1.2 Ooba



#### Notice

© Do not install inverter with damaged or missing parts.

Or it may cause fire and human injury hazards.

When unpacking, please confirm carefully that there is damage during transportation, and that type and specification in the name plate is consistent with order requirement. If not consistent or any part is missing, contact factory or supplier as early as possible.

### 1.3 Description of inverter type

For description of inverter type, see Figure 1.1.

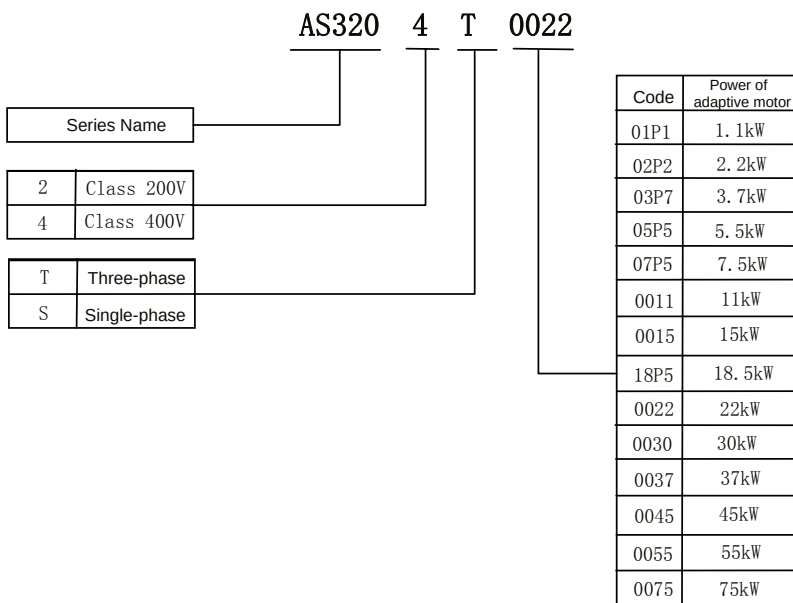


Fig.1.1 Description of inverter type

## 1.4 Description of inverter name plate

For inverter name plate, see Fig. 1.2. Name plate of inverter records inverter type, specification and batch No. etc.

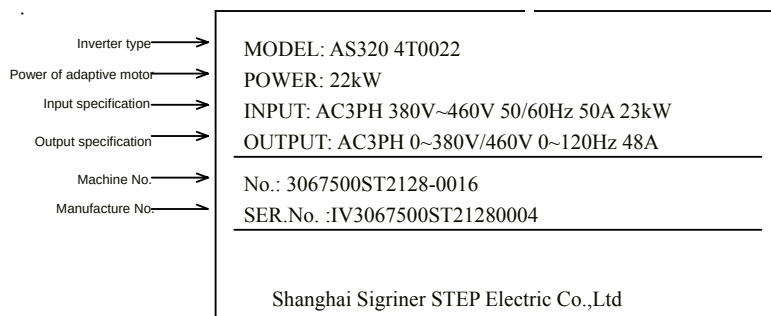


Fig.1.2 Description of inverter name plate

## 1.5 Safety notice



**Danger**

- ◎ **Please install the device on non-flammable materials such as metal.**  
Or it may cause fire hazard.
- ◎ **Do not install the device in the environment containing explosive gas.**  
Or it may cause explosion hazard.
- ◎ **Do not place flammable materials nearby.**  
Or it may cause fire hazard.



**Caution**

- ◎ **When carrying, please support the base of body.**  
Or if inverter body falls, it may cause hazard of human injury and inverter damage.
- ◎ **When installing, bearing capacity of the platform should be considered.**  
Or if inverter body falls, it may cause hazard of human injury and inverter damage.
- ◎ **Please do not install the device in the environment with pipe splashing drops.**  
Or it may cause hazard of inverter damage.
- ◎ **Do not fall foreign matters such as screw, gaskets and metal bar into inverter.**  
Or it may cause hazard of fire or inverter damage.



**Danger**

- ◎ **Before wiring, confirm that input power supply is disconnected totally.** Or it may cause electric shock.
- ◎ **Wiring task must be carried out by professional engineer.** Or it may cause electric shock.
- ◎ **Protective grounding terminal E of inverter must be grounded reliably.** Or it may cause electric shock.
- ◎ **Do not confuse input terminal for main loop of inverter with output terminal.** Or it may damage inverter or cause explosion.
- ◎ **Do not short terminal ○+○+○+○+⊕1/○+○+○+○+⊕2 to ○-○-○-○-⊖.** Or it may cause fire and explosion hazard.
- ◎ **Cover plate must be covered properly before power on.** Or it may cause electric shock or explosion.
- ◎ **Do not operate inverter if your hands are wet.** Or it may cause electric shock.
- ◎ **When connecting safety loop of emergency stop, please check its wiring carefully after operation.** Or it may cause hazard.



- ◎ **For inverter with storage period over 2 years, power should be supplied slowly by voltage regulator during power on.** Or it may cause electric shock or explosion.
- ◎ **When inverter is running, do not mis-operate.** Or it may cause HV electric shock.
- ◎ **Within a period after power off, dangerous high voltage still exists inside the inverter, therefore, do not open the cover plate or touch terminal.** Or it may cause HV electric shock.
- ◎ **Only qualified professional staff having training can maintain the inverter.** Or it may cause inverter damage or electric shock.
- ◎ **Before maintenance staff works, all metal articles such as watch and ring must be taken off. During working, operator must use clothes and tools conforming to insulation requirement.**  
Or it may cause electric shock or explosion.

## 1.6 Notice in use

When using AS320 series inverter, following things should be noticed.

### 1.6.1 Selection of braking resistor

Elevator is potential-energy loading and four-quadrant running, braking power status exists. Therefore, it should consider selection of braking component, or overvoltage may occur, resulting in tripping. AS320 series inverters are all equipped with built-in braking unit, only braking resistor must be prepared externally. For specification of external braking resistor of inverter, see Table 1.1.

**Table 1.1 Configuration table of braking resistor for AS320 series elevator-used inverters**

| Inverter type<br>AS320- | Adaptive<br>motor (kW) | Min.<br>Value<br>(Ω) | Max.<br>value (Ω) | Recommended<br>value (Ω) | Total power of recommended resistor<br>(W) |              |
|-------------------------|------------------------|----------------------|-------------------|--------------------------|--------------------------------------------|--------------|
|                         |                        |                      |                   |                          | Synchronous                                | Asynchronous |
| 200V inverter           |                        |                      |                   |                          |                                            |              |
| 2S01P1                  | 1.1                    | 26                   | 72                | 64                       | 1000                                       | 1000         |
| 2S02P5                  | 2.2                    | 26                   | 58                | 50                       | 1000                                       | 1000         |
| 2S03P7                  | 3.7                    | 26                   | 39                | 30                       | 1600                                       | 1200         |
| 400V inverter           |                        |                      |                   |                          |                                            |              |
| 4T02P2                  | 2.2                    | 56                   | 210               | 100                      | 1000                                       | 1000         |
| 4T03P7                  | 3.7                    | 56                   | 144               | 80                       | 1600                                       | 1200         |
| 4T05P5                  | 5.5                    | 56                   | 100               | 70                       | 2000                                       | 1600         |
| 4T07P5                  | 7.5                    | 56                   | 72                | 64                       | 3200                                       | 2000         |
| 4T0011                  | 11                     | 34                   | 48                | 40                       | 4000                                       | 3200         |
| 4T0015                  | 15                     | 34                   | 41                | 36                       | 5000                                       | 4000         |
| 4T18P5                  | 18.5                   | 17                   | 31                | 24                       | 6400                                       | 5000         |
| 4T0022                  | 22                     | 17                   | 27                | 20                       | 8000                                       | 6400         |
| 4T0030                  | 30                     | 11                   | 20                | 15                       | 10000                                      | 8000         |
| 4T0037                  | 37                     | 8                    | 16                | 12                       | 12000                                      | 10000        |
| 4T0045                  | 45                     | 5                    | 10                | 9                        | 18000                                      | 15000        |
| 4T0055                  | 55                     | 5                    | 8                 | 8                        | 22000                                      | 18000        |
| 4T0075                  | 75                     | 5                    | 6                 | 6                        | 30000                                      | 25000        |

### 1.6.2 Absorber is prohibited at output side

Because inverter output is pulse wave, if capacitor for power factor improvement or anti-lightning VDR etc. is installed at output side, all of them may cause inverter tripping or part damage. This must be considered during line design. In case of old elevator reconstruction, capacitor or VDR connected originally at output side must be removed.

Do not connect capacitor to output side of inverter, for the schematic, see Fig. 1.3.

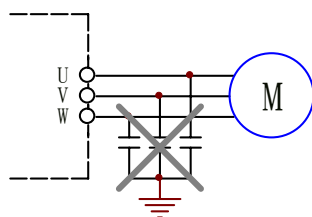


Fig.1.1 Schematic of not connect capacitor to output side of inverter

### 1.6.3 Service voltage of inverter

AS320 series inverter only suits to work within its rated voltage range, if voltage is different from its rated voltage, voltage regulator is required for transformation.

### 1.6.4 2-phase input is not proper

It is not proper to change 3-phase input into 2-phase input, or fault may occur.

### 1.6.5 User control of output contactor

When output contactor is controlled with user application, to ensure output contactor opens and closes without current, it will be better to close contactor before transmitting running order to inverter, disconnect contactor following a period after elevator stops signal output.

### 1.6.6 Altitude and de-rating use

In the area with altitude over 1000m, thin air will cause radiation effect of inverter poor, in this case, it is necessary to use inverter by de-rating. In case of de-rating use for inverter, relation curve between its rated current and altitude, see Fig. 1.4.

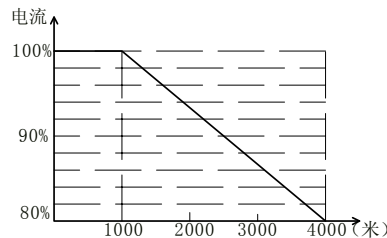


Fig.1.2 Relation diagram between rated output current and altitude of inverter

### 1.6.7 Ambient temperature and de-rating use

Normal operating temperature of this inverter is  $-10\sim+45^{\circ}\text{C}$ , when it exceeds  $45^{\circ}\text{C}$ , 10% should be derated for each  $5^{\circ}\text{C}$  higher, and it can be up to  $60^{\circ}\text{C}$ .

### 1.6.8 Synchronous star delay

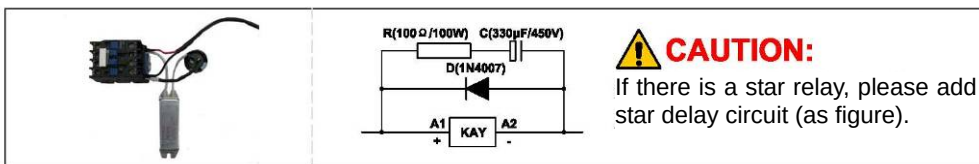


Figure 1.5 Hint sign of star delay circuit

### 1.6.9 Low Voltage directive complies

That our products comply with EN61800-5-1:2007 standards, which comply with the Low Voltage Directive (Low Voltage Directive 2006/95/EC). If the inverter as a component integration in the entire electrical system, make sure the entire system meets the requirements of EC Directives. Note:

- ① Machine make sure the ground, and make sure the ground terminal is a separate ground.
- ② Inverter prohibited for use in of  $\Delta$  ground and IT power systems.
- ③ If you are loading the cabinet, make sure that the cabinet ground.

- ④ Use CE certified circuit breaker, electromagnetic contactor, and other accessories. Leakage circuit breakers, please use the B-type leakage circuit breakers.
- ⑤ Inverter in the overvoltage catalog III, Pollution Degree II, under the conditions of use. The level of protection of the inverter protection class I

## 1.7 Reject notice

When rejecting inverter, it should be treated as industrial rubbish.

### 1.7.1 Capacitor treatment

Electrolytic capacitors in main loop and on printed board may explode when they are buried. Therefore, it is prohibited to burn capacitor.

### 1.7.2 Treatment of plastic piece

There are many plastic pieces on the inverter, plastic piece burning will produce poisonous gas. Therefore it is prohibited to burn poisonous gas.

## Chapter 2 Type and specification

This chapter provides type, specification and installation dimension of **AS320** series inverter.

### 2.1 Inverter type

For type of **AS320** series inverter, see Table 2.1.

Table 2.1 Type of **AS320** series inverter

| Inverter type<br><b>AS320-</b> | Rated capacity (kVA) | Rated output current<br>(A) | Adaptive motor<br>(kW) |
|--------------------------------|----------------------|-----------------------------|------------------------|
| 2S01P1                         | 2.3                  | 6.0                         | 1.1                    |
| 2S02P2                         | 4.6                  | 12                          | 2.2                    |
| 2S03P7                         | 6.9                  | 18                          | 3.7                    |
| 4T02P2                         | 4.7                  | 6.2                         | 2.2                    |
| 4T03P7                         | 6.9                  | 9                           | 3.7                    |
| 4T05P5                         | 8.5                  | 13                          | 5.5                    |
| 4T07P5                         | 14                   | 18                          | 7.5                    |
| 4T0011                         | 18                   | 27                          | 11                     |
| 4T0015                         | 24                   | 34                          | 15                     |
| 4T18P5                         | 29                   | 41                          | 18.5                   |
| 4T0022                         | 34                   | 48                          | 22                     |
| 4T0030                         | 50                   | 65                          | 30                     |
| 4T0037                         | 61                   | 80                          | 37                     |
| 4T0045                         | 74                   | 97                          | 45                     |
| 4T0055                         | 98                   | 128                         | 55                     |
| 4T0075                         | 130                  | 165                         | 75                     |

### 2.2 Technical index and specification of inverter

For technical index and specification of **AS320** series inverter, see Table 2.2.

Table 2.2 Technical index and specification of **AS320** series inverter

|                                        |                                                      | 2S01P1                                                                                                                                                    | 2S02P2 | 2S03P7 | 4T02P2 | 4T03P7 | 4T05P5 | 4T07P5 | 4T0011 | 4T0015 | 4T0018 | 4T0022 | 4T0030 | 4T0037 | 4T0045 | 4T0055 | 4T0075 |
|----------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Maximum applicable motor capacity (kW) |                                                      | 1.1                                                                                                                                                       | 2.2    | 3.7    | 2.2    | 3.7    | 5.5    | 7.5    | 11     | 15     | 18.5   | 22     | 30     | 37     | 45     | 55     | 75     |
| Rated output                           | Rated capacity (kVA)                                 | 2.3                                                                                                                                                       | 4.6    | 6.9    | 4.7    | 6.9    | 8.5    | 14     | 18     | 24     | 29     | 34     | 50     | 61     | 74     | 98     | 130    |
|                                        | Rated current (A)                                    | 6.0                                                                                                                                                       | 12     | 18     | 6.2    | 9      | 13     | 18     | 27     | 34     | 41     | 48     | 65     | 80     | 97     | 128    | 165    |
|                                        | Maximum output voltage (V)                           | 200V: single phase 220~240 (corresponding input voltage)<br>400V: 3-phase 380/400/415/440/460V (corresponding input voltage)                              |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
| Input power supply                     | Phase number, voltage, frequency                     | 3-phase 380/400/415/440/460V 50/60Hz                                                                                                                      |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Allowable voltage variation                          | -15%~+10%                                                                                                                                                 |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Allowable frequency variation                        | -5%~+5%                                                                                                                                                   |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Reduced bearing capacity of instantaneous voltage    | AC300V above, continuous running<br>When de-rating from rated input status to below AC300V, perform undervoltage protection after 15ms continuous running |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
| Controlling characteristics            | Controlling way                                      | With PG card vector control, voltage vector, V/F, open-loop vector, torque control                                                                        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Starting moment                                      | 150% 0Hz (with PG card vector control), 120% 0.5Hz (voltage vector V/F), 150% 0.5Hz (open-loop vector)                                                    |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Speed controlling range                              | 1:1000 (with PG vector control), 1: 200 (without PG vector control)                                                                                       |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Precision of speed control                           | ±0.02% (with PG vector control 25±10℃), ±0.2% (without PG vector control 25±10℃)                                                                          |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Moment limit                                         | Yes (with parameter setting)                                                                                                                              |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Moment precision                                     | ±5%                                                                                                                                                       |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Frequency controlling range                          | 0~120Hz                                                                                                                                                   |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Frequency precision (temp. variation)                | ±0.01% (digital command -10~+45℃), ±0.1%(analog command 25±10℃)                                                                                           |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Frequency setting resolution                         | 0.01Hz (digital command), ±0.06Hz/120Hz (analog command 11bit + no symbol)                                                                                |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Output frequency resolution (calculation resolution) | 0.01Hz                                                                                                                                                    |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Overloading capacity                                 | Zero speed is 150%, when < 3Hz, 160%, when > 3Hz, 200%                                                                                                    |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Braking moment                                       | 150%(connecting external braking resistor), built-in braking unit                                                                                         |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Time of ACC/DEC                                      | 0.01~600s                                                                                                                                                 |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Carrier frequency                                    | 2~11kHz                                                                                                                                                   |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | Speed setting                                        | Digital setting; analog setting; panel setting                                                                                                            |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | LV bus running                                       | During power cutoff, batteries are relied on to enable elevator to run at low speed to nearly floor.                                                      |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
| PG card interface signal               | PG card power supply                                 | 5V, 12V, 300mA                                                                                                                                            |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | PG card signal                                       | Collector open、push-pull, differential, SIN/COS, Endat absolute value type                                                                                |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|                                        | PG card frequency dividing output                    | OA, OB orthogonality, frequency dividing coefficient 1~128                                                                                                |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
| Control I/O                            | Analog voltage input                                 | 2-way, -10~+10VDC, precision of 0.1%                                                                                                                      |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |



|                     |                                             |                                                                                                                    |
|---------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| signal              | Analog voltage output                       | 2-way, -10~+10VDC, precision of 0.1%                                                                               |
|                     | Optoelectronic isolated input               | 8-way. Input function can be defined                                                                               |
|                     | Open collector output                       | 4-way. Output function can be defined                                                                              |
|                     | Programmable relay output                   | 2-way. NO, NC dual-contact, contact capacity: resistor type, 5A 250VAC or 5A 30VDC; Output function can be defined |
|                     | RS485 communication interface               | 1-way                                                                                                              |
|                     | RS232 communication interface               | 1-way, used for operating device or PC                                                                             |
| Protective function | Motor overloading protection                | Protective curve of motor through parameter setting                                                                |
|                     | Inverter overloading                        | When < 3Hz, 160%, 5s; when > 3Hz, 185%, 10s                                                                        |
|                     | Short protection                            | In case of overcurrent caused by short of any 2 phases at output side, protect the inverter                        |
|                     | Input open-phase protection during running  | During running, in case of input open-phase, cut off output to protect the inverter                                |
|                     | Output open-phase protection during running | During running, in case of output open-phase, cut off output to protect the inverter                               |
|                     | Overvoltage threshold                       | Bus voltage 410V (200V series) , 810V (400V series)                                                                |
|                     | Undervoltage threshold                      | Bus voltage 180V (200V series) 、 380V (400V series)                                                                |
|                     | Instantaneous power outage compensation     | Protection above 15ms                                                                                              |
|                     | Radiator fan overheating                    | Protect by thermo-sensitive resistor                                                                               |
|                     | Stall out prevention                        | Stall out protection that speed offset is greater than 30% of rated speed during running                           |
|                     | Pulse encoder trouble                       | PG disconnection                                                                                                   |
|                     | Braking unit protection                     | Check that braking unit is abnormal automatically, protect                                                         |
|                     | Module protection                           | Overcurrent, short, overheating protection                                                                         |
|                     | Current sensor protection                   | Self-check at power on                                                                                             |
|                     | Speed reverse protection                    | Inspect with encoder                                                                                               |
|                     | I <sup>2</sup> t protection                 | Inspect with 3-phase inspection                                                                                    |
|                     | Input overvoltage protection                | 400V grade is greater than 725V, 200V grade is greater than 360V, inspect after stop                               |
|                     | Output grounding protection                 | Any phase is shorted to ground during running, cut off output to protect inverter                                  |
|                     | Output unbalance protection                 | Unbalance of output 3-phase current is detected during running, cut off output to protect inverter                 |
|                     | Short protection for braking resistor       | Inspect at braking                                                                                                 |

|                     |                      |                                                         |
|---------------------|----------------------|---------------------------------------------------------|
|                     | Encoder interruption | Evaluate interruption degree of encoder and alarm       |
|                     | EEPROM trouble       | Self-check at power on                                  |
| Display             | LCD in Chinese       | Various levels of menus                                 |
| Environment         | Ambient temperature  | -10~+45℃                                                |
|                     | Humidity             | Below 95%RH (without condensate)                        |
|                     | Storage temperature  | -20~+60℃ (short-time temperature during transportation) |
|                     | Location             | Indoor (no corrosive gas or dust)                       |
|                     | Altitude             | Below 1000m                                             |
| Structure           | IP                   | IP20                                                    |
|                     | Cooling way          | Forced air cooling                                      |
| Installation manner |                      | Inside the cabinet                                      |

## 2.3 Installation dimension and quality of inverter

For installation dimension and quality of inverter, see Fig. 2.1 and 2.3.

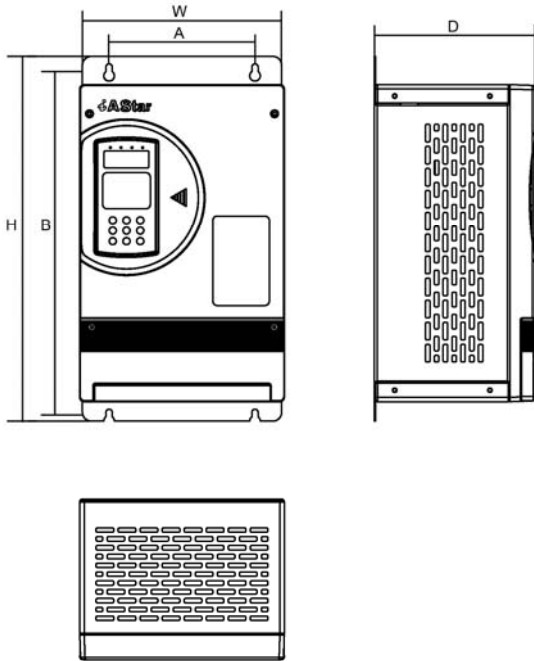


Fig. 2.1 Installation dimension and quality of inverter

Table 2.3 Installation dimension and mass of AS320 series inverter

| Inverter type<br>AS320- | A<br>(mm) | B<br>(mm) | H<br>(mm) | W<br>(mm) | D<br>(mm) | Installation<br>n diameter<br>Φ(mm) | Installation |     |        | Tightening<br>torque (Nm) | Mass (kg) |
|-------------------------|-----------|-----------|-----------|-----------|-----------|-------------------------------------|--------------|-----|--------|---------------------------|-----------|
|                         |           |           |           |           |           |                                     | Bolt         | Nut | Washer |                           |           |
| 2S01P1                  | 100       | 288.5     | 300       | 160       | 166       | 5.0                                 |              | 4M4 | 4Φ4    | 2                         | 4.5       |
| 2S02P2                  |           |           |           |           |           |                                     | 4M4          |     |        |                           |           |

|        |       |     |     |     |     |      |     |     |      |    |      |
|--------|-------|-----|-----|-----|-----|------|-----|-----|------|----|------|
| 2S03P7 |       |     |     |     |     |      |     |     |      |    |      |
| 4T02P2 |       |     |     |     |     |      |     |     |      |    |      |
| 4T03P7 |       |     |     |     |     |      |     |     |      |    |      |
| 4T05P5 |       |     |     |     |     |      |     |     |      |    |      |
| 4T07P5 | 165.5 | 357 | 379 | 222 | 192 |      |     |     |      |    | 8.2  |
| 4T0011 |       |     |     |     |     |      |     |     |      |    |      |
| 4T0015 |       |     |     |     |     | 7.0  | 4M6 | 4M6 | 4Φ6  | 3  |      |
| 4T18P5 | 165.5 | 392 | 414 | 232 | 192 |      |     |     |      |    | 10.3 |
| 4T0022 |       |     |     |     |     |      |     |     |      |    |      |
| 4T0030 |       |     |     |     |     |      |     |     |      | 6  |      |
| 4T0037 | 200   | 512 | 530 | 330 | 290 | 9.0  | 4M8 | 4M8 | 4Φ8  | 9  | 30   |
| 4T0045 |       |     |     |     |     |      |     |     |      |    |      |
| 4T0055 | 200   | 587 | 610 | 330 | 310 | 10.0 | 4M1 | 4M1 |      |    | 42   |
| 4T0075 | 260   | 707 | 730 | 430 | 330 | 10.0 | 0   | 0   | 4Φ10 | 14 | 50   |

## 2.4 Dimension of operating device

For dimension of inverter's operating device, see Fig. 2.2.

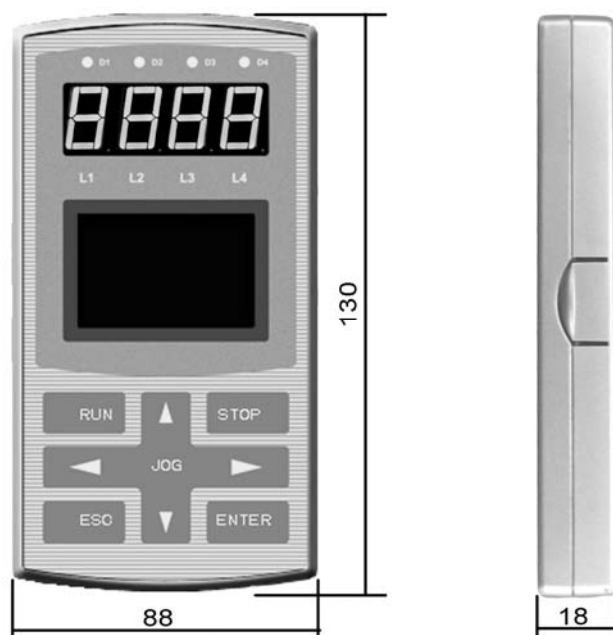


Fig. 2.2 Dimension of inverter's operating device



## Chapter 3 Inverter installation

This chapter describes installation requirements of inverter, notices, removal and installation of inverter panel etc.

### Installation location of inverter



**Danger**

- ⊙ **Please install the device on non-flammable materials such as metal.**  
Or it may cause fire hazard.
- ⊙ **No flammable material nearby**  
Or it may cause fire hazard.
- ⊙ **Do not install the device in the environment containing explosive gas.**  
Or it may cause explosion hazard.
- ⊙ **Enclosure installed with the device should conform to EN50178 standard.**



**Caution**

- ⊙ **During transport, do not lift operating panel or cover plate**  
Or it may cause hazard of inverter falling or damage.
- ⊙ **When installing, bearing capacity of the platform should be considered.**  
Or it may cause hazard of inverter falling or damage.
- ⊙ **It is prohibited to install the machine where drop may splash.**  
Or it may cause hazard of inverter damage.
- ⊙ **Do not fall foreign matters such as screw, gasket and metal bar into inverter.**  
Or it may damage inverter or cause explosion.
- ⊙ **If inverter is damaged or with missing part, do not install or operate it.**  
Or it may cause hazard of inverter damage.
- ⊙ **Do not install the machine in direct sunlight location**  
Or it may cause inverter overheating or accident.

Installation location of inverter must meet following conditions.

- a) Clean location without oil mist, dust, or floating matters should not invade into closed cabinet.
- b) Location where metal powder, oil or water will not enter into inverter interior.
- c) Location without flammable materials such as wood.
- d) Location without radioactive substances.

- e) Location without harmful gas or liquid.
- f) Location with little vibration.
- g) Location with less salt.
- h) Location without direct sunlight.
- i) Location where temperature is not easy to rise.

When installing in closed enclosure, please install cooling fan and cooling air conditioner, temperature should be below 40°C.

### 3.2 Installation direction and spacing requirement of inverter

To not reduce cooling effect of inverter, it should be installed at well-ventilated location. Generally, it is installed vertically. For spacing requirement for installation, see Fig. 3.1.

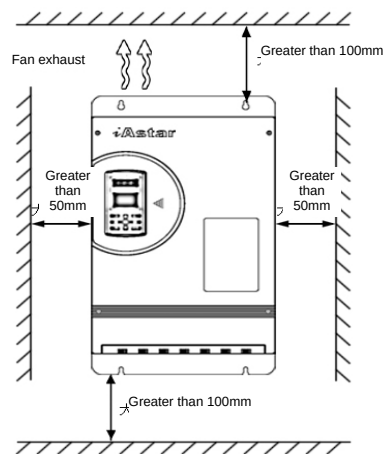


Fig. 3.1 Schematic of spacing of inverter installation

### 3.3 Inverter installation

Order of inverter installation is as follow:

- 1) Confirm 4 installing holes on the inverter, install 2 screws at upper first according to Fig. 2.1 – Installation dimension and quality of inverter, notice: do not tighten and leave blank of a few mm;
- 2) Hang 2 pear-shaped mounting holes at the upper of inverter to installed screw;
- 3) Install 2 screws at the upper and the lower, and tighten all 4 screws.



#### Important

**Fastener must be equipped with anti-vibration part such as spring washer;**

**4 inverter screws must be tightened.**

For inverter installation order, see Fig. 3.2.

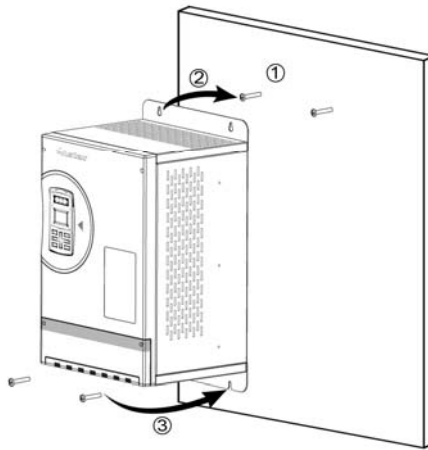


Fig. 3.2 Diagram of inverter installation order

### 3.4 Assembly and disassembly of inverter housing parts

#### 3.4.1 Inverter outline and part name

3.4.1 For inverter outline and part name, see Fig. 3.3.

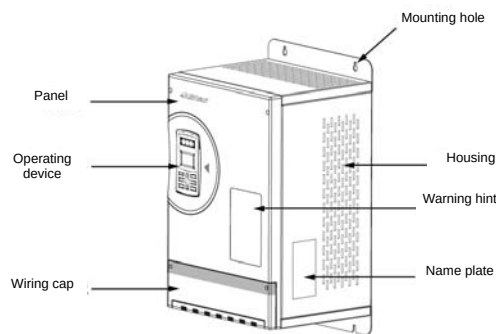


Fig. 3.3 Inverter outline and part name

#### 3.4.2 Assembly and disassembly of operating device

##### Take off operating device

- 1) Press latch springs at both sides of operating device at the same time so that they disengage from the panel, then take off operating device.
- 2) There is a connecting line at back of operating device, its plug should be pull off from operating device. Note: do not apply force on connecting line when pulling, or it may damage the line.

For assembly and disassembly of operating device, see Fig. 3.4.



Fig.3.4 Assembly and disassembly of operating device

### Install operating device

Insert connecting line plug into the socket at back of operating device, then embed a latch spring at one side of operating device into side groove of panel, then press the operating device to panel until a “crack” is heard, latch springs at both sides will be embedded into the panel.

### 3.4.3 Opening and closing of wiring cap

When connecting line to main loop or removing the panel, wiring cap is required to be opened.

#### Open wiring cap

- 1) Loose 2 screws on wiring cap;
- 2) Open the wiring cap downwards.

For opening operation of wiring cap, see Fig. 3.5

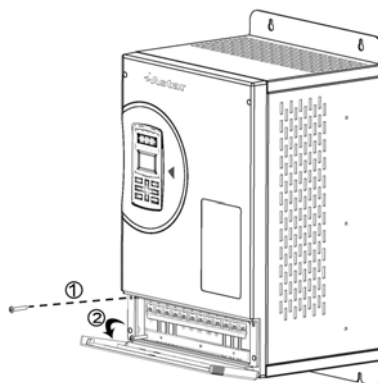


Fig. 3.5 Opening wiring cap

#### Close wiring cap

Operate in a reserve order of wiring cap opening, close the wiring cap and tighten 2 “fastening” screws on wiring cap.

### 3.4.4 Assembly and disassembly of front panel

When controlling loop wiring, it requires removing front panel. To facilitate wiring of main loop, it also allows front panel removal.

#### Remove front panel



Remove front panel as following steps.

- ① Take off operating device. See Chapter 3 “Assembly and disassembly of operating device”.
- ② Open wiring cap. See Chapter 3 “3.5 Opening and closing of wiring cap”.
- ③ Loose 2 screws at the upper of panel and 2 screws in wiring cap, then take off the panel.

For removal of front panel, see Fig. 3.6.

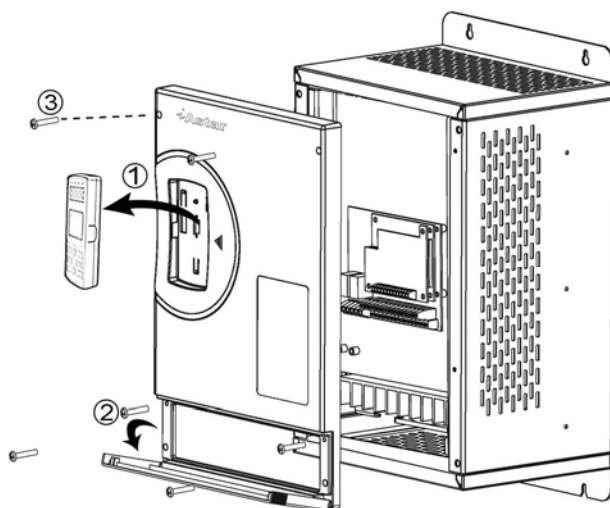


Fig. 3.6 Removal of front panel

#### Install front panel

Install front panel in a reserve order of removal.



## Chapter 4 Inverter wiring

This chapter details connection of inverter to peripheral equipment, overview of inverter terminal wiring, wiring of main loop terminal, wiring of control loop terminal and wiring of PG card terminal.



### Danger

- ◎ **Before wiring, confirm that input power supply is disconnected completely.**  
Or it may cause electric shock.
- ◎ **Wiring task must be carried out by professional engineer.**  
Or it may cause electric shock.
- ◎ **Grounding terminal E must be grounded reliably.**  
Or it may cause electric shock.
- ◎ **Do not touch terminal by hand directly, and outgoing line of inverter should not contact with outer cover.**  
Or it may cause electric shock.
- ◎ **Do not connect power supply to output terminal U, V and W.**  
Or it may cause hazard of inverter damage.
- ◎ **Do not short terminal  $\bigcirc + \bigcirc + \oplus 1 / \bigcirc + \bigcirc + \oplus 2$  to  $\bigcirc - \bigcirc - \oplus$ .**  
Or it may cause explosion hazard.



### Caution

- ◎ **Please confirm that voltage of power supply to main loop is consistent with rated voltage of inverter.**  
Or it may cause fire and human injury hazards.
- ◎ **Please connect braking resistor correctly as wiring diagram.**  
Or it may cause fire hazard.
- ◎ **Connection must be secure between main loop terminal and conductor or between conductor and its crimp type terminal.**  
Or it may cause hazard of inverter damage.

### 4.1 Connection of inverter to peripheral equipments

#### 4.1.1 Connection diagram of inverter to peripheral equipments

For connection diagram of inverter to peripheral equipments, see Fig. 4.1.

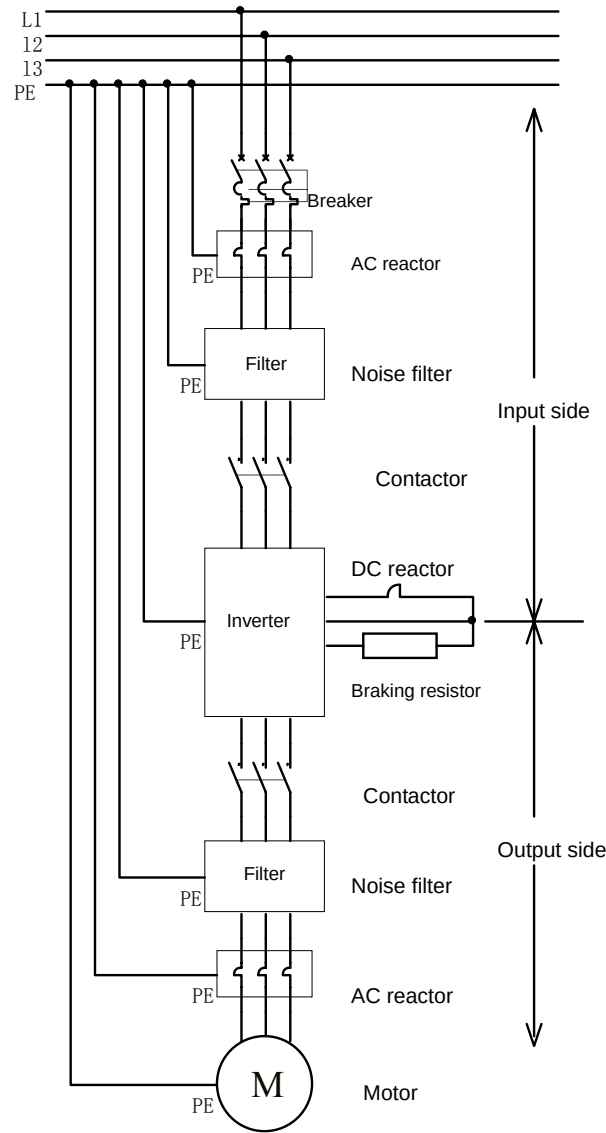


Fig. 4.1 Connection diagram of inverter to peripheral equipments

Note: The diagram adopts 3-phase power supply input for illustration.

#### 4.1.2 Connection of inverter to peripheral equipments

##### 4.1.2.1 Input power connection



**Danger**

Do not run inverter beyond rated voltage range of incoming line. Over-voltage may cause permanent damage to inverter.

Technical requirements of input power are as follow.

| Technical requirements of input power connection (main circuit) |                                                                                                                |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Input voltage                                                   | Voltage is 380/400/415/440/460V AC 3-phase, -15%~+10%                                                          |
| Short current (IEC60909 standard)                               | If incoming cable of inverter has proper fuse protection, maximum allowable short current is 100 kA within 1s. |
| Frequency                                                       | 50/60 $\pm$ 5% Hz                                                                                              |
| Cable temperature                                               | Allow working at 90°C for long term                                                                            |

## Input protection

Input protection includes breaker, fuse and emergency equipment etc.

### Breaker

Inverter does not contain breaker. Therefore, breaker must be installed between AC input power supply and inverter. This breaker must ensure that:

- ◎ Model should conform to safety regulation in actual application, including but not limited to national and local electric regulation.
- ◎ During inverter installation and maintenance, breaker must remain at closed position and locked.

Breaker is not allowed to control motoring start and stop. Buttons on operating device or command from I/O terminal should be used to control motor.

Breaker capacity should be selected as 1.5~2 times of rated inverter current.

Time characteristics of breaker should consider that of inverter overheating protection fully (150% of rated output current in 1 minute).

### Fuse

Final user must provide loop protection, and this protection model should be conform to national and local electric regulation. Table below gives recommended fuse type used to provide short protection for incoming power of inverter.

| AS320- | Input current (A) | Main fuse  |                |               |
|--------|-------------------|------------|----------------|---------------|
|        |                   | IEC gG (A) | UL grade T (A) | Bussmann type |
| 4T07P5 | 19                | 20         | 20             | CT20          |
| 4T0011 | 28                | 35         | 30             | FE35          |
| 4T0015 | 35                | 35         | 40             | FE40          |
| 4T18P5 | 42                | 45         | 50             | FE45          |
| 4T0022 | 49                | 50         | 50             | FE50          |

## Emergency equipment

Overall design and installation of equipment must include emergency equipment and other necessary safety equipments. Controlling motor with buttons on inverter operating device or command from I/O terminal can not ensure:

- ◎ Emergency motor stop

Separate inverter from dangerous voltage.

### 4.1.20.2 Input power cable/connection

Input cable can be connected in any one of following ways:

- ◎ 4-core cable (3-phase and grounding protection line)
- ◎ 4-core insulated conductor is installed in conduit.

Select proper power cable according to local safety regulation, input voltage grade and loading current of inverter. Conductor must be smaller than the maximum limit defined for terminal dimension (see Chapter 4 “4.5.4 Specification of main loop connecting conductor”). Table below lists cable type of copper-core cable under different loading current. Types recommended are only suitable for situations listed in upper part. It is not recommended to use aluminum-core cable.

| IEC                                                                                                                                                                                                                                                  | NEC                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Based on:<br>◎ EN 60204-1 and IEC 60364-5-2/2001 standard<br>◎ PVC insulation<br>◎ Ambient temp. 30 °C<br>◎ Surface temp. 70 °C<br>◎ Symmetrical cable shielded with copper net<br>◎ Cable aligned in the same cable tray should not exceed 9 pieces | Based on:<br>◎ For copper-core cable, see NEC table 310-16<br>◎ Cable insulation 90 °C<br>◎ Ambient temp. 40 °C<br>◎ Carrying lines in the same cable groove, cable pitch or of buried cable should not exceed 3 pieces<br>◎ Copper-core cable shielded with copper net |

| Max. loading current (A) | Copper-core cable (mm <sup>2</sup> ) | Max. loading current (A) | Type of copper-core cable (AWG/kcmil) |
|--------------------------|--------------------------------------|--------------------------|---------------------------------------|
| 14                       | 3x1.5                                | 22.8                     | 14                                    |
| 20                       | 3x2.5                                | 27.3                     | 12                                    |
| 27                       | 3x4                                  | 36.4                     | 10                                    |
| 34                       | 3x6                                  | 50.1                     | 8                                     |
| 47                       | 3x10                                 | 68.3                     | 6                                     |
| 62                       | 3x16                                 | 86.5                     | 4                                     |
| 79                       | 3x25                                 | 100                      | 3                                     |
| 98                       | 3x35                                 | 118                      | 2                                     |
| 119                      | 3x50                                 | 137                      | 1                                     |
| 153                      | 3x70                                 | 155                      | 1/0                                   |
| 186                      | 3x95                                 | 178                      | 2/0                                   |

#### 4.1.2.3 Grounding connection of input power cable

To ensure human safety, correct operation and reduction of electromagnetic radiation, inverter and motor must be grounded at mounting location.

- ◎ Conductor diameter must meet requirements in safety regulation.
- ◎ Shield of power cable must be connected to PE terminal of inverter to meet safety rules.
- ◎ Only when specification of power cable shield meets requirements in safety regulation, this shield can be used as grounding line of the equipment.

When installing more than one inverter, do not connect inverter terminals in series.

#### 4.1.20.4 Output power cable/connection

##### Motor connection



**Hazar**

Never connect incoming power supply to inverter output end: U, V and W.  
Connecting incoming power supply to output end will lead to permanent damage to inverter unit.



**Notice**

Do not connect motor with rated voltage less than half of rated input voltage of inverter to the inverter.



**Notice**

Before performing dielectric strength test or insulation resistance test on motor or motor cable, it is a must to disconnect inverter from motor cable. Do not perform those tests mentioned above on inverter.

Technical requirements of motor connection

| Technical requirements of output power (motor) |                                              |
|------------------------------------------------|----------------------------------------------|
| Output voltage                                 | 0 ~ input voltage, symmetric 3-phase voltage |

|                                                             |                                                                          |
|-------------------------------------------------------------|--------------------------------------------------------------------------|
| Current                                                     | See Chapter 2 “2.2 Technical index and specification of inverter”        |
| Switch frequency                                            | Can be defined: 2 ~ 11 kHz                                               |
| Rated cable temp.                                           | Allow working at 90℃ for long term                                       |
| Relation between length of motor cable and switch frequency | See Chapter 4 “4.4.4 Relation between line length and carrier frequency” |

## Grounding and wiring

### Motor cable shielding

Motor cable is required to be shielded with wire conduit, armored cable or shielded cable.

#### 1) Wire conduit

- ① Each end of wire conduit is required to install a bridge connection with grounded conductor.
- ② Wire conduit is fixed on housing.
- ③ Lay motor cable with a single wire conduit line (and input power cable and controlling cable should also be laid dividedly).
- ④ Each inverter uses a single wire conduit line.

#### 2) Armored cable

- ① Each end of wire conduit is required to install a bridge connection with grounded conductor.
- ② 6 pieces of conductors should be used (3 for power supply line and 3 for grounding line), MC continuous wave aluminous armored cable with symmetrical grounding line.
- ③ Armored motor cable and input power cable use a common cable tray, but armored motor cable can not share the same cable tray with controlling cable.

#### 3) Shielded cable

It is recommended that user should use cable with symmetrically-structured PE conductor meeting CE or C-Tick standard.

## Grounding

See **Grounding connection of input power cable** above.

#### 4.1.2.5 AC reactor at input side

AC reactor can be equipped at input side to improve power factor of input-side power supply and reduce high-order harmonic current.

#### 4.1.2.6 Interference filter at input side

Interference filter can be equipped at input side to suppress high-frequency noise interference of power line of inverter to power supply.

#### 4.1.2.7 Contactor at input side

To protect power supply and prevent trouble from extending, control power supply to inverter by means of opening and closing the contactor at input side.

Please do not use this contactor to control motor start and stop.

#### 4.1.2.8 Contactor at output side

To meet the requirement that current should not pass through motor at motor stop defined in GB7588-2003

“National Elevator Safety Standard”, a contactor should be installed at output side.

#### 4.1.2.9 Interference filter at output side

Special output-side interference filter can be equipped to suppress interference noise and drain current of conductor produced at output side of inverter.

#### 4.1.2.10 AC reactor at output side

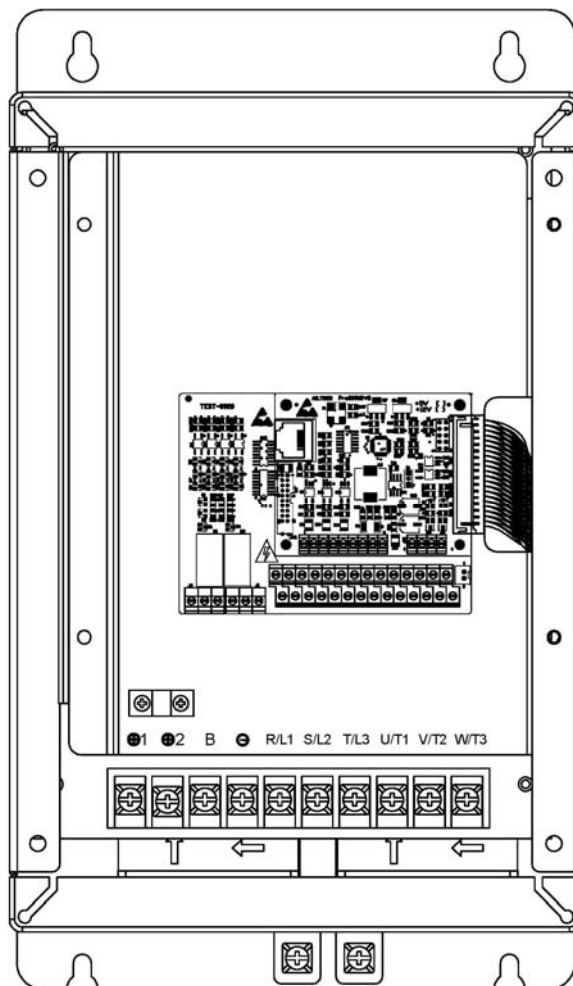
Output-side AC reactor can be equipped to suppress radio interference from inverter.

If the connecting line is too long between inverter and motor (>20m), output-side AC reactor can prevent inverter overcurrent due to distributed capacitance of conductor.

#### 4.1.2.11 DC reactor

DC reactor can be equipped to improve power factor.

## 4.2 Wiring of inverter terminal



For internal view of inverter, see Fig. 4.2.

Fig. 4.2 Internal view of inverter

Note: Except that position and arrangement of power input/output terminals are different slightly, inverter terminals of various power levels are all the same. In the figure, 11KW is used for illustration.



### 4.2.1 Wiring diagram of inverter terminal

For wiring diagram of inverter terminal, see Fig. 4.3.

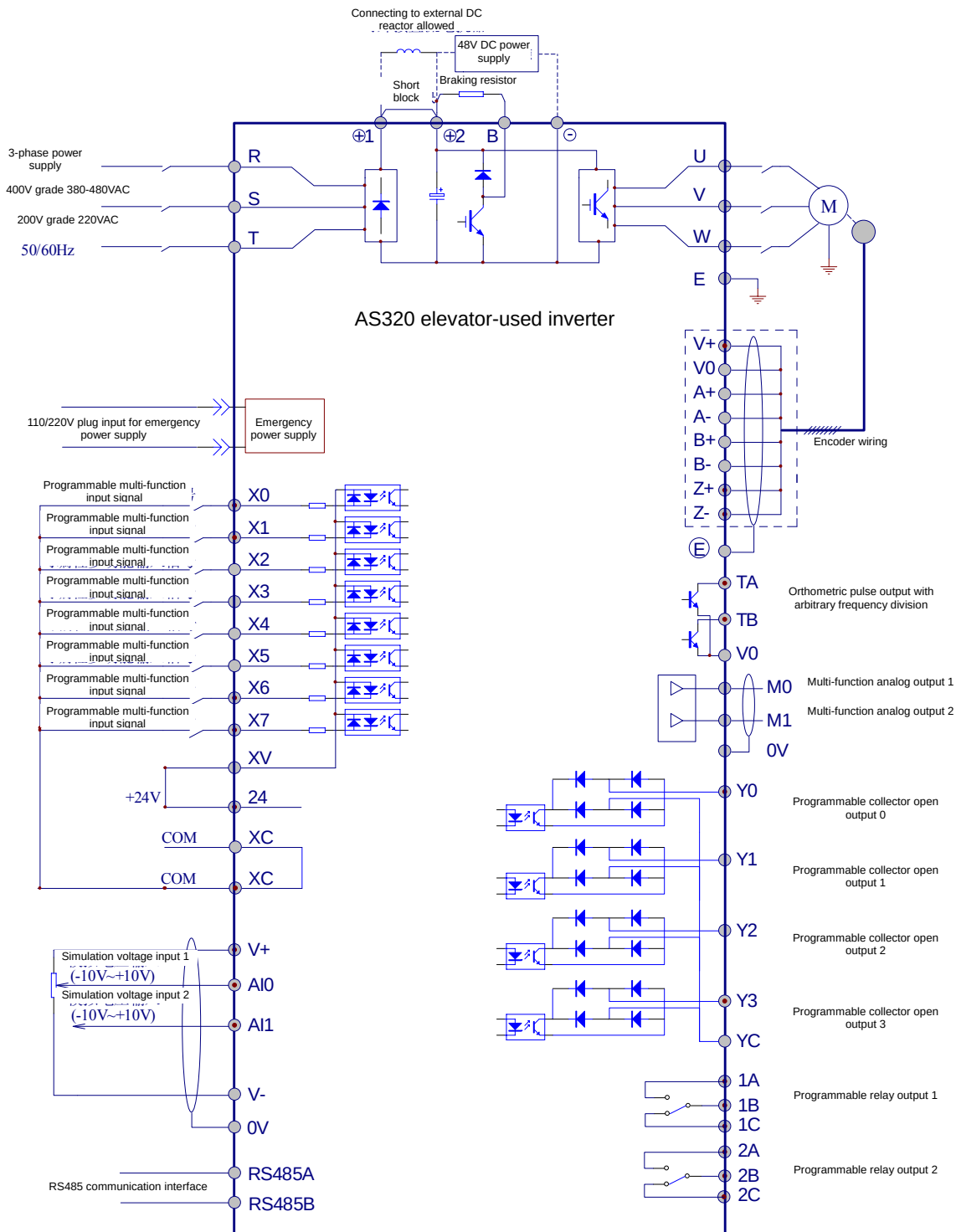


Fig. 4.3 Wiring diagram of inverter terminals

### 4.2.2 Wiring Precautions



## IMPORTANT

- a) The connection shall conform to relevant electrical engineering standards.

b) Check the wiring and its reliability after wiring. The following items shall be checked:

Is all wiring correct?

Have any wire clippings or screws been left inside the Inverter?

Is any screw loosened?

Does any stripped bare wire at terminal end contact with other terminals?

c) Although **AS320** series elevator-used inverter is equipped with a braking unit, an external braking resistor is still necessary. The braking resistor shall be installed between Terminal B and Terminal  $\oplus 2$ , and not anywhere else, or the resistor and the Inverter may be damaged.

d) The DC reactor shall be connected between Terminals  $\oplus 1$  and  $\oplus 2$ , and the short-circuit bar between them shall be removed.

e) When bus low-voltage running is needed, an emergency power of 220 V shall be connected between Terminals RO and TO, and a DC 48V shall be put between Terminals R and S. These may be saved if no bus-voltage running is required.

f) It is recommended that the grounding wire PE of the Inverter be connected to a special grounding terminal and the grounding resistor shall have its impedance below 10  $\Omega$ .

g) The grounding cable shall be as short as possible.

h) When there is need for wiring changes after powering on, the power shall be cut off first. Since it takes some time for the main circuit charge capacitor to discharge, subsequent procedures may be taken only after the charging indicator extinguishes and the DC voltage across the capacitor is measured through a DC voltmeter to be below 24 VDC safety level.

i) “ $\bigcirc$ ” in the figure stands for terminals of the main circuit, and “ $\odot$ ” for terminals of the control circuit.

## 4.3 Wiring Main Circuit Terminals

### 4.3.1 Alignment of main circuit terminals



|            |            |   |           |      |      |      |      |      |      |
|------------|------------|---|-----------|------|------|------|------|------|------|
| $\oplus 1$ | $\oplus 2$ | B | $\ominus$ | R/L1 | S/L2 | T/L3 | U/T1 | V/T2 | W/T3 |
|------------|------------|---|-----------|------|------|------|------|------|------|

### 4.3.2 Symbols and Functions of Main Circuit Terminals

The functions of main circuit terminals are listed in Table 4.1.

Table 4.1 Functions of main circuit terminals

| Terminal symbol | Function                                     |
|-----------------|----------------------------------------------|
| $\oplus 1$      | To connect DC reactor, shorting for ex works |

|           |                                                            |
|-----------|------------------------------------------------------------|
| $\phi 2$  |                                                            |
| $\phi 2$  | External braking resistor connection                       |
| B         |                                                            |
| $\ominus$ | Negative output of DC bus                                  |
| R/L1      | AC power for the main circuit, to 3-phase input            |
| S/L2      |                                                            |
| T/L3      |                                                            |
| U/T1      | Inverter output, to 3-phase synchronous/asynchronous motor |
| V/T2      |                                                            |
| W/T3      |                                                            |

### 4.3.3 Wire sizes of main circuit

600V plastic copper conductors or other insulated conductors for power supply may be used. Cable specifications and tightening torques are listed in Table 4.2.

Table 4.2 Cable specifications and tightening torques

| Model: <b>AS320-</b> | Permissible cable size (mm <sup>2</sup> ) | Recommended cable size (mm <sup>2</sup> ) | Tightening torque (N.m) |
|----------------------|-------------------------------------------|-------------------------------------------|-------------------------|
| 2S01P1               | 2~6                                       | 2.5                                       | 1.5                     |
| 2S02P2               | 2~6                                       | 4                                         | 1.5                     |
| 2S03P7               | 2~6                                       | 4                                         | 1.5                     |
| 4T02P2               | 2~6                                       | 4                                         | 1.5                     |
| 4T03P7               | 2~6                                       | 4                                         | 1.5                     |
| 4T05P5               | 2~6                                       | 4                                         | 1.5                     |
| 4T07P5               | 4~8                                       | 6                                         | 2.5                     |
| 4T0011               | 4~8                                       | 6                                         | 2.5                     |
| 4T0015               | 4~8                                       | 6                                         | 2.5                     |
| 4T18P5               | 8~16                                      | 16                                        | 4.0                     |
| 4T0022               | 8~16                                      | 16                                        | 4.0                     |
| 4T0030               | 14~25                                     | 25                                        | 9                       |
| 4T0037               | 35~100                                    | 35                                        | 9                       |
| 4T0045               | 35~100                                    | 50                                        | 9.0                     |
| 4T0055               | 60~100                                    | 60                                        | 18.0                    |
| 4T0075               | 80~125                                    | 80                                        | 18.0                    |



## IMPORTANT

The wire sizes are determined at an ambient temperature of 50°C and a permissible temperature of 75°C.

The main circuit of Inverter adopts open terminal connection, for which round crimp terminals shall be used. The selection of round crimp terminals may be found in Table 4.3.

Table 4.3 Round crimp terminals

| Cross section (mm <sup>2</sup> ) | Screw | Terminal |
|----------------------------------|-------|----------|
| 0.5                              | M3.5  | 1.25/3.5 |
|                                  | M4    | 1.25/4   |
| 0.75                             | M3.5  | 1.25/3.5 |
|                                  | M4    | 1.25/4   |
| 1.25                             | M3.5  | 1.25/3.5 |
|                                  | M4    | 1.25/4   |
| 2                                | M3.5  | 2/3.5    |
|                                  | M4    | 2/4      |
|                                  | M5    | 2/5      |
|                                  | M6    | 2/6      |
|                                  | M8    | 2/8      |
| 3.5/5.5                          | M4    | 5.5/4    |
|                                  | M5    | 5.5/5    |
|                                  | M6    | 5.5/6    |
|                                  | M8    | 5.5/8    |
| 8                                | M5    | 8/5      |
|                                  | M6    | 8/6      |
|                                  | M8    | 8/8      |
| 14                               | M6    | 14/6     |
|                                  | M8    | 14/8     |
| 22                               | M6    | 22/6     |
|                                  | M8    | 22/8     |
| 30/38                            | M8    | 38/8     |
| 50/60                            | M8    | 60/8     |
|                                  | M10   | 60/10    |
| 80                               | M10   | 80/10    |
| 100                              |       | 100/10   |



## IMPORTANT

Sufficient attention shall be paid to the voltage drop along the line to determine cable cross section.

Typically, the voltage shall be maintained below 2% of the rated value. If the drop is too heavy, a larger cross section shall be used. The voltage drop may be calculated as follows:

$$\text{Line-to-line voltage drop (V)} = \sqrt{3} * \text{line resistance } (\Omega) * \text{current (A)}$$

### 4.3.4 Main Circuit Configurations

The main circuit configurations are shown in Figure 4.4.

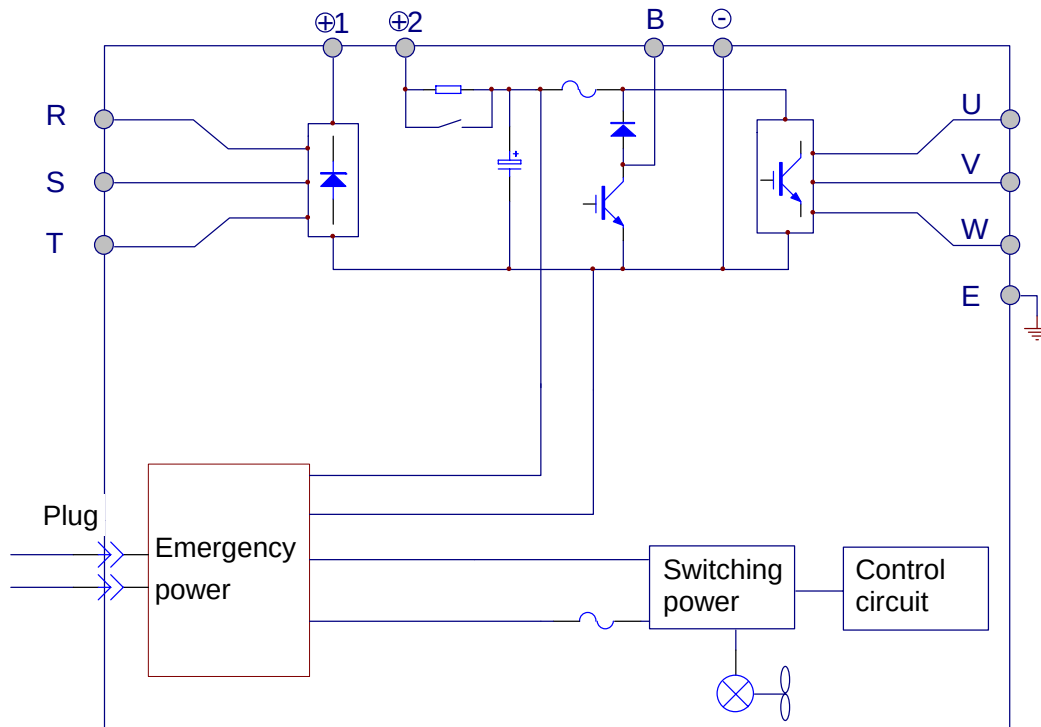


Figure 4.4 Main circuit configurations

### 4.3.5 Illustration of Main Circuit Wiring

#### 4.3.5.1 Grounding Terminal (E)/(PE)

- It is recommended to connect the grounding terminal to a specialized grounding electrode. Reliable connection shall be ensured. The grounding resistance shall be lower than 10  $\Omega$ .
- The grounding conductor may not be shared with welding machines or other power devices.
- Always use a grounding conductor that complies with the technical standards on the electrical equipment and minimize the length of the wire. Long distance between the grounding conductor and the grounding electrode may lead to leakage current of the Inverter which causes instability in grounding terminal potential.
- Multi-strand copper lines over 3.5 mm<sup>2</sup> shall be used for the grounding wire. It is recommended to use specific green-yellow grounding wires.
- It is recommended not to loop the grounding wire when more than one Inverter is to be grounded in order to avoid grounding loop.

The method to ground more than one Inverter is shown in Figure 4.5.

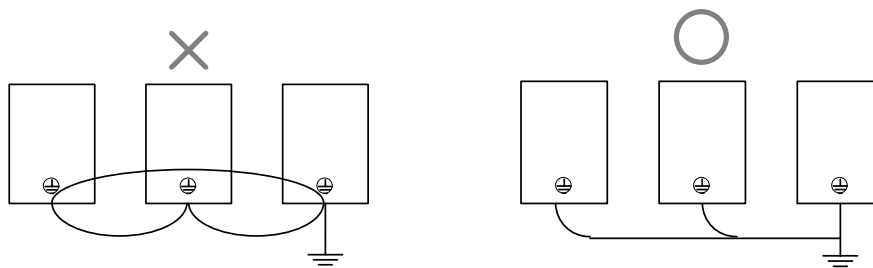


Figure 4.5 Grounding method of more than one Inverter

### 4.3.5.2 +48V DC Bus Terminal Block

- At power grid failure, storage batteries connected to Terminals R and S may be used to supply a direct low-voltage power to the Inverter to enable the elevator to run at a low speed leveling at the nearest floor.
- The connection of UPS and storage battery is shown in Figure 4.6.

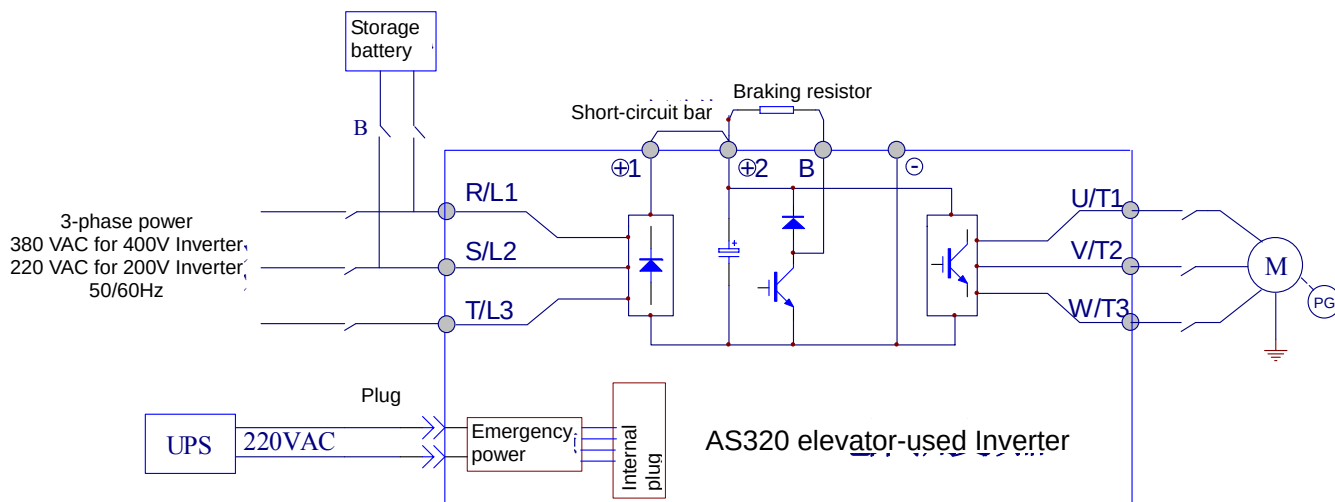


Figure 4.6 Emergency power and storage battery connection

### 4.3.5.3 Power Supply Input Terminals for the Main Circuit (R/L1, S/L2, T/L3)

- A 3-phase AC power supply may be connected through a breaker to any one of Terminals R/L1, S/L2, and T/L3. The phase sequence of the input power supply is irrelevant to the sequence of R/L1, S/L2, and T/L3.
- A noise filter may be installed on the power supply side in order to reduce transmission and radiation interferences of the Inverter caused to the input power supply. The noise filter may reduce the electromagnetic interference both from the power line to the Inverter and vice versa.



**Special caution: please use only noise filters specifically for inverters.**

Figure 4.7 shows the correct setting of a noise filter on the power supply side.

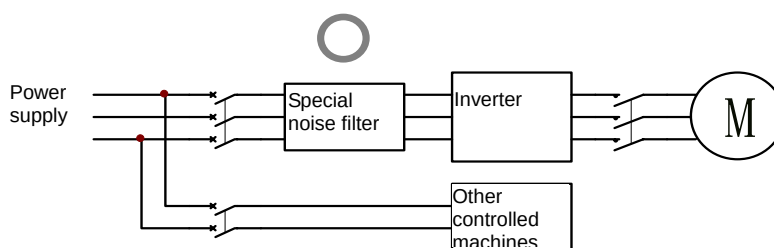


Figure 4.7 Noise filter on the power supply side

Examples of incorrect settings of noise filter on the power supply side are given in Figure 4.8 and Figure 4.9.

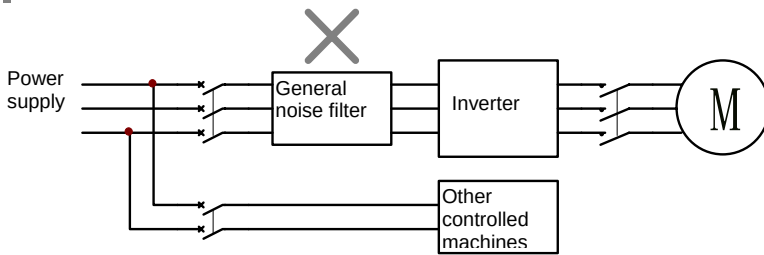


Figure 4.8 Example 1 of incorrect noise filter setting on the power supply side

In Figure 4.8, the general noise filter on the power supply side may not satisfy expected requirements and thus shall be avoided.

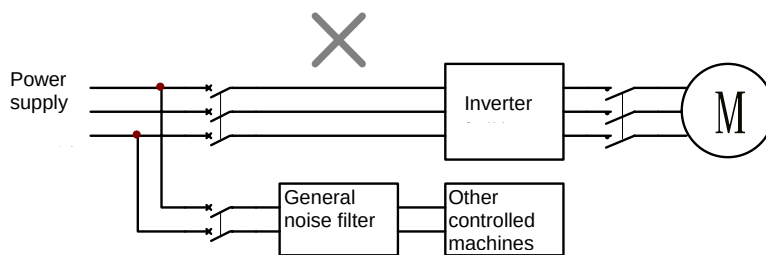


Figure 4.9 Example 2 of incorrect noise filter setting on the power supply side

In Figure 4.9, the general noise filter on the power supply side may not satisfy expected requirements and thus shall be avoided.

#### 4.3.5.5 External DC Reactor Terminals (⊕1, ⊕2)

- An external DC reactor may be added to improve the power factor. Remove the short-circuit bar between Terminals ⊕1 and ⊕2 pre-wired at the factory when connecting a DC reactor to the Inverter.
- If no DC reactor is used, please do not remove the short-circuit bar, or the Inverter will not work normally.

The wiring of the short-circuit bar is shown in Figure 4.10.

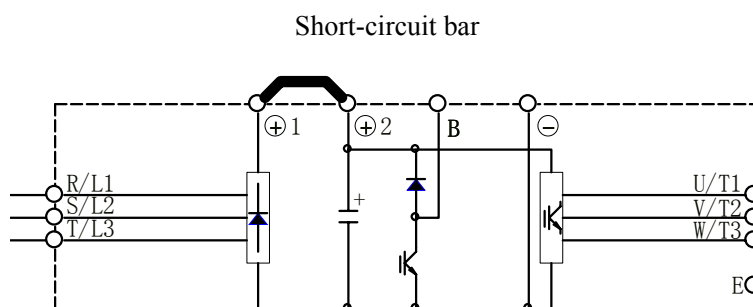


Figure 4.10 Wiring diagram of short-circuit bar

The wiring of the DC reactor is shown in Figure 4.11.

External DC reactor

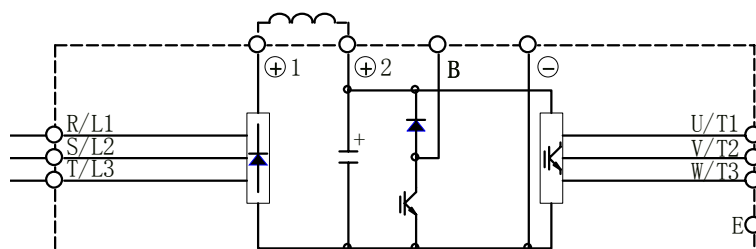


Figure 4.11 Wiring of the external DC reactor

#### 4.3.5.6 Connecting the External Braking Resistor Terminals (+2, B)

- Since each AS320 Inverter is equipped with a built-in braking unit, an external braking resistor is required to absorb the energy released during braking. The types of braking resistors are listed in Table 1.6.1 Braking Resistors Configuration Table for 400V Inverters in Chapter I.
- The braking resistor is put between Terminals +2 and B.
- Sufficient attention shall be paid to heat dissipation and ventilation in order to maintain good performance of the braking resistor.
- The wire connecting the braking resistor may not be longer than 5 m.

The wiring of external braking resistor is shown in Figure 4.12.

External braking resistor

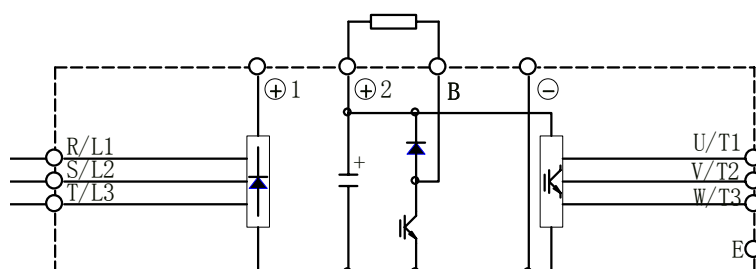


Figure 4.12 External braking resistor wiring

#### 4.3.5.7 Inverter Output Terminals (U/T1, V/T2, W/T3)

- Connect Inverter output Terminals U/T1, V/T2, W/T3 to motor lead wires U, V and W respectively. Change any two of the output terminals of the Inverter or the motor when the motor is not in the desired rotation direction.
- Never connect the power supply to the Inverter output Terminals U/T1, V/T2 and W/T3.
- The output terminals may never be grounded or shorted.
- Never connect a capacitor and/or a surge filter on the Inverter output side, since the Inverter may be thus over-heated or damaged due to its higher harmonics.

Figure 4.13 shows that capacitor shall never be connected on the output side of the Inverter.

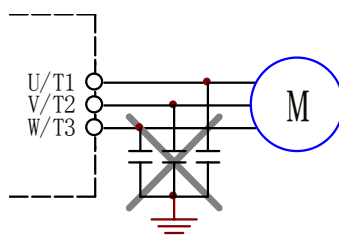




Figure 4.13 Never connect capacitor on the output side of the Inverter

## 4.4 Countermeasures against Noise

### 4.4.1 Install a Special Noise Filter on the Output Side

A special noise filter may be installed on the Inverter output side to restrain the noise from this side. The connection is shown in Figure 4.14.

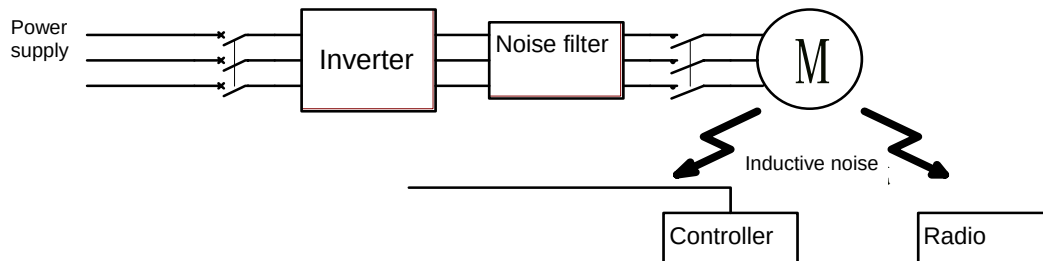


Figure 4.14 Connection of noise filter on the output side of the Inverter

### 4.4.2 Main circuit wiring

The main circuit and the control circuit shall be separately wired in order to improve the resistance to inductive noises from the output side. Cables of the main circuit may be routed through a grounded metal pipe at least 10 cm from the signal line. The wiring of the main circuit is shown in Figure 4.15.

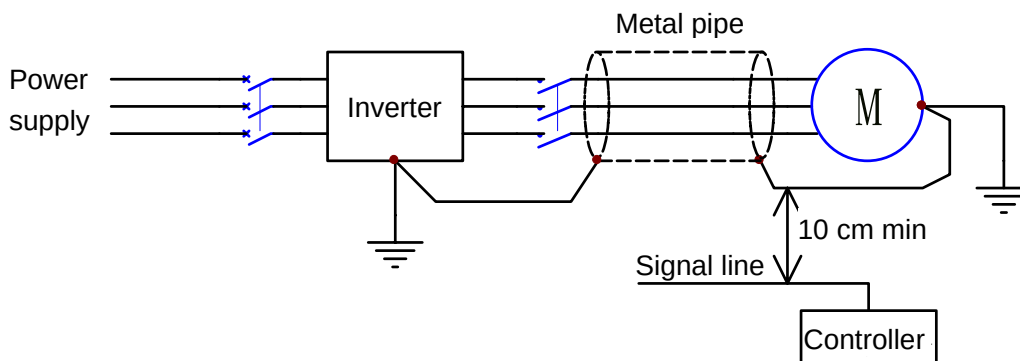


Figure 4.15 Main circuit wiring

### 4.4.3 Better Countermeasures against Noise

To reduce noises more effectively, a noise filter shall be installed on both the input and the output side of the Inverter and the Inverter shall be enclosed in a steel box, as shown in Figure 4.16.

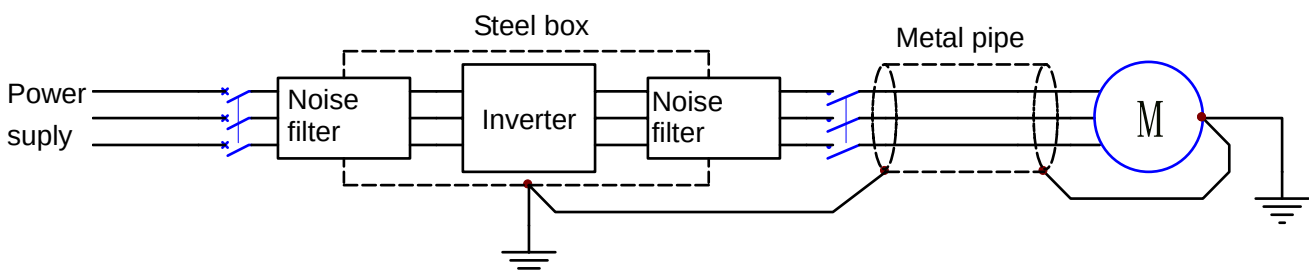


Figure 4.16 Better countermeasures against noise

#### 4.4.4 Relationship between Cable Length and Carrier Frequency

If the cable linking the Inverter and the motor is too long, the high order harmonic leakage current may increase due to distributed capacitance, which may trigger over-current protection of the Inverter output and thus causes negative impacts on surrounding equipment and motors. Therefore, the cable between the Inverter and the motor shall be not longer than 100 m. Otherwise, please adjust carrier frequency PO2.14 and select a noise filter and reactor for the output side according to the following table.

|                   |                 |                  |            |
|-------------------|-----------------|------------------|------------|
| Cable length      | 50m and shorter | 100m and shorter | Over 100m  |
| Carrier frequency | Below 11kHz     | Below 8kHz       | Below 5kHz |

### 4.5 Wiring the Control Circuit Terminals

#### 4.5.1 Control Circuit Terminals

Terminals of the control circuit are shown in Figure 4.17.

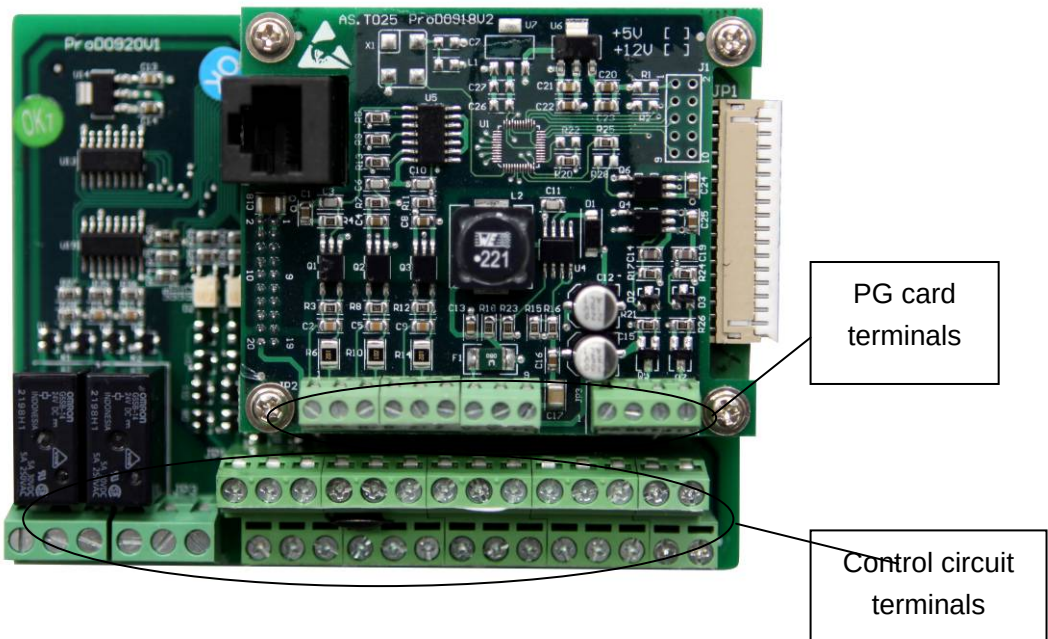


Figure 4.17 Control circuit terminals

Note: the PG card in the above figure is an ABZ incremental PG card.

#### 4.5.2 Terminal Symbols of Control Circuit

The terminal symbols of the control circuit are shown in Figure 4.18.

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|    |    |    |    |    |    | Y1 | Y3 | YC | 24 | XV | X1 | X3 | X5 | X7 | SC | 0V | 0V | A0 | A1 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 1A | 1B | 1C | 2A | 2B | 2C |    |    |    |    |    |    |    |    |    |    |    |    |    |    | Y0 | Y2 | XC | XC | X0 | X2 | X4 | X6 | A+ | B- | M0 | M1 | V+ | V- |

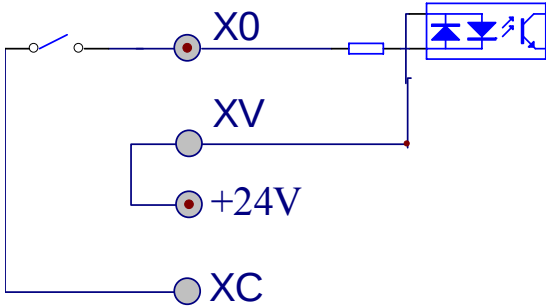
Figure 4.18 Terminal symbols of control circuit

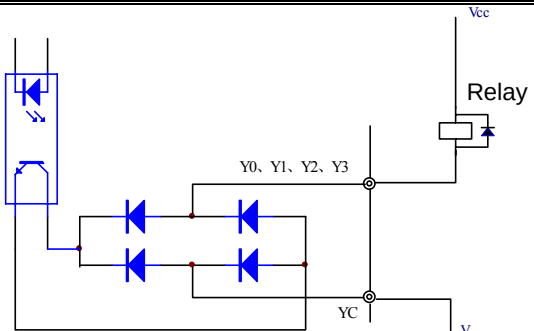
#### 4.5.3 Control Circuit Terminal Functions

The functions of the control circuit terminals are shown in Table 4.5.

Table 4.5 Control circuit terminal functions

| Name | Terminal | Signal | Remarks |
|------|----------|--------|---------|
|------|----------|--------|---------|

|                                      |                                                                                                                             |                       |                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------|-------------------|-----------------------|--------------------------------------|---------------------------------|---------------|----------------|
| Digital terminal                     | input                                                                                                                       | X0                    | Multi-function input (function code: P05.00)                                                                                | Effective when it is closed. The function of each terminal is selected by parameters P05.00~P05.07.<br>Circuit configuration of digital input is shown as follows:<br><table><tr><td>Internal power supply</td><td>+24VDC</td></tr><tr><td>Max. load current</td><td>20mA</td></tr></table>                                                                  | Internal power supply | +24VDC        | Max. load current | 20mA                  |                                      |                                 |               |                |
|                                      |                                                                                                                             | Internal power supply | +24VDC                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | Max. load current     | 20mA                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | X1                    | Multi-function input (function code: P05.01)                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | X2                    | Multi-function input (function code: P05.02)                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | X3                    | Multi-function input (function code: P05.03)                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | X4                    | Multi-function input (function code: P05.04)                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | X5                    | Multi-function input (function code: P05.05)                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | X6                    | Multi-function input (function code: P05.06)                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | X7                    | Multi-function input (function code: P05.07)                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | 24                    | Internal +24VDC power output                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | XV                    | Input common end 24V                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
| XC                                   | Input common end 0V                                                                                                         |                       |                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
| Analog terminal                      | input                                                                                                                       | A0                    | Multi-function analog input (function code: P07.01)                                                                         | External analog voltage input, ranging from -10 to +10 V, for the input of analog speed reference signal.                                                                                                                                                                                                                                                                                                                                      |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | A1                    | Multi-function analog input (function code: P07.07)                                                                         | External analog voltage input, ranging from -10 to +10 V, for the input of preload weighing signal.                                                                                                                                                                                                                                                                                                                                            |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | V+                    | +10V power output                                                                                                           | +10 VDC power output terminal for analog input, max. permissible current 50mA                                                                                                                                                                                                                                                                                                                                                                  |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | V-                    | -10V power output                                                                                                           | -10 VDC power output terminal for analog input, max. permissible current 50mA                                                                                                                                                                                                                                                                                                                                                                  |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | 0V                    | Reference ground for analog input                                                                                           | Reference ground for analog input signal                                                                                                                                                                                                                                                                                                                                                                                                       |                       |               |                   |                       |                                      |                                 |               |                |
| Relay terminal                       | output                                                                                                                      | 1A<br>1B<br>1C        | Programmable relay output: (function code: P06.00)<br>1A-1B: NO contact (make contact)<br>1B-1C: NC contact (break contact) | The output functions of the programmable relay may be selected by the function parameter of P06.<br>One pair of switching contacts are configured as follows: <table><tr><td>Item</td><td>Specification</td></tr><tr><td>Rated capacity</td><td>5A/250VAC<br/>5A/30VDC</td></tr><tr><td>Switching frequency<br/>120 times/min</td><td>Failure rate P level<br/>10mA/5V</td></tr><tr><td>Response time</td><td>Less than 10ms</td></tr></table> | Item                  | Specification | Rated capacity    | 5A/250VAC<br>5A/30VDC | Switching frequency<br>120 times/min | Failure rate P level<br>10mA/5V | Response time | Less than 10ms |
|                                      |                                                                                                                             | Item                  | Specification                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
| Rated capacity                       | 5A/250VAC<br>5A/30VDC                                                                                                       |                       |                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
| Switching frequency<br>120 times/min | Failure rate P level<br>10mA/5V                                                                                             |                       |                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
| Response time                        | Less than 10ms                                                                                                              |                       |                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
| 2A<br>2B<br>2C                       | Programmable relay output: (function code: P06.01)<br>2A-2B: NO contact (make contact)<br>2B-2C: NC contact (break contact) |                       |                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
| Transistor open collector terminal   | open output                                                                                                                 | Y0                    | Programmable open collector output 1 (function code: P06.02)                                                                | The functions of the programmable open collector outputs may be selected by the function parameter of P06.                                                                                                                                                                                                                                                                                                                                     |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | Y1                    | Programmable open collector output 2 (function code: P06.03)                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |
|                                      |                                                                                                                             | Y2                    | Programmable open collector output 3 (function code: P06.04)                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |               |                   |                       |                                      |                                 |               |                |

|                            |    |                                                              |                                                                                                                                                                                                   |
|----------------------------|----|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | Y3 | Programmable collector output 4 open (function code: P06.05) |  <p>Drive capacity: no more than DC30V, 50 mA</p>                                                               |
|                            | YC | Programmable collector output common end                     |                                                                                                                                                                                                   |
| Analog output terminal     | M0 | Programmable analog output 1 (function code: P08.00)         | <p>The functions of the programmable analog outputs may be selected by the function parameters of P08.00 and P08.01. These may be used for the inputs of output monitoring and other devices.</p> |
|                            | M1 | Programmable analog output 2 (function code: P08.01)         |                                                                                                                                                                                                   |
|                            | 0V | Reference ground for analog outputs                          | Reference ground for analog outputs                                                                                                                                                               |
| 485 communication terminal | A+ | 485 communication signal +                                   | For 485 communication signals                                                                                                                                                                     |
|                            | B- | 485 communication signal -                                   |                                                                                                                                                                                                   |
|                            | SC | Signal ground                                                | 485 communication signal ground                                                                                                                                                                   |

Note: a short circuit must be for 24V and XV

#### 4.5.4 Cable Specifications of Control Circuit Wiring

600V plastic insulated copper cable is used for the control circuit. Cable specifications and tightening torque are listed in Table 4.6.

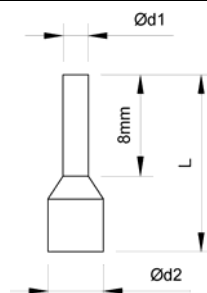
Table 4.6 Cable specifications and tightening torque

| Model | Permissible cable, mm <sup>2</sup> | Recommended cable, mm <sup>2</sup> | Tightening torque (N.m) |
|-------|------------------------------------|------------------------------------|-------------------------|
| AS320 | 0.75~1                             | 0.75                               | 1.5                     |

The size of the conductor is determined at an ambient temperature of 50 °C and a permissible temperature of 75 °C.

It is recommended that bar-like terminals be used for the control circuit. The specifications of bar-like terminals are listed in Table 4.7.

Table 4.7 Bar-like terminals

| Conductor cross section, mm <sup>2</sup> (AWG) | d1 (mm) | d2 (mm) | L (mm) | Illustration                                                                          |
|------------------------------------------------|---------|---------|--------|---------------------------------------------------------------------------------------|
| 0.25 (24)                                      | 0.8     | 2       | 12.5   |  |
| 0.5 (20)                                       | 1.1     | 2.5     | 14     |                                                                                       |
| 0.75 (18)                                      | 1.3     | 2.8     | 14     |                                                                                       |
| 1.5 (16)                                       | 1.8     | 3.4     | 14     |                                                                                       |
| 2 (14)                                         | 2.3     | 4.2     | 14     |                                                                                       |

## 4.5.5 Control Circuit Terminal Wiring

### 4.5.5.1 Analog Input Terminals

The Inverter is equipped with two analog voltage input ports. The acceptable range of analog voltage signal is  $-10\text{V} \sim +10\text{V}$ . A0 is defaulted and defined as signal input for a speed reference; A1 as signal input for starting preload. If A0 and A1 are used for the same signal type, a conflict will occur at the time of use.

The cable connecting the analog signal and the inverter shall be as short as possible (no longer than 30m), and shielded conductors shall be used. The shield shall be grounded through 0V terminal on the analog input. Figure 4.18 shows the grounding of the analog signal shielded conductor.

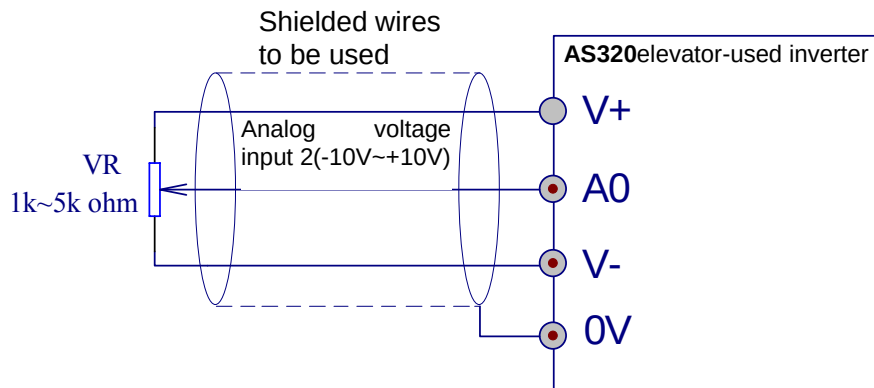


Figure 4.18 Analog input signal shielded wire wiring

In Figure 4.18, the analog voltage signal is provided by the Inverter, ranging from  $-10\text{V}$  to  $+10\text{V}$ . In most applications, the voltage signals for analog inputs are provided by a controller sending analog signals, and most of the voltage signals range from  $0\text{V}$  to  $10\text{V}$ . Figure 4.19 shows its wiring.

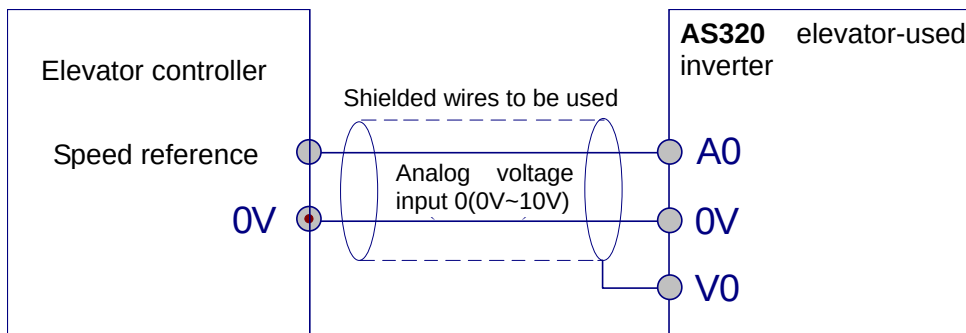


Figure 4.19 AIO wiring

When analog signal inputs are used, parameters PO7.00 to P07.11 may be used to set gain, offset, filtering time and other parameters for each input, so as to make full use of the analog ports. See 6.2.8 for more details.

### 4.5.5.2 Digital Input Terminals

The function codes P5.00 to P5.07 may be used to define the input function of each multi-function digital input terminal. The value of P5.00 to P5.07 ranges from 0 to 31, each standing for the following function:

- 00: no function (not used);
- 03: signal input of multi-speed port 0;
- 04: signal input of multi-speed port 1;
- 05: signal input of multi-speed port 2;

- 06: signal input of multi-speed port 3;
- 07: up signal input;
- 08: down signal input;
- 13: fault reset signal input;
- 14: external fault signal input
- 15: external self-learning input terminal, magnetic pole tuning input;
- 16: emergency power running signal input;
- 17: weighing compensation input (special for particular user)
- 18: NO input of base block signal;
- 19: light load switching signal input;
- 20: heavy load switching signal input;

Note: if 1 is added before the function definition, the input signal is NC input. For example, the function definition is set as 107, which indicates if the input signal disconnects, there is up signal input; if the input signal switches on, there is no up signal input.

#### 4.5.5.3 Digital Output Terminals

Digital output terminals include relay contact output terminals and open collector output terminals. The parameters of function code P06 may be used to set the functions of each digital output terminal. The setting data range is 0 to 31, each standing for a special output function:

- 0: no function (not used);
- 01: all normal signal after power on
- 02: fault signal
- 03: run signal
- 04: frequency/speed arriving signal
- 05: frequency/speed correspondency signal
- 06: null speed signal
- 07: DC bus voltage not less than 85% of rated voltage
- 08: over 5% of rated current during running, 10% of rated current when stopping
- 09: in self-tuning of encoder position
- 10: speed detection 1
- 11: speed detection 2
- 12: fault forecasting signal
- 13: self-tuning request of synchronous motor
- 14: zero-servo torque directin output

Note: if 1 is added before the function definition, the output signal is NC output. For example, the function

definition is set as 103, which indicates if the Inverter operates, output is disconnected; if the Inverter stops, the output signal is connected.

Note: “connected” here means pick-up of NO contact and release of NC contact of relays, and low level of outputs of open collectors. On the same basis, “disconnected” means release of NO contact and pick-up of NC contact of relays, and high resistance of open collectors.

Digital outputs consist of relay contact outputs and open collector outputs. The former is realized through idle contacts, including two pairs of switching contacts.

There are four channels for open collector outputs. The circuit is shown in Figure 4.20.

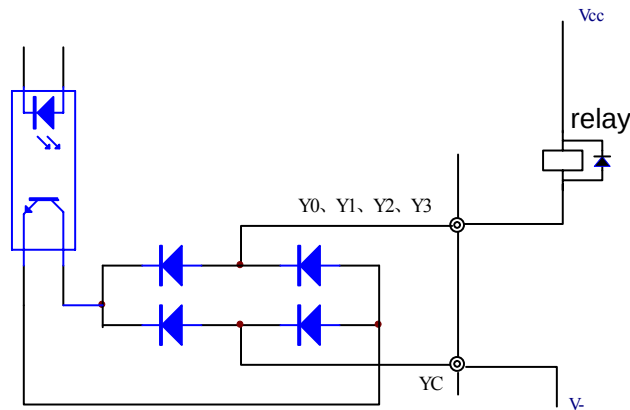


Figure 4.20 Circuit of open collector outputs

Open collector outputs adopt external power supplies. Polarization shall be noticed when the power is connected. The power supply may not exceed +30VDC, 50mA, or the output circuit may be damaged.

#### 4.5.5.4 Multi-function Analog Output Terminals

Parameters of P08.00 and P08.01 are used to define the functions of multi-function analog output terminals. The value of P08.00 and P08.01 ranges from 0 to 31, each standing for a special output function (P08.00 parameters corresponding to M0 outputs, and P08.01 corresponding to M1 outputs):

- 0: No definition
- 1: U phase current
- 2: V phase current
- 3: W phase current
- 6: reference of speed regulator
- 7: feedback of speed regulator
- 13: output of speed regulator
- 14: reference of current regulator IQ
- 15: reference of current regulator ID
- 30: output of current regulator IQ
- 32: DC bus voltage
- 44: speed deviation

See 6.2.9 Analog Output I Parameters for more information.

#### 4.5.5.5 Other precautions for wiring

Keep the control circuit away from the main circuit power line, or interference may cause wrong actions.

### 4.6 Wiring PG Cards

There are three types of PG cards to adapt to different encoder types, as shown in the following table.

| PG card              | Applicable motor         | Model    | Input signal                                                      | Remarks             |
|----------------------|--------------------------|----------|-------------------------------------------------------------------|---------------------|
| ABZ incremental 12V  | Synchronous/asynchronous | AS.T025  | Open collector signals, Push-Pull signals                         | Encoder voltage 12V |
| SIN/COS              | Synchronous              | AS.T024  | SIN/COS differential signals                                      |                     |
| ABZ incremental 5V   | Synchronous/asynchronous | AS.T041  | Open collector signals, Push-Pull signals and differential signal | Encoder voltage 5V  |
| Endat absolute value | Synchronous/asynchronous | AS.L06/L | endat output signal                                               |                     |

#### 4.6.1 ABZ Incremental 12 V PG Card

ABZ incremental 12V PG card (AS.T025) is able to receive two types of encoder output signals, and thus may be equipped with encoders with open collector signals or Push-Pull signals.

##### 4.6.1.1 Terminal Alignment of ABZ Incremental 12V PG Card

The terminal alignment of ABZ incremental 12V PG card (Model AS.T025) is shown in Figure 4.20.

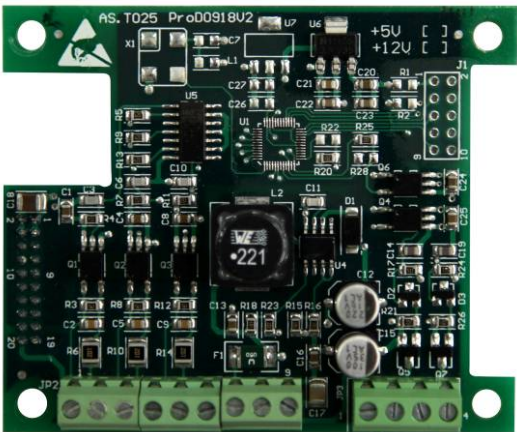


Figure 4.20 Terminal alignment of ABZ incremental 12V PG card

##### 4.6.1.2 Terminal Symbols of ABZ Incremental 12V PG Card

The terminal symbols of ABZ incremental 12V PG card are shown as follows:

JP3 divided-frequency output terminals:

|    |    |    |    |
|----|----|----|----|
| FA | V0 | FB | V0 |
|----|----|----|----|

JP2 input terminals:

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| A+ | A- | B+ | B- | Z+ | Z- | V+ | V- | PE |
|----|----|----|----|----|----|----|----|----|



### 4.6.1.3 Terminal Functions of ABZ Incremental 12V PG Card

The terminal functions of ABZ incremental 12V PG Card are listed in Table 4.8.

Table 4.8 Terminal functions of ABZ incremental 12V PG card

| Name                     | Pin number | Symbol | Function                         | Specification                                                      |
|--------------------------|------------|--------|----------------------------------|--------------------------------------------------------------------|
| Divided-frequency output | JP3.1      | FA     | Divided-frequency output phase-A | Open collector outputs of triodes (max. output frequency: 100 kHz) |
|                          | JP3.2      | 0V     | 24V GND                          |                                                                    |
|                          | JP3.3      | FB     | Divided-frequency output phase-B |                                                                    |
|                          | JP3.4      | 0V     | 24V GND                          |                                                                    |
| Encoder input            | JP2.1      | A+     | Phase-A signal + of encoder      | Open collector/Push-Pull, max. input frequency 100 kHz             |
|                          | JP2.2      | A-     | Phase-A signal - of encoder      |                                                                    |
|                          | JP2.3      | B+     | Phase-B signal + of encoder      |                                                                    |
|                          | JP2.4      | B-     | Phase-B signal - of encoder      |                                                                    |
|                          | JP2.5      | Z+     | Phase-Z signal + of encoder      |                                                                    |
|                          | JP2.6      | Z-     | Phase-Z signal - of encoder      |                                                                    |
|                          | JP2.7      | V+     | Anode of encoder power           | Voltage 12 VDC, max. output current 500 mA                         |
|                          | JP2.8      | V-     | Cathode of encoder power         |                                                                    |
|                          | JP2.9      | PE     | Shielded ground                  | Grounding through shielded wires                                   |

### 4.6.1.4 Wiring between Input Terminals of ABZ Incremental 12V PG card and Encoder Output Signals

The ABZ incremental 12V PG card may receive two types of encoder output signals, namely, open collector signals and Push-Pull signals.

Wiring with the open collector signals of the encoder is shown in Figure 4.22.

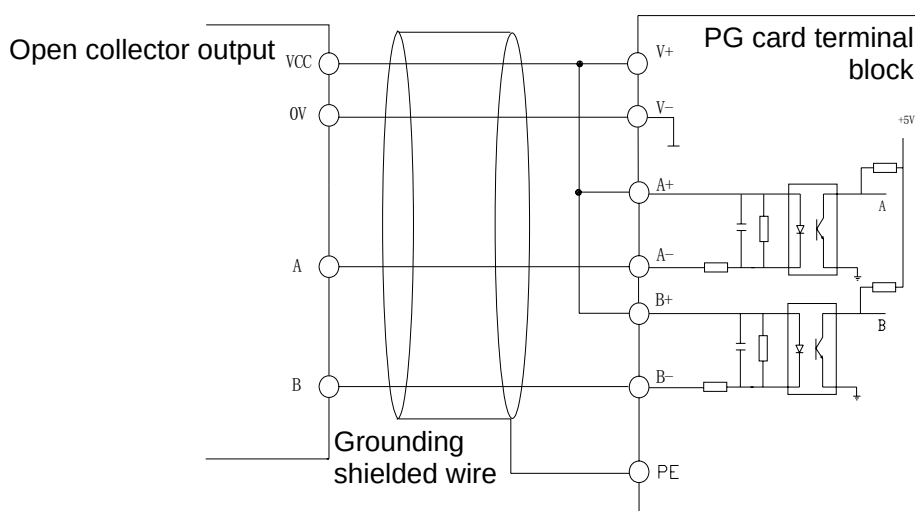


Figure 4.22 Wiring with the open collector signals of the encoder

Wiring with the Push-Pull signals of the encoder is shown in Figure 4.23.

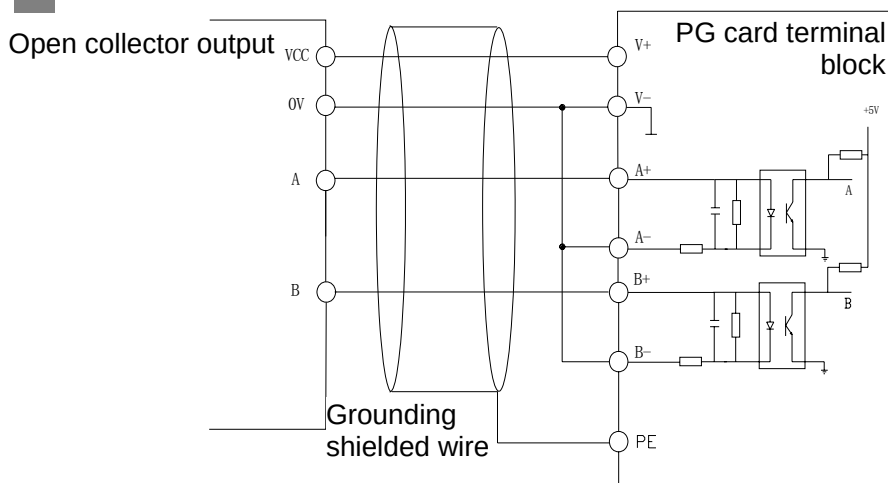


Figure 4.23 Wiring with the Push-Pull signals of the encoder

### 4.6.2 SIN/COS PG Card

The SIN/COS PG card (Model AS.T024) may receive SIN/COS differential signals from the encoder, and thus may be equipped with encoders with SIN/COS differential signals.

#### 4.6.2.1 Terminal Alignment of SIN/COS PG Card

The terminal alignment of the SIN/COS PG card is shown in Figure 4.24.

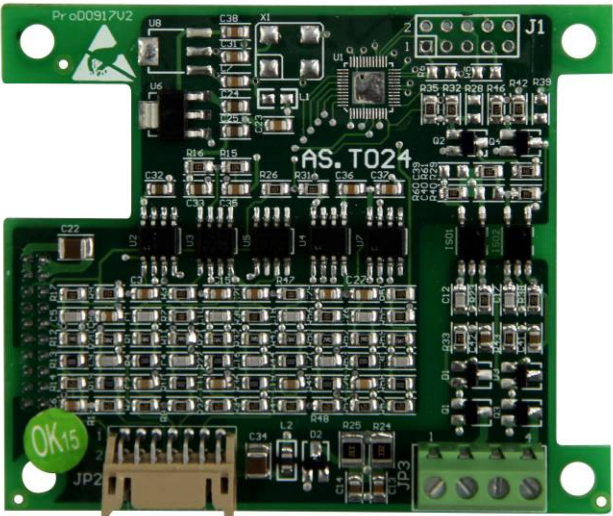


Figure 4.24 Terminal alignment of the SIN/COS PG card (AS.T024)

#### 4.6.2.2 Terminal Symbols of SIN/COS PG Card

The terminal symbols of the SIN/COS PG card (AS.T024) are shown as follows:

Symbols of JP3 terminal

|    |    |    |    |
|----|----|----|----|
| FA | V0 | FB | V0 |
|----|----|----|----|

Symbols of JP2 terminal (14-pin socket)

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| NC | NC | R- | R+ | B- | B+ | A- | A+ | D- | D+ | C- | C+ | 0V | V+ |

#### 4.6.2.3 Terminal Functions of SIN/COS PG Card

The terminal functions of the SIN/COS PG card are listed in Table 4.9.

Table 4.9 Terminal Functions of SIN/COS PG Card

| Name                  | Symbol | Function                                | Specification                                                      |
|-----------------------|--------|-----------------------------------------|--------------------------------------------------------------------|
| Open collector output | FA     | Divided-frequency signal output phase-A | Open collector outputs of triodes (max. output frequency: 100 kHz) |
|                       | 0V     | 24V GND                                 |                                                                    |
|                       | FB     | Divided-frequency signal output phase-B |                                                                    |
|                       | 0V     | 24V GND                                 |                                                                    |
| Encoder input         | A+,A-  | Phase-A of encoder                      | Differential signal, max. input frequency 100 kHz                  |
|                       | B+,B-  | Phase-B of encoder                      |                                                                    |
|                       | R+,R-  | Phase-Z of encoder                      |                                                                    |
|                       | C+,C-  | SIN signal of encoder                   |                                                                    |
|                       | D+,D-  | COS signal of encoder                   |                                                                    |
|                       | V+     | +5V                                     |                                                                    |
|                       | 0V     | +5V GND                                 |                                                                    |

#### 4.6.2.4 Wiring between Input Terminals of SIN/COS PG Card and Encoder Outputs

SIN/COS PG card may receive SIN/COS differential output signals from encoders.

The wiring with the encoder is shown in Figure 4.25.

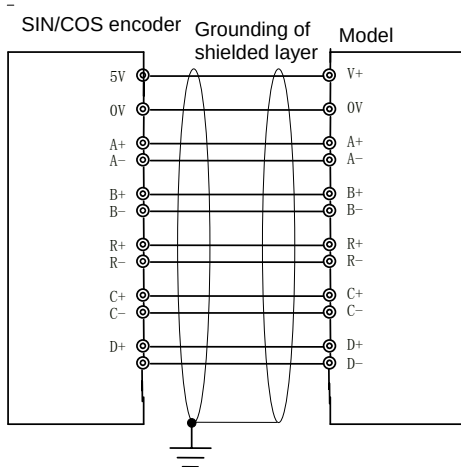


Figure 4.25 Wiring with SIN/COS differential signals of the encoder

#### 4.6.3 ABZ incremental 5V PG card

The ABZ incremental 5V PG card (AS.T041) could receive three types of encoder output signals, and thus may be equipped with the encoders with open collector signals or Push-Pull signals or differential signals.

##### 4.6.3.1 Terminals alignment of ABZ incremental 5V PG card

Refer to Figure 4.26 for terminals alignment of ABZ incremental 5V PG card (AS.T041).

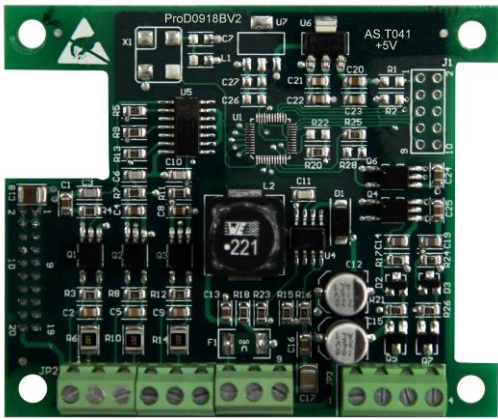


Figure 4.26 Terminals alignment of ABZ incremental 5V PG card

4.6.3.2 Terminal symbol of ABZ incremental 5V PG card

Terminal symbol of ABZ incremental 5V PG card is shown as the following:

|    |    |    |    |
|----|----|----|----|
| FA | V0 | FB | V0 |
|----|----|----|----|

JP2 input terminal:

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| A+ | A- | B+ | B- | Z+ | Z- | V+ | V- | PE |
|----|----|----|----|----|----|----|----|----|

4.6.3.3 Terminal functions of ABZ incremental 5V PG card

Terminal functions of ABZ incremental 5V PG card refer to Table 4.10.

Table 4.10 Terminal functions of ABZ incremental 5V PG card

| Name                     | Pin   | Symbol | Function                         | Specification                                                       |
|--------------------------|-------|--------|----------------------------------|---------------------------------------------------------------------|
| Divided-frequency output | JP3.1 | FA     | Divided-frequency output phase-A | Open collector outputs of triodes (max. output frequency: 100 kHz)  |
|                          | JP3.2 | 0V     | 24V GND                          |                                                                     |
|                          | JP3.3 | FB     | Divided-frequency output phase-B |                                                                     |
|                          | JP3.4 | 0V     | 24V GND                          |                                                                     |
| Encoder input            | JP2.1 | A+     | Phase-A signal + of encoder      | Open collector/Push-Pull/differential, max. input frequency 100 kHz |
|                          | JP2.2 | A-     | Phase-A signal - of encoder      |                                                                     |
|                          | JP2.3 | B+     | Phase-B signal + of encoder      |                                                                     |
|                          | JP2.4 | B-     | Phase-B signal - of encoder      |                                                                     |
|                          | JP2.5 | Z+     | Phase-Z signal + of encoder      |                                                                     |

|       |    |                             |                                           |
|-------|----|-----------------------------|-------------------------------------------|
| JP2.6 | Z- | Phase-Z signal - of encoder | Voltage 5 VDC, max. output current 500 mA |
| JP2.7 | V+ | Anode of encoder            |                                           |
| JP2.8 | V- | Cathode of encoder          |                                           |
| JP2.9 | PE | Shielded ground             | Grounding terminal through shielded wires |

#### 4.6.4 Endat absolute value PG card

Endat absolute value PG card (model AS.L06/L) can receive endat output signal of encoder, which mean it can install the encoder with endat output signal. For example the HEADS model: 1313 or 413 encoder

##### 4.6.4.1 Endat absolute value PG card terminal layout

Endat absolute value PG card ( model AS.L06/L ) terminal layout see fig 4.28



Fig .4.28 Endat absolute value PG card terminal layout

##### 4.6.4.2 Endat absolute value PG card terminal label

Endat absolute value PG card terminal label as follow:

JP3 terminal label

|    |    |    |    |     |
|----|----|----|----|-----|
| FA | V0 | FB | V0 | 12V |
|----|----|----|----|-----|

JP2 terminal label (14 pin plug)

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| NC | NC | NC | NC | B- | B+ | A- | A+ | D- | D+ | C- | C+ | 0V | V+ |

##### 4.6.4.3 Endat absolute value PG card terminal function description

Endat absolute value PG card terminal function description see table 4.11

Table 4.11 Endat absolute value PG card terminal function description

| name                         | Terminal label | Terminal function description            | specification                                                                        |
|------------------------------|----------------|------------------------------------------|--------------------------------------------------------------------------------------|
| Collector open signal output | FA             | Frequency-dividing signal output A phase | triode collective open output (max output frequency 100khz) max output current 50mA, |
|                              | 0V             | GND                                      |                                                                                      |
|                              | FB             | Frequency-dividing signal output B phase |                                                                                      |
|                              | 0V             | GND                                      |                                                                                      |
|                              | +12V           | 12V power output                         |                                                                                      |
| Encoder input                | A+,A-          | Encoder A phase signal                   | differential signal, max input frequency 100kHz                                      |
|                              | B+,B-          | Encoder B phase signal                   |                                                                                      |
|                              | C+,C-          | Encoder clock signal                     |                                                                                      |
|                              | D+,D-          | Encoder data signal                      |                                                                                      |
|                              | V+             | +5V                                      |                                                                                      |
|                              | 0V             | +5V 的 GND                                |                                                                                      |

#### 4.6.5 PG Card Terminal Wiring Precautions



### IMPORTANT

Keep encoder signal wiring away from the main circuit and other power lines. Never lay wires closely in parallel. Shielded wires shall be used for encoder wiring, with the shielded layer clipping with the earthing casing PE.

## Chapter 5 Digital Operator

The digital operator is the basic tool of Inverter operation for observing the status and fault codes of the Inverter and setting and viewing the parameters. This chapter describes basic operations of the operator in detail.

### 5.1 Function of Digital Operator Components

The components of the digital operator and their functions are shown in Figure 5.1.

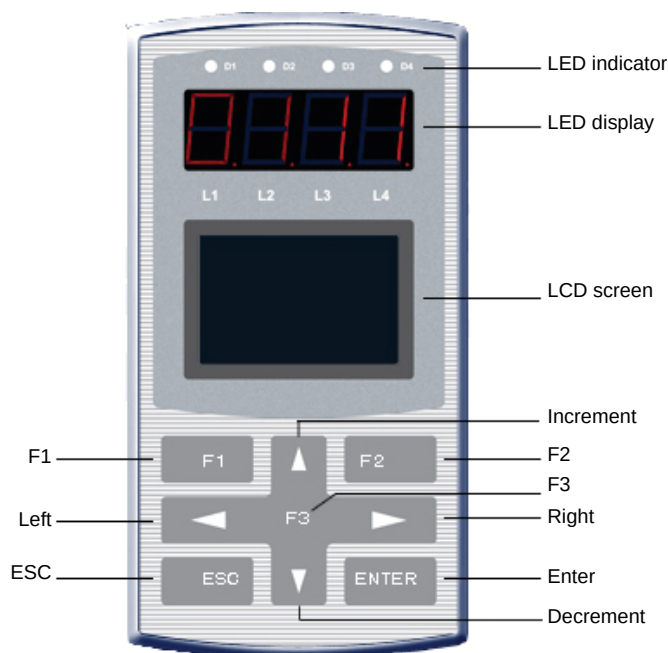


Figure 5.1 Components and their functions of the digital operator

#### 5.1.1 LED Indicator

At the top of the front cover there are four LED indicators showing the four status of the elevator, namely D1 (RUN), D2 (UP/DOWN), D3 (LOC/REMOTE) and D4 (FAULT). The functions of these indicators are shown in Table 5.1.

Table 5.1 Elevator status indicated by the four indicators

| Elevator status | D1 (RUN) | D2 (UP/DOWN) | D3 (LOC/REMOTE) | D4 (FAULT) |
|-----------------|----------|--------------|-----------------|------------|
| UP              | ON       | ON           | OFF             | OFF        |
| DOWN            | ON       | OFF          | OFF             | OFF        |
| FAULT/WARNING   | OFF      | Not related  | Not related     | Flashing   |
| Panel operation | ON       | ON/OFF       | ON              | OFF        |

#### 5.1.2 LED Digital Tube

Below the LED indicators there are 4 LED digital tubes showing real-time running speed of the elevator. The displayed contents may be selected by parameters.

#### 5.1.3 LCD Display

At the middle of the operator there is an LCD display for setting Inverter parameters, showing elevator running parameters and viewing Inverter codes.

### 5.1.4 Keyboard

The functions of the nine keys at the bottom of the operator are shown in Table 5.2.

Table 5.2 Key functions

| Key                                                                                 | Name      | Function                                                                                                                                                            |
|-------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Right     | To select the next function group under <b>【Function Select】</b> mode;<br>To move the cursor to the right bit under <b>【Parameter setting】</b> mode.                |
|    | Left      | To select the previous function group under <b>【Function Select】</b> mode;<br>To move the cursor to the left bit under <b>【Parameter setting】</b> mode.             |
|    | Increment | To select the previous function code under <b>【Function Select】</b> mode;<br>To increase the value of the selected parameter under <b>【Parameter setting】</b> mode. |
|    | Decrement | To select the next function code under <b>【Function Select】</b> mode;<br>To decrease the value of the selected parameter under <b>【Parameter setting】</b> mode.     |
|    | Enter     | Enter the Function Select interface under <b>【Monitoring State】</b> ;<br>Enter the selected function interface under <b>【Function Select】</b> .                     |
|    | ESC       | To go back to <b>【Monitoring State】</b> from <b>【Function Select】</b> mode;<br>To go back to <b>【Function Select】</b> from each function operation interface.       |
|    | F1        | To darken the display under <b>【Monitoring State】</b> mode.<br>To be RUN function under LOCAL state.                                                                |
|  | F2        | To brighten the display under <b>【Monitoring State】</b> mode.<br>To be STOP function under LOCAL state.                                                             |
|  | F3        | To switch between operator (LOCAL) run mode and control circuit terminal (REMOTE) run mode.                                                                         |

## 5.2 Operation

### 5.2.1 Display after Power on

“Monitoring State” is displayed 5 seconds later after power on. The speed reference (Vref), feedback speed (Vfbk) and current state (Irms) recorded currently are displayed on this interface by default.

### 5.2.2 【Monitoring State】

On “Monitoring State” interface, press  and  keys or  and  keys to switch the interfaces in monitoring state. Under “Monitoring State”, 10 real time data for elevator running are displayed by default. These data can be displayed only but not be modified.

Table 5.3 Comparison of default running state data

| Display | Designation       | Explanation                                                  | Setting Range | Unit | Factory Setting | Remarks |
|---------|-------------------|--------------------------------------------------------------|---------------|------|-----------------|---------|
| Vref    | Speed reference   | Display the speed reference instructions of the motor        | ×             | rpm  | ×               |         |
| Vfbk    | Feedback speed    | Display the feedback speed of the motor                      | ×             | rpm  | ×               |         |
| Vdev    | Speed deviation   | Display the deviation of feedback speed from speed reference | ×             | rpm  | ×               |         |
| Irms    | Output current    | Display the output current                                   | ×             | A    | ×               |         |
| Torq    | Output torque     | Display the output torque                                    | ×             | %    | ×               |         |
| Tzero   | Zero-servo torque | Display the zero-servo torque at starting                    | ×             | %    | ×               |         |



|      |                                |                                                                                                                                               |   |    |   |  |
|------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|--|
| Udc  | DC bus voltage                 | Display the DC voltage of the main circuit in the Inverter                                                                                    | × | V  | × |  |
| Uout | Output voltage                 | Display the output voltage of the Inverter                                                                                                    | × | V  | × |  |
| AI0  | A0 input voltage               | Display the Inverter analog voltage input 0 (A0)                                                                                              | × | V  | × |  |
| AI1  | A1 input voltage               | Display the Inverter analog voltage input 1 (A1)                                                                                              | × | V  | × |  |
| AI2  | A2 input current               | Display the Inverter analog current input 2 (A2)                                                                                              | × | mA | × |  |
| DI   | Input status X0-X7             | Display the input status of terminals X0-X7, in "XXXXXXXX", where "X" = 0, indicating no input, while "X" = 1, indicating input.              | × | ×  | × |  |
| DO   | Output Y0-Y3 and K1, K2 status | Display the input status of terminals Y0-Y3 and K1, K2, in "XXXXXXXX", where "X" = 0, indicating no output, while "X" = 1, indicating output. | × | ×  | × |  |

### 5.2.3 【Panel Control】

On the "Monitoring State" interface, press **F3** to switch between "Monitoring State" and "Panel control", and the LED indicator D3 on the operator becomes on under "Panel Control" mode; then, press **F1**, control the Inverter to enter RUN state, and the LED indicator D1 on the operator becomes on; press **F2**, control the Inverter to enter STOP state, and the LED indicator D1 on the operator becomes off. On the "Panel Control" interface, press  and  to switch the monitored items, and there are 2 parameters controlling running and 4 real time data displaying elevator running, of which panel operation speed Vref and elevator running direction Vdir may be modified, and other 4 data can be displayed but not be modified.

Table 5.4 Comparison of panel control data

| Display | Designation                | Explanation                                                | Setting Range | Unit | Factory Setting | Remarks |
|---------|----------------------------|------------------------------------------------------------|---------------|------|-----------------|---------|
| Vref    | Panel operation speed      | Set the speed reference of Inverter at panel operation     | 0.00~50.00    | Hz   | 5.00            |         |
| Vfbk    | Feedback speed             | Display the feedback speed of the motor                    | ×             | Hz   | ×               |         |
| Irms    | Output current             | Display the output current                                 | ×             | A    | ×               |         |
| Vdir    | Elevator running direction | Set elevator UP or DOWN                                    | 0~1           | ×    | 1               |         |
| Udc     | DC bus voltage             | Display the DC voltage of the main circuit in the Inverter | ×             | V    | ×               |         |
| Uout    | Output voltage             | Display the output voltage of the Inverter                 | ×             | V    | ×               |         |

### 5.2.4 Operation Mode

The digital operator has four operation modes, namely 【Parameter Setting】, 【Motor Tuning】, 【Fault Inspection】 and 【Parameter Processing】. In any monitoring state, press **ENTER** to enter the following "Function Select" interfaces.

- \* 1: parameter setting
- 2: motor tuning
- 3: fault detect
- 4: parameter processing

#### 5.2.4.1 【Parameter Setting】

Modify parameters under 【Parameter Setting】 mode. The setting range of parameters refers to Chapter 6.

Under 【Parameter Setting】 mode, select parameter group by pressing  or , and select parameter code of each group by pressing  or . After the parameter is selected, press , and a cursor indicating modification presents at the place of parameter to be modified. Increase or decrease the parameter value by pressing  or , and press  to confirm modification. If  is not pressed, the modification is invalid.

Press  to return to the previous menu.

#### 5.2.4.2 【Motor Tuning】

Under 【Motor Tuning】 mode, self learn the parameters of motor (asynchronous) and encoder phase angle (synchronous motor) manually, and select the corresponding self-learning mode by modifying X value in ATun = X. Press , and a cursor indicating modification presents at the place of parameter to be modified; press  or  to select self-learning item, and press  to confirm. Self-tuning selection parameters have 6 modes, defined as follows:

- 0: normal running mode
- 1: encoder static self-learning
- 2: encoder dynamic self-learning
- 3: end of encoder self-learning
- 4: motor static self-learning
- 5: motor dynamic self-learning
- 6: motor static advanced learning
- 7: encoder dynamic self-learning

Press  to return to the previous menu.

#### 5.2.4.3 【Fault Inspection】

Under 【Fault Inspection】, view the recent 8 faults and the voltage, current, speed reference, and feedback speed status recorded while the fault occurs. On main state interface, press  to shown ER0=X, press  or  to change from ER0 to ER7, of which ER0 represents the serial number of latest fault, and ER7 for the farthest one, X for fault code of current number; at the same time, the meaning of fault code will be shown below in Chinese. Under fault code display mode, press  again, the recorded DC bus voltage (Udc), output current (Irms), speed reference (Vref), and feedback speed (Vfbk) for the current fault are shown, and press  again to return to fault code display mode. Press  to return to the previous menu.

#### 5.2.4.4 【Parameter Processing】

Under 【Parameter Processing】 mode, upload, download, initialize the parameters, and eliminate all faults. Select the relevant operation mode by modifying X value in Init = X. Press , and a cursor indicating modification presents at the place of parameter to be modified (X place); press  or  to select corresponding operation mode, and press  to confirm. Parameter processing selection parameters have 4 modes, defined as follows:

- 1: parameter upload to operator
- 2: parameter download to Inverter
- 7: Reset parameter
- 8: Reset fault

Press  to return to the previous menu.

### 5.3 Fault indication

When a fault occurs to the Inverter, the fault indicator D4 flashes on the top of operator. LED digital tubes show the current fault code. Fault codes and types are listed in Table 5.5.

Table 5.5 Fault codes and names

| Fault code | Fault indication                          | Fault code | Fault indication                                 |
|------------|-------------------------------------------|------------|--------------------------------------------------|
| 1          | Module overcurrent protection             | 2          | ADC fault                                        |
| 3          | Heatsink overheat                         | 4          | Brake unit fault                                 |
| 5          | Fuse blown fault                          | 6          | Output torque overload                           |
| 7          | Speed deviation                           | 8          | DC bus over-voltage protection                   |
| 9          | DC bus undervoltage                       | 10         | Output phase loss                                |
| 11         | Motor low speed overcurrent               | 12         | Encoder fault                                    |
| 13         | Current detected during elevator stopping | 14         | Reversed speed direction detected during running |
| 15         | Speed detected during elevator stopping   | 16         | Wrong motor phase                                |
| 17         | Fwd overspeed                             | 18         | Rev overspeed                                    |
| 19         | Wrong phase sequence of UVW encoder       | 20         | Encoder communication fault                      |
| 21         | Abc overcurrent                           | 22         | Brake detection trouble                          |
| 23         | Input overvoltage                         | 24         | UVW encoder disconnected                         |
| 25         | Standby                                   | 26         | Encoder not self-learning                        |
| 27         | Output overcurrent                        | 28         | Sincos encoder fault                             |
| 29         | Input phase loss                          | 30         | Overspeed protection                             |
| 31         | Motor high speed overcurrent              | 32         | Ground protection                                |
| 33         | Aging capacitor                           | 34         | External fault                                   |
| 35         | Output unbalance                          | 36         | Wrong parameter setting                          |
| 37         | Current sensor fault                      | 38         | Braking resistor short circuit                   |
| 39         | Instantaneous current too large           |            |                                                  |



## Chapter 6 Table of Functional Parameters

This chapter describes in details all the function codes and related information of the elevator-used inverter for reference.

### 6.1 Classification of functional groups

Function codes are grouped according to their functional performance. See Table 6.1 for functional grouping.

Table 6.1 Table of Functional Grouping

| Functional group | Designation of functional group                                  |
|------------------|------------------------------------------------------------------|
| P00              | Password parameters and basic control mode                       |
| P01              | Motor and encoder parameters, self-learning command              |
| P02              | PID regulator and start & braking regulating parameters          |
| P03              | Speed reference parameter                                        |
| P04              | Torque reference and compensation parameters                     |
| P05              | Definition of digital input                                      |
| P06              | Definition of digital output                                     |
| P07              | Definition of analog input                                       |
| P08              | Definition of analog output and display selection of LCD and LED |
| P09              | Other information reference                                      |

### 6.2 List of functions and its description

#### 6.2.1 Password parameters and basic control mode

Login, setting and modification of password as well as parameter protection mode are involved in P00 parameter group. In addition, selection of the basic control mode of the inverter is included.

| Function code | Designation                | Content                                                                                                                                                                                                             | Setting range | Unit | Factory setting | Remarks |
|---------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|---------|
| P00.00        | Login password             | It is the login password of the inverter, the operator shall have the right to change the parameter only that it pass the correct password (the same password as the last one set at P00.01)                        | 0 ~ 65535     | ×    | 0               |         |
| P00.01        | Modify or set the password | Protecting password of the inverter can be set or modified through this parameter. If it is set as 0, meaning there isn't password protection. This parameter is invisible, which can't be displayed after setting. | 0 ~ 65535     | ×    | 0               |         |
| P00.02        | Basic control mode         | Setting of the basic modes for the inverter:<br>0: Voltage vector V/F control mode<br>1: Vector control without speed sensor<br>2: Torque control with speed sensor<br>3: Vector control with speed sensor          | 0 / 1 / 2 / 3 | ×    | 3               |         |
| P00.03        | Run command given way      | Setting run command given way<br>0: panel<br>1: terminal                                                                                                                                                            | 0/1           | ×    | 1               |         |
| P00.04        | Language option            | Setting language :<br>0: Chinese<br>1: English                                                                                                                                                                      | 0/1           | ×    | 0               |         |

**Note 1: password login**

1) Login the password through P00.00 parameter before each parameter setting or modification. If you have set the same data at P00.00 as the password value set at P00.01 last time, then you can successfully login, to set and modify other parameters of the inverter. Otherwise you will fail to login, and can't set or modify any parameter of the inverter.

2) P00.01 will be set as 0 when the inverter leaves the factory, so there isn't password protection for the inverter. Therefore the password isn't required when the inverter is used for the first time.

### Note 2: Modification and setting of the password

1) No password protection is available when the inverter leaves the factory due to that P00.01 was set as 0, yet password protection is required by setting a password through parameter P00.01.

2) As an invisible parameter, P00.01 can't be checked once having being set. So the password set must be remembered, otherwise you can't login during the next operation, so as not to set or modify the parameters.

3) P00.01 password parameter can be modified. It can be modified after successfully login the password.

4) If you want to cancel password protection, you can set P00.01 password parameter as 0.

### Note 3: Basic control mode

1) P00.02 is a setting parameter for basic control mode of the inverter. As the elevator-used inverter, vector control mode with speed sensor must be adopted during normal use. Then this parameter is used as default 3 during the normal operation. But during the debugging, P00.02 can be set as 0 temporarily because the encoder hasn't been installed well and the elevator overhaul must be activated slowly, and the inverter is set as voltage vector V/F control mode. It is needed to reminder that before the elevator enters high speed operation, wiring for the encoder must be completed, and P00.02 must be recovered to 3, making the inverter to be run under vector control mode with speed sensor.

## 6.2.2 Motor and encoder parameters, self-learning commands

Different parameters of the motor and encoder can be set in group P01. And self-learning commands of the motor are contained in this group.

| Function code | Designation              | Content                                                                                                                   | Setting range | Unit | Factory setting                 | Remarks                    |
|---------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------|------|---------------------------------|----------------------------|
| P01.00        | Type of motor            | Set the traction motor to be synchronous or asynchronous; asynchronous; synchronous<br>0: asynchronous;<br>1: synchronous | 0 / 1         | ×    | 0                               |                            |
| P01.01        | Rated power of motor     | Set the rated power of traction motor                                                                                     | 0.40 ~ 160.00 | KW   | As per the inverter parameter s | As per the motor nameplate |
| P01.02        | Rated current of motor   | Set the rated current of traction motor                                                                                   | 0.0 ~ 300.0   | A    | As per the inverter parameter s | As per the motor nameplate |
| P01.03        | Rated frequency of motor | Set the rated frequency of traction motor                                                                                 | 0.00 ~ 120.00 | Hz   | 50.00                           | As per the motor nameplate |
| P01.04        | Rated speed of motor     | Set the rated speed of traction motor                                                                                     | 0 ~ 3000      | rpm  | 1460                            | As per the motor nameplate |
| P01.05        | Rated voltage of motor   | Set the rated voltage of traction motor                                                                                   | 0 ~ 460       | V    | As per the inverter parameter s | As per the motor nameplate |

|        |                                               |                                                                                                                                                    |                 |        |                           |                                                                                                                                         |
|--------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| P01.06 | Number of poles of motor                      | Set number of poles of traction motor                                                                                                              | 2 ~ 128         | ×      | 4                         | As per the motor nameplate                                                                                                              |
| P01.07 | Rated slip frequency of motor                 | Set the rated slip frequency of traction motor                                                                                                     | 0 ~ 10.00       | Hz     | 1.40                      | As per the motor nameplate                                                                                                              |
| P01.08 | Phase sequence of motor                       | Set the phase sequence of input voltage of traction motor, so as to adjust its running direction<br>1: Positive direction<br>0: Negative direction | 0 / 1           | ×      | 1                         |                                                                                                                                         |
| P01.09 | Rated current coefficient of motor at no load | Set the proportion value between no load current and rated current of traction motor                                                               | 0.00 ~ 60.00    | %      | 32.00                     | No setting in general                                                                                                                   |
| P01.10 | Resistance of the motor stator                | Resistance value of traction motor stator                                                                                                          | 0.000 ~ 65.000  | Ω      | As per the inverter power | Only for the asynchronous motor                                                                                                         |
| P01.11 | Resistance of the motor rotor                 | Resistance value of traction motor rotor                                                                                                           | 0.000 ~ 65.000  | Ω      | As per the inverter power | Only for the asynchronous motor                                                                                                         |
| P01.12 | Inductance of the motor stator                | Inductance value of the traction motor stator                                                                                                      | 0.0000 ~ 6.0000 | H      | As per the inverter power | Only for the asynchronous motor                                                                                                         |
| P01.13 | Inductance of the motor rotor                 | Inductance value of the traction motor rotor                                                                                                       | 0.0000 ~ 6.0000 | H      | As per the inverter power | Only for the asynchronous motor                                                                                                         |
| P01.14 | Mutual inductance of the motor                | Mutual inductance value of traction motor                                                                                                          | 0.0000 ~ 6.0000 | H      | As per the inverter power | Only for the asynchronous motor                                                                                                         |
| P01.15 | Type of encoder                               | Set the type of encoder used to detect the speed of traction motor:<br>0: Incremental encoder<br>1: Sine/cosine encoder<br>2: Endat encoder        | 0/1/2           | ×      | 0                         |                                                                                                                                         |
| P01.16 | Number of pulses of encoder                   | Number of pulses once the encoder begins to operate                                                                                                | 500 ~ 16000     | PPr    | 1024                      |                                                                                                                                         |
| P01.17 | Phase angle of encoder                        | Phase angle value of encoder                                                                                                                       | 0.0 ~ 360.0     | degree | 0.0                       | This data is obtained automatically when the inverter operates for the first time after energization.<br>Only for the synchronous motor |
| P01.18 | Filtering time of encoder                     | Set the filtering time constant for feedback speed input of encoder                                                                                | 1 ~ 30          | ms     | 0                         |                                                                                                                                         |
| P01.19 | Feedback direction of encoder                 | Set the feedback speed direction of encoder<br>1: positive sequence<br>0: negative sequence                                                        | 0 / 1           | ×      | 1                         |                                                                                                                                         |
| P01.20 | Input voltage of inverter                     | Set the input voltage of inverter                                                                                                                  | 0 ~ 460         | V      | 380                       | Initialization is unavailable after setting                                                                                             |

**Note 1: Number of poles of motor**

P01.06 is used to set the number of poles of motor according to the motor nameplate.

If there isn't any parameter on number of poles of the motor, it can be calculated according to the following formula:

$$\text{Number of poles} = (120 \times f) \div n.$$

Where: n is rated speed, and f is rated frequency.

The integral number from the calculated value is the number of poles.

#### **Note 2: Setting of slip frequency**

If there isn't slip frequency data on the motor nameplate, P01.07 setting value can be obtained from the following formula:

Set: rated frequency as f (P01.03), rated speed as n (P01.04) and number of poles of the motor as p (P01.06),

Then: slip frequency =  $f - ((n \times p) \div 120)$ .

For example: rated frequency is 50Hz, rated speed is 1430rpm and number of poles of the motor is 4,

Then setting value of P01.07 is  $50 - ((1430 \times 4) \div 120) = 2.33\text{Hz}$ .

#### **Note 3: Phase sequence of motor**

P01.08 is set as 1 typically, but if the running direction of the motor is opposite to the required direction, then P01.08 can be changed from 1 to 0, to realize the negative direction for the motor.

#### **Note 4: Internal parameters of the motor and self-learning**

The five parameters such as P01.10, P01.11, P01.12, P01.13 and P01.14 are only valid for the asynchronous motor, which are the internal parameters of the motor and can be automatically obtained through the self-learning operation by the inverter to the motor. For the asynchronous motor, if the parameter settings are correct, no self-learning operation is needed for the motor. But if the accurate motor parameters can't be understood on site, or in order to guarantee that the inverter can more accurately carry out torque control to the motor, then the inverter shall be experienced with a self-learning operation for the motor firstly after the elevator installation is completed, making the internal resistance, inductance and other characteristic parameters can be automatically and correctly obtained. The specific steps are shown as the following:

- 1) All wiring related to the inverter and encoder is correctly completed;
- 2) Connect the inverter and set the parameters required be set in P01 group;
- 3) Try to operate the contactor between control inverter and motor (if there are two contactors, then both of them shall be operated), making the inverter and the motor to be well connected, and it will guarantee that the internal contracting brake of the traction motor can't be opened;
- 4) Choose "2 Motor Tuning" on the main screen of operator, press down ENTER to enter Self-learning screen;
- 5) "ATun = 0" is displayed on Self-learning screen, of which, data on the right of equal sign can be changed, from 0 to 6, making the motor to be in the static advanced learning mode. Then press down ENTER, self-learning of the motor is started.

At this time, 9, 8, 7, 6, 5, 4, 3, 2 and 1 will be displayed on the screen in turn, until 0, showing that self-learning operation is finished. If Failed Self-learning is shown, please find out the reasons and carry out self-learning operation again.

#### **Note 5: Phase angle of encoder**

P01.17 is phase angle data of the encoder, which is valid only for the synchronous motor. It isn't a setting



parameter, but a characteristic of the motor and encoder automatically obtained when an inverter operates for the first time.

#### Note 6: Feedback direction of encoder

Feedback direction of the encoder can be chosen by parameter P01.19, whose default value is 1 and won't be changed in general. But if the feedback direction is opposite to the actual direction due to error wiring of the encoder, then the feedback direction can be adjusted by modifying P01.19.

### 6.2.3 PID regulator parameters and starting & braking adjusting parameters

Parameters in group P02 are the setting parameters of PID regulator and starting & braking adjustment. Parameters used to adjust PWM carrier frequency are contained.

| Function code | Designation                          | Content                                                                                             | Setting range | Unit | Factory setting | Remarks                                                                                                     |
|---------------|--------------------------------------|-----------------------------------------------------------------------------------------------------|---------------|------|-----------------|-------------------------------------------------------------------------------------------------------------|
| P02.00        | Zero servo gain P0                   | Gain value of PID regulator acting under zero servo                                                 | 0.00 ~ 655.35 | ×    | 130.00          | See the following explanations<br>Recommended adjusting range: minimum: 1/2 default; maximum: twice default |
| P02.01        | Zero servo integral I0               | Integral value of PID regulator acting under zero servo                                             |               |      | 80.00           | See the following explanations<br>Recommended adjusting range: minimum: 1/2 default; maximum: twice default |
| P02.02        | Zero servo differential D0           | Differential value of PID regulator acting under zero servo                                         |               |      | 0.50            | See the following explanations<br>Recommended adjusting range: minimum: 1/2 default; maximum: twice default |
| P02.03        | Gain at low speed section P1         | Gain value of PID regulator acting when reference speed is less than switching frequency F0         |               |      | 70.00           | See the following explanations<br>Recommended adjusting range: minimum: 1/2 default; maximum: twice default |
| P02.04        | Integral at low speed section I1     | Integral value of PID regulator acting when reference speed is less than switching frequency F0     |               |      | 30.00           | See the following explanations<br>Recommended adjusting range: minimum: 1/2 default; maximum: twice default |
| P02.05        | Differential at low speed section D1 | Differential value of PID regulator acting when reference speed is less than switching frequency F0 |               |      | 0.50            | See the following explanations<br>Recommended adjusting range: minimum: 1/2 default; maximum: twice default |

|        |                                               |                                                                                                                                                                                                                                                                           |                |     |        |                                                                                                             |
|--------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|--------|-------------------------------------------------------------------------------------------------------------|
| P02.06 | Gain at intermediate speed section P2         | Gain value of PID regulator acting whose reference speed is between the switching frequency F0 and F1                                                                                                                                                                     |                |     | 120.00 |                                                                                                             |
| P02.07 | Integral at intermediate speed section I2     | Integral value of PID regulator acting whose reference speed is between the switching frequency F0 and F1                                                                                                                                                                 |                |     | 25.00  |                                                                                                             |
| P02.08 | Differential at intermediate speed section D2 | Differential value of PID regulator acting whose reference speed is between the switching frequency F0 and F1                                                                                                                                                             |                |     | 0.20   |                                                                                                             |
| P02.09 | Gain at high speed section P3                 | Gain value of PID regulator acting when the reference speed is higher than switching frequency F1                                                                                                                                                                         |                |     | 140.00 | See the following explanations<br>Recommended adjusting range: minimum: 1/2 default; maximum: twice default |
| P02.10 | Integral at high speed section I3             | Integral value of PID regulator acting when the reference speed is higher than switching frequency F1                                                                                                                                                                     |                |     | 5.00   | See the following explanations<br>Recommended adjusting range: minimum: 1/2 default; maximum: twice default |
| P02.11 | Differential at high speed section D3         | Differential value of PID regulator acting when the reference speed is higher than switching frequency F1                                                                                                                                                                 |                |     | 0.10   | See the following explanations<br>Recommended adjusting range: minimum: 1/2 default; maximum: twice default |
| P02.12 | Switching frequency at low speed point F0     | Set the parameters of switching frequency at slow speed point of PID regulator according to rated frequency percentage. For example, the rated frequency is 50Hz, the required switching frequency F0 is 100Hz, because 10Hz is 20% of 50Hz, this data shall be set as 20 | 0. ~ 100.0     | %   | 1.0    |                                                                                                             |
| P02.13 | Switching frequency at high speed point F1    | Set the parameters of switching frequency at high speed point of PID regulator according to rated frequency percentage. For example, the rated frequency is 50Hz, the required switching frequency F1 is 40Hz, because 40Hz is 80% of 50Hz, this data shall be set as 80  | 0.0 ~ 100.0    | %   | 50.0   |                                                                                                             |
| P02.14 | Excitation time                               | The inverter will send run signal after it receives run command and passes excitation of this time, then the elevator can operate.                                                                                                                                        | 0.0 ~ 10.0     | s   | 0.3    | Inoperative when controlling the synchronous motor                                                          |
| P02.15 | Zero servo time                               | The elevator begins to accelerate after the inverter sends run signal and the torque is kept during this time                                                                                                                                                             | 0.0 ~ 30.0     | s   | 0.5    |                                                                                                             |
| P02.16 | Break releasing time                          | Mechanical acting time of internal contracting brake                                                                                                                                                                                                                      | 0.00 ~ 30.00   | s   | 0.25   |                                                                                                             |
| P02.17 | PWM carrier frequency                         | Set the value of PWM carrier frequency of inverter                                                                                                                                                                                                                        | 1.100 ~ 11.000 | kHz | 6.000  | Typically this parameter shall not be adjusted                                                              |

|        |                        |                                                                                                                                  |               |     |       |                                                |
|--------|------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------|-----|-------|------------------------------------------------|
| P02.18 | PWM carrier width      | Set the change width of PWM carrier of inverter                                                                                  | 0.000 ~ 1.000 | kHz | 0.000 | Typically this parameter shall not be adjusted |
| P02.19 | Current ramp down time | Period from run command cancellation of the inverter to reduction of its output current to 0                                     | 0.00 ~ 10.00  | s   | 0.00  |                                                |
| P02.20 | Mode of regulator      | 0: Mode of rapid regulator<br>1: Mode of standard regulator<br>2: Mode of relatively slow regulator<br>3: Mode of slow regulator | 0/1/2/3       | ×   | 1     |                                                |

**Note 1:** Starting adjustment

No-load sensor starting compensation technology is developed in this inverter, the elevator has the excellent starting comfort without the measuring device installed. The main parameters to adjust starting comfort include: P02.00, P02.01, P02.02, P02.14, P02.15 and P02.16.

P02.00, P02.01 and P02.02 respectively refer to the proportion, integral and differential parameter of speed loop PID regulation which is operative when the elevator is started. They will be operative within the zero servo duration (setting parameter P02.15). Where, P02.00 is P value (proportional parameter) of PID regulator; P02.01 is I value (integral parameter) of PID regulator and P02.02 is D value (differential parameter) of PID regulator.

P02.14 is excitation time adjusting parameter. After the inverter receives the run direction (and enable) signal from the controller, then sends run response signal to the controller, so that the controller can release the brake. If the excitation time is properly increased, it is helpful to guarantee the output of torque when the elevator is started. But if it is too long to quicken the elevator, and the operating efficiency will be influenced. This parameter will be valid only for controlling the asynchronous motor.

P02.15 is zero servo time adjusting parameter. Zero servo means a period that the inverter outputs a zero speed holding torque between completion of excitation and speed reference. This parameter also will decide the action time of such three zero servo PID parameters as P02.00, P02.01 and P02.02. Zero servo action time is shown as Figure 6.1.

P02.16 is an adjusting parameter for brake mechanical open time, which will be set according to the actual releasing mechanical action time.

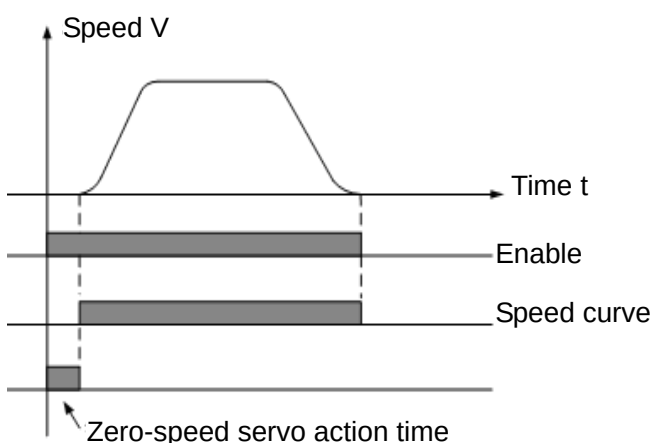


Figure 6.1 Zero Servo Action Time

Functions of proportional constant P in PID regulator: increase of P can improve the response follow capability of the system, but if it is too big, overshoot and oscillation are easily produced. Influence of P on the feedback tracking is shown as Figure 6.2: integral constant I will affect the response time of system, bigger the value is, the faster the response will be. If the system is found out to have too big overshoot or too slow dynamic response, I can be properly enlarged. But if it is too big, system oscillation will be easily caused. Influence of I on feedback

tracking is shown as Figure 6.3; differential constant D will have an effect on the system response sensitivity, and increase of D can realize more sensitive system response, but system oscillation will be caused if D is too big.

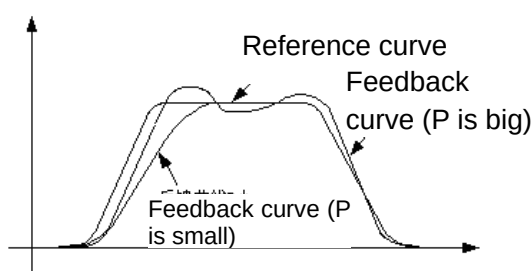


Figure 6.2 Influence of proportional constant P on feedback tracking

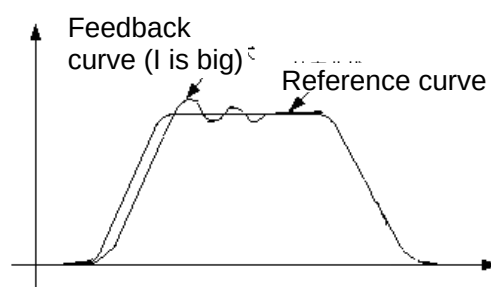


Figure 6.3 Influence of integral constant I on feedback tracking

During adjustment of PID constant, proportional constant P will be adjusted firstly. P will be increased as far as possibly with a precondition that no system oscillation will be caused, then adjust integral constant I, making that the system not only has the rapid response feature, but not too big overshoot. Differential constant D can be turned up properly when P and I are adjusted not to improve the system sensitivity.

**Note 2:** Comfort adjustment during running

Function codes P02.03 ~ P02.13 are to adjust the different parameters of PID regulator at each section (see Figure 6.4) during running, comfort of each section of the elevator during running can be improved by means of adjusting the parameters P02.03 ~ P02.13.

P02.03, P02.04 and P02.05 respectively are P1, I1 and D1 of the PID acting at slow speed section (shown as Figure 6.4). Functions of each parameter have been described in Note 1 above. P02.06, P02.07 and P02.08 respectively are P2, I2 and D2 of the PID acting at intermediate speed section (shown as Figure 6.4), while P02.09, P02.10 and P02.11 respectively are P3, I3 and D3 of the PID acting at high speed section (shown as Figure 6.4). P02.12 and P02.13 refer to the two switching frequencies (or thresholds) used to divide the low speed section, intermediate speed section and high speed section during running curve. Areas whose speed is less than P02.12 (f1) is low speed section, those whose speed is higher than P02.13 (f2) are defined as high speed section, while those between f1 and f2 are intermediate speed section.

**Note 3:** Carrier frequency and width

Parameter P02.17 is used to set PWM carrier frequency of the inverter. The higher the carrier frequency is, the smaller the motor noise will be, but loss of the inverter also will be increased. Generally, the user shall not set it and its default value (6KHz) is ok. If the carrier frequency shall be increased because the motor noise is really required to be reduced on site, then loss of the inverter will be increased with increase of the carrier frequency, therefore the inverter must reduce 5% for each increase of 1KHz after the carrier frequency exceeds the default.

Parameter P02.18 is the width of carrier frequency, typically the user only press down its default, without any adjustment. Function of this parameter is to make carrier frequency to be automatically changed within this width scope, in some cases, it will lighten the motor noise. For example, setting value of P02.17 is 6KHz, P02.18 is set as 0.4KHz, then actual carrier frequency of the inverter will automatically be changed within 5.8 ~ 6.2KHz.

**Note 4:** Current ramp down time

Parameter P02.19 is to set current dropping time after the inverter receives output stop command until the actual output current is dropped to 0. Typically, its default will be applied, only in some special cases, urgent release of inverter current will lead to greater noise of the motor when the elevator stops, so this parameter can be properly increased. But it shall not be greater than the delay time released by the main contactor, otherwise arc discharge will be produced at the contact when the contactor's contact suffers from charged release, so as to influence the life of the contactor. Additionally, the loop disconnect after the contactor releases, the inverter can't output any

current.

**Note 5:** Mode of regulator

Parameter P02.20 is PID regulator mode parameter, typically its default 1 will applied: mode of standard regulator.

## 6.2.4 Speed reference parameters

Set the different parameters related to speed reference in group P03.

| Function code | Designation                   | Content                                                                                                                                                                                                                                                                                                                                                                                                | Setting range | Unit | Factory setting | Remarks                                |
|---------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|----------------------------------------|
| P03.00        | Speed reference selection     | The parameter is used to determine the running speed reference selection of the elevator:<br>0: panel setting<br>1: speed reference for multi section controlled by switching value<br>4: speed reference for AI0 analog<br>6: speed reference for AI1 analog                                                                                                                                          | 0 / 1 / 4 / 6 | ×    | 4               | Invalid when P00.02 is set as 2        |
| P03.01        | Acceleration time             | The parameter is used to determine the acceleration gradient of elevator (namely the constant acceleration), it refers to the acceleration time required of the elevator from zero speed to the maximum speed at constant acceleration. It must be noted that: it differs from the mean acceleration. Besides the data, the mean acceleration is related to the size of the two accelerate rounds.     | 0.10 ~ 60.00  | s    | 2.50            | Used only at the multi speed reference |
| P03.02        | Deceleration time             | The parameter is used to determine the deceleration gradient of elevator (namely the constant deceleration), it refers to the deceleration time required of the elevator from the maximum speed to the zero speed at constant deceleration. It must be noted that: it differs from the mean deceleration. Besides the data, the mean deceleration is related to the size of the two decelerate rounds. | 0.10 ~ 60.00  | s    | 2.50            | Used only at the multi speed reference |
| P03.03        | Time for acceleration round 0 | Set the time for acceleration round at starting section in S curve, the longer the time is, the bigger the round will be.                                                                                                                                                                                                                                                                              | 0.00 ~ 10.00  | s    | 1.30            | Used only at the multi speed reference |
| P03.04        | Time for acceleration round 1 | Set the time for acceleration round at constant speed section in S curve, the longer the time is, the bigger the round will be.                                                                                                                                                                                                                                                                        | 0.00 ~ 10.00  | s    | 1.30            | Used only at the multi speed reference |
| P03.05        | Time for deceleration round 0 | Set the time for deceleration round at the section begin to decelerate in S curve, the longer the time is, the bigger the round will be.                                                                                                                                                                                                                                                               | 0.00 ~ 10.00  | s    | 1.30            | Used only at the multi speed reference |
| P03.06        | Time for deceleration round 1 | Set the time for deceleration round at the section where deceleration ends in S curve, the longer the time is, the bigger the round will be.                                                                                                                                                                                                                                                           | 0.00 ~ 10.00  | s    | 1.30            | Used only at the multi speed reference |
| P03.07        | Speed reference 1             | Set the speed reference 1 of digital at multi speed reference, in Hz                                                                                                                                                                                                                                                                                                                                   | 0.00 ~ 60.00  | Hz   | 2.50            | Used only at the multi speed reference |
| P03.08        | Speed reference 2             | Set the speed reference 2 of digital at multi speed reference, in Hz                                                                                                                                                                                                                                                                                                                                   | 0.00 ~ 60.00  | Hz   | 1.20            | Used only at the multi speed reference |

|        |                    |                                                                       |              |    |       |                                        |
|--------|--------------------|-----------------------------------------------------------------------|--------------|----|-------|----------------------------------------|
| P03.09 | Speed reference 3  | Set the speed reference 3 of digital at multi speed reference, in Hz  | 0.00 ~ 60.00 | Hz | 1.50  | Used only at the multi speed reference |
| P03.10 | Speed reference 4  | Set the speed reference 4 of digital at multi speed reference, in Hz  | 0.00 ~ 60.00 | Hz | 5.00  | Used only at the multi speed reference |
| P03.11 | Speed reference 5  | Set the speed reference 5 of digital at multi speed reference, in Hz  | 0.00 ~ 60.00 | Hz | 25.00 | Used only at the multi speed reference |
| P03.12 | Speed reference 6  | Set the speed reference 6 of digital at multi speed reference, in Hz  | 0.00 ~ 60.00 | Hz | 40.00 | Used only at the multi speed reference |
| P03.13 | Speed reference 7  | Set the speed reference 7 of digital at multi speed reference, in Hz  | 0.00 ~ 60.00 | Hz | 50.00 | Used only at the multi speed reference |
| P03.14 | Speed reference 8  | Set the speed reference 8 of digital at multi speed reference, in Hz  | 0.00 ~ 60.00 | Hz | 0.00  | Used only at the multi speed reference |
| P03.15 | Speed reference 9  | Set the speed reference 9 of digital at multi speed reference, in Hz  | 0.0 ~ 60.0   | Hz | 0.0   | Used only at the multi speed reference |
| P03.16 | Speed reference 10 | Set the speed reference 10 of digital at multi speed reference, in Hz | 0.0 ~ 60.0   | Hz | 0.0   | Used only at the multi speed reference |
| P03.17 | Speed reference 11 | Set the speed reference 11 of digital at multi speed reference, in Hz | 0.0 ~ 60.0   | Hz | 0.0   | Used only at the multi speed reference |
| P03.18 | Speed reference 12 | Set the speed reference 12 of digital at multi speed reference, in Hz | 0.0 ~ 60.0   | Hz | 0.0   | Used only at the multi speed reference |
| P03.19 | Speed reference 13 | Set the speed reference 14 of digital at multi speed reference, in Hz | 0.0 ~ 60.0   | Hz | 0.0   | Used only at the multi speed reference |
| P03.20 | Speed reference 14 | Set the speed reference 14 of digital at multi speed reference, in Hz | 0.0 ~ 60.0   | Hz | 0.0   | Used only at the multi speed reference |
| P03.21 | Speed reference 15 | Set the speed reference 15 of digital at multi speed reference, in Hz | 0.0 ~ 60.0   | Hz | 0.0   | Used only at the multi speed reference |

**Note 1:** Speed reference selection

P03.00 can choose speed reference selection and the commonly used one is 4 (AI0 analog reference) and 1 (digital multi speed reference). Only when parameter P03.00 is set as 1 (digital multi speed reference), the parameters P03.01 ~ P03.21 are valid.

**Note 2:** Analog speed reference

Correspondence relationship between analog signal at analog speed reference and speed reference signal is shown as Figure 6.5.

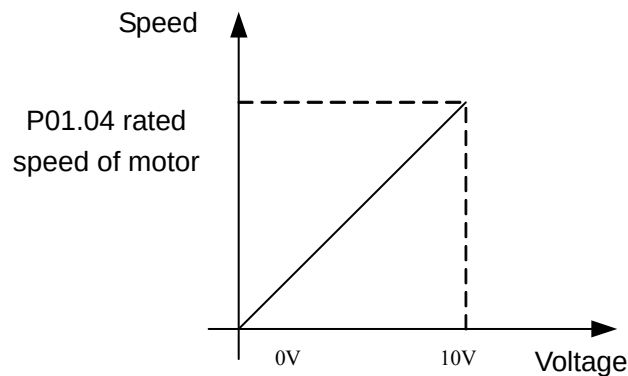


Figure 6.5 Correspondence relationship between analog and speed

**Note 3:** Multi speed curve is shown as Figure 6.6:

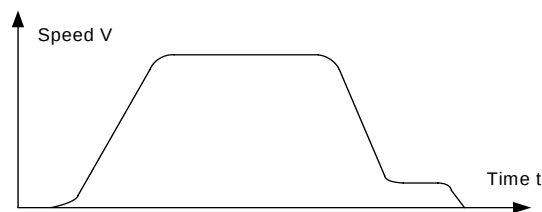


Figure 6.6 Multi speed operation curve

**Note 4: Parameters adjusting multi speed curve**

P03.01 ~ P03.06 are used to set S curve parameters for elevator operation at digital multi speed reference. They appointed acceleration time (P03.01), deceleration time (P03.02), acceleration round time (P03.03 and P03.04) and deceleration round time (P03.05 and P03.06), which directly affect the characteristics of S curve, therefore are directly related to the operation efficiency and comfort of elevator. Specific location of the above parameters in elevator operation S speed curve is shown as Figure 6.7.

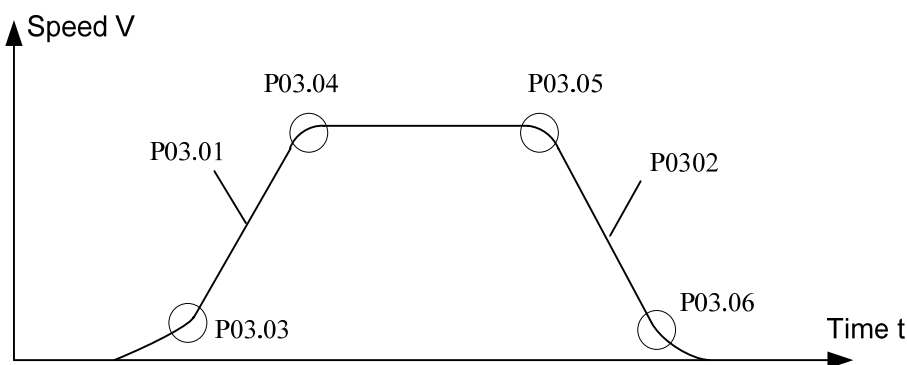


Figure 6.7 Location of S curve in elevator operation



## Important

1) Acceleration time P03.01 and deceleration time P03.02 of S curve can be adjusted within the limits. If the value is turned down, acceleration (deceleration) will be turned up, making to improve the operating efficiency.

But it is harmful to comfort, so it must be considered reasonably.

2) At the acceleration section, acceleration round time P03.03 at start section and P03.04 at end section can be adjusted within the limits. If the value is turned down, acc jerk will be turned up, making to improve the operating efficiency. But the comfort at two rounds will be reduced during the deceleration, so it must be considered reasonably.

3) At the deceleration section, deceleration round time P03.05 at start section and P03.06 at end section can be adjusted within the limits. If the value is turned down, dec jerk will be turned up, making to improve the operating efficiency. But the comfort at two rounds will be reduced during the deceleration, so it must be considered reasonably.

**Note 5:** Influence of the above S curve parameters on the elevator operation curve is shown as Figure 6.8. P03.01 and P03.02 will determine the gradient of speed curve (S curve), the smaller the value is, the steeper the gradient will be; P03.03 ~ P03.06 will determine the round of four speed curves, the smaller the value is, the smaller the round will be (the bigger the curvature will be)

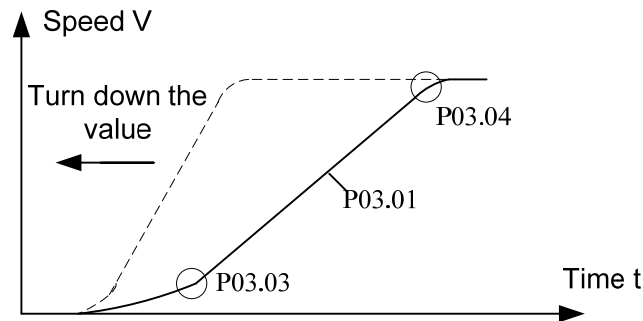


Figure 6.8 Influence of curve parameters on elevator operation curve

**Note 6:** P03.07 ~ P03.21 respectively define the speed command values of 15 multi speed from speed reference 1 to speed reference 15. 16 kinds of status are combined by four input point binary codes of digital multi speed reference 0-3, which are separately corresponding to 15 speed reference commands and 0 speed reference (when the combined code is 0) aforementioned of P03.07 ~ P03.21. Corresponding relationship between multi speed input port signal and speed reference is shown in Table 6.2 following:

Table 6.2 Corresponding relationship between multi speed input combination and speed reference

| Multi speed combined code | Multi speed reference 3 | Multi speed reference 2 | Multi speed reference 1 | Multi speed reference 0 | Speed reference            |
|---------------------------|-------------------------|-------------------------|-------------------------|-------------------------|----------------------------|
| 0                         | 0                       | 0                       | 0                       | 0                       | Speed reference is 0       |
| 1                         | 0                       | 0                       | 0                       | 1                       | Speed reference 1 (P03.07) |
| 2                         | 0                       | 0                       | 1                       | 0                       | Speed reference 2 (P03.08) |
| 3                         | 0                       | 0                       | 1                       | 1                       | Speed reference 3 (P03.09) |
| 4                         | 0                       | 1                       | 0                       | 0                       | Speed reference 4 (P03.10) |
| 5                         | 0                       | 1                       | 0                       | 1                       | Speed reference 5 (P03.11) |
| 6                         | 0                       | 1                       | 1                       | 0                       | Speed reference 6 (P03.12) |
| 7                         | 0                       | 1                       | 1                       | 1                       | Speed reference 7          |



|    |   |   |   |   |                             |
|----|---|---|---|---|-----------------------------|
|    |   |   |   |   | (P03.13)                    |
| 8  | 1 | 0 | 0 | 0 | Speed reference 8 (P03.14)  |
| 9  | 1 | 0 | 0 | 1 | Speed reference 9 (P03.15)  |
| 10 | 1 | 0 | 1 | 0 | Speed reference 10 (P03.16) |
| 11 | 1 | 0 | 1 | 1 | Speed reference 11 (P03.17) |
| 12 | 1 | 1 | 0 | 0 | Speed reference 12 (P03.18) |
| 13 | 1 | 1 | 0 | 1 | Speed reference 13 (P03.19) |
| 14 | 1 | 1 | 1 | 0 | Speed reference 14 (P03.20) |
| 15 | 1 | 1 | 1 | 1 | Speed reference 15 (P03.21) |

In the above table, status 0 means no input signal at this input port, status 1 means input signal at this input port. We will further describe the table with an example: if there is input signal for speed reference 0 and 1, but no any input signal for speed reference 2 or 3, then binary coding will be “0011” = 3, and the corresponding speed reference is speed reference 3, whose value will be appointed by parameter P03.09. In general, 7 speed references are enough for elevator driving (elevator speed doesn't exceed 2.5m/s), therefore input port of speed reference 3 won't be defined. Only such three input points from speed reference 0 to speed reference 2 will be applied. Typically, parameters P03.14 - P03.21 may not be used.

### 6.2.5 Torque reference and torque compensation parameters

Parameters related to torque reference and torque compensation will be set in group P04.

| Function code | Designation                        | Content                                                                                                                                                                                                               | Setting range | Unit | Factory setting | Remarks                                                                                                                                                                                                                                         |
|---------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P04.00        | Torque reference selection         | Set the torque reference selection:<br>0: panel reference<br>1: AI0 analog torque reference<br>2: AI1 analog torque reference                                                                                         | 0/1/2         | ×    | 0               | In most cases, torque reference selection isn't required, this parameter is set as 0. If torque reference selection is adopted, then speed reference selection won't be used (P03.00 is set as 0).<br>It is valid only when P00.02 is set as 2. |
| P04.01        | Torque compensation reference mode | Set the compensation torque reference selection:<br>0: no torque compensation<br>1: provide compensation according to light and heavy load switch<br>2: AI0 analog torque reference<br>3: AI1 analog torque reference | 0/1/2/3       | ×    | 0               |                                                                                                                                                                                                                                                 |
| P04.02        | Torque compensation direction      | Set the torque compensation direction<br>0: positive direction<br>1: negative direction                                                                                                                               | 0/1           | ×    | 0               |                                                                                                                                                                                                                                                 |

|        |                                            |                                                                                    |             |   |       |                                              |
|--------|--------------------------------------------|------------------------------------------------------------------------------------|-------------|---|-------|----------------------------------------------|
| P04.03 | Torque compensation gain                   | Set the torque compensation gain                                                   | 0.0 ~ 200.0 | % | 100.0 | It is valid only when P04.01 is set as 2 ~ 3 |
| P04.04 | Torque compensation offset                 | Set torque compensation offset                                                     | 0.0 ~ 100.0 | % | 0.0   | valid only P04.01 is set as 2 ~ 3            |
| P04.05 | Compensation quantity of light load switch | Compensation quantity of downward torque when set the actions of light load switch | 0.0 ~ 100.0 | % | 0.0   | It is valid only when P04.01 is set as 1     |
| P04.06 | Compensation quantity of heavy load switch | Compensation quantity of upward torque when set the actions of heavy load switch   | 0.0 ~ 100.0 | % | 0.0   | It is valid only when P04.01 is set as 1     |
| P04.07 | Output torque limit                        | Set the limit of output torque, which is percentage data of rated torque           | 0 ~ 200     | % | 175   |                                              |

**Note 1:** Torque reference selection

In elevator control system, torque reference control mode won't be used generally, so P04.00 basically is applied as its factory default 0. If it is adopted by the system, P04.00 is recommended to be set as 1, taking AI0 analog input port as torque reference input point. At this time, speed reference will be used no longer, so P03.00 is required to be set as 0.

**Note 2:** Torque compensation

**AS320** series elevator-used inverter has the function of no weighing preload compensation start. For the synchronous gearless elevator, if SIN/COS encoder is applied, then the best starting comfort can be realized without preload torque compensation to be increased; for the asynchronous gear elevator, the ideal start effect can be reached without increasing the preload torque compensation when starting the elevator. If A, B and Z incremental encoder is used by the synchronous gearless elevator, then preload torque compensation function will be properly increased to further improve the comfort when the elevator is started.

P04.01 is a parameter to choose preload torque compensation mode. When it is set as 0, showing that torque compensation function won't be used; if it is set as 1, showing to choose compensation mode of light and heavy load switch; it is set as 2 or 3, the system will adopt the accurate compensation mode of analog input. If it is set as 2, AI0 analog port will be chosen as compensation quantity input; while it is set as 3, then AI1 analog port will be chosen as compensation quantity input. Under the general circumstances, because AI0 is used as input port of speed reference, then if torque compensation mode of analog input is used, it is recommended to set P04.01 as 3, and AI1 analog port will be chosen as its input point.

P04.03 and P04.04 are the parameters to adjust the torque compensation when P04.01 is set as 2 or 3. P04.04 is compensation offset, which won't be adjusted generally and set as 0 (default). P04.03 is compensation gain, compensation will be increased under the same analog port compensation input quantity if it is turned up; on the contrary, the compensation will be reduced. If there is downward impact (slip during up and too fast during down) for heavy load start and upward impact (slip during down and too fast during up) for light load, showing the compensation is inadequate, and compensation gain of P04.03 shall be turned up; on the contrary, if there is upward impact (slip during down and too fast during up) for heavy load start and downward impact (slip during up and too fast during down) for light load, showing the compensation is too big, and compensation gain of P04.03 shall be properly turned down.

P04.05 and P04.06 are two adjusting parameters based on simple torque compensation of light and heavy load switch. When this simple torque compensation method is adopted, the weighing device accurately installed isn't required, and only two weighing switches simply mounted are required: light load and heavy load switch. Generally, the light load switch can be activated when the car load is adjusted to be less than 25% rated load;

while the heavy load switch can be activated when the car load is adjusted to be greater than 75% rated load. These two switches are connected to digital input point of the inverter. P04.05 is a adjusting parameter to adjust the torque compensation of simple light load when light load switch is activated. If there is upward impact (slip during down and too fast during up) for light load start, showing the light load compensation is inadequate, and P04.05 shall be turned up; on the contrary, if there is downward impact (slip during up and too fast during down) for light load start, showing the light load compensation is too big, and P04.05 shall be properly turned down. Similarly, P04.06 is an adjusting parameter to adjust the torque compensation of simple heavy load when heavy load switch is activated. If there is downward impact (slip during up and too fast during down) for heavy load start, showing the heavy load compensation is inadequate, and P04.06 shall be turned up; on the contrary, if there is upward impact (slip during down and too fast during up) for heavy load start, showing the heavy load compensation is too big, and P04.06 shall be properly turned down.

P04.02 is an adjusting parameter to adjust the torque compensation direction. Generally, it is set as the factory default 0. But if the torque compensation direction of the system makes opposite due to other reasons, simply modify P04.02 from 0 to 1 for solution. The method to judge whether the torque compensation direction is correct is shown as: under the light load (or light load switch action) condition, if the upward impact will be reduced or downward impact will be increased when the elevator is started with increase of P04.03 (or P04.05), showing the compensation direction is right; on the contrary, if downward impact will be reduced or upward impact will be increased when the elevator is started with increase of P04.03 (or P04.06), showing the compensation direction is wrong, which is required to be corrected. Similarly, under the heavy load (or heavy load switch action) condition, if downward impact will be reduced or upward impact will be increased when the elevator is started with increase of P04.03 (or P04.06), showing the compensation direction is right; on the contrary, if upward impact will be reduced or downward impact will be increased when the elevator is started with increase of P04.03 (or P04.06), showing the compensation direction is wrong, which is required to be corrected.

AS320 elevator-used inverter has developed new study on the starting technology of permanent magnetic synchronous gearless elevator, with the unique advantages. If SIN/COS encoder is adopted, the stable starting features can be obtained without any torque compensation. If A, B and Z incremental encoders of 8192 pulses are used, then the satisfactory starting comfort can be realized similarly with the aid of simple torque compensation method of light and heavy load switch. Comparing to SIN/COS encoder, A, B and Z incremental encoders have the obvious advantage in price, with less wiring and strong anti-interference capability. Therefore adoption of simple torque compensation method of light and heavy load switch has a lot of advantages.

### 6.2.6 Definition parameters of digital input

In group P05, functions of digital input terminal and other elements related to the features of input terminal are set.

| Function code | Designation                               | Content                                                                                          | Setting range       | Unit | Factory setting | Remarks                                                                        |
|---------------|-------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------|------|-----------------|--------------------------------------------------------------------------------|
| P05.00        | Definition of X0 terminal input function  | Set the function codes for digital input port:<br>0: no functions (the related port is invalid); | 0 ~ 20<br>103 ~ 120 | ×    | 0               | Factory settings:<br>P05.02 = 3: X0 terminal input is multi speed reference 0; |
| P05.01        | Definition of X1 terminal input function  | 3: digital multi speed reference 0;                                                              |                     |      | 0               | P05.03 = 4: X1 terminal input is multi speed reference 1;                      |
| P05.02        | Definition of X2 terminal input function  | 4: digital multi speed reference 1;<br>5: digital multi speed reference 2;                       |                     |      | 3               | P05.04 = 5: X2 terminal input is multi speed reference 2;                      |
| P05.03        | Definition of X3 terminal input function. | 6: digital multi speed reference 3;                                                              |                     |      | 4               | P05.06 = 7: X6 terminal input is up command signal;                            |
| P05.04        | Definition of X4 terminal input           | 7: up command;                                                                                   |                     |      | 5               | P05.07 = 8: X7 terminal                                                        |

|        |                                          |                                                                                                                                                                                                                                                                                                                                                    |  |  |    |                                                                                      |
|--------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----|--------------------------------------------------------------------------------------|
|        | function                                 | 8: down command;<br>13: external reset signal;<br>14: external fault signal;<br>15: command to set phase angle of external encoder ;<br>16: emergency power running;<br>17: weighing compensation input (specially used by special users);<br>18: base block signal;<br>19: light load compensation switch;<br>20: heavy load compensation switch. |  |  |    | input is down command signal;<br>P05.05 = 18: X5 terminal input is base block signal |
| P05.05 | Definition of X5 terminal input function |                                                                                                                                                                                                                                                                                                                                                    |  |  | 18 |                                                                                      |
| P05.06 | Definition of X6 terminal input function |                                                                                                                                                                                                                                                                                                                                                    |  |  | 7  |                                                                                      |
| P05.07 | Definition of X7 terminal input function |                                                                                                                                                                                                                                                                                                                                                    |  |  | 8  |                                                                                      |

**Note 1:** Detailed descriptions for function codes

a) If the function code is set as 0, showing no definition for the corresponding port, that is to say this input terminal can't be used.

b) if the function code is set as 3 (or 103), the corresponding port is defined as digital multi-speed reference 0 input; if the function code is set as 4 (or 104), the corresponding port is defined as digital multi-speed reference 1 input; if the function code is set as 5 (or 105), the corresponding port is defined as digital multi-speed reference 2 input; while if the function code is set as 6 (or 106), the corresponding port is defined as digital multi-speed reference 3 input; all of them are valid only when P03.00 is set as 1 (speed reference selection is digital multi-speed reference). Specific meaning of the commands of combined multi-speed reference refers to the explanations in Table 6.2.

c) When the function code is set as 15 (or 115), the corresponding port is defined as a command to set the phase angle of external encoder.

This command is not required because the inverter has the function of automatically setting the phase angle of encoder. It is only prepared for the special users.

d) When the function code is set as 16 (or 116), the corresponding port is defined as input port for run signal of emergency power.

This code is used to define the input port for receiving the run signal of emergency power. This input port must be defined when there is emergency running function.

When the signal is input from the related input port, the elevator will enter into emergency power running mode, at this time, the inverter can operate at low speed under a condition of low voltage bus.

e) When the function code is set as 7 (107) and 8 (108), the corresponding ports are defined as input port for up and down signal.

This code is used to define the input port for receiving the up and down signal of elevator, up and down signal must be provided for the inverter under each speed reference selection (digital multi-speed reference and analog voltage speed reference). Function code 7 (107) and 8(108) respectively is used to define a port, other than redefine multi port.

f) If the function code is set as 18(118), the corresponding port is defined as input port for base block signal.

If base block signal input is found at this input port, the inverter shall cut off the output from power module at once.

g) If the function code is set as 19 (119) and 20 (120), the corresponding ports are defined as light and heavy load switch input.

When the light and heavy switch simple torque compensation mode (P04.01 is set as 1) is used by the system, input ports of these two functions shall be defined.

**Note 2:** several input ports with special input function are defined as follows when leaving the factory:

a) Input port to control the run signals of inverter

P05.06 = 7, X6 is defined as input port for up command signal;

P05.07 = 8, X7 is defined as input port for down command signal;

P05.05 = 18, X5 is defined as input port for base block.

b) Input port for multi-speed reference command

P05.00 = 3, X0 is defined as input port for multi-speed reference 0;

P05.01 = 4, X1 is defined as input port for multi-speed reference 1;

P05.02 = 5, X2 is defined as input port for multi-speed reference2.

Speed of the elevator doesn't exceed 2.5m/s in elevator control system, so three multi-speed reference input ports (which can be used to combine 7 speed reference commands) are enough, only three multi-speed reference import ports are defined when leaving factory. When the speed of elevator reached above 3m/s, if digital multi-speed reference selection is adopted, then input port for multi-speed reference 3 will be assigned by defining function code 6 (or 106).

**Note 3:** setting of normally open and closed of input contacts

In order to apply digital input port conveniently, input contact of each input port can be set as normally open or normally closed. Normally open means the input signal is valid between input port and XC (common port). Otherwise it means there isn't any input signal. On the contrary, when input port is defined as normally closed, the input signal is valid when it is disconnected between input port and XC (common port), otherwise it means there isn't any input signal. Function code 1-20 are used to set the input contacts as normally open ones, while function code 101-120 are used to set the input contacts as normally closed ones. Among the function codes, those whose the last two digits are the same have the same functions, for example, input functions defined by 106 and 6 (6 also is 06, the top digit 0 here is omitted) are the same, both of them are multi-speed reference 3 with the difference that the former is defined as normally closed contact and the latter is defined as normally open contact input.

## 6.2.7 Definition parameters of digital output

Functions of digital output terminals and other elements related to features of output terminals are set in group P06.

| Function code | Designation                                      | Content                                                                                                                                  | Setting range       | Unit | Factory setting | Remarks                                                                                                                                                                               |
|---------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P06.00        | Definition of output function of K1 port (relay) | Set functions of different digital output terminals:<br>When it is set as 0: showing that the corresponding input port hasn't been used. | 0 ~ 15<br>101 ~ 115 | ×    | 0               | The three relay output terminals corresponding to K1 are 1A, 1B and 1C respectively, contacts between 1A and 1B are normally closed, while those between 1B and 1C are normally open. |
| P06.01        | Definition of output function of K2 port (relay) | It is set as other value, refer to Note 1 the following for its meaning.                                                                 |                     |      | 0               | The three relay output terminals corresponding to K2 are 2A, 2B and 2C respectively, contacts between 2A and 2B are normally closed, while those between 2B and 2C are normally open. |

|        |                                          |                                                                                                      |            |   |   |                                                        |
|--------|------------------------------------------|------------------------------------------------------------------------------------------------------|------------|---|---|--------------------------------------------------------|
| P06.02 | Definition of output function of Y0 port |                                                                                                      |            |   | 3 | Y0 terminal is defined as run signal of the inverter   |
| P06.03 | Definition of output function of Y1 port |                                                                                                      |            |   | 2 | Y1 terminal is defined as fault signal of the inverter |
| P06.04 | Definition of output function of Y2 port |                                                                                                      |            |   | 0 |                                                        |
| P06.05 | Definition of output function of Y3 port |                                                                                                      |            |   | 0 |                                                        |
| P06.06 | Output delay of K1 terminal              | Set the time for delayed action of output status of K1 output terminal after the actual signal is ON | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.07 | Reset delay of K1 terminal               | Set the time for delayed reset of output status of K1 output terminal after the actual signal is OFF | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.08 | Output delay of K2 terminal              | Set the time for delayed action of output status of K2 output terminal after the actual signal is ON | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.09 | Reset delay of K2 terminal               | Set the time for delayed reset of output status of K2 output terminal after the actual signal is OFF | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.10 | Output delay of Y0 terminal              | Set the time for delayed action of output status of Y0 output terminal after the actual signal is ON | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.11 | Reset delay of Y0 terminal               | Set the time for delayed reset of output status of Y0 output terminal after the actual signal is OFF | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.12 | Output delay of Y1 terminal              | Set the time for delayed action of output status of Y1 output terminal after the actual signal is ON | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.13 | Reset delay of Y1 terminal               | Set the time for delayed reset of output status of Y1 output terminal after the actual signal is OFF | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.14 | Output delay of Y2 output                | Set the time for delayed action of output status of Y2 output terminal after the actual signal is ON | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.15 | Reset delay of Y2 terminal               | Set the time for delayed reset of output status of Y2 output terminal after the actual signal is OFF | 0.0 ~ 60.0 | s | 0 |                                                        |
| P06.16 | Output delay of Y3 terminal              | Set the time for delayed action of output status of Y3 output terminal after the actual signal is ON | 0.0 ~ 60.0 | s | 0 |                                                        |

| P06.17 | Reset delay of Y3 terminal                      | Set the time for delayed reset of output status of Y3 output terminal after the actual signal is OFF                                                                                                                                                                      | 0.0 ~ 60.0   | s  | 0    |                                            |
|--------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----|------|--------------------------------------------|
| P06.18 | Non-zero current detection threshold at parking | Set non-zero current detection threshold while parking. When the inverter has current while parking, and the current exceeds this setting value, non-zero current detection signal is valid. It is a percentage data, whole actual value = this data X rated current/100. | 0.0 ~ 100.0  | %  | 2.0  |                                            |
| P06.19 | Speed detected at any frequency                 | It is a frequency detection reference data, and used with P06.20                                                                                                                                                                                                          | 0.00 ~ 60.00 | Hz | 1.00 | Refer to Note 7 the following for details. |
| P06.20 | Width detected at any frequency                 | It is a frequency detection width data, and used with P06.19                                                                                                                                                                                                              | 0.00 ~ 20.00 | Hz | 0.20 | Refer to Note 7 the following for details. |

**Note 1:** Setting of such 6 parameters from P06.00 to P06.05 defines the functions of such 6 output ports from K1 to K2 and Y0 to Y3. Value range set and functions of the related output ports are described as follows:

0: No function

1 or 101: Readiness for operation of the inverter is completed (RDY)

1: The inverter is normal while self-checking, without any fault. The related output points are connected, otherwise disconnected.

101: The inverter is normal while self-checking and the related output points are connected, otherwise disconnected.

2 or 102: inverter fault

2: when the inverter is in a state of shutdown due to fault, the related output points must be connected, otherwise disconnected;

102: when the inverter is in a state of shutdown due to fault, the related output points must be disconnected, otherwise connected;

3 or 103: Run signal of inverter (RUN)

3: when the inverter can normally run responding to run command, the related output points must be connected, otherwise disconnected;

103: when the inverter can normally run responding to run command, the related output points must be disconnected, otherwise connected;

6 or 106: zero speed run of the inverter

6: if output frequency of the inverter is 0 during its running, the related output points must be connected, otherwise disconnected;

106: If output frequency of the inverter is 0 during its running, the related output points must be disconnected, otherwise disconnected;

7 or 107: voltage of DC bus isn't less than 85% rated value

7: when bus voltage of the inverter isn't less than 85% rated value, the related output points must be connected, otherwise disconnected;

107: when bus voltage of the inverter isn't less than 85% rated value, the related output points must be disconnected, otherwise connected;

8 or 108: 5% rated current is exceeded during running and 10% rated current is exceeded while stopped

8: when the above conditions are met, the related output points must be connected, otherwise disconnected;

108: when the above conditions are met, the related output points must be disconnected, otherwise connected;

9 or 109: during the self-learning

9: when the inverter is in a state of self-learning, the related output points must be connected, otherwise disconnected;

109: when the inverter is in a state of self-learning, the related output points must be disconnected, otherwise connected;

10 or 110: Frequency detected 1

When output frequency of the inverter reaches or exceeds the total of frequency detected speed (P06.19) and frequency detected width (P06.20), frequency detected 1 will be triggered; after the related point is activated and when output frequency of the inverter falls to frequency detected speed (P06.19), frequency detected 1 will reset.

10: when frequency detected 1 is activated, the related output point must be disconnected;

110 when frequency detected 1 is activated, the related output point must be connected;

11 or 111: Frequency detected 2

When output frequency of the inverter reaches or exceeds frequency detected speed (P06.19), frequency detected 2 will be triggered; after output frequency of the inverter falls to the value that frequency detected speed (P06.19) minus frequency detected width (P06.20), frequency detected 2 will reset.

11: when frequency detected 2 is activated, the related output point must be connected;

111: when frequency detected 2 is activated, the related output point must be disconnected;

12 or 112: fault forecast

12: when forecasting the fault, the related output point must be connected, otherwise disconnected;

112: when forecasting the fault, the related output point must be disconnected, otherwise connected.

13 or 113: inverter alarm

13: the related output point shall be connected when the inverter is in state of alarm other than shutdown due to fault, otherwise disconnected;

113: the related output point shall be disconnected when the inverter is in state of alarm other than shutdown due to fault, otherwise connected.

14 or 114: zero servo torque direction judgment (used at emergency leveling when the elevator is in power failure)

14: when the inverter has heavy testing load and light counterweight, the related output point must be connected, other disconnected;

114: when the inverter has heavy testing load and light counterweight, the related output point must be disconnected, other connected.

15 or 115: zero current detection



15: output current of the inverter is greater than non-zero current detection threshold (P06.18 setting) while it stops, the related output point must be connected, otherwise disconnected;

115: output current of the inverter is greater than non-zero current detection threshold (P06.18 setting) when its stops, the related output point must be connected, otherwise disconnected.

Note: “connected” above means: for relay output, the normally open contacts (1B and 1C, 2B and 2C) shall be connected; while the normally closed ones (1B and 1A, 2B and 2A) shall be disconnected; for open collector output, it means the output point is in low level state. Similarly, “disconnected” above means: for relay output, the normally open contacts (1B and 1C, 2B and 2C) shall be disconnected, and normally closed ones (1B and 1A, 2B and 2A) shall be connected; for open collector output, it means output point is in high resistance state.

**Note 2:** For factory setting, P06.03 = 3 is used to assign Y0 port as the output port of run signal (RUN); P06.03 = 2 is used to assign Y1 port as the output port for fault signal of the inverter.

**Note 3:** Providing of run signal (RUN)

The inverter will provide run signal (RUN) when it receives up/down command signal, without base block.

**Note 4:** time sequence of fault signals

Fault signal will be output when the inverter has any fault. At the same time, run signal will be cleared. If the fault signal is latched, it can be eliminated by means of inputting reset signal externally, or reset operation on the operator, or power failure or delay time set internally.

Time sequence of fault signals is shown as Figure 6.9.



Figure 6.9 Time Sequence of Fault Signal

**Note 5:** Setting of signal output delay and reset delay at output end

P06.06 ~ P06.17 are the time constants to set action delay and reset delay for such 6 signals from K1 to K2 and from Y0 to Y3 at output end. Related delay time of the actual signal for output state at each output end can be set according to these parameters. And delay of the above output state can respectively set the delay time whether the signal is being triggered or reset.

**Note 6:** Non-zero current detection threshold of the inverter

Parameter P06.18 is to set non-zero current detection threshold of the inverter. When the inverter current is greater than this threshold while stopping, the related output end set by function code 15 (or 115) will be activated. It is a necessary function for the system whose main circuit only has one contactor. GB7588 specifies that if there is only one contactor to cut off the current of traction motor, then the system must be provided with a monitoring device, to monitor whether the motor current flows or not when the elevator stops, and once the elevator stops, there is still current flowing, so as to control the release of contactor and prevent the elevator from being restarted. Requirements specified in GB7588 can be easily met if the above mentioned non-zero current detection function is applied. Refer to 7.9.4 in Chapter 7 for the specific method.

**Note 7:** Frequency detected

P06.19 and P06.20 are two parameters used for frequency detected: frequency detected speed and frequency detected width. Combination of these two parameters are used for the functions of frequency detected 1 and 2, mainly is to determine whether output frequency of the inverter is within a certain designated frequency range. In

frequency detected 1, when output frequency of the inverter reaches or exceeds the total of frequency detected speed (P06.19) and frequency detected width (P06.20), frequency detected 1 will be triggered; after the related output point is activated and when the output frequency of the inverter falls to frequency detected speed (P06.19) again, frequency detected 1 will reset. Frequency detected 1 is negative logic, whose related output state will be OFF when triggering and ON when resetting.

In frequency detected 2, when output frequency of the inverter reaches or exceeds frequency detected speed (P06.19), frequency detected 2 will be triggered; after the related output point is activated and when the output frequency of the inverter falls to frequency detected speed (P06.19) again, frequency detected 2 will reset. Frequency detected 2 is positive logic, whose related output state will be ON when triggering and OFF when resetting. Figure 6.10 and 6.11 are the schematics of frequency detected 1 and 2.

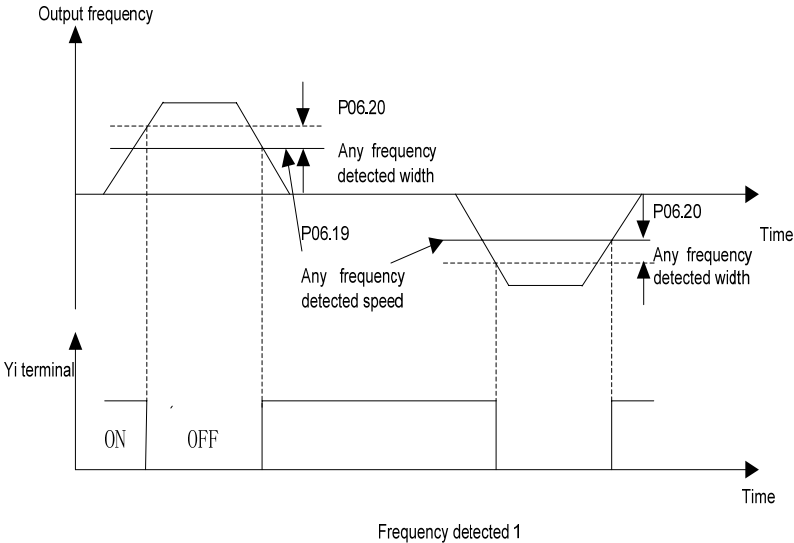


Figure 6.10 Schematic of Frequency Detected 1

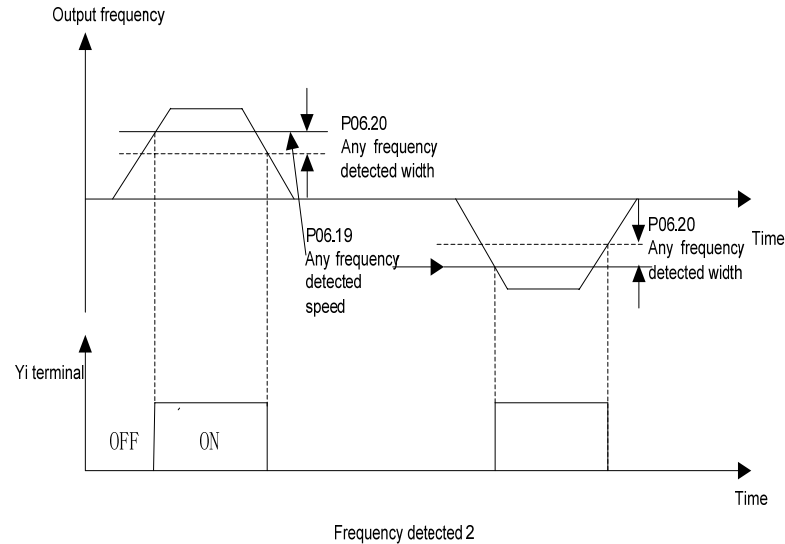


Figure 6.11 Schematic of Frequency Detected 2

6.2.8 Definitions parameters of analog input function

Functions of analog input terminals and related features can be set in group P07.

| Function code | Designation                                 | Content                                                                                                                                                                                                                                                | Setting range  | Unit | Factory setting | Remarks                                                                        |
|---------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------|-----------------|--------------------------------------------------------------------------------|
| P07.00        | Definition of type of AI0 analog input      | Set the type of analog input ports from AI0 to AI1, meaning of each parameter is shown as:<br>0: 0 ~ 10V<br>1: -10V ~ 10V                                                                                                                              | 0 / 1          | ×    | 1               |                                                                                |
| P07.06        | Definition of type of AI1 analog input      |                                                                                                                                                                                                                                                        |                |      | 1               |                                                                                |
| P07.01        | Definition of functions of AI0 analog input | Set the functions of analog input ports from AI0 to AI1, meaning of each parameter is shown as:<br>0: invalid (this input point can not be used)<br>2: analog speed reference<br>3: analog torque reference<br>4: analog compensation torque reference | 0 / 2 / 3 / 4  | ×    | 2               | Delivery of AI0 analog input points is set as analog speed reference input     |
| P07.07        | Definition of functions of AI1 analog input |                                                                                                                                                                                                                                                        |                |      | 0               | Delivery of AI1 analog input points is set as analog torque compensation input |
| P07.02        | Offset of AI0 analog input                  | Set offset voltage value for AI0 analog input,                                                                                                                                                                                                         | 0.000 ~ 20.000 | V    | 10.000          |                                                                                |
| P07.03        | Gain of AI0 analog input                    | Set the gain for AI0 analog input, which is a percentage data                                                                                                                                                                                          | 0.0 ~ 100.0    | %    | 100.0           |                                                                                |
| P07.04        | Filtering time of AI0 analog input          | Set filtering time for AI0 analog input signal                                                                                                                                                                                                         | 0 ~ 30         | ms   | 10              |                                                                                |
| P07.05        | Voltage limit of AI0 analog input           | Set voltage limit value for AI0 analog input                                                                                                                                                                                                           | 0.000 ~ 10.000 | V    | 10.000          |                                                                                |
| P07.08        | Offset of AI1 analog input                  | Set offset voltage value for AI1 analog input                                                                                                                                                                                                          | 0.000 ~ 20.000 | V    | 10.000          |                                                                                |
| P07.09        | Gain of AI1 analog input                    | Set the gain for AI1 analog input, which is a percentage data                                                                                                                                                                                          | 0.0 ~ 100.0    | %    | 100.0           |                                                                                |
| P07.10        | Filtering time of AI1 analog input          | Set filtering time for AI1 analog input signal                                                                                                                                                                                                         | 0 ~ 30         | ms   | 10              |                                                                                |
| P07.11        | Voltage limit of AI1 analog input           | Set voltage limit value for AI1 analog input                                                                                                                                                                                                           | 0.000 ~ 10.000 | V    | 10.000          |                                                                                |

**Note 1:** Definition of type of analog input port

P07.00 and P07.06 are definition parameters to define type of analog input port AI0 and AI1. Type of analog input port will be determined by the hardware, type -10V ~ 10V is applied in this inverter. So default 1 will be used by these two parameters, without modification.

**Note 2:** Definition of functions of analog input port

P07.01 is a definition parameter for functions of the first analog input AI0, and P07.07 is for the second one AI1. In most special control systems for elevator, AI0 is set as analog speed reference input port, so factory default of P07.01 is set as 2. It must note that in order to successfully make AI0 become analog speed reference input port, it isn't enough to set P07.01 as 2, also P03.00 must be set as 4 (choose analog speed reference selection).

**Note 3:** Setting of related adjusting parameters of AI0 analog input

P07.02 is to calibrate the zero offset of AI0 analog input, whose setting can be calculated with the following formula:

$P07.02 = 10.000 - \text{actual zero offset of AI0 analog input (minimum input value)}$

For example: zero offset of AI0 analog input is 0.010V, then

$P07.02 = 10.000 - 0.010 = 9.990$ .

Generally, because the minimum value of AI0 analog input is 0, factory default of P07.02 will be 10.000.

P07.03 is a gain parameter to set input signal of AI0 analog input. If the actual speed is required to be dropped to 90% rated speed on site, then  $P07.03 = 90.0$ .

P07.04 is filtering time of AI0 analog input. It indicates there is 10ms filtering when its default is 10. If input signal is disturbed, filtering time can be properly increased to effectively control the disturbance from analog input signal, but filtering time won't be too long, otherwise lagging will be caused to input signal.

P07.05 is amplitude limit of input voltage signal of AI0 analog input. Analog voltage input signals of AI0 is treated with offset and gain aforementioned, if the data exceeds the limiting data of P07.05, then they will be treated as per limit data. It has no any meaning if limiting data of P07.05 is greater than 10V because 10V of AI0 analog voltage input signal after being treated with offset and gain is corresponding to the maximum value of related signals (such as rated speed of elevator). In addition, it must note that default 10V is adopted for this data in the special control system for elevator, without modification.

**Note 3:** Setting of related adjusting parameters of AI1 analog input is the same as AI0

### 6.2.9 Definition of analog output functions and selection parameters of displayed contents on LCD & LED

Functions and related features of analog output are set in group P08, also displayed contents on LCD & LED can be chosen.

| Function code | Designation                                 | Content                                                                                                                                                                                                                                                                                                                                                                                                                                 | Setting range | Unit | Factory setting | Remarks |
|---------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|---------|
| P08.00        | Definition of functions of MO analog output | Set the functions of different analog output from M0 to M1, definition of different parameters are shown as:<br>0: no definition<br>1: U phase current<br>2: V phase current<br>3: W phase current<br>6: speed reference<br>7: feedback speed<br>13: speed regulator output<br>14: current regulator IQ reference<br>15: current regulator ID reference<br>30: current regulator IQ output<br>32: DC bus voltage<br>44: speed deviation | 0 ~ 44        | ×    | 1               |         |
| P08.01        | Definition of functions of M1 analog output |                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |      | 2               |         |

Continued:

| Function code | Designation             | Content                                | Setting range  | Unit | Factory setting | Remarks |
|---------------|-------------------------|----------------------------------------|----------------|------|-----------------|---------|
| P08.02        | M0 analog output offset | Set voltage offset of MO analog output | 0.000 ~ 20.000 | V    | 15.000          |         |
| P08.03        | M0 analog output gain   | Set gain of M0 analog output           | 0.0 ~ 1000.0   | %    | 100.0           |         |
| P08.04        | M1 analog output offset | Set voltage offset of M1 analog output | 0.000 ~ 20.000 | V    | 15.000          |         |
| P08.05        | M1 analog output gain   | Set gain of M1 analog output           | 0.0 ~ 1000.0   | %    | 100.0           |         |

|        |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |   |    |  |
|--------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|----|--|
| P08.06 | Choose U01 data display content on LCD | The operator is provided with LCD and LED, the latter only displays a data, while the former can set 8 data display U01 ~ U08. The parameters set assign the signal content of this data displayed, their corresponding relationship is shown as:<br><br>0: no definition<br>1: feedback speed (rpm)<br>2: speed reference (Hz)<br>3: feedback speed (Hz)<br>4: output current (A)<br>5: output voltage (V)<br>6: output torque (%)<br>7: bus voltage (V)<br>8: signals of analog input 1 (V)<br>9: signals of analog input 2 (V)<br>13: static self-learning magnetic pole angle<br>16: zero-servo torque (%)<br>17: Number of disturbance of Z phase of encoder<br>18: Number of disturbance of A and B phase of encoder<br>23: Load compensation torque (%)<br>24: speed reference (rpm)<br>25: speed deviation (rpm) | 0 ~ 31 | × | 24 |  |
| P08.07 | Choose U02 data display content on LCD |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |   | 1  |  |
| P08.08 | Choose U03 data display content on LCD |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |   | 25 |  |
| P08.09 | Choose U04 data display content on LCD |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |   | 4  |  |
| P08.10 | Choose U05 data display content on LCD |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |   | 6  |  |
| P08.11 | Choose U06 data display content on LCD |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |   | 16 |  |
| P08.12 | Choose U07 data display content on LCD |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |   | 7  |  |
| P08.13 | Choose U08 data display content on LCD |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |   | 5  |  |
| P08.14 | Choose data display content on LED     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |   | 1  |  |

**Note 1:** Definition of the functions of analog output

P08.00 and P08.01 respectively are to define output functions for M0 and M1 analog output. The specific meaning of each function code is described in the above table. Voltage range of M0 and M1 analog output is -10V ~ 10V. Typically, voltage output of analog output is 0V when the corresponding signal is 0; while that is 10V when it is the maximum value.

**Note 2:** Selection of analog output offset and gain

P08.02 and P08.04 are offset parameters of M0 and M1 analog output, which are defined by hardware. The default 15V is adopted, without modification.

P08.03 and P08.05 are gain parameters of M0 and M1 analog output respectively, and the default 100% is adopted. If it is less than 100%, analog output voltage can't reach 10V, then output voltage width of analog port can't be used fully; while if it is greater than 100%, analog output voltage has reached the maximum value 10V when the signal still doesn't reach the maximum, therefore analog output signal can't display the state of entire corresponding signals.

**Note 3:** Setting of LCD data display of the operator

P08.06 ~ P08.13 are respectively to set the display signal contents of 8 data display on LCD. Three data will be displayed simultaneously on LCD: U01 ~ U03 for the first screen, U04 ~ U06 for the second screen and U07 ~ U08 for the last screen.

**Note 4:** Setting of LED data display of the operator

P08.14 is to set LED data display content of the operator, whose function codes are the same as those of 8 data display on LCD.

### 6.2.10 Other information access

Parameters in group P09 are protection parameters of inverter, which are used to access other information on the inverter.

| Function code | Designation                                                     | Factory setting | Setting range | Unit      | Remarks                                                                                                                                                                                                                                                              |
|---------------|-----------------------------------------------------------------|-----------------|---------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P09.00        | Accumulated on time                                             |                 |               | <b>h</b>  | Read-only parameters                                                                                                                                                                                                                                                 |
| P09.01        | Accumulated run time                                            |                 |               | <b>H</b>  | Read-only parameters                                                                                                                                                                                                                                                 |
| P09.02        | Highest temperature of heatsink                                 |                 |               | <b>℃</b>  | Read-only parameters                                                                                                                                                                                                                                                 |
| P09.03        | Hardware version                                                |                 |               | <b>×</b>  | Read-only parameters                                                                                                                                                                                                                                                 |
| P09.04        | Software version of control panel                               |                 |               | <b>×</b>  | Read-only parameters                                                                                                                                                                                                                                                 |
| P09.05        | Rated power of inverter                                         |                 |               | <b>KW</b> | Read-only parameters                                                                                                                                                                                                                                                 |
| P09.14        | Electric current loop given prop filter                         |                 |               | <b>ms</b> | Don't usually do the modification                                                                                                                                                                                                                                    |
| P09.36        | Frequency dividing coefficient of encoder<br>(PG card required) | 0               | 0~7           |           | 0: ( no frequency dividing),<br>1:( 2 frequency dividing), 2: (4 frequency dividing), 3:(8 frequency dividing),<br>4: (16 frequency dividing), 5 (32 frequency dividing), 6:(64 frequency dividing), 7: (128 frequency dividing)<br><b>Note: ( PG card required)</b> |
| P09.39        | Synchronous motor study angle or not when power on              | 1               | 0/1           |           | Determine whether synchronous motor conduct angle self-study or not when power on , 0 for no study, 1 for study                                                                                                                                                      |
| P09.40        | Current gain when self-study                                    | 150             | 0~400         | <b>%</b>  | Current gain when synchronous motor conduct angle self-study                                                                                                                                                                                                         |
| P09.41        | Confirmation threshold of Encoder phase CD failure              | 300             | 0~65535       |           | No 28 failure reported in case that the D-value of the absolute position and computing position of encoder exceed the setting value.                                                                                                                                 |
| P09.43        | Protection threshold of ABZ encoder disconnection               | 20              | 0~100         |           | Protection in case of speed feedback deviation of synchronous motor confirmed for more than the setting value                                                                                                                                                        |

**P09.36**—Frequency dividing coefficient of encoder. This parameter selects the frequency dividing output

coefficient of PG card, the default value is 0. It requires the PG card with frequency dividing output function. 0: ( no frequency dividing), 1:( 2 frequency dividing), 2: (4 frequency dividing),3:(8 frequency dividing),4: (16 frequency dividing),5 (32 frequency dividing),6:(64 frequency dividing), 7: (128 frequency dividing)

**P09.39**—self study selection or not when power on . The default is 1: self-study every time when power on. If it changes to 0, never conduct another self-study when power on once the self-study has been done. (Only for sincos encoder and Endata encoder) .user can manually change the phase angel P01.17 data. If P01.17 value is 0, the integrated unit will conduct self-study automatically. Please note: once P9.39 set as 0 and main unit or rotary encoder has been replaced, P01.17 should be set as 0 to do a self-study. Otherwise, the wrong phase angel will cause the vibration of motor.

**P09.40**—current gain when conducting self-study. The default value is 150, which mean the default phase self-study is done with 1.5 times rated current  
When using special main unit (such as BOMA motor) on site, phase self-study is required for several times. After the self-study is completed, there is the need to run main unit once. The phase angle position obtained can be seen at F01.17. Change F01.17 to 0, the main unit can do the self-study even without power disconnection. Comparing the P01.17 values obtained from self-study each time, increase the F09.40 to redo the self-study until the deviation of self-study every time is less than 10 if the variation range is more than  $+10 \sim -10$ . F09.40 value should not be too high, otherwise the main unit running at the first time after the self-study will cause noise. So it is ok to set the value below 300. For BOMA main unit on site, the value should be set as 250 to do the self-study. Its phase angle deviation is within 8.

**P09.43** — **ABZ** encoder disconnection protection threshold. The default value is 20%. When the synchronous motor adopted the ABZ encoder, the encoder disconnection is identified once the feedback speed deviation exceeds the protection threshold. The No 12 fault is reported with the frequency converter.





## Chapter 7 Application of Elevator

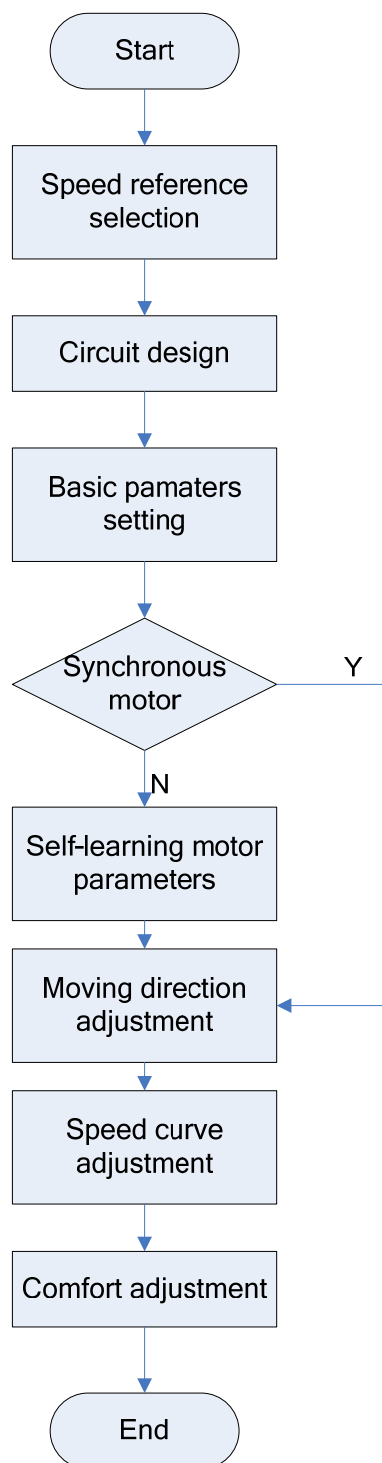
### 7.1 Introduction

AS320 series inverter is specially designed for elevator driving. It adopts advanced vector control technology and has excellent speed governing performance. Owing to its special design for elevator, AS320 series inverter's adjustment and operation is very convenient. What's more, the elevator's comfortability in all operating process will be much easier to be adjusted to the best conditions.

The internal control block diagram of AS320 series inverter is referred to Figure 7.1.

The total elevator electrical control system may be divided to control system and drive system. The control system mainly manages the sequential control of elevator's starting and braking, as well as all the elevator's automatic operational functions, and some control systems also give speed command curve to the elevator's operation; the drive system mainly manages the elevator traction motor's speed control to ensure smooth speed changing during the whole operating process, so as to provide favorable riding comfort to the users. The drive system may also calculate the speed command curve, and if under multi-speed reference mode, the drive system will automatically calculate speed curve based on the preset speed of various sections given by the control system. Under any mode of speed reference, the main task of the drive system is to control the motor to operate according to the speed curve strictly. AS320 series inverter is a typical elevator drive system, thus, it must be combined with a special control system of elevator to form a whole electrical control system. The elevator control systems which may match with it include SMART COM elevator-used control system of STEP Company and other similar elevator control systems.

To apply AS320 series inverter to the whole elevator control system, firstly the speed reference mode (analog reference or multi-speed reference) must be chosen based on the elevator controller's characteristics; and then the circuit shall be designed according to the signal communication requirements between the inverter and the elevator controller; and then to set basic parameters based on requirements; for asynchronous motor, it need to run to self-learning motor parameters; and in the end, the moving direction, speed curve and riding comfort shall be adjusted on site. The above-mentioned process is as shown as the following diagram:



The following sections of this chapter introduce all the details of above-mentioned process. And the last section introduces the application method of several optional functions for users' reference.

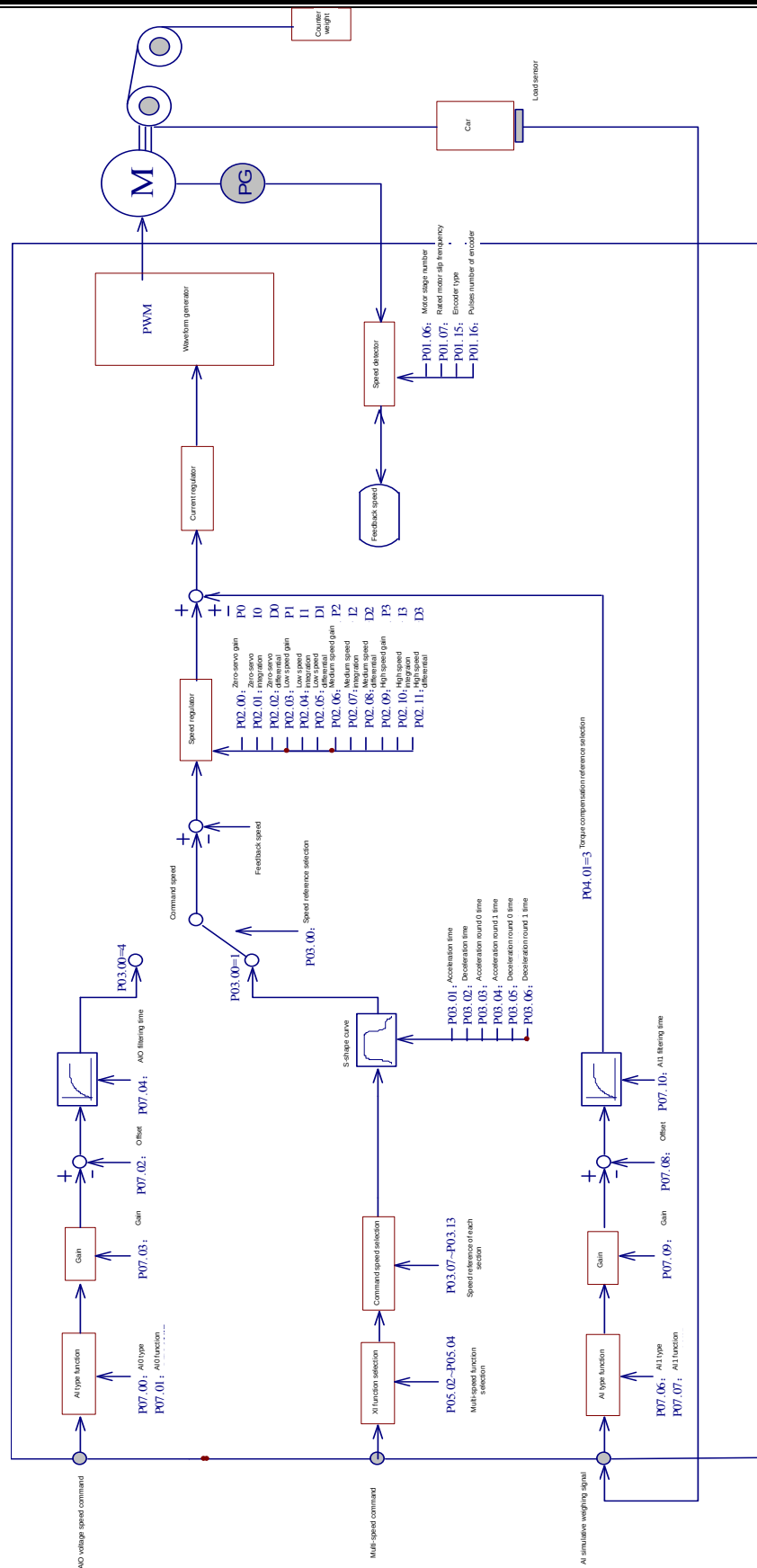


Figure 7.1 Internal Control Block Diagram of Inverter

## 7.2 Speed Reference Selection Introduction

AS320 series elevator-used inverter has two kinds of speed reference selections: analog speed reference selection

and multi-speed reference selection. In the analog speed reference selection, the elevator controller may produce speed command curve and transmit the signal to the inverter's analog input port by the analog output port, thus in every moving of the elevator, the inverter will instantaneously gain elevator's speed reference command based on the input signal strength of the analog input signal; while in the multi-speed reference selection, the elevator need not to calculate detailed speed curve and it need only to produce a binary code signal by the on-off output signal and output a section speed command (target speed command) to the inverter, and the inverter will automatically produce an S-shape curve transitting from former section speed to new section speed after recived the new section speed and then control the elevator to operate strictly according to the S-shape curve speed command. The analog speed reference selection of AS320 series elevator-used inverter is a speed reference selection based on analog voltage signal and the inverter inputs 0~10V voltage signals through AIO input port which are corresponding to the speed reference command from zero speed to the highest speed.

Typically, if the elevator controller may produce speed command curve and it has analog output which may match with AS320 series elevator-used inverter, we recommend to use analog speed reference selection. But if the adopted elevator controller has no analog output maching with AS320 series elevator-used inverter or it can't produce speed command cureve of the elevator moving, we can only use multi-speed reference selection. Compared with analog speed reference selection, the multi-speed reference selection is more complicated for adjusting the speed cureve; in addition, if the elevator controller may adopt distance deceleration principle during the elevator's deceleration process, except convenient adjustment, analog reference selection may also make the elevator stop directly so as to increase elevator's operating efficiency.

In the multi-speed reference selection, under any circumstance, after the inverter accepted a section speed, it will produce an S-shape speed command curve which takes the section speed as goal and takes time as variable based on the acceleration and deceleration gradient (acceleration time and deceleration time paramters) and S-shape rounds parameter (acceleration round time and deceleration round time). In this mode, the deceleration distance of each moving speed is assigned by the elevator controller parameters. If the actual moving speed of elevator may totally coincide with the S-shape speed curve without any error, then the elevator's every deceleration may also realize direct stop, namely the elevator will exactly stop at floor level when its speed just decelerate to zero based on the speed curve, so the elevator's operation will reach the optimal efficiency. But in fact, because the elevator's loading conditions and moving directions will continuously change and no inverter can make the actual speed to totally coincide the speed reference under various conditions, so the elevator's actual deceleration distance will be different under different conditions, therefore, to ensure the elevator not to overstep the floor level at every stop (which will be complained by customers), in actual adjustment, the deceleration distance will be adjusted to the maximal deceleration distance under various conditions. Thus under some conditions, if the actual deceleration distance is shorter, a momentary creep will be produced which will influence the elevator's efficiency. If deceleration based on distance is adopted, when speed follow phenomena under various conditions are caused, the speed reference curve will be finely adjusted according to actual conditions in real time so as to ensure direct stop of the elevator under all conditions and make the operating efficiency optimal.

### 7.3 Connection between Elevator Controller and Inverter

There are mainly two kinds of signal cables between the elevator controller and AS320 series elevator-used inverter as follows: one kind is signals from the elevator controller to the inverter, which mainly include up and down direction command signals and the speed reference command signals; and the other kind is signals from the inverter to the elevator, which include operating signals, fault signals and encoder's pulse signals after processed by PG card. Because there are two kinds of speed reference selections and the signal cable connection in different speed reference selection is a little different, the following two sections introduce the connection reference diagrams of the two kinds of speed reference selections respectively. The encoder signal has also different cable connection modes for different PG card, therefore if the adopted PG card is neither the same with any example of the following reference diagrams, please refer to the wiring of PG card terminals in section 4.6.

### 7.3.1 Reference diagram of cable connection in analog speed reference selection

The cable connection in analog speed reference selection is as shown as figure 7.2.

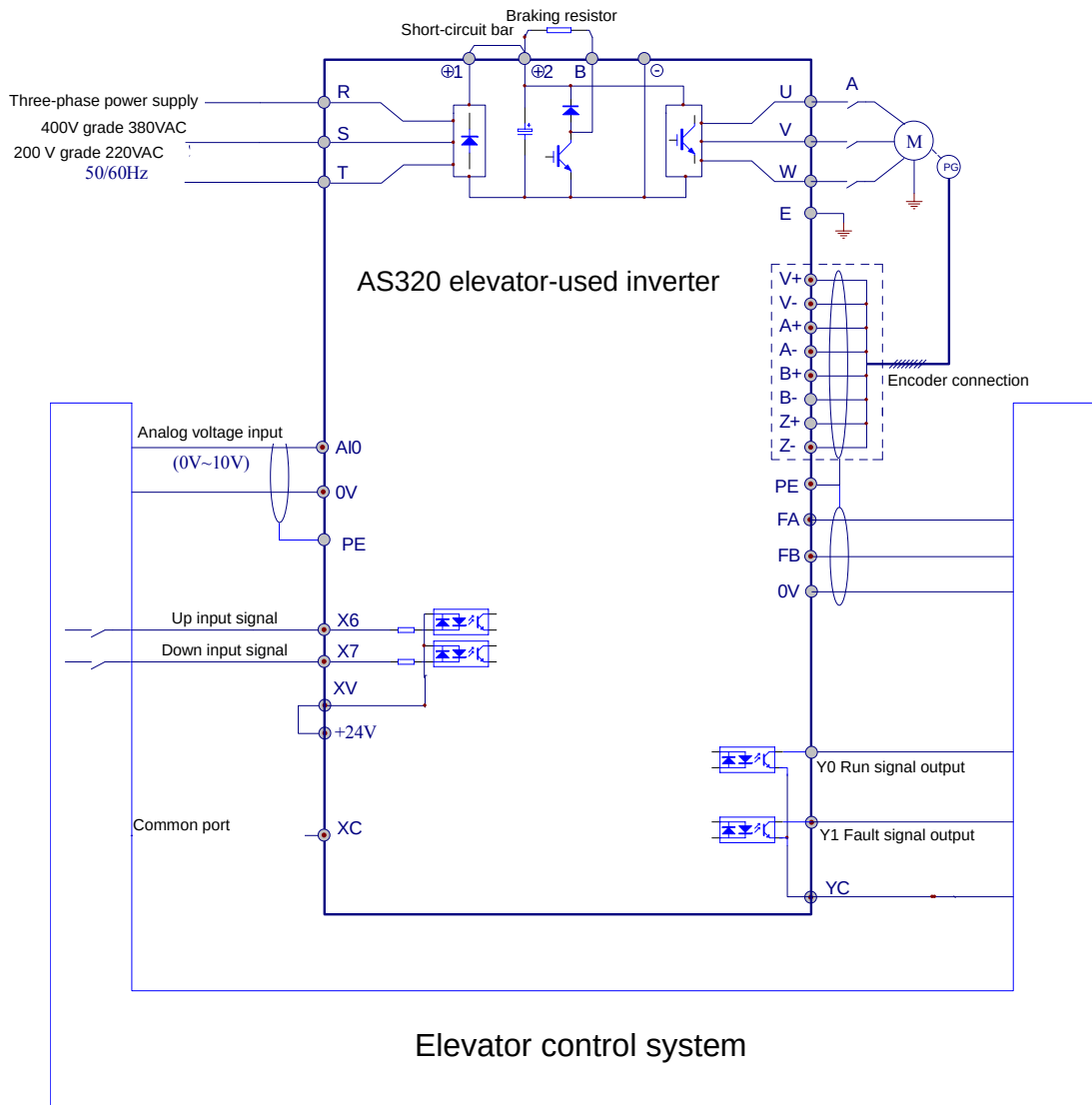


Figure 7.2 Reference Diagram of Basic Circuit in Analog Speed Reference selection Based on Voltage Signal

The terminals definition in the figure is referred to table 7.1.

Table 7.1 Inverter Terminals Definition in Analog Speed Reference selection

| Terminals    | Name                       | Signal Type   |
|--------------|----------------------------|---------------|
| X6 (Default) | Up signal                  | Input signal  |
| X7 (Default) | Down signal                | Input signal  |
| A0           | Analog voltage input 0~10V | Input signal  |
| 0V           | Analog 0V                  | Input signal  |
| Y0 (Default) | Run signal                 | Output signal |
| Y1 (Default) | Fault signal               | Output signal |

**Note:** The input terminals of up & down signals and the output terminals of run & fault signals are defined as above table in the factory and under general conditions, the parameters need not to be changed to redefine the ports. For special requirements, the related input ports may be redefined by digital input (refer to section 6.2.6) or

the related output ports may be redefined by digital output (refer to section 6.2.7). After the ports are redefined, the information of above table shall also be changed correspondingly.

### 7.3.2 Reference diagram of cable connection in multi-speed reference selection

The cable connection in multi-speed reference selection is as shown as figure 7.3.

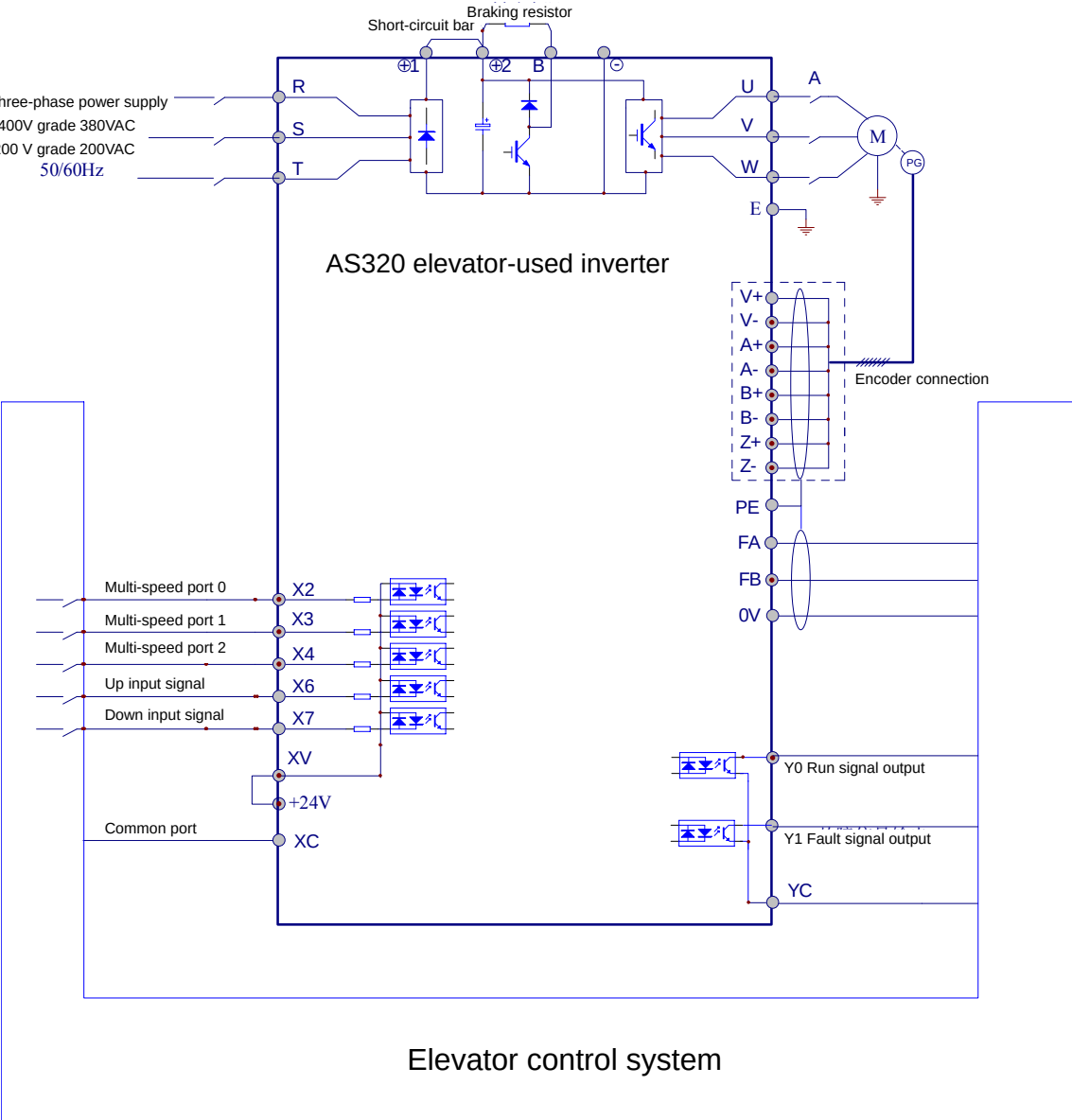


Figure 7.3 Reference Diagram of Multi-speed Reference Selection

The terminals definition in the figure is referred to table 7.2.

Table 7.2 Inverter Terminals Definition in Multi-Speed Reference Selection

| Terminals      | Name                         | Signal Type  |
|----------------|------------------------------|--------------|
| X2 ( Default ) | Multi-speed reference port 0 | Input signal |
| X3 ( Default ) | Multi-speed reference port 1 | Input signal |

|                |                              |               |
|----------------|------------------------------|---------------|
| X4 ( Default ) | Multi-speed reference port 2 | Input signal  |
| X6 ( Default ) | Up signal                    | Input signal  |
| X7 ( Default ) | Down signal                  | Input signal  |
| Y0 ( Default ) | Run signal                   | Output signal |
| Y1 ( Default ) | Fault signal                 | Output signal |

**Note:** The input terminals of up & down signals and the multi-speed input port 0~2 and the output terminals of run & fault signals are defined as above table in the factory and under general conditions, the parameters need not to be changed to redefine the ports. For special requirements, the related input ports may be redefined by digital input (refer to section 6.2.6) or the related output ports may be redefined by digital output (refer to section 6.2.7). After the ports are redefined, the information of above table shall also be changed correspondingly.

The multi-speed ports conditions input by the digital input terminals may compose of 8 conditions. The eight conditions are corresponding to the 8 reference speeds of 0 and P03.07~P03.13, namely 8 section speed reference commands. The relationship between the multi-speed input terminals and the speed references is as the table below.

| Combination Code of Multi-speed | Multi-speed Reference Port 2 | Multi-speed Reference Port 1 | Multi-speed Reference Port 0 | Speed Reference                          |
|---------------------------------|------------------------------|------------------------------|------------------------------|------------------------------------------|
| 0                               | 0                            | 0                            | 0                            | 0                                        |
| 1                               | 0                            | 0                            | 1                            | Speed reference 1 (Function Code P03.07) |
| 2                               | 0                            | 1                            | 0                            | Speed reference 2 (Function Code P03.08) |
| 3                               | 0                            | 1                            | 1                            | Speed reference 3 (Function Code P03.09) |
| 4                               | 1                            | 0                            | 0                            | Speed reference 4 (Function Code P03.10) |
| 5                               | 1                            | 0                            | 1                            | Speed reference 5 (Function Code P03.11) |
| 6                               | 1                            | 1                            | 0                            | Speed reference 6 (Function Code P03.12) |
| 7                               | 1                            | 1                            | 1                            | Speed reference 7 (Function Code P03.13) |

## 7.4 Basic Parameters Setting

For every actual project, the basic parameters of inverter must be correspondingly set according to the specification of the elevator traction motor and encoder. Because **AS320** series inverter is elevator-used inverter and it has taken full consideration of elevator's actual operating requirements in designing, the factory setting (default) of most parameters need not to be changed before operation. The parameters need to be changed and adjusted on site is far less than the introduced, thus, the parameter setting of inverter on site is very convenient. The parameters introduced in this section are the basic parameters which must be set according to the specification of the elevator traction motor and encoder. The motor can self-learn only after these parameters are set, and then the trial run and running performance adjustment of the elevator may be done. The next section will introduce how to adjust the parameters. It must be emphasized that the inverter shall be reset firstly before setting the parameters to ensure all parameters are default value. The method to reset parameters is: enter parameter processing interface by the operator and it shows "Init=0", change the 0 after equal sign to 7, press "Enter" button and input reset password (same as logging on password), and then the interface will show "Successful Reset" which proves the system has reset all parameters to default value and the reset operation is over.

### 7.4.1 Parameter Setting of Speed Reference Selection

There is only one parameter to set the speed reference selection: P03.00. As shown as below table, three reference selections of analog speed reference selection through AI0 input port, analog speed reference selection through AI1 input port and multi-speed reference selection may be selected by parameter P03.00. In actual application, the analog quantity speed reference selection through AI0 input port is generally not selected.

| Function code | Designation               | Content                                                                                                                                                                                                                                      | Setting Range | Unit | Factory Setting | Remarks |
|---------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|---------|
| P03.00        | Speed Reference Selection | This parameter decides the speed reference selection of the elevator's running speed:<br>0: Set in panel<br>1: Multi-speed reference selection controlled by digital value<br>4: AI0 Analog speed reference<br>6: AI1 Analog speed reference | 0 / 1 / 4 / 6 | ×    | 4               |         |

### 7.4.2 Parameters setting of traction motor

All parameters in group P01 are related with the specification of elevator traction motor and its encoder. These parameters are very important to the inverter, if they are improperly set, it might cause vibration or fault of the elevator and even damage the equipment or other accident. So every parameter in the following table must be carefully set according to nameplate specification. Under general conditions, only the parameters in the following table of group P01 need to be set; some of other parameters are generally achieved by the motor's self-learning or use default value so need not to be manually set; and some parameters shall be considered to be changed or not according to requirements during operating adjustment process. Whenever the inverter is replaced (for asynchronous motor), if the customer doesn't want to redo motor self-learning, the related parameters of group P1 may also be read from the old inverter which has accomplished self-learning and then input the values to the new inverter.

| Function code | Designation                | Content                                                                                    | Setting Range | Unit | Factory Setting                 | Remarks                      |
|---------------|----------------------------|--------------------------------------------------------------------------------------------|---------------|------|---------------------------------|------------------------------|
| P01.00        | Motor type                 | Synchronous motor or asynchronous motor selection:<br>0: Asynchronous<br>1: Synchronous    | 0 / 1         | ×    | 1                               |                              |
| P01.01        | Rated motor power          | Set rated power of traction motor                                                          | 0.40~160.00   | KW   | Based on inverter specification | Set based on motor nameplate |
| P01.02        | Rated motor current        | Set rated current of traction motor                                                        | 0.0~300.0     | A    | Based on inverter specification | Set based on motor nameplate |
| P01.03        | Rated motor frequency      | Set rated frequency of traction motor                                                      | 0.00~120.00   | Hz   | 50.00                           | Set based on motor nameplate |
| P01.04        | Rated motor rotation speed | Set rated speed of traction motor                                                          | 0~3000        | rpm  | 1460                            | Set based on motor nameplate |
| P01.05        | Rated motor voltage        | Set rated voltage of traction motor                                                        | 0~460         | V    | Based on inverter specification | Set based on motor nameplate |
| P01.06        | Pole number of motor       | Set pole number of traction motor                                                          | 2~128         | ×    | 4                               | Set based on motor nameplate |
| P01.07        | Rated motor slip frequency | Set rated slip frequency of traction motor                                                 | 0~10.00       | Hz   | 1.40                            | Set based on motor nameplate |
| P01.15        | Encoder type               | Set encoder type used to detect traction motor's rotation speed:<br>0: Incremental encoder | 0/1/2         | ×    | 0                               |                              |



|        |                      |                                               |           |     |      |  |
|--------|----------------------|-----------------------------------------------|-----------|-----|------|--|
|        |                      | 1: Sine/cosine encoder<br>2: Endat encoder    |           |     |      |  |
| P01.16 | Encoder pulse number | Pulse number of one revolution of the encoder | 500~16000 | PPr | 1024 |  |

**Note 1: Pole number of motor**

P01.06 is used to set pole number of motor according to nameplate.

If there is no pole number of motor on the nameplate, it may be calculated by the following equation:

$$\text{Pole number} = (120 \times f) \div n.$$

Where: n is rated rotation speed; f is rated frequency.

The even round number of the calculation result is the pole number.

**Note 2: Set slip frequency**

If there is no slip frequency data on the nameplate, the set value of P01.07 may be calculated by the following equation:

Assume that: rated frequency is f (P01.03), rated rotation speed is n (P01.04) and pole number of motor is p (P01.06),

$$\text{Then: slip frequency} = f - ((n \times p) \div 120).$$

For example: rated frequency is 50Hz, rated rotation speed is 1430rpm and pole number of motor is 4,

$$\text{Then P01.07 value} = 50 - ((1430 \times 4) \div 120) = 2.33\text{Hz}.$$

**7.5 Motor Parameters Self-learning Introduction**

By virtue of the initial technology of this inverter, for synchronous motor, if only the parameters are set as above-mentioned, the inverter will automatically achieve the encoder's phase angle, the motor need not to do self-learning. Namely, if the elevator's traction motor is synchronous motor, this section may be skipped and the reader may directly enter the next section.

It shall be noticed that: if this inverter is used to control synchronous motor, it will automatically capture encoder information at the first operation whenever it is powered, which need 2 seconds. So the run signal will be output a little later than ordinary. Please take consideration of this detail to avoid unnecessary fault.

For asynchronous motor, if the motor parameters are accurately set, the motor need not to self-learning too. But if the motor specification can't be exactly known on site, or to ensure more exactly torque control of the inverter to the motor, after the elevator is installed, it is recommended to make the inverter to do once self-learning to exactly achieve the motor's internal resistance, inductance and other specifications so as to make the elevator run more smoothly and provide better comfort to the users.

The self-learning method of this inverter to asynchronous motor is the very simple static self-learning and the car need not to be lifted. The detailed operation method is as follows:

- 1) Correctly complete all connection with inverter and encoder
- 2) Switch on inverter and set all parameters of group P01;
- 3) Close the contactor between the control inverter and motor (if there are two contactors, both of them must be closed) to ensure good connection between the inverter and motor;
- 4) In the main interface of operator, choose "2 motor tuning", press "Enter" button and enter "Self-learning"

interface;

5) In the “Self-learning” interface, there is display of “ATUN=0” and the data on the right side of the equal sign may be changed, change “0” to “6” and press “Enter” button again, and then the motor will start self-learning;

The interface will display 9, 8, 7, 6, 5, 4, 3, 2, 1 and 0 in turn and when 0 is displayed, it stands for self-learning accomplishment.

## 7.6 Adjustment of Moving Direction of Elevator

The elevator must confirm that the moving direction is right before running. The factors influencing moving direction mainly include: the up & down direction command signal cable connection from the elevator to the inverter, the connection of U, V and W phase power between the inverter and the traction motor and the connection of the phase A and phase B of encoder's feedback signal (for synchronous motor). The adjustment steps are as follows:

- 1) In the analog speed reference selection, the moving speed of elevator inspection must be set through the elevator controller. The inspection speed is recommended to be around 0.2m/s.
- 2) In multi-speed reference selection, the section speed parameter (one of P03.07~P03.13) of the moving speed of elevator maintenance must be set firstly in the group P03 parameters of the inverter and its value is also recommended to be around 0.2m/s after converted to elevator speed.
- 3) Make the elevator move upward or downward and observe the actual operational conditions. The following process gives the method of adjusting elevator moving direction.

The adjustment procedures of upward moving direction in elevator's upward inspection operation are as the following Figure 7.4.

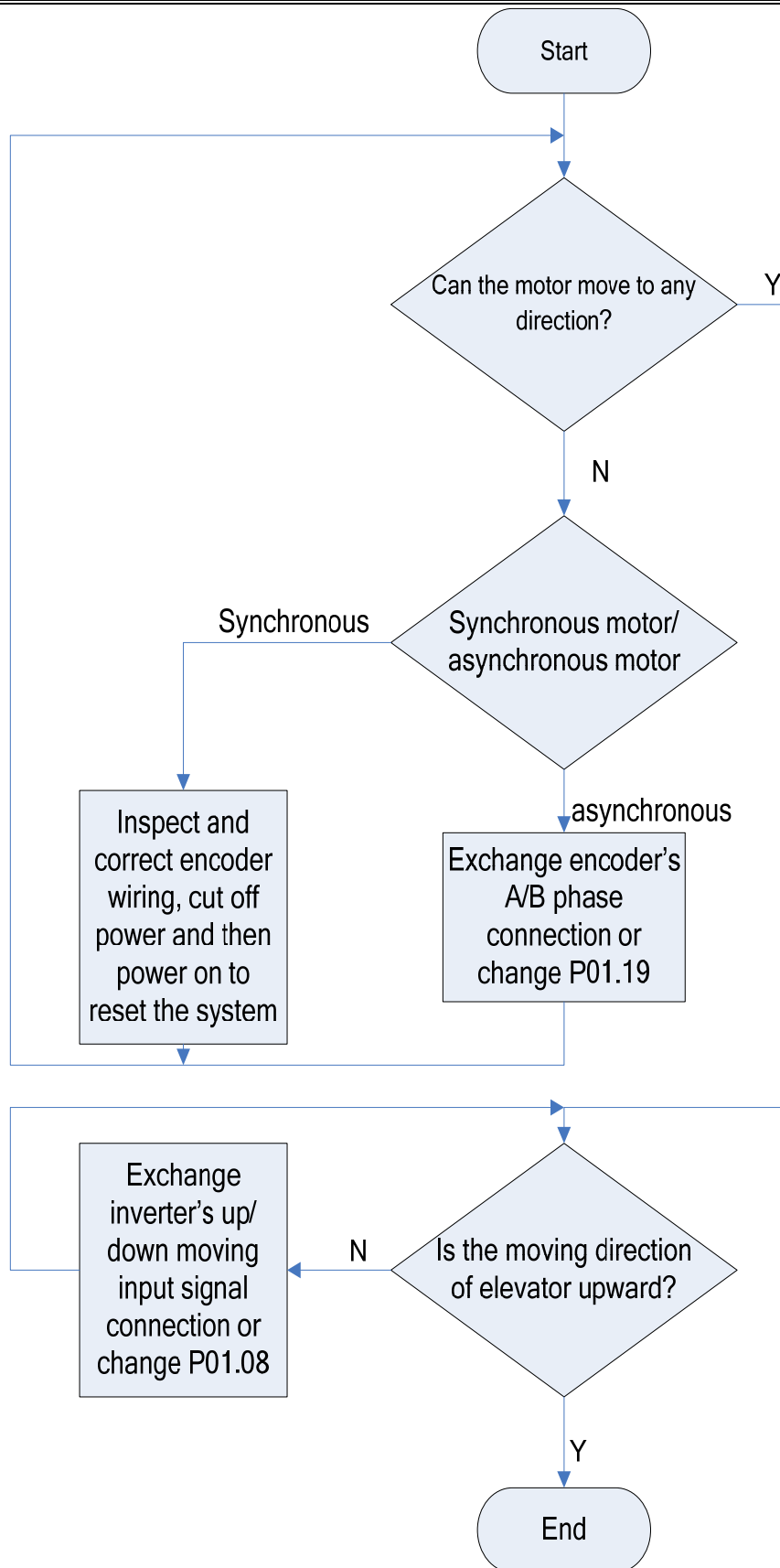


Figure 7.4 Adjustment Procedures of Upward Moving Direction

4) The adjustment procedures of downward moving direction in elevator's downward inspection operation are as the following Figure 7.5.

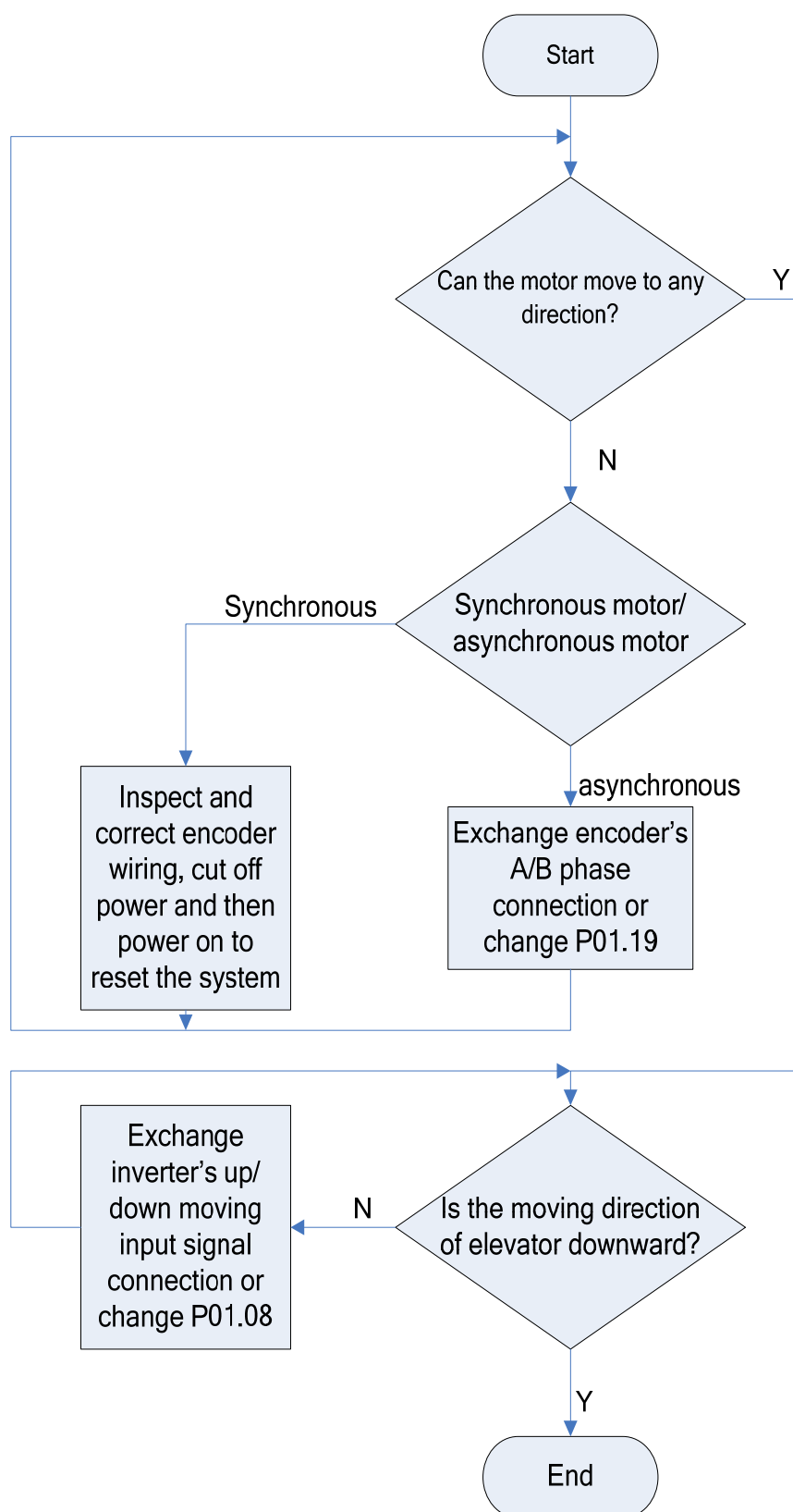


Figure 7.5 Adjustment Procedures of Downward Moving Direction

## 7.7 Speed Curve Adjustment

The adjustment of elevator's speed reference curve directly influences the efficiency and comfortability of the elevator operation. If the elevator may move at high speed it will be necessary to properly adjust elevator's speed

curve.

### 7.7.1 Adjustment method of analog speed reference selection

In analog speed reference selection, the value of parameter P03.00 of speed reference selection must be the same as the following table.

| Speed Reference Selection           | Value of P03.00 |
|-------------------------------------|-----------------|
| Speed reference value of analog AI0 | 4               |

In analog speed reference selection, the speed curve is provided by the elevator controller, so the adjustment of speed curve can only be adjusted by regulating the elevator controller's parameters. The adjustment contents of the speed curve include: acceleration, deceleration and four S-shape curve's rounds parameter (acceleration value or S-shape round time): starting round, full speed round, deceleration round and stopping round. The increase of acceleration (or deceleration) will make curve steeper, shorten acceleration (or deceleration) time and increase operating efficiency of elevator; but at the same time the comfortability will be reduced. Similarly, reducing the transit time of S-shape curve's round will increase the elevator's operating efficiency but will also lower the elevator's comfortability. Therefore, the adjustment of speed curve must take consideration of the operating efficiency and comfortability at the same time. The principles are: 1. complying with requirements of related national standards, for elevator with speed over 1m/s, the average acceleration and deceleration can't be lower than 0.5m/s, and plus the influence of S-shape curve round time, the preset acceleration (deceleration) value can't be lower than 0.6m/s; 2. adjusting the speed curve based on customers' requirement, if the customers prefer efficiency to comfortability, the parameter setting shall be inclined to efficiency increase; however, if the customers prefer comfortability to efficiency, the parameter setting shall be inclined to comfortability.

### 7.7.2 Adjustment method of multi-speed reference selection

In multi-speed reference selection, the value of parameter P03.00 of speed reference selection must be the same as the following table.

| Speed Reference selection       | Value of P03.00 |
|---------------------------------|-----------------|
| Multi-speed reference selection | 4               |

In multi-speed reference selection, the elevator controller output the reference value of target speed to the inverter by digital signal. Generally, the speed of elevator adopting multi-speed reference selection will not exceed 2.5m/s which may be satisfied by three digital input terminals. Because three digital signals may form 8 conditions by binary code (including a zero speed), so the controller may give at most 8 speed commands to the inverter. The detailed speed command curve (S-shape curve) is calculated by inverter so the acceleration time, deceleration time and four S-shape curve round times mentioned in former section are all preset in the inverter; and the inverter shall also provide speed parameters of every section speed. The range of parameters setting and adjustment is as the following table.

| Function Code | Designation               | Content                                                                                                                                                                                                                                          | Setting Range | Unit | Factory Setting | Remarks                                       |
|---------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|-----------------------------------------------|
| P03.00        | Speed reference selection | This parameter decides the speed reference selection of elevator's running speed.<br>0: Set in panel<br>1: Multi-speed reference selection<br>4: Analog speed reference selection through AI0<br>6: Analog speed reference selection through AI1 | 0 / 1 / 4 / 6 | ×    | 4               | Invalid if P00.02 is set to 2.                |
| P03.01        | Acceleration time         | This parameter decides the elevator's acceleration slope (namely constant acceleration), which is the required acceleration time of elevator from 0 speed to the highest speed. It shall be noticed that it's not                                | 0.1~60.0      | s    | 2.5             | Only valid in multi-speed reference selection |

|        |                           |                                                                                                                                                                                                                                                                                                                                                                                 |            |    |      |                                     |
|--------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----|------|-------------------------------------|
|        |                           | same with average acceleration and average acceleration is related not only with this data, but also with the two acceleration rounds' value.                                                                                                                                                                                                                                   |            |    |      |                                     |
| P03.02 | Deceleration time         | This parameter decides the elevator's deceleration slope (namely constant deceleration), which is the deceleration time of elevator from the highest speed to 0 speed at constant deceleration. It shall be noticed that it's not same with average deceleration and average deceleration is related not only with this data, but also with the two deceleration rounds' value. | 0.1 ~ 60.0 | s  | 2.5  | Only valid in multi-speed reference |
| P03.03 | Acceleration round time 0 | Decide the acceleration round time of S-shape curve's starting section and the longer the time, the smaller the acceleration, and conversely, the higher the acceleration.                                                                                                                                                                                                      | 0.0 ~ 10.0 | s  | 1.3  | Only valid in multi-speed reference |
| P03.04 | Acceleration round time 1 | Decide the acceleration round time of S-shape curve's constant speed section and the longer the time, the smaller the acceleration, and conversely, the higher the acceleration.                                                                                                                                                                                                | 0.0 ~ 10.0 | s  | 1.3  | Only valid in multi-speed reference |
| P03.05 | Deceleration round time 0 | Decide the deceleration round time of S-shape curve's starting section of speed reduction and the longer the time, the smaller the deceleration, and conversely, the higher the deceleration.                                                                                                                                                                                   | 0.0 ~ 10.0 | s  | 1.3  | Only valid in multi-speed reference |
| P03.06 | Deceleration round time 1 | Decide the deceleration round time of S-shape curve's end section of speed reduction and the longer the time, the smaller the deceleration, and conversely, the higher the deceleration.                                                                                                                                                                                        | 0.0 ~ 10.0 | s  | 1.3  | Only valid in multi-speed reference |
| P03.07 | Speed reference 1         | Set digital speed reference 1 in Hz of multi-speed reference.                                                                                                                                                                                                                                                                                                                   | 0.0 ~ 60.0 | Hz | 2.5  | Only valid in multi-speed reference |
| P03.08 | Speed reference 2         | Set digital speed reference 2 in Hz of multi-speed reference.                                                                                                                                                                                                                                                                                                                   | 0.0 ~ 60.0 | Hz | 1.2  | Only valid in multi-speed reference |
| P03.09 | Speed reference 3         | Set digital speed reference 3 in Hz of multi-speed reference.                                                                                                                                                                                                                                                                                                                   | 0.0 ~ 60.0 | Hz | 1.5  | Only valid in multi-speed reference |
| P03.10 | Speed reference 4         | Set digital speed reference 4 in Hz of multi-speed reference.                                                                                                                                                                                                                                                                                                                   | 0.0 ~ 60.0 | Hz | 5.0  | Only valid in multi-speed reference |
| P03.11 | Speed reference 5         | Set digital speed reference 5 in Hz of multi-speed reference.                                                                                                                                                                                                                                                                                                                   | 0.0 ~ 60.0 | Hz | 25.0 | Only valid in multi-speed reference |
| P03.12 | Speed reference 6         | Set digital speed reference 6 in Hz of multi-speed reference.                                                                                                                                                                                                                                                                                                                   | 0.0 ~ 60.0 | Hz | 40.0 | Only valid in multi-speed reference |
| P03.13 | Speed reference 7         | Set digital speed reference 7 in Hz of multi-speed reference.                                                                                                                                                                                                                                                                                                                   | 0.0 ~ 60.0 | Hz | 50.0 | Only valid in multi-speed reference |

Note 1: speed reference setting

- 1) Firstly confirm P03.00 to be 1 (set the speed reference selection as digital multi-speed reference).
- 2) Because under most conditions, the rated elevator speed doesn't exceed 2.5m/s and the elevator with rated speed higher than 2.5m/s doesn't recommend digital multi-speed reference mode. This section only introduces the multi-speed reference mode with three digital inputs. The three digital signals form 8 multi-speed commands (including a zero speed) and P03.07~P03.13 are corresponding to the parameters of seven non-zero speed references (speed reference 1 ~ speed reference 7). These parameters are the section speed command of the seven

speed references respectively and each section speed is a speed command of the elevator under various conditions. Generally (the rated elevator speed is lower than 2.5m/s), the elevator operation need the section speeds as following:

| Section Name                      | Description                                                                                                                                                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inspection speed                  | Running speed of inspectioin or automatically leveling process.                                                                                                                                                            |
| Relevelling speed                 | Running speed of relevelling.                                                                                                                                                                                              |
| Inspection half speed             | Running speed reaching terminal floor (deceleration switch activated) under inspection condition or automatically leveling process.                                                                                        |
| High speed 1 (one floor speed)    | Speed of automatically running at high speed and the distance is one-floor height; when the elevator speed doesn't exceed 1m/s it is also double-floor speed and multi-floor speed.                                        |
| High speed 2 (double-floor speed) | Speed of automatically running at high speed and the distance is two-floor height; when the elevator speed doesn't exceed 1.75m/s it is also the multi-floor speed. If the elevator doesn't exceed 1m/s, it is not needed. |
| High speed 3 (multi-floor speed)  | Speed of automatically running at high speed and the distance is three-floor or more than three-floor height. If the elevator doesn't exceed 1.75m/s, it is not needed.                                                    |
| Leveling speed (creep speed)      | Speed of the last section during elevator's deceleration or the speed after elevator enters door area during automatically leveling process.                                                                               |

To correctly set the section speed parameters, how the elevator controller defines the speed reference code must be known firstly. Namely: what conditions the three digital outputs are as the speed reference signal corresponding to each section speed in above table. Next the setting method of section speed parameters is introduced through an elevator controller example.

The relationship between the speed reference code and the section speed name is shown as the following table:

| Speed Reference Code | Section Speed Name                |
|----------------------|-----------------------------------|
| 1                    | Inspection half speed             |
| 2                    | Relevelling speed                 |
| 3                    | Leveling (creep) speed            |
| 4                    | Inspection speed                  |
| 5                    | High speed 1 (one-floor speed)    |
| 6                    | High speed 2 (double-floor speed) |
| 7                    | High speed 3 (multi-floor speed)  |

The speed reference codes in above table are composed of various conditions of the three digital inputs and the details may refer to table 6.2 of section 6.2.4. Assume that the rated elevator speed is 2m/s, then the parameters setting of the section speed may be shown as the following table.

| Parameters                 | Section Speed Name             | Value                        |
|----------------------------|--------------------------------|------------------------------|
| No                         | Zero speed                     | 0                            |
| P03.07 (speed reference 1) | Inspection half speed          | 3Hz (elevator speed 0.12m/s) |
| P03.08 (speed reference 2) | Relevelling speed              | 0.75 Hz (0.03m/s)            |
| P03.09 (speed reference 3) | Leveling (creep) speed         | 1.25 Hz (0.05m/s)            |
| P03.10 (speed reference 4) | Inspection speed               | 6 Hz (0.24m/s)               |
| P03.11 (speed reference 5) | High speed 1 (one-floor speed) | 25 Hz (1m/s)                 |

|                            |                                   |                |
|----------------------------|-----------------------------------|----------------|
| P03.12 (speed reference 6) | High speed 2 (double-floor speed) | 40 Hz (1.6m/s) |
| P03.13 (speed reference 7) | High speed 3 (multi-floor speed)  | 50 Hz (2m/s)   |

If the section speed code's definition of the elevator controller is not the same as above example, the customer may process it by himself based on understanding of above example.

**Note 2:** adjustment of parameters of acceleration (deceleration) time and S-shape curve rounds

P03.01 and P03.02 are the adjustable parameters of elevator's acceleration time and deceleration time and P03.03 ~ P03.06 are the adjustable parameters of the four S-shape curve rounds. The meaning and adjustment method of these parameters is basically the same as that of the analog speed reference which has been introduced. The difference is that in analog speed reference these parameters are adjusted in the elevator controller, however in multi-speed reference these parameters are adjusted in the inverter. And what else to be explained is that: the parameters of P03.03 ~ P03.06 stand for the transit time of an S-shape curve round respectively, so the smaller the parameter value, the higher the round acc jerk and the higher the elevator's operating efficiency, but the worse the elevator's comfortability. On the contrary, the higher the parameter value, the lower the round acc jerk and the lower the elevator's operating efficiency, but the better the elevator's comfortability.

The following figure 7.6 provides the position of each parameter of P03.01, P03.02 and P03.03 ~ P03.06 on the operating speed curve for reference in adjustment.

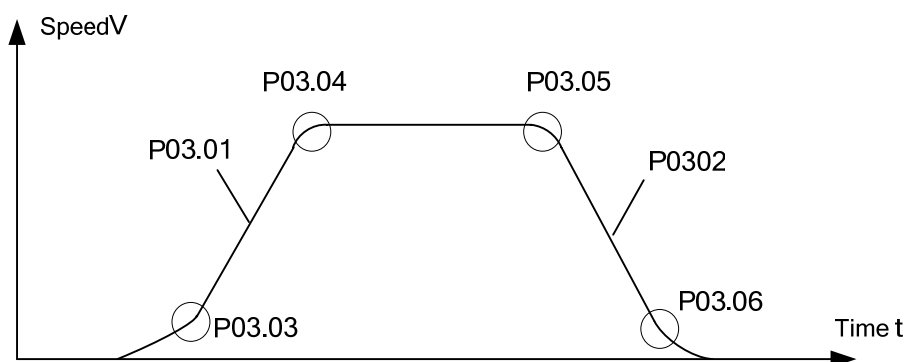


Figure 7.6 Position of Each Parameter on the Speed Adjustment Curve

## 7.8 Adjustment for comfort

Riding comfort is an important index measuring the elevator performance. There are many factors that can influence the riding comfort of elevator. The mechanical factors involve: Quality of guide rail and guide shoe installation and adjustment, performance of traction motor, etc.; The electrical factors involve: performance of speed curve, electromagnetic interference degree of analog speed reference signal (if the analog speed reference selection is used), quality of encoder feedback signal and driving performance of inverter. As the instruction of the inverter, our next discussion will be based on the fact that other factors influencing comfort have been adjusted well and address how to improve the driving performance of inverter by adjusting the appropriate parameters of inverter, in order to improve the riding comfort of elevator.

### 7.8.1 Adjustment for starting comfort

This inverter adopts the unique no-load sensor starting compensation technology, so the excellent starting comfort can be achieved by adjusting parameters even though there is no starting compensation of preload device.

#### 7.8.1.1 Adjustment method of general start comfort

Generally, start comfort of elevator can be improved effectively by adjusting parameters of inverter (zero-servo



PID and excitation time for example). Related adjustment parameters are listed below.

| Function code | Designation                | Content                                                                                                                                            | Setting range | Unit | Factory Setting | Remarks                             |
|---------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|-------------------------------------|
| P02.00        | Zero servo gain P0         | PID regulator gain value active under zero servo mode                                                                                              | 0.00~655.35   | ×    | 130.00          |                                     |
| P02.01        | Zero servo integral I0     | PID regulator integral value active under zero servo mode                                                                                          |               |      | 80.00           |                                     |
| P02.02        | Zero servo differential D0 | PID regulator differential value active under zero servo mode                                                                                      |               |      | 0.50            |                                     |
| P02.14        | Excitation time            | After receiving the operation command, the inverter will give the operation signal after this excitation time, the elevator can release the brake. | 0.0~10.0      | s    | 0.3             | Only controlling asynchronous motor |
| P02.15        | Zero servo time            | From the moment inverter gives operation signal and over the time of holding torque, the accelerating start begins.                                | 0.0~30.0      | s    | 0.5             |                                     |

**Note 1:** About excitation time

If the traction motor is an asynchronous motor, it needs a period of excitation before it can output the torque. More power the motor outputs, longer time the excitation needs. P02.14 is a parameter for adjusting excitation time. If P02.14 is too small, motor has not output enough torque when elevator releases the brake, which causes the poor start comfort. But if P02.14 is too large, bad results such as the too slow start of elevator will be caused.

**Note 2:** Adjustment of speed loop PID regulator at start point

P02.15 is a time parameter of zero servo, which can be used to adjust the delay time given by the speed curve of control system. The delay time here mentioned is also the acting time of zero servo (or zero speed) PID regulator P0, I0 and D0. The detailed action sequence chart is shown as below.

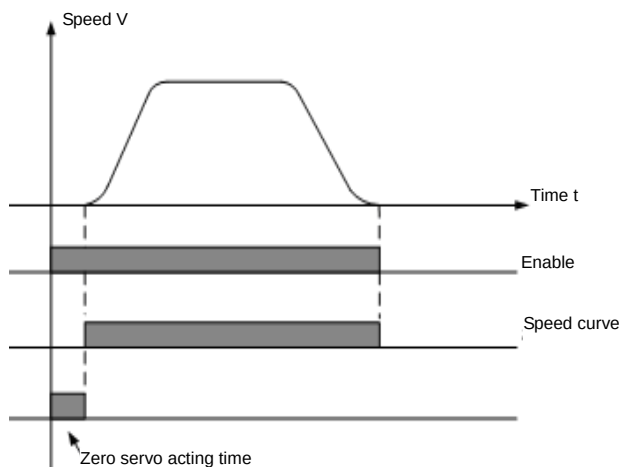


Figure 7.7 Zero servo time sequence

What needs to be cleared is the following content: If the speed is given by using the multi-speed speed mode, the speed curve will be given naturally after zero servo time is over, because the speed curve is generated by inverter itself. But if the analog speed reference selection, speed curve will be given by control system and not be controlled by inverter. So, the time of speed curve given by control system must match with the zero servo time of inverter. If the time of speed curve given by control system is earlier than the end time of inverter zero servo, here the speed curve is invalid because the inverter will ignore any speed reference at zero servo time. Therefore this leads to the abrupt change of physical and valid speed curve. Finally the uncomfortable start of elevator may occur.

If the time of speed curve given by control system is later than the end time of inverter zero servo, the inverter will continue the moment keeping state (actually the zero servo state) after the given zero servo time is over and when the speed reference is still 0. So, under the condition above mentioned, the real zero servo time will be expanded until the real speed given way starts.

P02.00, P02.01 and P02.02 are zero servo regulator gain (P0), integral constant (I0) and differential constant (D0) respectively. During adjustment, firstly set P0 value to a small value, at this point the elevator will run downwards with no-load and the reverse pulling phenomenon will occur when elevator starts. Gradually increase P0 value until the reverse pulling cannot be felt when elevator starts downwards. Too big P0 will cause the oscillation up and down when elevator starts. Therefore P0 value shall be decreased if there is oscillation up and down during start of elevator. I0 is an integral constant of PID regulator at zero speed when elevator starts. The I0 is bigger, the response time is quicker. If I0 value is too small, the P0 will have no time to work; If I0 value is too big, it is easy to generate the oscillation of higher frequency. D0 is helpful to the system response speed. The D0 is bigger, the response time is quicker; But if D0 is too big, the oscillation can also be caused.

### 7.8.1.2 Improve the start comfort by adjusting time sequence

The start time sequence so called refers to the sequence coordination of several actions during start of elevator, such as main contactor closing, issue of inverter upwards or downwards command (or enable signal), brake releasing, issue of speed reference signal. Generally, when elevator starts, the main contactor closes firstly, next the inverter enable signal is issued, finally the brake releasing and speed reference is issued. The sequence coordination between speed reference and brake releasing can influence start comfort of elevator greatly. The ideal coordination point is: When brake mechanism acts (releases really), the speed reference is issued simultaneously. Because there is brake contactor delay and mechanical delay for brake itself, it is not easy to give the accurate data to make both actions meet the ideal requirements. Time sequence can be adjusted according to the following principle: If the downwards start has the obvious feel of reverse pulling when elevator runs with no load, the brake release time can be adjusted to be later (or the issue time of speed reference can be adjusted to be earlier); If the feel of reverse pulling is very weak for the downwards start but too strong for the upwards start, the brake release time can be adjusted to be earlier (or the issue time of speed reference can be adjusted to be later). For the time sequence chart of start and stop, see Figure 7.8 below.

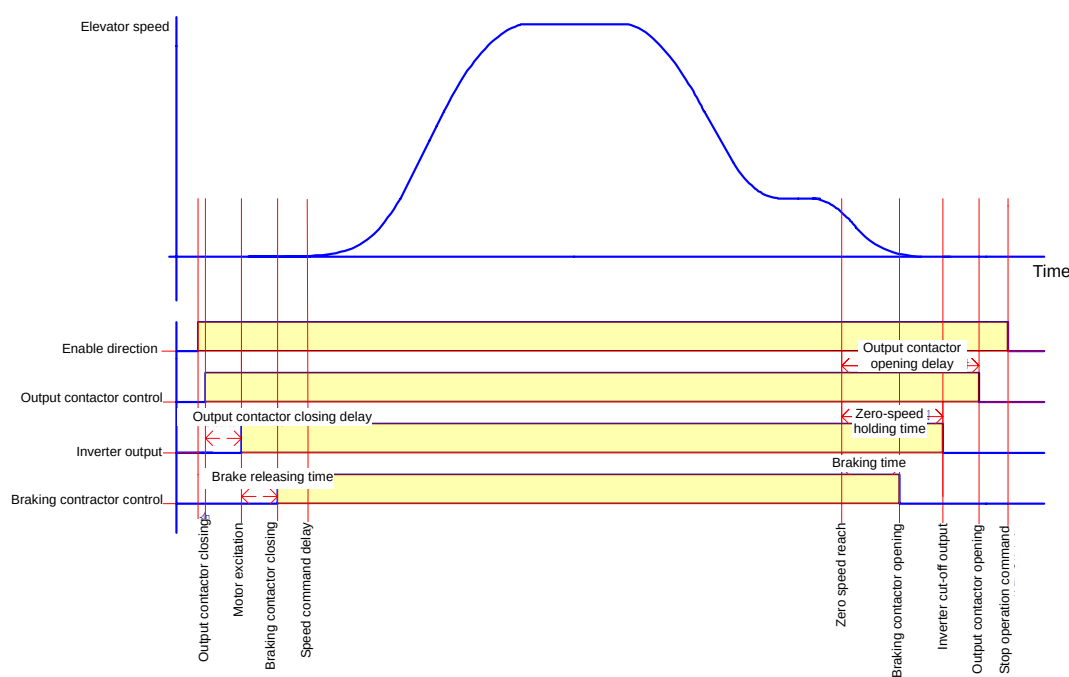


Figure 7.8 Adjustable time sequence chart

### 7.8.2 Adjustment of comfort feel during operation process

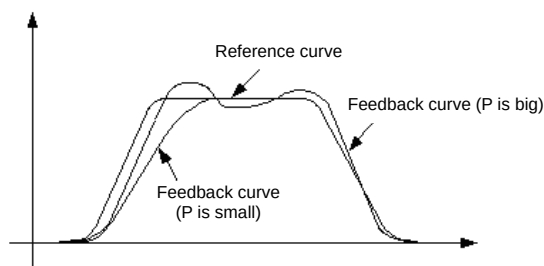
Comfort feel can be improved during the elevator operation process by adjusting PID regulator's parameters in each speed section of operation process. The adjustable parameters are shown in the following table:

| Function code | Designation                             | Content                                                                                                                                                                                                                                                                                              | Setting range | Unit | Factory Setting | Remarks                                                                                                                                                                                 |
|---------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P02.03        | Low speed section gain P1               | Functional PID regulator gain value in the case of speed reference lower than switching frequency F0                                                                                                                                                                                                 |               |      | 70.00           | See the explanation below                                                                                                                                                               |
| P02.04        | Low speed section integral I1           | Functional PID regulator integral value in the case of speed reference lower than switching frequency F0                                                                                                                                                                                             |               |      | 30.00           | See the explanation below                                                                                                                                                               |
| P02.05        | Low speed section differential D1       | Functional PID regulator differential value in the case of speed reference lower than switching frequency F0                                                                                                                                                                                         |               |      | 0.50            | See the explanation below                                                                                                                                                               |
| P02.06        | Medium speed section gain P2            | PID regulator gain value between switching frequency F0 and F1 for speed reference                                                                                                                                                                                                                   |               |      | 120.00          |                                                                                                                                                                                         |
| P02.07        | Medium speed section integral I2        | PID regulator integral value between switching frequency F0 and F1 for speed reference                                                                                                                                                                                                               |               |      | 25.00           |                                                                                                                                                                                         |
| P02.08        | Medium speed section differential D2    | PID regulator differential value between switching frequency F0 and F1 for speed reference                                                                                                                                                                                                           |               |      | 0.20            |                                                                                                                                                                                         |
| P02.09        | High speed section gain P3              | Functional PID regulator gain value in the case of speed reference higher than switching frequency F1                                                                                                                                                                                                |               |      | 140.00          |                                                                                                                                                                                         |
| P02.10        | High speed section integral I3          | Functional PID regulator integral value in the case of speed reference higher than switching frequency F1                                                                                                                                                                                            |               |      | 5.00            |                                                                                                                                                                                         |
| P02.11        | High speed section differential D3      | Functional PID regulator differential value in the case of speed reference higher than switching frequency F1                                                                                                                                                                                        |               |      | 0.10            |                                                                                                                                                                                         |
| P02.12        | Low speed point switching frequency F0  | Set the parameters of switching frequency of low speed point in PID regulator subsection, based on the percentage of the rated frequency. If the rated frequency is 50Hz, the switching frequency F0 required is 10Hz. Because 10Hz accounts for 20% of 50Hz, this parameter should be set to be 20  | 0.~100.0      | %    | 1.0             | See the explanation below.<br>For the medium speed between F0 and F1, the adjustment data of PID is generated by system according to PID data of low speed and high speed automatically |
| P02.13        | High speed point switching frequency F1 | Set the parameters of switching frequency of high speed point in PID regulator subsection, based on the percentage of the rated frequency. If the rated frequency is 50Hz, the switching frequency F1 required is 40Hz. Because 40Hz accounts for 80% of 50Hz, this parameter should be set to be 80 | 0.0~100.0     | %    | 50.0            | See the explanation below.<br>For the medium speed between F0 and F1, the adjustment data of PID is generated by system according to PID data of low speed and high speed automatically |

Parameters P02.03~P02.05 refer to the values of P, I and D of PID regulator of low speed section (P1, I1, D1), and parameters P02.06~P02.08 the values of P, I and D of PID regulator of medium speed section (P2, I2, D2), and parameters P02.09~P02.11 the values of P, I and D of PID regulator of high speed section (P3, I3, D3). They work in each section of operation curve during the whole operation process of elevator (See Figure 7.9). Parameters P02.12 and P02.13 are switching frequency working in subsection (See Figure 7.9). By adjusting parameters P02.03~P02.05, P02.06~P02.08 and P02.09~P02.11 as well as P02.12 and P02.13, the comfort of

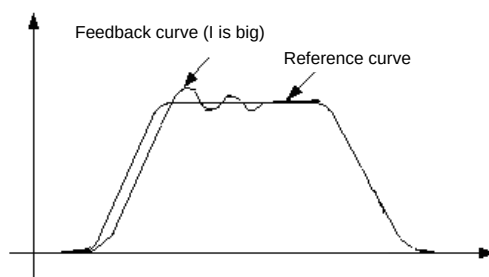
each section of elevator operation process can be improved.

Increasing the proportionality constant  $P$  can enhance the dynamic response capability of system. But if  $P$  is too big, the system is easy to produce the overshooting and oscillation. Influence of  $P$  to feedback tracking is shown as following figure.



Influence of proportionality constant  $P$  to feedback tracking

Increasing integral constant can make the system dynamic response speed faster. If system overshoot is found to be too large or dynamic response is too slow,  $I$  can be increased appropriately. But  $I$  can not be too big, or system may produce oscillation. The figure below demonstrates how integral parameter influences the feedback speed.



Influence of integral constant  $I$  to feedback tracking

Likewise, increasing differential constant  $D$  can increase the sensibility of system. But too big  $D$  may make system too sensitive so as to produce oscillation.

When adjusting parameters of PID regulator, generally the proportionality constant  $P$  should be adjusted firstly. Increase  $P$  value as large as possible on condition that system won't oscillate, then adjust integral constant to make system have the fast response characteristics and not overshoot too large. Then adjust  $D$  value properly only when adjusting  $P$  and  $I$  can not acquire the ideal results.

Subsection range of PID regulator in elevator operation curve is shown as in Figure 7.9 below.

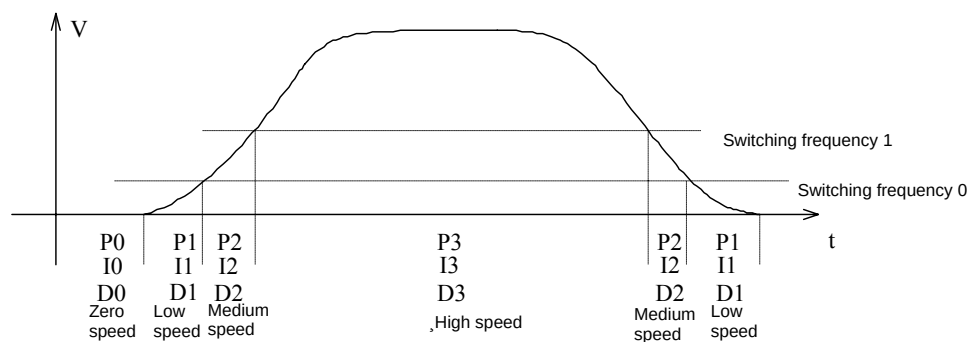


Figure 7.9 PI control diagram in subsection of elevator operation curve

As we can see from figure above, this inverter's PID regulator can execute the adjustment in three speed levels respectively, which can facilitate the adjustment work greatly. If the comfort of high speed section is not good, then only adjust the PID parameters in high speed section, and this won't influence the comfort of other two speed levels; If the comfort of medium speed section or low speed section is not good, then only adjust PID parameters of medium speed section or low speed section. Because PID parameters required by different speed levels to acquire the ideal comfort are various, each speed level can reach its best comfort feel after adjusting PID in different level.

### 7.8.3 Adjustment of stop comfort

There are two factors that can influence the comfort of elevator stop: The first factor: PID value in low speed level, which can be decreased according to the introduction of last section to make comfort of stop better. The second factor: Time sequence in the case of elevator stop, which mainly refers to the sequence coordination of stop speed reference and braking operation. Ideal state is: Braking of elevator is just over when the speed reference of elevator becomes zero. Principle of adjustment is: If elevator rushes suddenly in the case of stop, this means that the braking is too early; On the contrary, if elevator has the sliding phenomenon, this means that the braking is too late.

## 7.9 Additional function description

The above sections of this chapter introduce the inverter adjustment during the general elevator commissioning process. This section will introduce the usage of several additional functions for the reference of user.

### 7.9.1 Introduction to preload weighing compensation method of analog input

This inverter has the advanced no-load sensor starting compensation technology, so the outstanding start comfort can be achieved after using this technology even if the elevator has no preload weighing device installed. Their starting characteristic is shown in Figure 7.10.

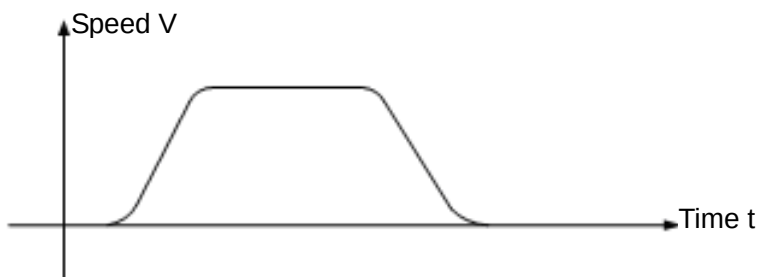


Figure 7.10 No-load sensor starting compensation characteristics diagram

Though AS320 series inverter does not need the preload weighing device, the analog weighing device has been installed in some applications in order to capture the overload and full load signals. Or some elevator users have extremely high requirements on start comfort of elevator, so the start compensation of preload weighing device has to be available. Another case is that: In the case of applications using gearless traction motor, the encoder used does not accord to the requirement of no preload compensation start, then the elevator should install the preload weighing device additionally and inverter adopts the torque compensation technology of start. When using the preload starting compensation technology, the weighing device is demanded to output the analog DC voltage signal featuring the excellent linear characteristics. The analog voltage signal of weighing device is directed to A1 analog input of inverter, as shown in Figure 7.11 wiring diagram.

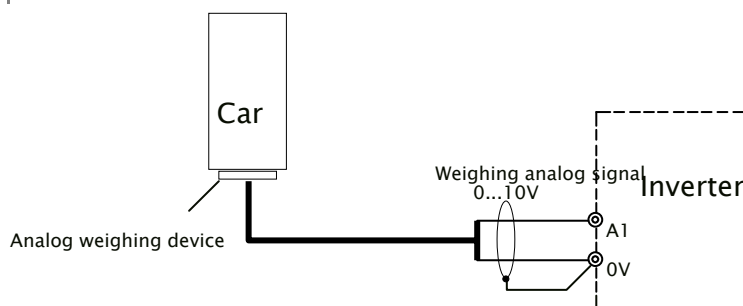


Figure 7.11 Wiring diagram of weighing analog signal

Use the starting compensation technology of preload weighing device, the parameters that need to be set and adjusted are listed below:

| Function code | Designation                             | Content                                                                                                                                                                                                                                         | Setting range  | Unit | Factory Setting | Remarks                                                                            |
|---------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------|-----------------|------------------------------------------------------------------------------------|
| P04.01        | Torque compensation reference selection | Setting compensation torque reference selection:<br>0: No torque compensation<br>1: Giving the compensation according to the light load or heavy load switch<br>2: AI0 analog torque reference<br>3: AI1 analog torque reference                | 0/1/2/3        | ×    | 0               |                                                                                    |
| P04.02        | Torque compensation direction           | Setting torque compensation direction:<br>0: Forward direction<br>1: Backward direction                                                                                                                                                         | 0/1            | ×    | 0               |                                                                                    |
| P04.03        | Torque compensation gain                | Setting torque compensation gain                                                                                                                                                                                                                | 0.0 ~ 200.0    | %    | 100.0           | Valid only when P04.01 is set to be 2 ~ 3                                          |
| P04.04        | Torque compensation offset              | Setting torque compensation offset                                                                                                                                                                                                              | 0.0 ~ 100.0    | %    | 0.0             | Valid only when P04.01 is set to be 2 ~ 3                                          |
| P07.07        | AI1 analog input Function definition    | Set the function of AI0 ~ AI1 analog input, and the meaning of each parameter is as follows:<br>0: Invalid (this input point is not used)<br>2: Analog speed reference<br>3: Analog torque reference<br>4: Analog compensation torque reference | 0 / 2 / 3 / 4  | ×    | 0               | AI1 analog input point can be set to be analog torque compensation input if needed |
| P07.08        | AI1 analog input Offset                 | Set the offset voltage of AI1 analog input                                                                                                                                                                                                      | 0.000 ~ 20.000 | V    | 10.000          |                                                                                    |
| P07.09        | AI1 analog input Gain                   | Set gain value of AI1 analog input, which is in percentage                                                                                                                                                                                      | 0.0 ~ 100.0    | %    | 100.0           |                                                                                    |
| P07.10        | AI1 analog input Filtering time         | Set filtering time of AI1 analog input signal                                                                                                                                                                                                   | 0 ~ 30         | ms   | 10              |                                                                                    |

**Note 1:** Setting of parameters necessary for load compensation method of analog input preload

Firstly, P04.01 must be set to be 3, and the torque compensation reference mode must be defined as the mode of AI1 analog input torque compensation. The corresponding result is: P07.07 should be set to be 4, and AI1 analog

port should be appointed as the analog signal input port of preload torque compensation.

**Note 2:** Usage of adjustment parameters

P04.02 is the setting parameter of preload compensation direction. If the torque compensation direction of system is reversed by some reason, this problem can be fixed by change the P04.02 parameter from 0 to 1 simply. The method of judging the correctness of torque compensation direction is as follows: In the case of light load, if increasing P04.03 parameter can decrease the upward shock or increase the downward shock during elevator start, the compensation direction is correct; On the contrary, if increasing this data can decrease the downward shock or increase the upward shock during elevator start, the compensation direction is wrong, and this problem can be fixed by changing P04.02 parameter. Likewise, in the case of heavy load, if increasing P04.03 parameter can decrease the upward shock or increase the downward shock during elevator start, the compensation direction is correct; On the contrary, if increasing this data can decrease the upward shock or increase the downward shock during elevator start, the compensation direction is wrong, and this problem can be fixed by changing P04.02 parameter.

P07.08 is AI1 analog input and the offset parameter of preload torque compensation. If AI1 analog voltage input signal is not equal to 0 when balancing the load, the parameter of P07.08 should be set. Or else, value of P07.08 should be the default factory value 10.00.

Setting formula of P07.08 is shown below:  $P07.08 = 10.000 - \text{AI1 analog input port's real zero offset (AI1 analog voltage input signal when balancing load)}$ .

For example: If AI1 analog input's voltage zero offset is 0.100V, then:

$$P07.08 = 10.000 - 0.100 = 9.900.$$

When actually performing adjustment, the following method can also be used: When elevator is balancing the load, set the elevator's inspection speed to be 0, and adjust P07.08 parameter until the car can keep still when elevator runs in inspection state.

P07.09 is AI1 analog input and the gain parameter of preload torque compensation. P07.09 can be adjusted by using the following mode on condition that P04.02 (preload compensation direction) has been adjusted well: Because increasing this parameter will increase the compensation effect under the conditions of the same analog port compensation input; Compensation effect will be decreased vice versa. So, if there is downward shock phenomenon (i.e. slip when ascending and too fast when descending) in the case of heavy load start or upward shock phenomenon (i.e. slip when descending and too fast when ascending) in the case of light load, this means that compensation is not enough, and the gain parameter of P07.09 can be increased properly; On the contrary, if there is upward shock phenomenon (i.e. slip when descending and too fast when ascending) in the case of heavy load start or downward shock phenomenon (i.e. slip when ascending and too fast when descending) in the case of light load, this means that compensation is too great, and the gain parameter of P07.09 can be decreased properly.

P04.03 is the parameter of torque compensation gain, and its function is as same as P07.09 during the process of torque compensation adjustment of preload start. So the torque compensation effect of preload start above can also be improved by adjusting P04.03 parameter instead of P07.09. Likewise, P04.04 is the parameter of torque compensation offset, and its function is as same as P07.07 during the process of torque compensation adjustment of preload start. For detailed information of parameter P04.03 and P04.04, [see section 6.2.5](#).

P07.10 is the filtering time parameter of AI1 analog port input. Generally, it can be set to be 10 (default value). If the interference is too intense on site and this problem can not be fixed through the hardware processing, a bit of filtering time can be added to enhance the interference immunity of analog port input signal. But the filtering time should not be too long, or else the response capability and effect of preload start compensation will be affected.

### 7.9.2 Starting compensation method for using light and heavy switch

**AS320** elevator-used inverter can also adopt another easy starting compensation mode in addition to the starting compensation of analog input's preload: Starting compensation mode for using light and heavy switch. When this type of starting compensation is used, encoder can be the 8192 pulse's A, B, Z phase incremental encoder, and no accurate weighing device is needed. Only two inching switches should be added in car platform. For the elevator with synchronous gearless traction motor, if the no load compensation start mode is used, the SIN/COS encoder with higher resolution must be used. By comparing SIN/COS encoder with A, B, Z phase incremental encoder, SIN/COS encoder is costly and has much wiring and weak interference immunity. So, comparing with no weighing compensation start mode, the start compensation mode by using light and heavy switch is cheap, has less wiring and strong interference immunity. Comparing with the preload starting compensation mode of analog input, it has the benefits of low cost, easy installation and convenient commissioning, etc. because of eliminating the installation of an accurate weighing device. So, for those customers who use **AS320** elevator-used inverter, the starting compensation mode by using light and heavy load switch is recommended.

If starting compensation mode by using light and heavy load switch is adopted, the car platform should add one light load switch and one heavy load switch. Its light load switch will work when the car load is less than 25% of related load, and its heavy load switch will work when car load is more than 75% of related load. These two switches' input signal should be directed to the corresponding X input point of inverter.

Main parameters that should be adjusted when using the light and heavy switch starting compensation are shown in the table below.

| Function code | Designation                             | Content                                                                                                                                                                                                             | Setting range | Unit | Factory Setting | Remarks                                               |
|---------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|-------------------------------------------------------|
| P04.01        | Torque compensation reference selection | Setting torque reference selection:<br><br>0: No torque compensation<br><br>1: Giving the compensation according to the weighing switch<br><br>2: AI0 analog torque reference<br><br>3: AI1 analog torque reference | 0/1/2/3       | ×    | 0               |                                                       |
| P04.02        | Torque compensation direction           | Setting torque compensation direction:<br><br>0: Forward direction<br><br>1: Backward direction                                                                                                                     | 0/1           | ×    | 0               |                                                       |
| P04.05        | Light load switch compensation Quantity | The downwards torque's compensation amount when setting the actuation of light load switch                                                                                                                          | 0.0 ~ 100.0   | %    | 0.0             | Valid only when P04.01 is set to be 1                 |
| P04.06        | Heavy load switch compensation Quantity | The upwards torque's compensation amount when setting the actuation of heavy load switch                                                                                                                            | 0.0 ~ 100.0   | %    | 0.0             | Valid only when P04.01 is set to be 1                 |
| P05.00        | X0 terminal input                       | <u>Setting X0 input function. For detailed meaning of function</u>                                                                                                                                                  |               | ×    | 0               | It is recommended to set P05.00 to be 19, in order to |



|        |                                           |                                                                                     |  |   |   |                                                                                                |
|--------|-------------------------------------------|-------------------------------------------------------------------------------------|--|---|---|------------------------------------------------------------------------------------------------|
|        | function<br>Definition.                   | code, see section 6.2.6                                                             |  |   |   | define X0 as the light load switch input                                                       |
| P05.01 | X1 terminal input function<br>Definition. | Setting X1 input function. For detailed meaning of function code, see section 6.2.6 |  | × | 0 | It is recommended to set P05.01 to be 20, in order to define X1 as the heavy load switch input |

**Note 1:** Parameter setting that is a must for load compensation method of heavy and light load

Firstly, P04.01 must be set to be 1, and the torque compensation reference mode must be defined as the torque compensation mode of light and heavy load switch signal input; Meanwhile, P05.00 should be set to be 19, and X0 should be defined to be light load input point; P05.01 should be set to be 20, and X1 to be heavy load switch input point (for the input point selection of light and heavy load switch, see further instruction of Note 3 below).

**Note 2:** Usage of adjustment parameters

P04.02 is the setting parameter of preload compensation direction. If the torque compensation direction of system is reversed by some reason, this problem can be fixed by change the P04.02 parameter from 0 to 1 simply. The method of judging the correctness of torque compensation direction is as follows: In the case of light load (light load switch acts), if increasing P04.05 parameter can decrease the upward shock or increase the downward shock during elevator start, the compensation direction is correct; On the contrary, if increasing this data can decrease the downward shock or increase the upward shock during elevator start, the compensation direction is wrong and should be corrected. Likewise, in the case of heavy load (heavy load switch acts), if increasing P04.06 parameter can decrease the downward shock or increase the upward shock during elevator start, the compensation direction is correct; On the contrary, if increasing this data can decrease the upward shock or increase the downward shock during elevator start, the compensation direction is wrong and should be corrected.

P04.05 and P04.06 are two adjustment parameters based on the simple torque compensation of load and heavy load switch. These two switches are connected to the digital input point of inverter (X0 and X1). P04.05 is the adjustment parameter for simple light load's torque compensation amount when the light load switch acts. If there is upward shock phenomenon (i.e. slip when descending and too fast when ascending) in the case of light load start, this means the light load compensation is not enough, and P04.05 can be increased to fix this problem; On the contrary, if there is downward shock phenomenon (i.e. slip when ascending and too fast when descending) in the case of light load start, this means the light load compensation is too large, and P04.05 can be decreased properly to fix this problem; Likewise, P04.06 is the adjustment parameter for simple heavy load's torque compensation amount when the heavy load switch acts. If there is downward shock phenomenon (i.e. slip when ascending and too fast when descending) in the case of heavy load start, this means the heavy load compensation is not enough, and P04.06 can be increased to fix this problem; On the contrary, if there is upward shock phenomenon (i.e. slip when descending and too fast when ascending) in the case of heavy load start, this means the heavy load compensation is too large, and P04.06 can be decreased properly to fix this problem;

**Note 3:** Selection of light and heavy load switch input point

According to the application features of this inverter in elevator-used control system, we recommend in Note 1 above to use X0 as the light load switch signal input point (P05.00 is set to be 19) and X1 as the heavy load switch signal input point (P05.01 is set to be 20). In addition to X0 and X1, customer can also define other input points as the input points of light and heavy load switch signals. In this inverter, if the speed reference selection of analog voltage signal input is used, then the three input points of X2 ~ X4 can be redefined; In addition, if the inverter base block signal is not used, then X5 can also redefine the input function. So, in addition to X0 and X1, you can select two terminals from four points of X2 ~ X5 as the light and heavy load switch signal input point provided that the conditions above are met. Here, define the corresponding parameter of one of the input points to be 19 (setting it as light load switch input), and the corresponding parameter of the other to be 20 (setting it as heavy

load switch input), according to the coincidence relation of the input points and definition function parameters in the table above.

### 7.9.3 Bus low voltage operation mode used for emergency leveling

If there is an emergency power off during the operation of elevator, the elevator may be stuck in the shaft and the passenger may be trapped in the car until the related personnel get informed and rescue the passenger by manual barring gear with great care. This certainly will bring a lot of trouble for the passengers. So, one power off emergency leveling device should be added to the elevator in many cases. But an independent emergency leveling device is costly. So, the inverter low voltage and low speed operation function should be added (generally 4 batteries of 48V or 220V UPS power supply can work as the emergency power to power elevator) and special treatment should be done to elevator controller, then a simple emergency leveling can be made. Thus the passengers can be rescued in case of emergency power off and the cost can be reduced.

When the bus runs in low voltage, a digital input point must be defined as the input port of the emergency power run signal (bus low voltage operation) firstly. Therefore the parameter range that should be set is shown in the table below.

| Function code | Designation                            | Content                                                                             | Setting range | Unit | Factory Setting | Remarks                                                                                                     |
|---------------|----------------------------------------|-------------------------------------------------------------------------------------|---------------|------|-----------------|-------------------------------------------------------------------------------------------------------------|
| P05.00        | X0 terminal input function Definition. | Setting X0 input function. For detailed meaning of function code, see section 6.2.6 |               | ×    | 0               |                                                                                                             |
| P05.01        | X1 terminal input function Definition. | Setting X1 input function. For detailed meaning of function code, see section 6.2.6 |               | ×    | 0               |                                                                                                             |
| P05.02        | X2 terminal input function Definition. | Setting X2 input function. For detailed meaning of function code, see section 6.2.6 |               | ×    | 3               |                                                                                                             |
| P05.03        | X3 terminal input function Definition. | Setting X3 input function. For detailed meaning of function code, see section 6.2.6 |               | ×    | 4               |                                                                                                             |
| P05.04        | X4 terminal input function Definition. | Setting X4 input function. For detailed meaning of function code, see section 6.2.6 |               | ×    | 5               |                                                                                                             |
| P05.05        | X5 terminal input function Definition. | Setting X5 input function. For detailed meaning of function code, see section 6.2.6 |               | ×    | 18              | It is recommended to set P05.05 to be 16 and define X5 as the signal input of low voltage operation command |

Actually, if one of parameters in the table above is set, i.e. selecting one input port as the input point of the low voltage run signal, it is OK. The recommended practice is: Set P05.05 to be 16 and define X5 as the input port of emergency power run signal (provided that system does not use the base block signal). At this point, when X5 switches on, the inverter will enter into the bus low voltage operation mode, i.e. the inverter can also work even under the low voltage down to 48V. There are two types of bus low voltage operation mode to power the inverter, i.e. the battery mode in case of emergency power off and UPS mode. Its main loop's basic circuit is shown as in Figure 7.10. Voltage of battery is 48V (general 4 batteries of 12V in series) and its current is not less than 20Ah. Specially note that: Emergency power must be added if this mode is adopted.

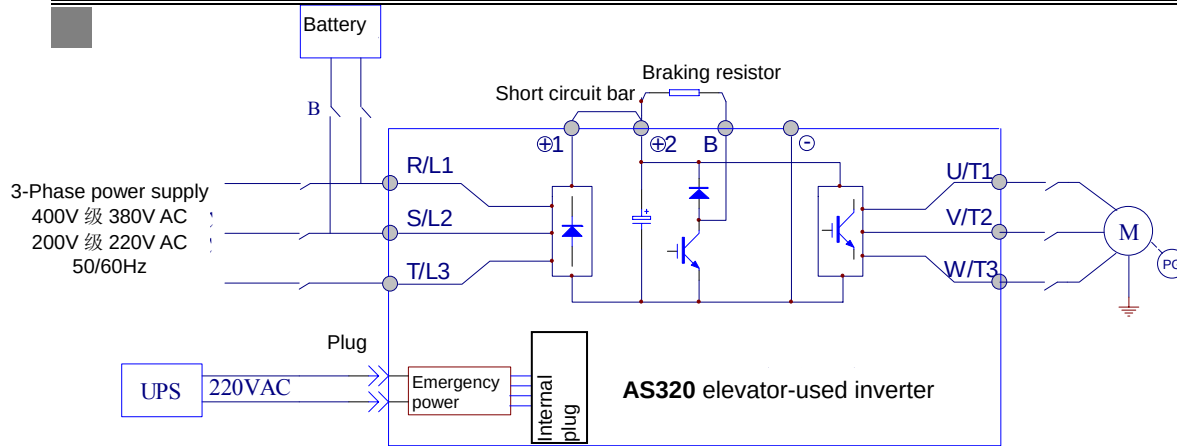


Figure 7.12 Basic circuit diagram of bus low voltage operation powered by both UPS and battery

When the power is off in the second mode, either the drive power or control power is powered by UPS, so the battery is not needed. Its main loop's basic circuit is shown as in Figure 7.13.

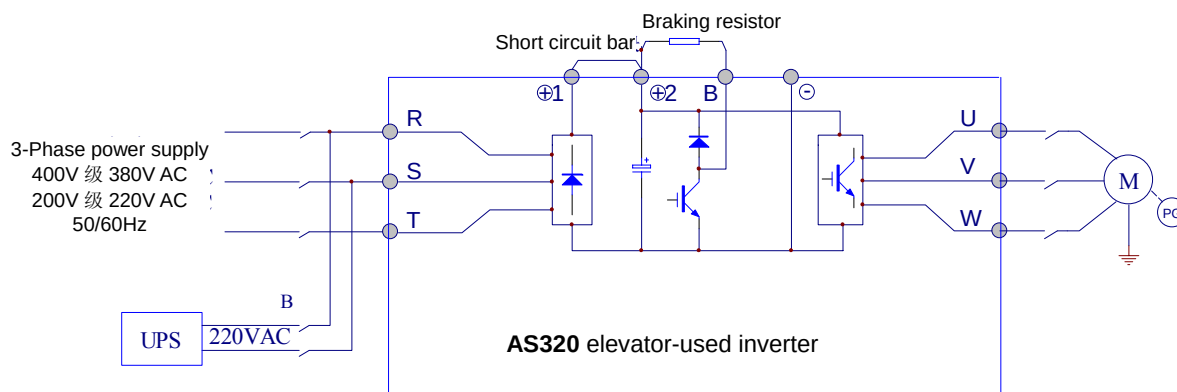


Figure 7.13 Basic circuit diagram of bus low voltage operation powered by UPS only

What needs to be cleared is the following content: Contactor B in the diagram above only closes in the case of emergency power off, or else different power conflict will rise up leading into serious fault.

During the operation process of bus low voltage used as function of emergency leveling, both UPS and battery (if any) will together power the inverter and elevator controller in the case of emergency power off. When the elevator controller identifies the emergency power off state through a special contact detection device, a switching on signal will be output to X5 (that has been set as the emergency power run signal) input point of inverter to point out the emergency leveling operation state available. Then control the inverter to reach the nearest leveling position in certain direction with low speed and stop, then the emergency leveling process is over after the door is opened.

#### 7.9.4 Wiring and parameter setting method in the main loop of one contactor

According to GB-7588 national elevator safety standard, the main loop must conform to one of following two clauses:

- There are two independent contactors, each of which can cut the traction motor loop, and if one of the contactors has main contact closed when elevator stops, the car must be prevented from operating again till the next operation indirection changes at the latest. Or:
- There is one contactor used for cutting traction motor loop and at the same time there is a control device that can block the current flow in the static element, and also a monitoring device that can inspect the current flow blockage each time elevator stops, and the regulation that: If the static element fails to block the current flow

during the normal stop, the control device should make the contactor release and prevent the elevator operating again.

If inverter fails to issue the timely and accurate signal of judging whether the elevator can block current flow or not during the stop, the control system of elevator doesn't have the monitoring device stated in (b), then method of using two contactors in main loop should be adopted. The **AS320** series inverter can accurately detect whether the current flow can be blocked or not during the stop of elevator and can output the detection results timely. So control system can feature the monitoring device stated in (b) above only considering **AS320** series inverter into the design. Thus a contactor can be used in main loop in order to reduce the elements and cost.

Therefore, firstly define a digital output point to output the current detection signal above, at the same time set the threshold parameter of zero current detection. Related parameters are shown below.

| Function code | Designation                                      | Content                                                                                                                                                                                                                                                                                                                           | Setting range | Unit | Factory Setting | Remarks                                                                                                     |
|---------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|-------------------------------------------------------------------------------------------------------------|
| P06.18        | Non-zero current detection threshold during stop | Set the non-zero current detection threshold during stop. When inverter has the current flow larger than this threshold, then the non-zero current detection signal is effective. It is a data of percentage and the real value is the result that this data is multiplied by related current then the product is divided by 100. | 0.0 ~ 100.0   | %    | 2.0             | P06.18                                                                                                      |
| P06.00        | K1 terminal input function Definition.           | Setting K1 input function. For detailed meaning of function code, see section 6.2.7                                                                                                                                                                                                                                               |               | ×    | 0               | It is recommended that P06.00 is set to be 15 and K1 is defined as non-zero current detection output signal |
| P06.01        | K2 terminal input function Definition.           | Setting K2 input function. For detailed meaning of function code, see section 6.2.7                                                                                                                                                                                                                                               |               | ×    | 0               |                                                                                                             |

The recommended practice is: If P06.00 is set to be 15, then K1 (relay) output can be defined as output non-zero current detection signal. Or do not use K1, but define K2 as the output of non-zero current detection signal (set P06.01 to be 15). In addition, it is necessary to set P06.18 parameter (non-zero current detection threshold). P06.18 can be set to be 2% or so. This data can not be set to be too large or too small. If too large, the protection effect is not adequate and system is not safe enough; but if too small, the system will be too sensitive, so it is easy to generate the error protection and the reliability is poor. Figure 7.14 gives a wiring example.

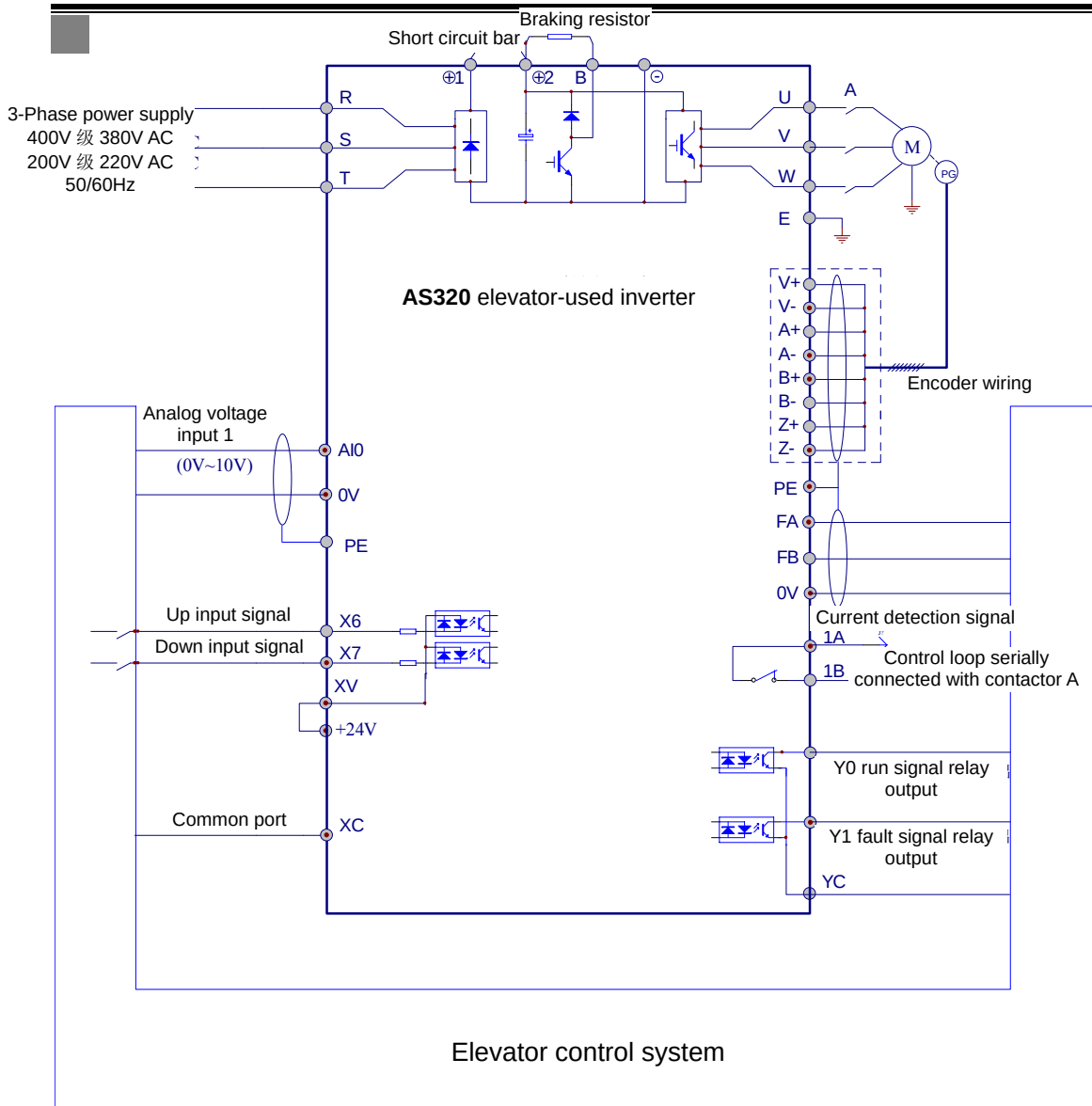


Figure 7.14 Wiring example of only using one contactor

In Figure 7.14, the inverter output K1 relay's NC contact is serially connected to the coil loop of contactor A in main loop. If the inverter cannot detect the current flow after the elevator stops, K1 relay closes and its NC contact is opened, then the power of contactor A in main loop is cut off, so the contactor A cannot close (or release) and elevator cannot start. So, it can meet the requirements of (b) above mentioned.



## Chapter 8 Fault Check

This chapter addresses the detailed fault, fault code, content, reason and countermeasures during the operation of inverter, and gives the analysis process for all fault phenomena during elevator commissioning and operation.



**Danger**

⊙ **Maintenance operation can be done 10 minutes after input power is off, when the charging indicator is off completely or DC bus voltage is below 24VDC.**

Or else there will a danger of electric shock.

⊙ **Never retrofit inverter by yourself.**

Or else there will be a danger of electric shock and/or personnel injury.

⊙ **Only the qualified electrical mechanics are allowed to perform the maintenance. It is prohibited to leave the stub or metal inside the inverter.**

Or else, there may be a danger of fire hazard.



**Attention**

⊙ **Never change wiring or connect or disconnect the terminal under energized state.**

Or else there will a danger of electric shock.

### 8.1 Protection and check functions

When inverter fails to work, the fault indicator LED above digital operator lights up and LED digital tube display the current fault code in real time.

Inverter has 39 fault codes in total. For fault reasons and countermeasures corresponding to the fault code, see Table 8.1 Fault List.

Table 8.1 Fault list

| Fault code | Fault display                  | Possible reasons                 | Countermeasures                                                                                                    |
|------------|--------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1          | Module over current protection | DC terminal voltage is too high  | Check grid power, and check whether there is mass load and non-power consumption braking fast stop.                |
|            |                                | The peripheral has short circuit | Check that motor and its output connection have no short circuit and that the grounding wire has no short circuit. |
|            |                                | Output has phase loss            | Check that motor and its output connection are not loosened.                                                       |
|            |                                | Encoder is fault                 | Check that encoder has no damage and wiring is correct                                                             |
|            |                                | Encoder phase fault              | Check encoder phase                                                                                                |

| Fault code | Fault display                                | Possible reasons                             | Countermeasures                                                                 |
|------------|----------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------|
|            |                                              | Motor phase fault                            | Check motor phase                                                               |
|            |                                              | Phase angle self-study fault                 | Restudy phase angle                                                             |
|            |                                              | Lack current when phase angle self-studying  | Increase current gain                                                           |
|            |                                              | Hardware has poor contact or is damaged      | Ask the professional personnel to maintain it                                   |
|            |                                              | Inverter's internal plug-in part is loosened | Ask the professional personnel to maintain it                                   |
|            |                                              | Induction motor slip setting unreasonable    | Adjust the induction motor slip                                                 |
|            |                                              | No-load current coefficient set unreasonable | Adjust the no-load current coefficient                                          |
| 2          | ADC fault                                    | Current sensor is damaged                    | Replace the current sensor                                                      |
|            |                                              | There is a problem in current sampling loop  | Replace the control board                                                       |
| 3          | Heatsink overheat                            | Ambient temperature is too high              | Reduce the ambient temperature and enhance the ventilation and heat elimination |
|            |                                              | Air duct is blocked                          | Clean the air duct dust and flocks, etc.                                        |
|            |                                              | Fan is abnormal                              | Check that fan power cable is OK or replace it with the fan of same model       |
|            |                                              | Temperature detection circuit failed to work | Ask the professional personnel to maintain it                                   |
| 4          | Braking unit failed to work                  | Braking unit is damaged                      | Replace the related drive module                                                |
|            |                                              | External braking resistor has short circuit  | Check the braking resistor wiring                                               |
| 5          | Fuse blown fault                             | Over current made the fuse blown             | Check the fuse loop is connected well and the connection point is not loosened  |
| 6          | Output over large torque                     | Input power voltage is too low               | Check input power supply                                                        |
|            |                                              | Motor is locked or load is changed abruptly  | Prevent the motor from being locked and reduce the load abrupt change           |
|            |                                              | Encoder is fault                             | Check that encoder has no damage and wiring is correct                          |
|            |                                              | Output has phase loss                        | Check that motor and its output connection are not loosened.                    |
| 7          | Speed deviation                              | Acceleration time is too short               | Extend the Acceleration time                                                    |
|            |                                              | Load is too heavy                            | Reduce the load                                                                 |
|            |                                              | Current limitation is too low                | Increase the current limit value within the proper range                        |
| 8          | (in accelerated running) DC bus over-voltage | Input power supply is abnormal               | Check input power supply                                                        |
|            |                                              | Motor quickly starts again during its        | Motor starts again after stop                                                   |



| Fault code | Fault display                                              | Possible reasons                                                                              | Countermeasures                                                                                                            |
|------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|            | protection                                                 | high speed rotating                                                                           |                                                                                                                            |
|            | (in deceleration running) DC bus over-voltage protection   | The load moment of inertia is too large                                                       | Use the proper power consumption braking component                                                                         |
|            |                                                            | Deceleration time is too short                                                                | Extend the deceleration time                                                                                               |
|            |                                                            | Braking resistance is too large or no braking resistor is connected                           | Connect the proper braking resistor                                                                                        |
|            | (in constant speed running) DC bus over-voltage protection | Input power supply is abnormal                                                                | Check input power supply                                                                                                   |
|            |                                                            | The load moment of inertia is too large                                                       | Use the proper power consumption braking component                                                                         |
|            |                                                            | Braking resistance is too large or no braking resistor is connected                           | Connect the proper braking resistor                                                                                        |
| 9          | DC bus undervoltage                                        | Supply voltage is under the minimum operating voltage of equipment                            | Check input power supply                                                                                                   |
|            |                                                            | Instantaneous power off occurs                                                                | Check the input power supply and then restart after reset when the input voltage becomes normal                            |
|            |                                                            | Voltage fluctuation of input power supply is too large                                        |                                                                                                                            |
|            |                                                            | Power terminal is loosened                                                                    | Check the input wiring                                                                                                     |
|            |                                                            | Internal switch power is abnormal                                                             | Ask the professional personnel to maintain it                                                                              |
|            |                                                            | There exists a load of high starting current in the same power supply system                  | Change the power system to adapt the specification                                                                         |
| 10         | Output has phase loss                                      | Inverter output side has the abnormal wiring, grounding missing or broken wire                | Check inverter output side wiring and eliminate the grounding missing and broken wire according to the operation practice. |
|            |                                                            | Output terminal is loosened                                                                   |                                                                                                                            |
|            |                                                            | Motor power is too small, just below 1/20 of the inverter's applicable maximum motor capacity | Adjust the inverter capacity or motor capacity                                                                             |
|            |                                                            | Output three-phase is imbalanced                                                              | Check that motor wiring is in good order                                                                                   |
|            |                                                            |                                                                                               | Cut off the power then check that inverter output side terminal has the same characteristics as DC side terminal           |
| 11         | Motor has over current at low speed (in acceleration run)  | Grid voltage is too low                                                                       | Check input power supply                                                                                                   |
|            |                                                            | Motor parameter setting is not correct                                                        | Set the motor parameters properly                                                                                          |
|            |                                                            | Motor quickly starts directly during its operation                                            | Motor starts again after stop                                                                                              |
|            | Motor has over current at low speed (in deceleration run)  | Grid voltage is too low                                                                       | Check input power supply                                                                                                   |
|            |                                                            | The load moment of inertia is too large                                                       | Use the proper power consumption braking component                                                                         |
|            |                                                            | Motor parameter setting is not correct                                                        | Set the motor parameters properly                                                                                          |

| Fault code | Fault display                                                           | Possible reasons                                                             | Countermeasures                                                                  |
|------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|            | Motor has over current at low speed (in constant speed run)             | Deceleration time is too short                                               | Extend the deceleration time                                                     |
|            |                                                                         | Load changes abruptly during motor operation                                 | Reduce the frequency and amplitude of abrupt load change                         |
|            |                                                                         | Motor parameter setting is not correct                                       | Set the motor parameters properly                                                |
| 12         | Encoder is fault                                                        | Encoder connection is not correct                                            | Change encoder wiring                                                            |
|            |                                                                         | Encoder has no signal output                                                 | Check the order of encoder and the power supply condition                        |
|            |                                                                         | Encoder wiring is broken                                                     | Fix the broken wire                                                              |
|            |                                                                         | Function code setting is abnormal                                            | Confirm that the setting of related function code of inverter encoder is correct |
| 13         | Current is detected when the motor stops                                | Current flow is not cut effectively when motor stops                         | Synchronous motor has slip phenomenon                                            |
|            |                                                                         |                                                                              | Ask the professional personnel to maintain it                                    |
| 14         | Speed is reversed during the running                                    | Speed is converse when motor is running                                      | Check whether the external load changes abruptly                                 |
|            |                                                                         | Encoder does not accord to the motor phase sequence                          | Change the phase sequence of motor or encoder                                    |
|            |                                                                         | Motor rotates reversely when starting, and current reaches the limited value | Current limitation is too low or motor does not match                            |
| 15         | Speed is detected when motor stops                                      | Brake is loosened and elevator slips                                         | Check the brake                                                                  |
|            |                                                                         | Encoder is interfered or encoder is loosened                                 | Fasten the encoder and eliminate the interference                                |
| 16         | Phase sequence of motor is wrong                                        | Motor has the reversely connected wire                                       | Correctly connect the wire or adjust the parameters                              |
| 17         | Overspeed in the same direction (within maximum speed range allowed)    | Synchronous motor runs away under loss of excitation                         | Check the motor                                                                  |
|            |                                                                         | Synchronous motor's angle self-learning is not correct                       | Do the self-learning again                                                       |
|            |                                                                         | Encoder parameter setting is wrong or interfered                             | Check loop of encoder                                                            |
|            |                                                                         | Forward load is too large or there is the abrupt change of load              | Check the external reasons of abrupt change of load                              |
| 18         | Overspeed in the reverse direction (within maximum speed range allowed) | Synchronous motor runs away under loss of excitation                         | Check the motor                                                                  |
|            |                                                                         | Synchronous motor's angle self-learning is not correct                       | Do the self-learning again                                                       |
|            |                                                                         | Encoder parameter setting is wrong or interfered                             | Check loop of encoder                                                            |
|            |                                                                         | Reverse load is too large or there is                                        | Check the external reasons of abrupt                                             |

| Fault code | Fault display                                  | Possible reasons                                                                                | Countermeasures                                                                                                                        |
|------------|------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
|            |                                                | the abrupt change of load                                                                       | change of load                                                                                                                         |
| 19         | UVW encoder phase sequence is wrong            | Encoder wiring has a problem or parameter setting is wrong                                      | Check the wiring or change the parameter                                                                                               |
| 20         | Encoder communication is fault                 | Encoder is fault                                                                                | Check the wiring of encoder and redo the encoder self-learning                                                                         |
| 21         | abc over current (3-phase instantaneous value) | Motor single phase is shorted to earth                                                          | Check motor and the output line loop                                                                                                   |
|            |                                                | Encoder is fault                                                                                | Check that encoder has no damage and wiring is correct                                                                                 |
|            |                                                | Encoder phase fault                                                                             | Check encoder phase                                                                                                                    |
|            |                                                | Motor phase fault                                                                               | Check motor phase                                                                                                                      |
|            |                                                | Phase angle self-study fault                                                                    | Restudy phase angle                                                                                                                    |
|            |                                                | Lack current when phase angle self-studying                                                     | Increase P09.40 current gain                                                                                                           |
|            |                                                | Drive board test loop is fault                                                                  | Replace the drive board                                                                                                                |
|            |                                                | Induction motor slip setting unreasonable                                                       | Adjust the induction motor slip                                                                                                        |
|            |                                                | No-load current coefficient set unreasonable                                                    | Adjust the no-load current coefficient                                                                                                 |
| 22         | Brake detector is fault                        | Output relay does not work                                                                      | Check the relay control loop                                                                                                           |
|            |                                                | When relay acts, the brake is not released                                                      | Check that the brake power line is not loosened or broken                                                                              |
|            |                                                | Feedback element does not detect the signal                                                     | Adjust the feedback element                                                                                                            |
| 23         | Input over-voltage                             | Incoming voltage is too high                                                                    | Check whether incoming voltage matches inverter or not                                                                                 |
|            |                                                | There a problem in the switching power supply's voltage detection loop                          | Ask the professional personnel to maintain it                                                                                          |
| 24         | UVW encoder has broken wire                    | Encoder wiring loop has a problem                                                               | Terminal is loosened or there is damage and breaking in the line                                                                       |
| 25         | Standby                                        |                                                                                                 |                                                                                                                                        |
| 26         | Encoder does not do self-learning              | Synchronous motor does not learn encoder angle                                                  | Do the encoder self-learning                                                                                                           |
| 27         | Output over current (effective value)          | When the over load state lasts very long time, the larger the load is, the shorter the time is. | If the stop occurs again during the running after lasting a period of time, check whether the load is within the range allowed or not. |
|            |                                                | Motor is blocked                                                                                | Check the motor or brake                                                                                                               |
|            |                                                | Motor coil is shorted                                                                           | Check the motor                                                                                                                        |
|            |                                                | Encoder phase fault                                                                             | Check encoder phase                                                                                                                    |
|            |                                                | Motor phase fault                                                                               | Check motor phase                                                                                                                      |

| Fault code | Fault display                                                       | Possible reasons                                                           | Countermeasures                                                                                                            |
|------------|---------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|            |                                                                     | Phase angle self-study fault                                               | Restudy phase angle                                                                                                        |
|            |                                                                     | Lack current when phase angle self-studying                                | Increase current gain                                                                                                      |
|            |                                                                     | Output is shorted                                                          | Check the wiring or motor                                                                                                  |
| 28         | Sincos encoder is fault                                             | Encoder is damaged or its connection is wrong                              | Check encoder and its wiring                                                                                               |
| 29         | Input phase loss                                                    | Input side voltage abnormal                                                | Check the grid voltage                                                                                                     |
|            |                                                                     | Input voltage has phase loss                                               |                                                                                                                            |
|            |                                                                     | Input side terminal is loosened                                            | Check the input side terminal connection                                                                                   |
| 30         | Overspeed protection (exceeding the maximum speed protection limit) | Encoder parameter setting is wrong or interfered                           | Check loop of encoder                                                                                                      |
|            |                                                                     | Abrupt change of load                                                      | Check the external reasons of abrupt change of load                                                                        |
|            |                                                                     | Overspeed protection parameter setting is wrong                            | Check the parameter                                                                                                        |
| 31         | Motor has over current at high-speed                                | Grid voltage is too low                                                    | Check input power supply                                                                                                   |
|            |                                                                     | Load changes abruptly during motor operation                               | Reduce the frequency and amplitude of abrupt load change                                                                   |
|            |                                                                     | Motor parameter setting is not correct                                     | Set the motor parameters properly                                                                                          |
|            |                                                                     | Encoder parameter setting is wrong or interfered                           | Check loop of encoder                                                                                                      |
| 32         | Grounding protection                                                | Wiring is wrong                                                            | Correct the wrong wiring by checking the user manual instruction                                                           |
|            |                                                                     | Motor is abnormal                                                          | Change the motor and test the insulation against ground firstly                                                            |
|            |                                                                     | Inverter output side has too large drain current against ground            | Ask the professional personnel to maintain it                                                                              |
| 33         | Capacitor is aged                                                   | Inverter capacitor is aged                                                 | Ask the professional personnel to maintain it                                                                              |
| 34         | External fault                                                      | External circuit has input fault signal                                    | Check the external fault reason                                                                                            |
| 35         | Output is imbalanced                                                | Inverter output side has abnormal wiring, grounding missing or broken wire | Check inverter output side wiring and eliminate the grounding missing and broken wire according to the operation practice. |
|            |                                                                     | Motor 3-phase is imbalanced                                                | Check the motor                                                                                                            |
| 36         | Parameter setting is wrong                                          | Parameter setting is incorrect                                             | Change inverter's parameters                                                                                               |
| 37         | Current sensor is fault                                             | Drive board has fault hardware                                             | Ask the professional personnel to maintain it                                                                              |
| 38         | Braking resistor is                                                 | External braking resistor has short                                        | Check the braking resistor wiring                                                                                          |

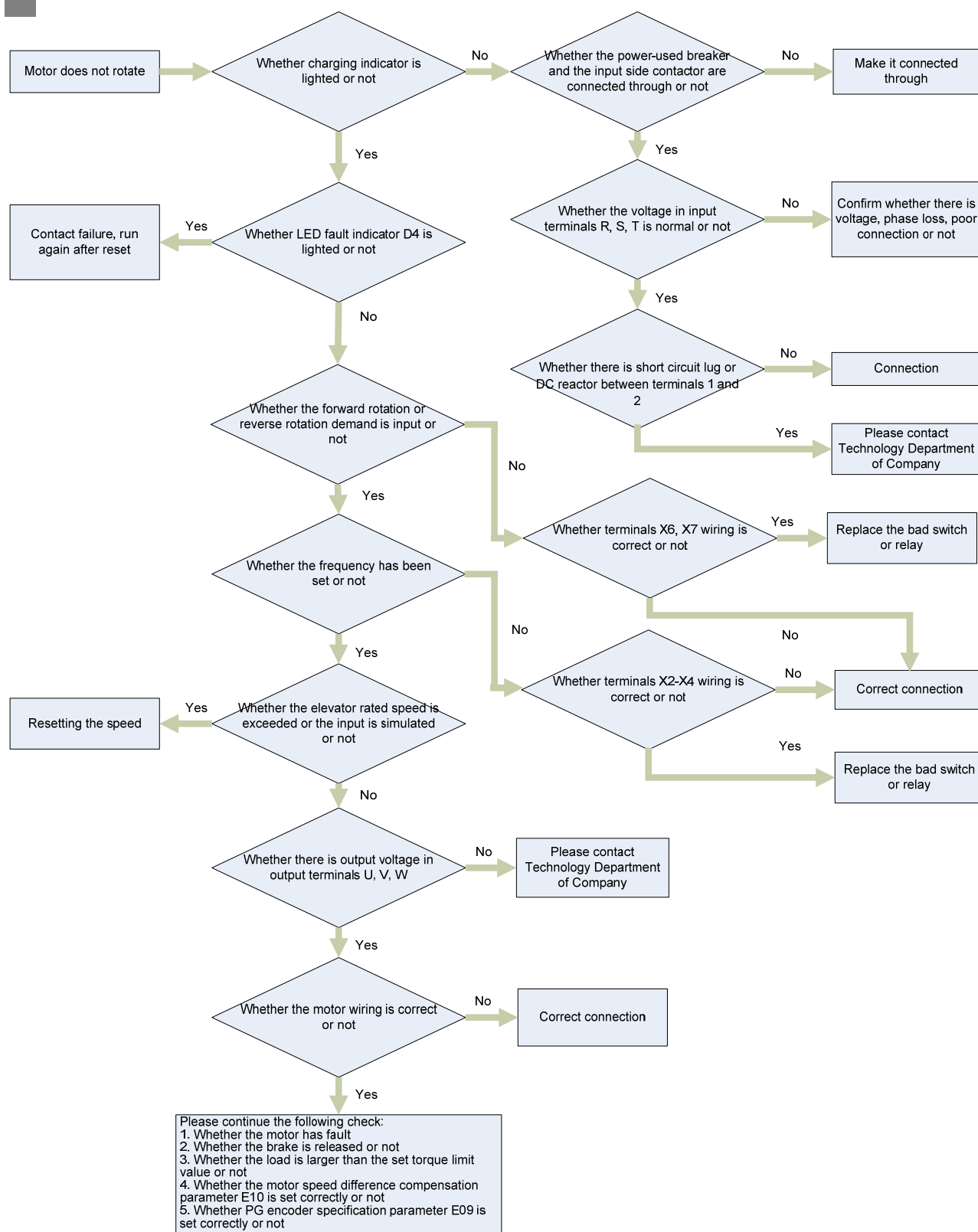
| Fault code | Fault display                           | Possible reasons                                                                                 | Countermeasures                                                                                        |
|------------|-----------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|            | shorted                                 | circuit                                                                                          |                                                                                                        |
| 39         | Current instantaneous value is too high | When Ia, Ib, Ic do not work, 3-phase current instantaneous value is too high and alarm is issued | Ask the professional personnel to maintain it                                                          |
| 42         | IGBT Short circuit protection           | There is short circuit in the circuit                                                            | Check if there is short circuit in motor or output wiring<br>Check if there is short circuit to ground |
| 44         | Voltage drop when running               | Input voltage drop when running due the the capacity of power                                    | Check the capacity of power                                                                            |
|            |                                         | Charge relay in inverter is damaged                                                              | Connect to STEP                                                                                        |

## 8.2 Fault diagnosis process

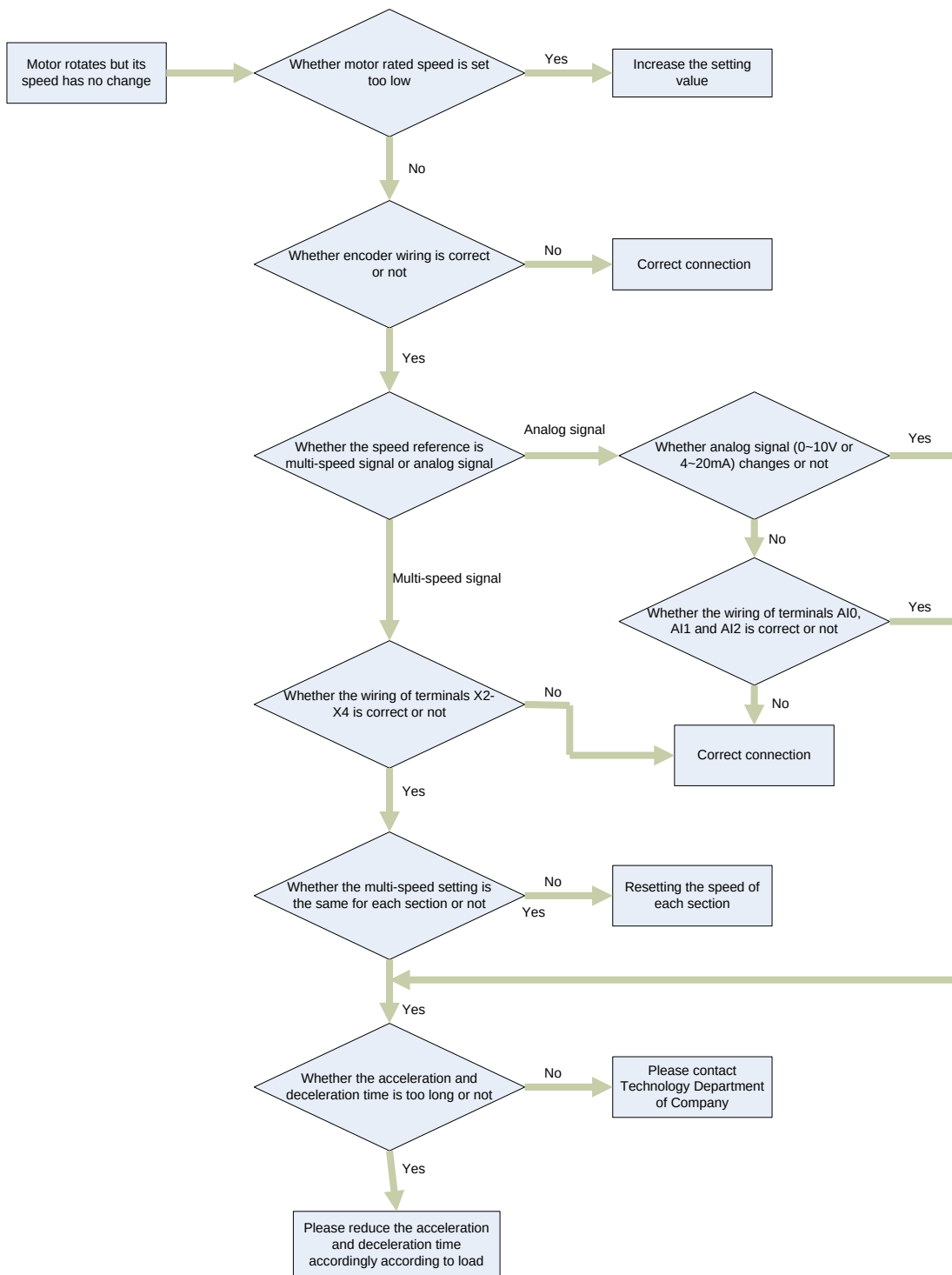
When system starts, the operation of inverter and motor may not accord to the setting sometimes due to the error of parameter setting or wiring. In this case, please analyze and address the problem with reference to the fault diagnosis process in this section.

Motor has abnormal operation:

- a) When the control terminal has operation demand, the motor does not rotate.



b) Motor rotates but its speed has no change







## Chapter 9 Service and Maintenance

This chapter introduces the general information about service and maintenance.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  <b>Danger</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| <p>⊙ <b>Maintenance operation can be done 10 minutes after input power is off, when the charging indicator is off completely or DC bus voltage is below 24VDC.</b></p> <p>Or else there will a danger of electric shock.</p> <p>⊙ <b>Never retrofit inverter by yourself.</b></p> <p>Or else there will be a danger of electric shock and/or personnel injury.</p> <p>⊙ <b>Only the qualified electrical mechanics are allowed to perform the maintenance. It is prohibited to leave the stub or metal inside the inverter.</b></p> <p>Or else, there may be a danger of fire hazard.</p> |  |
|  <b>Attention</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| <p>⊙ <b>Never change wiring or connect or disconnect the terminal under energized state.</b></p> <p>Or else there will a danger of electric shock.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |

### 9.1 Warranty Period

If inverter (main body) has the following condition, our company will provide the maintenance service:

If failure or damage is incurred under the normal operation, the manufacturer will be responsible for the repair and maintenance within the warranty period (since the date of leaving the factory); If the inverter is beyond the warranty period, the appropriate maintenance cost will be charged.

If the fault is caused by the following reasons, some cost will be charged even within the warranty period:

- 1) Problems caused by usage that fails to observe the instruction or by the unauthorized repair or retrofit.
- 2) Problems caused by usage beyond the requirements of standard specification.
- 3) Damage caused by falling or during the transportation process after being sold.
- 4) Damage caused by earthquake, fire, flood, lightning, abnormal voltage and/or other natural disaster and/or secondary disaster.

### 9.2 Product inquiry

If it is found that the product is damaged, failed to work or has other problems, please contact the office or customer service department of our company in the terms of the following items.

Inverter model

Serial number of production

Purchasing date

Problems worthy of contact include: Damage state, unclear question and the existing failure, etc.

### 9.3 Daily Check

Inverter enclosure can not be removed if energized or during operation, perform the external visual check to confirm that the operation state of inverter is normal. Daily check includes the following items:

- a) Whether ambient environment conforms to the standard specification or not;
- b) Whether the operation performance conforms to the standard specification or not;
- c) Whether there is abnormal noise, vibration and abnormality;
- d) Whether the cooling fan installed in inverter operates normally or not;
- e) Whether there is overheat phenomenon or not.

### 9.4 Regular check

When performing regular check, stop the operation firstly, then cut off the power supply, finally remove the enclosure. At this time, the charged capacitor in main circuit still contains the charging voltage, and the capacitor needs some time to discharge the electric energy completely. So please wait until the charge indicator off, then use the universal meter to confirm that the DC bus voltage is lower than safety value (below DC 24V) or not, then the check can be carried out.

If you contact the terminal immediately after the power supply is cut off, there may be a danger of electric shock.

For regular items, see Table 9.1.

Table 9.1 Regular check items

| Check parts                          |                                  | Check items                                                                                                                                                              | Check methods                                               | Judge standards                                                                                                                                                |
|--------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operation environment                |                                  | 1) Confirm the ambient temperature, humidity, vibration and dust, corrosive gas, oil mist and water drop, etc.<br>2) Confirm whether there is dangerous material around. | 1) Visual check, thermometer, hygrometer<br>2) Visual check | 1) Ambient temperature is below 40 °C . Other requirements, such as humidity, conforms to the environmental requirements.<br>2) There is no dangerous material |
| LCD                                  |                                  | 1) Whether LCD display is clear or not and backlight is even or not<br>2) Whether LCD display lacks numeric alphabetic                                                   | Visual check                                                | 1) Backlight is even<br>2) Display is normal                                                                                                                   |
| Connector assembly<br>Terminal, bolt |                                  | 1) Whether bolt is loosened or not<br>2) Whether connector assembly is loosened or not                                                                                   | 1) Tighten it<br>2) Visual check                            | 1) There is no abnormality<br>2) The installation is secured                                                                                                   |
| Main circuit                         | Conductor                        | 1) Whether shielded layer has been broken or has discoloration<br>2) Whether the connection copper bar has deformation or not                                            | Visual check                                                | There is no abnormality                                                                                                                                        |
|                                      | Electromagnetic contactor, relay | 1) Whether there is vibration and noise when working<br>2) Whether connecting points contact or are attracted                                                            | Audio check, visual check                                   | 1) Non<br>2) There is sound of contact pick-up                                                                                                                 |

|                 |                         |                                                                                                                                                              |                                                                                                                                                            |                                                        |
|-----------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
|                 | Charged capacitor       | 1) Whether there is liquid leakage, discoloration, crack and enclosure expansion<br>2) Whether the safety valve goes out or valve body has obvious expansion | Visual check                                                                                                                                               | There is no abnormality                                |
|                 | Heatsink fin            | 1) Whether there is dust accumulated or not<br>2) Whether the fan air duct is blocked or attached with foreign substance.                                    | Visual check                                                                                                                                               | There is no abnormality                                |
|                 | Cooling fan             | 1) Whether there is abnormal noise<br>2) Whether there is abnormal vibration<br>3) Whether there is discoloration and/or deformation caused by overheat      | 1) Perform audio check, visual check, rotate the fan blade manually after cutting the power supply.<br>2) Visual check<br>3) Visual check, olfaction check | 1) Rotating steadily<br>2), 3) There is no abnormality |
| Control circuit | Connection Plug-in unit | Whether the double-row connecting plug-in unit between control board and main circuit has dust accumulated and is attached with foreign substance.           | Visual check                                                                                                                                               | There is no abnormality                                |
|                 | Control board           | 1) Whether the control circuit board has discoloration and odor or not<br>2) Whether the control circuit board has crack, damage and deformation             | 3) Visual check, olfaction check<br>4) Visual check                                                                                                        | There is no abnormality                                |



## Appendix A Installation Guide to Inverter EMC

This appendix introduces the design and installation guide to inverter EMC for users' reference in the aspects such as noise suppression, wiring requirements, grounding, external equipment surge absorption, leakage current, installation area division and installation precautions, how to use power filters and radiation noise treatment.

### A.1 Noise suppression

The working principle of inverters makes them inevitable to produce certain noise whose influence on peripheral equipment is related to factors such as the type of noise, noise transmission path and the design, installation and wiring of the drive system.

#### A.1.1 Types of noise

The types of noise are shown in the following Figure A.1.

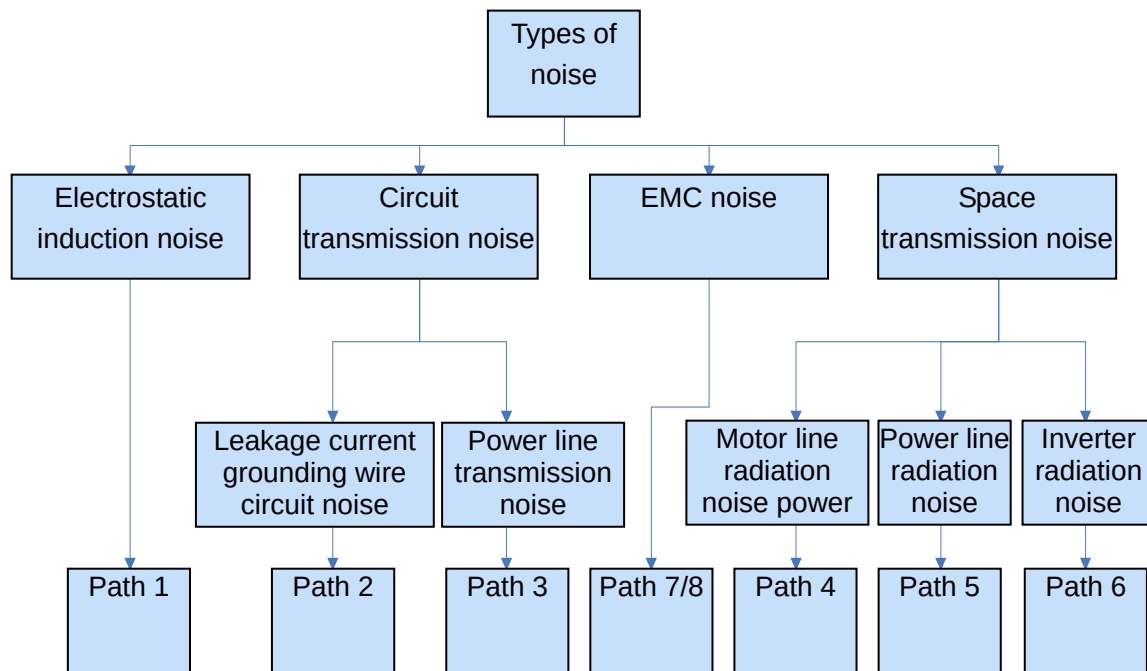


Figure A.1 Schematic of the types of noise

#### A.1.2 Noise transmission path

Noise transmission path is as shown in Figure A.2.

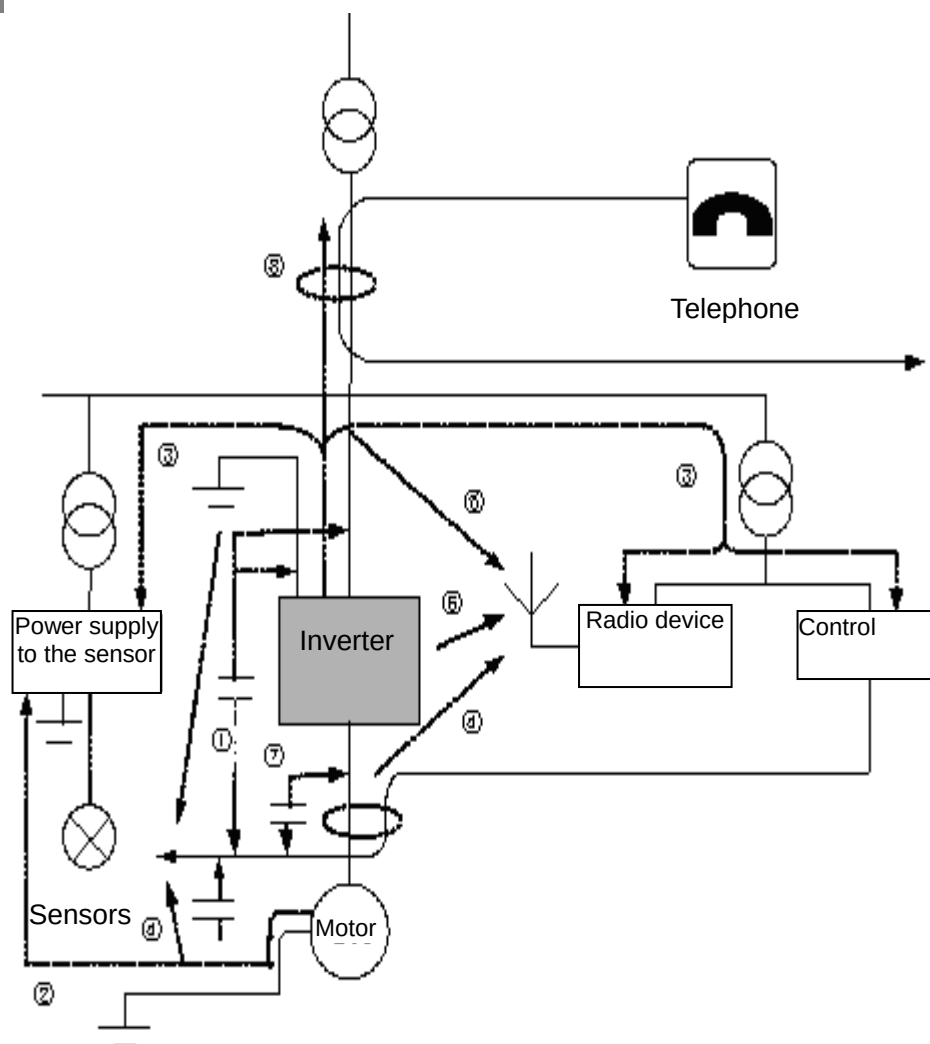


Figure A.2 Noise transmission schematic

### A.1.3 Basic countermeasures for noise suppression

The basic countermeasures for noise suppression are as shown in the attached table A.1.

Table A.1 Basic countermeasures for noise suppression

| No.         | Causes                                                                                                                                                                                                                                                                           | Countermeasures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ①<br>⑦<br>⑧ | If the signal line is laid in parallel to the power line or is laid by being bundled up together with the power line, the noise will transmit in the signal line due to EM induction and electrostatic induction, which will result in error action of the peripheral equipment. | 1. Signal line shall be prevented from being laid in parallel to the power line or from being bundled together with the power line;<br>2. Keep the peripheral equipment easy to be affected away from the inverters;<br>3. Keep the signal line easy to be affected away from the input and output cables of the inverter;<br>4. The signal line and power line use shielded wire. If they are respectively inserted into metallic tubes, the effect will be better (the metallic tubes shall be spaced at a distance of at least 20cm). |
| ②           | When the peripheral equipment forms a closed-loop circuit through wiring the inverter, the grounding wire leakage current of the inverter will lead to error action of the peripheral equipment.                                                                                 | At this time, if the peripheral equipment is not grounded, error action resulting from leakage current can be removed.                                                                                                                                                                                                                                                                                                                                                                                                                   |

|             |                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ③           | When the peripheral equipment and the inverter share a power supply system, error action may be produced on other peripheral equipment connected in the system since the noise produced by the inverter transmits along the power line.                                                                                               | Install a noise filter at the input end of the inverter or isolate the noise from other peripheral equipment with an isolation transformer/power filters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ④<br>⑤<br>⑥ | If weak current equipment such as control computers, measuring instruments, radio devices and sensors among the peripheral equipment and their signal lines are installed in the same control cabinet with the inverter and when the wires are laid near to the inverter, error action will be produced due to radiation disturbance. | <p>1. Peripheral equipment easy to be affected and their signal line shall be installed as far as possible away from the inverter. The signal lines shall use shielded cables with the shielded layer grounded and the cable inserted in the metallic tubes and shall be kept away from the inverter and their input and output cables. If the signal lines must go through the input and output cables of the inverter, they should be orthogonal;</p> <p>2. At the output and input sides of the inverter, respectively install radio noise filters or linear noise filters (ferrite common mode choke) which can suppress the noise radiation of the input and output cables of the inverter;</p> <p>3. Cables from the inverter to the motor shall be placed in a relatively thick barrier and can be placed in a tube more than 2mm long or be buried in a cement groove. The cables shall be bushed in metallic tubes and be shielded and grounded (the motor cables can use 4-core cables, one of which is grounded at the side of the inverter and connected with the motor shell at the other side.)</p> |

## A.2 Wiring requirements

### A.2.1 Requirements on cable laying

To avoid mutual coupling of disturbance, the control signal cables shall be laid separately from power cables and motor cables and shall be kept as far way as possible from them on the premise that enough distance can be ensured as shown in Figure A.3 (a); when the control signal cable must cross the power cable or motor cable, orthogonal cross shall be ensured between them as shown in Figure A.3 (b).

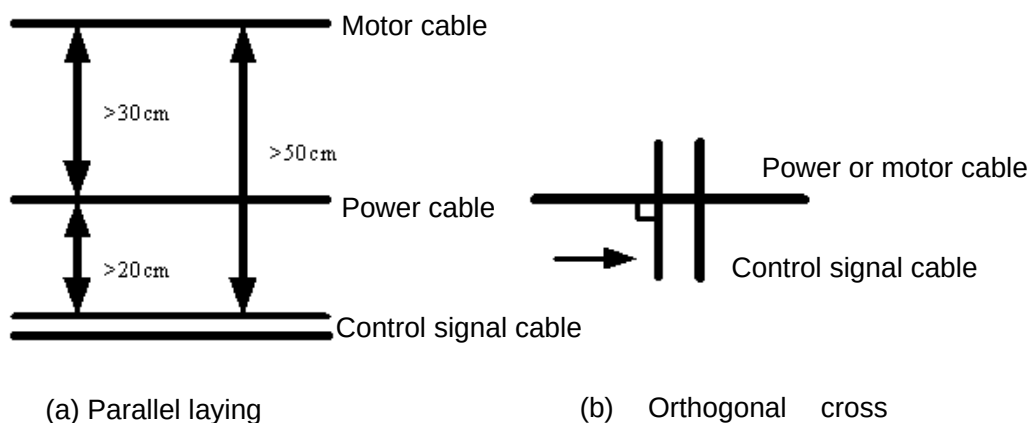


Figure A.3 Wiring requirement

### A.2.2 Requirement on the cross section of cables

Since the larger the cross section of cables is, the higher the earth capacitance will be and the higher the ground leakage current will also be, the motor cables shall be used with the ratings decreased to ensure the decrease in output current (for one level of increased cross section, the current will reduce by 5%), if the cross section of the motor cables is too high.

### A.2.3 Requirement on shielded cables

Shielded armored cables such as woven copper wire net and aluminum wire net with high frequency and low

impedance shall be adopted.

### A.2.4 Requirements on laying the shielded cables

Generally, control cables shall be shielded cables and the shielded metallic wire net must be connected with the metallic cabinets in a 360° ring-type connecting way via cable clamps as shown in the Figure A.4. The shielded grounding method shown in Figure A.5 is incorrect.

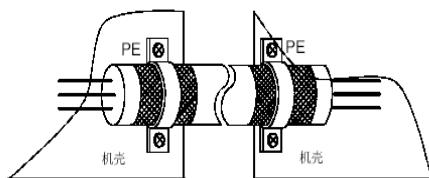


Figure A.4 Correct shielded grounding method

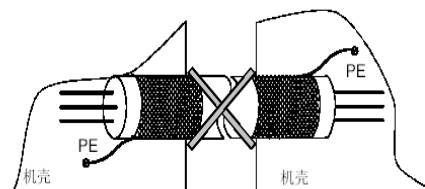


Figure A.5 Incorrect shielded grounding method

## A.3 Grounding

### A.3.1 Grounding methods

The grounding methods for the ground electrode are shown in Figure A.6.

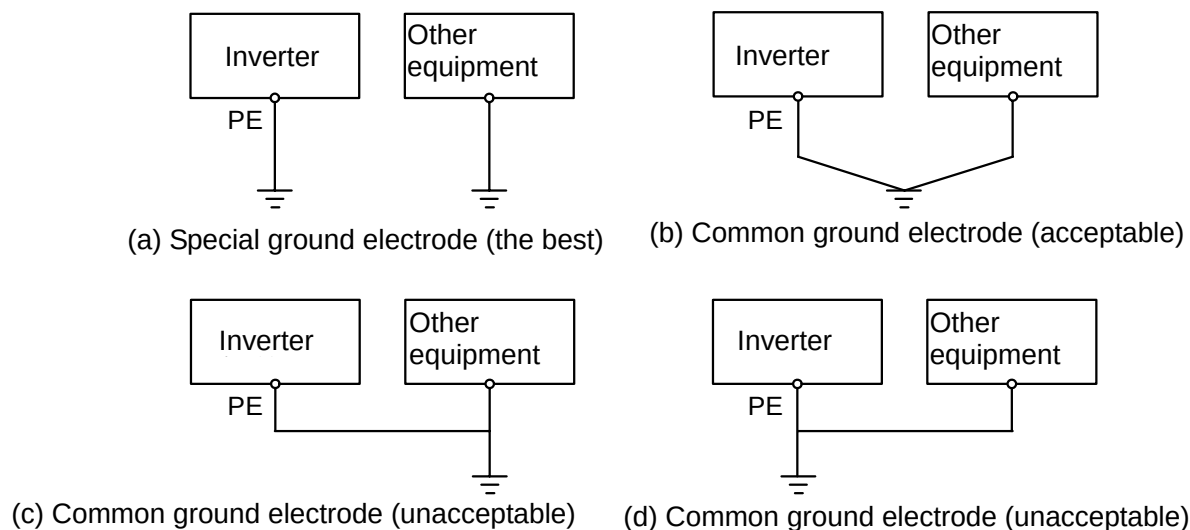


Figure A.6 Schematic for special ground electrode

Among the above four grounding methods, (a) is the best and the users are suggested using it.

### A.3.2 Precautions for ground wiring

- (1) Do best to adopt grounding cables with standard cross section to ensure the minimum grounding impedance; since flat cables have smaller high-frequency impedance than round conductors, flat cables will be a better choice if the cross sections are the same.
- (2) The grounding cable shall be as short as possible and the grounding point as close as possible to the inverter.
- (3) If four-core cables are adopted for motors, then one cable of the four-core cables must be grounded on the side of the inverter and the other side connected to the ground end of the motor. If the motors and inverters have their own special ground electrodes, the optimum grounding effect can be achieved.
- (4) When the grounding ends of all the parts in the control system are connected together, the noise source formed due to the ground leakage current will affect other peripheral equipment other than the inverters in the control



system. Therefore, in the same control system, the inverters and weak electrical equipment such as computers, sensors or audio equipment shall be grounded separately and can't be connected together.

(5) To acquire rather low high-frequency impedance, the fixing bolts of the equipment can be taken as the high-frequency terminal used to connect the cabinet and the rear panel. Please make sure to remove the insulation paint from the fixing points.

(6) The grounding cables shall be laid far away from the wiring of I/O for noise sensitive equipment and meanwhile the ground wire shall be made as short as possible.

## A.4 Surge absorber installation

Devices such as relays, contactors and EM brakes which produce large amount of noise must be equipped with surge absorbers even if they are installed outside the cabinet of the inverters as shown in Figure A.7.

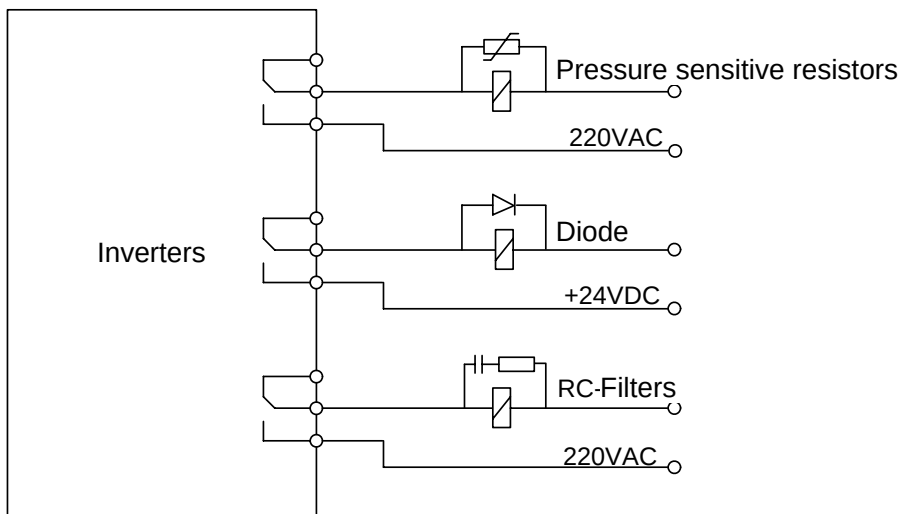


Figure A.7 Operation requirements on relays, contactors and EM brakes

## A.5 Leakage current and its countermeasures

The leakage current flows through the line capacitors and motor capacitors at the I/O side of the inverters, including the ground leakage current and the inter-line leakage current as shown in the Figure A.8. The size of the leakage current depends on the size of the carrier frequency and capacitance.

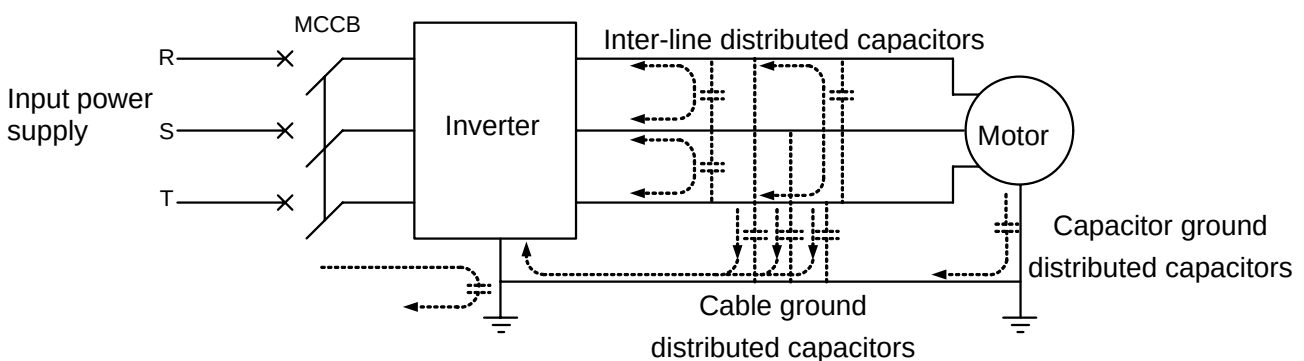


Figure A.8 Leakage current path

### A.5.1 Ground leakage current

The ground leakage current will not only flow into the inverters, but also can flow into other equipment through ground wires. It may lead to the error action of breakers, relays or other equipment leaking current. The higher the carrier frequency of the inverter is, the longer the motor cables and the higher the leakage current shall be.

Suppression measures: reduce the carrier frequency; make the motor cable as short as possible; use leakage breakers designed specially for the leakage of high harmonic/surge.

### A.5.2 Inter-line leakage

The high order harmonic of the leakage current flowing through the capacitors distributed among the cables at the output side of the inverters may lead to the error action of external thermal relay. Especially for inverters with a small capacity below 7.5kW, when the wires are very long (above 50m), the increased leakage current is easy to produce the error action of external thermal relays.

Suppression measures: reduce the carrier frequency; install AC output reactors at the output side; recommend using temperature sensors to monitor the temperature of the motors directly or replace the external thermal relay with the electronic thermal relay with overload protection functions for the motors of the inverters.

## A.6 Radiation emission suppression for inverters

Inverters are normally installed in a metallic control cabinet. The instruments and equipment outside the cabinet are subject to very small influence of the inverters' radiation emission and the cables for external connection are the major radiation emission source. Since the power cables, motor cables and control cables of the inverters as well as the keyboard cables all need to be led out of the shielded cabinet, special treatment shall be done at the outgoing locations, or the shield will become invalid.

In Figure A.9, part of the cables inside the shielded cabinet play the role as antenna which pick up the noise radiation inside the cabinet and then send it to the space outside the cabinet; in Figure A.10: connect the outlet of the cable shielded layer to the ground of the shielded cabinet case. Thus the noise radiation received by the cables inside the cabinet will flow into the ground directly via the shielded case so as to remove the influence on the environment.

When the shielded layer grounding method shown in Figure A.10 is adopted, the cable shielded layer shall be connected to the ground of the case as close as possible to the outlet, or the cable from the ground point to the outlet will still play the role of antenna and couple. The noise ground point shall keep a distance at the most 15cm (the smaller the better) from the outlet.

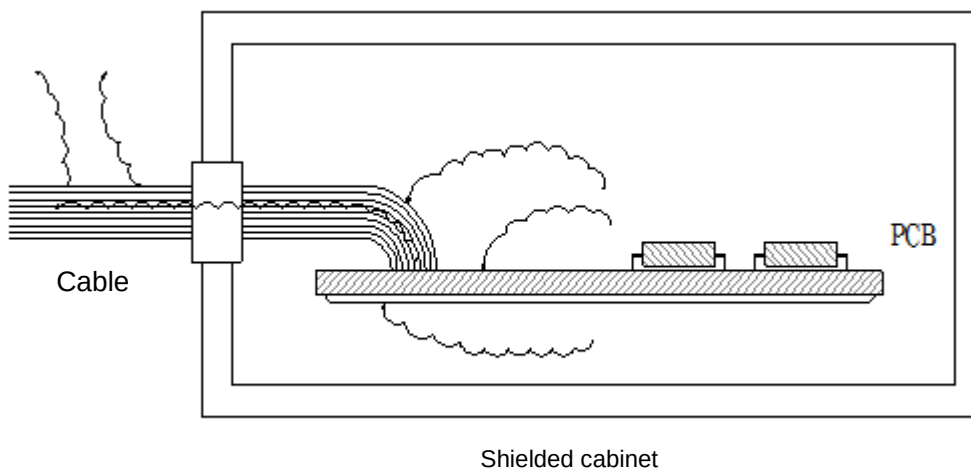


Figure A.9 Radiation resulting from the outgoing cables of the shielded cabinet

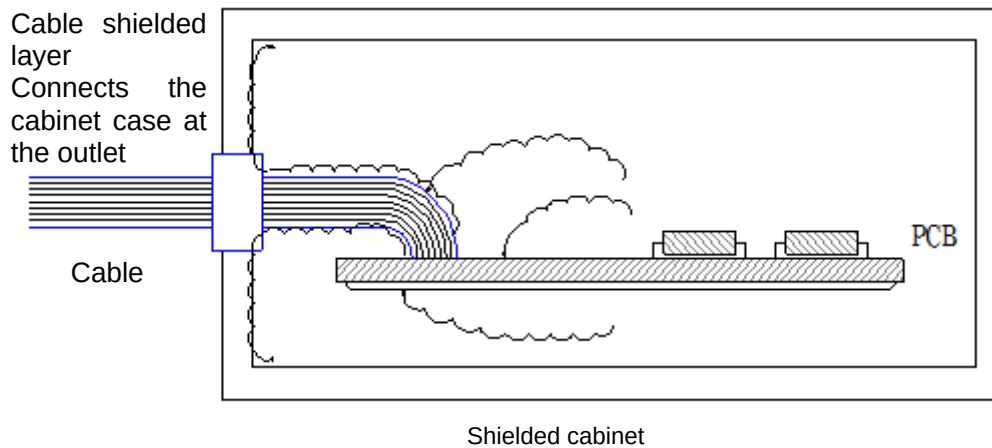


Figure A.10 Suppression of radiation by connecting the cable shielded layer to the ground of the case

## A.7 Users' guide to power line filters

Equipment which can produce strong disturbance and which is sensitive to external disturbance can use power line filters.

### A.7.1 Functions of the filters

- (1) The power line filters are dual low-pass filters which only allow direct current and current of 50Hz power frequency and refuse EM disturbance current with high frequency. Therefore, they can not only prevent the EM disturbance produced by the equipment itself from entering the power line, but also can restrain the disturbance on the power line from entering the equipment.
- (2) Power line filters can make the equipment satisfy the requirements in the conducted emission and conducted susceptibility EMC standards and meanwhile can suppress the radiation disturbance of the equipment.

### A.7.2 Precautions on power line filter installation

- (1) Inside the cabinet, the installation locations for the filters shall be as close as possible to the inlet end of the power line and the power input line of the filters shall be kept as short as possible inside the control cabinet.
- (2) If the input line and output line for the filters are laid too close to each other, high-frequency disturbance will bypass the filters and be coupled directly via the input line and output line of the filters to make the power filters defunct.
- (3) Typically there is a dedicated ground terminal on the shell of the filters. However, if a conductor is used to connect the terminal to the case of the cabinet, the filter can't play effective role of bypass and become useless due to the high-frequency impedance of the long conductor. Correct installation method is to apply the shell of the filters on the conductive plane of the metallic case and make the contact surface as large as possible. Make sure to remove the insulation paint at the time of installation and ensure sound electrical contact.

## A.8 Division of the installation area for the inverter's EMC

In the drive system made up of inverters and motors, the inverters and peripheral equipment such as the control devices and sensors are normally installed in the same control cabinet. The outside disturbance produced by the control cabinet can be suppressed by taking measures at the main connection, so a radio noise filter and an incoming AC resistor shall be installed at the incoming end of the control cabinet. To meet the EMC requirements, EMC shall also be realized inside the cabinet.

In the drive system made up of inverters and motors, the inverters, braking units and contactors are all strong noise sources which will influence the proper operation the noise sensitive peripheral equipment such as automation devices, encoders and sensors. The peripheral equipment can be installed in different EMC regions respectively according to their electrical characteristics so as to isolate the noise sources and noise receivers in space. This is the most effective measure to reduce disturbance. The installation area for the inverter's EMC is shown in Figure A.11.

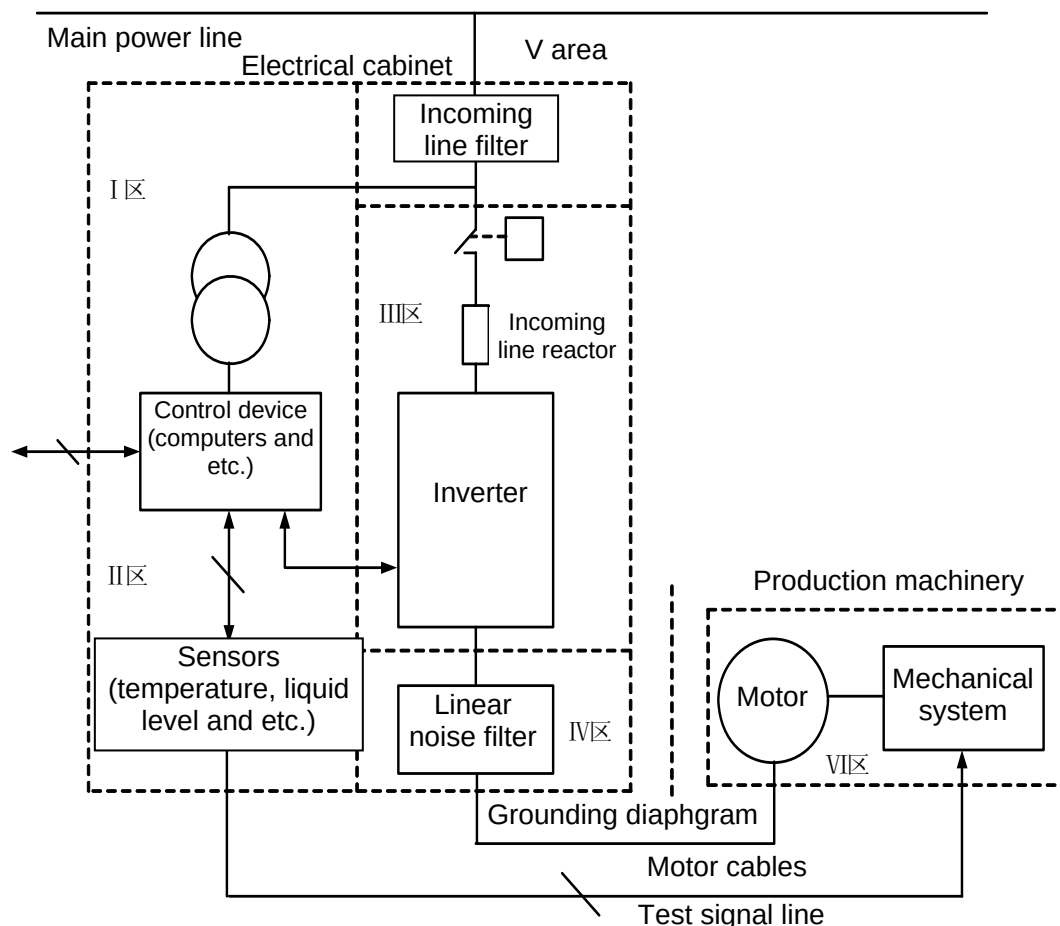


Figure A.11 Schematic of installation area for the inverter's EMC

The division of the above mentioned installation area is clarified as follows.

Area I : Control power transformer, control devices, sensors and etc.

Area II : Control signals and their cable interfaces require certain disturbance.

Area III: Main noise sources such as incoming line reactors, inverters, braking units and contactors.

Area IV: Output noise filters and other wiring parts.

Area V : Power supply (including radio noise filter wiring part).

Area VI: Motors and their cables.

The areas shall be separated at a minimum space of 20cm so as to realize EM decoupling. The areas had better decouple via a grounding diaphragm plate. Cables in different areas shall be placed in different cable ducts. When the filters are needed, they shall be installed at the connection of the areas. All the bus cable led out from the cabinet (such as RS485) and the signal cables must be shielded.

## A.9 Precautions for electrical installation of inverters

Electrical installation of inverters is shown in Figure A.12

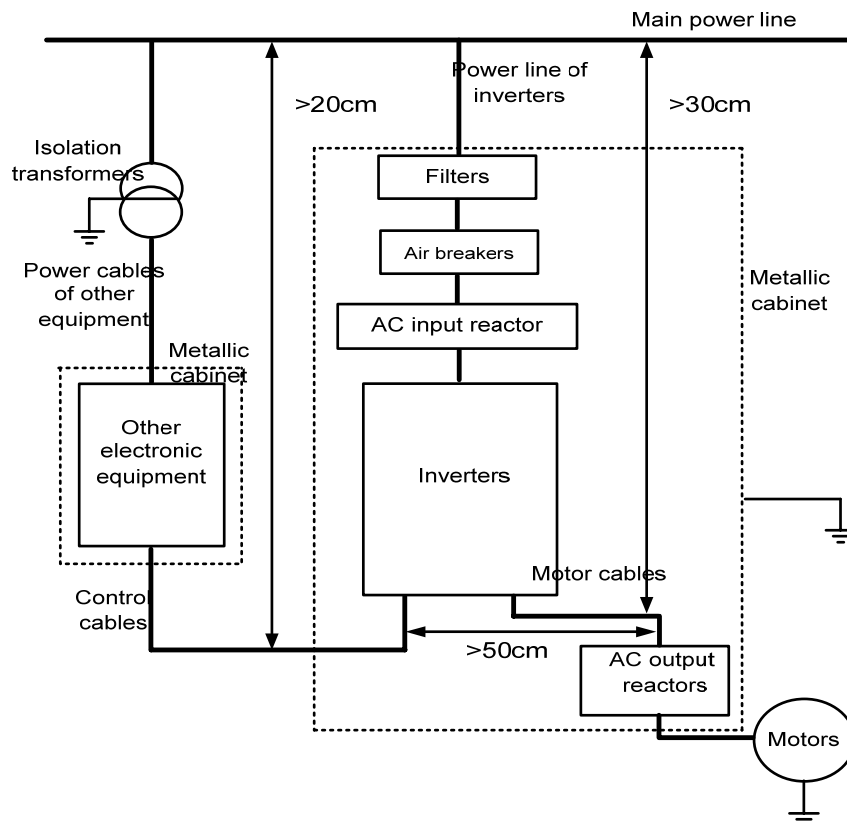


Figure A.12 Electrical installation schematic of inverters

To satisfy EMC requirements, pay attention to the following during installation:

- (1) The inverters shall be installed inside the cabinet and the shells of peripheral equipment such as the baseplate of inverters and the input filters shall be fixed on the backpanel of the control cabinet to ensure sound electrical contact with the backpanel. The distance between the inverters and the filters shall be kept less than 15cm so as to make the high-frequency impedance between the inverters and the input filters to the minimum and reduce high-frequency noise.
- (2) A wide grounding block shall be installed at the inlet of the control cabinet (not more than 5cm from the outlet) and the shielded layer of all the cables incoming and outgoing the cabinets shall be fixed on the ground block via a 360° ring connection way to ensure sound electrical contact.
- (3) The motor cables must be shielded cables and had better be cables shielded dually with screw metallic tape and metallic wire net. The shielded layer of the motor cables must be fixed on the backpanel of the cabinet with metallic cable clamps via a 360° ring connection way as shown in Figure A.4). There are two fixing locations: one is as close as possible to the inverter (better to be less than 15cm); the other is on the ground block. The shielded layer of the motor cables shall be connected with the motor's metallic shell via a 360° ring connection way when the motor goes through the motor terminal boxes. If there is any difficulty, the shielded layers can be stranded mutually into a plait which shall be connected to the ground terminal of the motor after it spread flat. The spread width shall be larger than 1/5 of the length of the plait. The cable cores of the motors and its PE flexible plait shall have an outgoing line as short as possible (better to be less than 5cm).
- (4) The terminal control cables must be shielded cables. The shielded layer shall be connected to the ground block at the inlet of the cabinet with metallic cable clamps via a 360° ring connection way. At the inverter end, metallic

cable clamps can be used to fix the shielded layer to the shell of the inverters. If there is any difficulty, the shielded layers can be stranded mutually into a wide but short plait which shall be connected to the PE terminal of the inverter after it spread flat. The exposed part of the cable cores and the length of the outgoing PE soft plait shall be kept as short as possible (better to be less than 15cm).

(5) The keyboard cables can't go out of the shielded cabinet.

(6) The size of the holes and seams in the shielded cabinet shall be as small as possible (the longest shall not be above 15cm).

## A.10 EMC standards to be satisfied by AS320 series elevator-used inverters

When AS320 series elevator-used inverters are equipped with proper input/output filters and AC reactors (for type selection, please refer to *accessories selection*) and are wired in reference to the above precautions, they can satisfy the EMC standards as shown in the Table A.2.

Table A.2 EMC performance overview of **AS320** series elevator-used inverters

| Items                                                      | Satisfied standards | Levels of the standards                                                                                                                                                                                                                                                         |
|------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conduction disturbance emission                            | EN12015.1998        | $0.15 \leq f < 0.50\text{MHz}$ , $100\text{dB}(\mu\text{V}/\text{m})$ quasi-peak value<br>$0.50 \leq f < 5.0\text{MHz}$ , $86\text{dB}(\mu\text{V}/\text{m})$ quasi-peak value<br>$5.0 \leq f < 30\text{MHz}$ , $90 \square 70\text{dB}(\mu\text{V}/\text{m})$ quasi-peak value |
| Radiation disturbance emission                             | EN12015.1998        | $30 \leq f < 230\text{MHz}$ , $40\text{dB}(\mu\text{V}/\text{m})$ quasi-peak value<br>$230 \leq f < 1000\text{MHz}$ , $47\text{dB}(\mu\text{V}/\text{m})$ quasi-peak value                                                                                                      |
| Static discharge disturbance immunity                      | EN12016.2004        | Criterion B (contact discharge 4000V, air discharge 8000V)                                                                                                                                                                                                                      |
| Radiation EM field disturbance immunity                    | EN12016.2004        | Level 3 Criterion A (3V/m)                                                                                                                                                                                                                                                      |
| Fast transient electrical pulse train disturbance immunity | EN12016.2004        | Level 4 Criterion B (strong current end $\pm$ 2KV/2.5kHz)                                                                                                                                                                                                                       |
| Surge disturbance immunity                                 | EN12016.2004        | Criterion B ( $\pm$ 1KV)                                                                                                                                                                                                                                                        |
| Conduction disturbance immunity                            | EN12016.2004        | Criterion A (3V, 0.15~80MHz)                                                                                                                                                                                                                                                    |

## Appendix B Summary of function parameters and failure record sheets

This appendix collects the function parameters, operating conditions and failure record sheets to facilitate the inverter users to consult and use.

### B.1 Function parameter sheet

| Function Code | Designation                                      | Content                                                                                                                                                                                                                                       | Setting Range | Unit                  | Factory Setting                                        | Remarks |
|---------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------|--------------------------------------------------------|---------|
| <b>F00</b>    | <b>Password parameter and basic control mode</b> |                                                                                                                                                                                                                                               |               |                       |                                                        |         |
| P00.00        | Logon password                                   | This is the logon password for the inverters. Only when the operators have logged on for this parameter with the correct password (the same password with that set in P00.01 the last time), can they have the right to modify the parameter. | 0~65535       | ×                     | 0                                                      |         |
| P00.01        | Modify or set the password                       | The password to protect the inverters can be set or modified by setting this parameter. If it is set to be 0, there will be no password protection. This is a concealed parameter and can't be displayed once it has been set.                | 0~65535       | ×                     | 0                                                      |         |
| P00.02        | Basic control mode                               | Set the basic modes for the inverters:<br>0: voltage vector V/F control mode<br>1: no speed sensor vector control<br>2: with speed sensor torque control<br>3: with speed sensor vector control\                                              | 0 / 1 / 2 / 3 | ×                     | 3                                                      |         |
| P00.03        | Run command given way                            | Setting run command given way<br>0:panel<br>1:terminal                                                                                                                                                                                        | P00.03        | Run command given way | Setting run command given way<br>0:panel<br>1:terminal | P00.03  |
| P00.04        | Language option                                  | Setting language :<br>0:Chinese<br>1:English                                                                                                                                                                                                  | P00.04        | Language option       | Setting language :<br>0:Chinese<br>1:English           | P00.04  |
| P00.05        | The version number                               | The version number for inverters                                                                                                                                                                                                              |               |                       | 104.02                                                 |         |
| P00.06        | Two-wire operation mode                          | 0:two-wire mode1<br>1:two-wire node2<br>2:third-wire mode1<br>3:third-wire mode2                                                                                                                                                              |               |                       | 0                                                      |         |
| P00.07        | Intertial stop mode                              | 0: Intertial stop                                                                                                                                                                                                                             |               |                       | 0                                                      |         |

|               |                                                            | 1: Decelerating stop<br>2: Decelerating and DC braking<br>3: Decelerating and keep excitation                                                                         |               |      |                                       |                                          |
|---------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|---------------------------------------|------------------------------------------|
| P00.08        | Parking keep frequency                                     |                                                                                                                                                                       |               |      | 0.00                                  |                                          |
| P00.09        | Time of parking keep frequency                             |                                                                                                                                                                       |               |      | 0.0                                   |                                          |
| P00.10        | parking excitation keep time                               |                                                                                                                                                                       |               |      | 0.0                                   |                                          |
| Function Code | Designation                                                | Content                                                                                                                                                               | Setting Range | Unit | Factory Setting                       | Remarks                                  |
| F01           | Parameters, self-learning commands for motors and encoders |                                                                                                                                                                       |               |      |                                       |                                          |
| P01.00        | Type of motor                                              | Set the traction motor to be asynchronous or synchronous:<br>0: asynchronous;<br>1: synchronous                                                                       | 0 / 1         | ×    | 0                                     |                                          |
| P01.01        | Motor rated power                                          | Set traction motor rated power                                                                                                                                        | 0.40~160.00   | KW   | In light of the inverters' parameters | Set according to the nameplate of motors |
| P01.02        | Motor rated current                                        | Set traction motor rated current                                                                                                                                      | 0.0~300.0     | A    | In light of the inverters' parameters | Set according to the nameplate of motors |
| P01.03        | Motor rated frequency                                      | Set traction motor rated frequency                                                                                                                                    | 0.00~120.00   | Hz   | 50.00                                 | Set according to the nameplate of motors |
| P01.04        | Motor rated speed                                          | Set traction motor rated speed                                                                                                                                        | 0~3000        | rpm  | 1460                                  | Set according to the nameplate of motors |
| P01.05        | Motor rated voltage                                        | Set traction motor rated voltage                                                                                                                                      | 0~460         | V    | In light of the inverters' parameters | Set according to the nameplate of motors |
| P01.06        | Number of poles of the motor                               | Set the number of poles for the traction motor                                                                                                                        | 2~128         | ×    | 4                                     | Set according to the nameplate of motors |
| P01.07        | Motor rated slip frequency                                 | Set the rated slip frequency for the traction motor                                                                                                                   | 0~10.00       | Hz   | 1.40                                  | Set according to the nameplate of motors |
| P01.08        | Motor phase sequence                                       | Set the phase sequence of the input voltage for the traction motor thus to adjust the running direction of the motor<br>1: positive direction<br>0: reverse direction | 0 / 1         | ×    | 1                                     |                                          |
| P01.09        | Motor no load rated current coefficient                    | Set the proportion of no load current to the rated current for the traction motor                                                                                     | 0.00~60.00    | %    | 32.00                                 | Normally no need of setting              |
| P01.10        | Resistance of motor stators                                | Resistance of the traction motor's stator                                                                                                                             | 0.000~65.000  | Ω    | In light of power of the inverters    | Only for the asynchronous                |
| P01.11        | Resistance for motor rotors                                | Resistance of the traction motor's rotor                                                                                                                              | 0.000~65.000  | Ω    | In light of power of the              | Only for the asynchronous                |



|               |                                                           |                                                                                                                                                        |               |        | inverters                          |                                                                                                                                               |
|---------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| P01.12        | Motor stator inductance                                   | Inductance of the traction motor's stator                                                                                                              | 0.0000~6.0000 | H      | In light of power of the inverters | Only for the asynchronous                                                                                                                     |
| P01.13        | Motor rotor inductance                                    | Mutual inductance of the traction motors                                                                                                               | 0.0000~6.0000 | H      | In light of power of the inverters | Only for the asynchronous                                                                                                                     |
| P01.14        | Motor mutual inductance                                   | Inductance of the traction motor's rotor                                                                                                               | 0.0000~6.0000 | H      | In light of power of the inverters | Only for the asynchronous                                                                                                                     |
| P01.15        | Type of encoder                                           | Set the type of encoders used to test the rotary speed of the traction motors:<br>0: incremental encoder<br>1: Sine/cosine encoder<br>2: Endat encoder | 0/1/2         | ×      | 0                                  |                                                                                                                                               |
| P01.16        | The number of pulses for encoders                         | The number of pulses for each revolution of encoder                                                                                                    | 500~16000     | PPr    | 1024                               |                                                                                                                                               |
| P01.17        | Phase angle of encoders                                   | The size of phase angle of the encoders                                                                                                                | 0.0~360.0     | Degree | 0.0                                | The data will be obtained automatically after the inverter is energized and runs the first time.<br>Only applicable to the synchronous motors |
| P01.18        | Filtering time of encoders                                | Set the constant for the filtering time when the feedback speed of the encoder is entered                                                              | 1~30          | ms     | 0                                  |                                                                                                                                               |
| P01.19        | Feedback direction of encoders                            | Set the feedback speed direction of the encoders<br>1: positive sequence<br>0: reverse sequence                                                        | 0 / 1         | ×      | 1                                  |                                                                                                                                               |
| P01.20        | Input voltage of inverters                                | Set the input voltage of inverters                                                                                                                     | 0~460         | V      | 380                                | Can be initialized after it is set                                                                                                            |
| Function Code | Designation                                               | Content                                                                                                                                                | Setting Range | Unit   | Factory Setting                    | Remarks                                                                                                                                       |
| F02           | PID regulators and starting/braking regulation parameters |                                                                                                                                                        |               |        |                                    |                                                                                                                                               |
| P02.00        | Zero servo gain P0                                        | The gain value of PID regulators when the zero servo works                                                                                             | 0.00~655.35   | ×      | 130.00                             | Refer to the following note:<br>Suggested adjustment scope: minimum: 1/2 of the default; maximum: 2 times of the default                      |
| P02.01        | Zero servo integral I0                                    | The integral value of PID regulators when the zero servo works                                                                                         |               |        | 80.00                              | Refer to the following note:<br>Suggested adjustment scope: minimum: 1/2 of the default; maximum: 2 times of the default                      |
| P02.02        | Zero servo differential D0                                | The differential value of PID regulators when the                                                                                                      |               |        | 0.50                               | Refer to the following note:                                                                                                                  |

|        |                                               |                                                                                                            |  |  |        |                                                                                                                          |
|--------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------|--|--|--------|--------------------------------------------------------------------------------------------------------------------------|
|        |                                               | zero servo works                                                                                           |  |  |        | Suggested adjustment scope: minimum: 1/2 of the default; maximum: 2 times of the default                                 |
| P02.03 | Gain P1 at low speed section                  | The gain value of PID regulators when the speed reference is lower than switching frequency F0             |  |  | 70.00  | Refer to the following note:<br>Suggested adjustment scope: minimum: 1/2 of the default; maximum: 2 times of the default |
| P02.04 | Integral I1 at low speed section              | The integral value of PID regulators when the speed reference is lower than switching frequency F0         |  |  | 30.00  | Refer to the following note:<br>Suggested adjustment scope: minimum: 1/2 of the default; maximum: 2 times of the default |
| P02.05 | Differential D1 at low speed section          | Differential value of PID regulators when the speed reference is lower than switching frequency F0         |  |  | 0.50   | Refer to the following note:<br>Suggested adjustment scope: minimum: 1/2 of the default; maximum: 2 times of the default |
| P02.06 | Gain P2 at intermediate speed section         | The gain value of PID regulators when the speed reference is between switching frequency F0 and F1         |  |  | 120.00 |                                                                                                                          |
| P02.07 | Integral I2 at intermediate speed section     | The integral value of PID regulators when the speed reference is between switching frequency F0 and F1     |  |  | 25.00  |                                                                                                                          |
| P02.08 | Differential D2 at intermediate speed section | The differential value of PID regulators when the speed reference is between switching frequency F0 and F1 |  |  | 0.20   |                                                                                                                          |
| P02.09 | Gain P3 at high speed section                 | The gain value of PID regulators when the speed reference is higher than switching frequency F1            |  |  | 140.00 | Refer to the following note:<br>Suggested adjustment scope: minimum: 1/2 of the default; maximum: 2 times of the default |
| P02.10 | Integral I3 at high speed section             | The integral value of PID regulators when the speed reference is higher than switching frequency F1        |  |  | 5.00   | Refer to the following note:<br>Suggested adjustment scope: minimum: 1/2 of the default; maximum: 2 times of the default |
| P02.11 | Differential D3 at intermediate speed section | The differential value of PID regulators when the speed reference is higher than switching frequency F1    |  |  | 0.10   | Refer to the following note:<br>Suggested adjustment scope: minimum: 1/2 of the default; maximum: 2 times of the default |

| P02.12        | Switching frequency F0 at a low speed  | Set the parameters for the switching frequency of PID regulators at segmented low speed points, expressed in the percentage of frequency rating. If the frequency rating is 50Hz, the required switching frequency F0 shall be 10Hz. Since 10Hz is 20% of 50Hz, the data shall be set to 20   | 0.0~100.0     | %    | 1.0             |                                                           |
|---------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|-----------------------------------------------------------|
| P02.13        | Switching frequency F1 at a high speed | Set the parameters for the switching frequency of PID regulators at segmented high speed points, expressed in the percentage of frequency rating. If the frequency rating is 50Hz, the required switching frequency F0 shall be 40Hz. Since 40Hz is 80% of 50Hz, the data shall be set to 80. | 0.0~100.0     | %    | 50.0            |                                                           |
| P02.14        | Excitation time                        | After the inverters receive the command to run, give the running command again after the excitation time elapses and the elevator can be switched off.                                                                                                                                        | 0.0~10.0      | s    | 0.3             | Out of work when it is used to control synchronous motors |
| P02.15        | Zero servo time                        | As of the time when the inverters issue the running command, accelerate to start when the time elapses.                                                                                                                                                                                       | 0.0~30.0      | s    | 0.5             |                                                           |
| P02.16        | Brake releasing time                   | Mechanical acting time of brake                                                                                                                                                                                                                                                               | 0.00~30.00    | s    | 0.25            |                                                           |
| P02.17        | PWM carrier frequency                  | Set the value of PWM carrier frequency for inverters                                                                                                                                                                                                                                          | 1.100~11.000  | kHz  | 6.000           | Normally the parameter is not to be adjusted              |
| P02.18        | PWM carrier width                      | Set the value of PWM carrier varying width for inverters                                                                                                                                                                                                                                      | 0.000~1.000   | kHz  | 0.000           | Normally the parameter is not to be adjusted              |
| P02.19        | Current Ramp Down Time                 | Time from withdrawing the running command of the inverters to the time when the output current of inverters drops to zero                                                                                                                                                                     | 0.00~10.00    | s    | 0.00            |                                                           |
| P02.20        | Modes of regulators                    | 0: fast regulator mode<br>1: standard regulator mode<br>2: relatively slow regulator mode<br>3: slow regulator mode                                                                                                                                                                           | 0/1/2/3       | ×    | 1               |                                                           |
| Function Code | Designation                            | Content                                                                                                                                                                                                                                                                                       | Setting Range | Unit | Factory Setting | Remarks                                                   |
| <b>F03</b>    | <b>Speed reference parameters</b>      |                                                                                                                                                                                                                                                                                               |               |      |                 |                                                           |
| P03.00        | Speed reference selection              | This parameter decides the way to set the running speed of elevators:<br>0: set on the panel<br>1: control speed setting at multi-speed sections via the                                                                                                                                      | 0 / 1 / 4 / 6 | ×    | 4               | Invalid when P00.02 is set to 2                           |

|        |                           |                                                                                                                                                                                                                                                                                                                                                            |                |    |      |                                                    |
|--------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----|------|----------------------------------------------------|
|        |                           | digital signal<br>4: AI0 analog speed reference<br>6: AI1 analog speed reference                                                                                                                                                                                                                                                                           |                |    |      |                                                    |
| P03.01 | Acceleration time         | This parameter decides the acceleration slope of elevators (i.e. the size of a constant acceleration). It is the time the elevator asks for to accelerate to the top speed from zero. Pay attention: it is different from the average acceleration which is related to the size of the two acceleration rounds besides this parameter.                     | 0.10~<br>60.00 | s  | 2.50 | Be used only under multi-speed reference selection |
| P03.02 | Deceleration time         | This parameter decides the deceleration slope of elevators (i.e. the size of a constant deceleration). It is the time the elevator asks for to decelerate at a constant speed from the top speed to zero. Pay attention: it is different from the average deceleration which is related to the size of the two deceleration rounds besides this parameter. | 0.10~<br>60.00 | s  | 2.50 | Be used only under multi-speed reference selection |
| P03.03 | Acceleration round 0 time | Set the time of the round for S curve starting segment. The longer the time is, the larger the round will be.                                                                                                                                                                                                                                              | 0.00~<br>10.00 | s  | 1.30 | Be used only under multi-speed reference selection |
| P03.04 | Acceleration round 1 time | Set the time of the round for S curve constant speed segment. The longer the time is, the larger the round will be.                                                                                                                                                                                                                                        | 0.00~<br>10.00 | s  | 1.30 | Be used only under multi-speed reference selection |
| P03.05 | deceleration round 0 time | Set the time of the round for S curve deceleration starting segment. The longer the time is, the larger the round will be.                                                                                                                                                                                                                                 | 0.00~<br>10.00 | s  | 1.30 | Be used only under multi-speed reference selection |
| P03.06 | deceleration round 1 time | Set the time of the round for S curve deceleration finishing segment. The longer the time is, the larger the round will be.                                                                                                                                                                                                                                | 0.00~<br>10.00 | s  | 1.30 | Be used only under multi-speed reference selection |
| P03.07 | Speed reference 1         | Take Hz as the unit and set the speed reference 1 under multi-speed reference selection for the digital signal                                                                                                                                                                                                                                             | 0.00~<br>60.00 | Hz | 2.50 | Be used only under multi-speed reference selection |
| P03.08 | speed reference2          | Take Hz as the unit and set the speed reference 2 under multi-speed reference selection for the digital signal                                                                                                                                                                                                                                             | 0.00~<br>60.00 | Hz | 1.20 | Be used only under multi-speed reference selection |
| P03.09 | speed reference3          | Take Hz as the unit and set the speed reference 3 under multi-speed reference                                                                                                                                                                                                                                                                              | 0.00~<br>60.00 | Hz | 1.50 | Be used only under multi-speed reference selection |

|        |                   |                                                                                                                 |            |    |       |                                                    |
|--------|-------------------|-----------------------------------------------------------------------------------------------------------------|------------|----|-------|----------------------------------------------------|
|        |                   | selection for the digital signal                                                                                |            |    |       |                                                    |
| P03.10 | speed reference4  | Take Hz as the unit and set the speed reference 4 under multi-speed reference selection for the digital signal  | 0.00~60.00 | Hz | 5.00  | Be used only under multi-speed reference selection |
| P03.11 | speed reference5  | Take Hz as the unit and set the speed reference 5 under multi-speed reference selection for the digital signal  | 0.00~60.00 | Hz | 25.00 | Be used only under multi-speed reference selection |
| P03.12 | speed reference6  | Take Hz as the unit and set the speed reference 6 under multi-speed reference selection for the digital signal  | 0.00~60.00 | Hz | 40.00 | Be used only under multi-speed reference selection |
| P03.13 | speed reference7  | Take Hz as the unit and set the speed reference 7 under multi-speed reference selection for the digital signal  | 0.00~60.00 | Hz | 50.00 | Be used only under multi-speed reference selection |
| P03.14 | speed reference8  | Take Hz as the unit and set the speed reference 8 under multi-speed reference selection for the digital signal  | 0.00~60.00 | Hz | 0.00  | Be used only under multi-speed reference selection |
| P03.15 | speed reference9  | Take Hz as the unit and set the speed reference 9 under multi-speed reference selection for the digital signal  | 0.0~60.0   | Hz | 0.0   | Be used only under multi-speed reference selection |
| P03.16 | speed reference10 | Take Hz as the unit and set the speed reference 10 under multi-speed reference selection for the digital signal | 0.0~60.0   | Hz | 0.0   | Be used only under multi-speed reference selection |
| P03.17 | speed reference11 | Take Hz as the unit and set the speed reference 11 under multi-speed reference selection for the digital signal | 0.0~60.0   | Hz | 0.0   | Be used only under multi-speed reference selection |
| P03.18 | speed reference12 | Take Hz as the unit and set the speed reference 12 under multi-speed reference selection for the digital signal | 0.0~60.0   | Hz | 0.0   | Be used only under multi-speed reference selection |
| P03.19 | speed reference13 | Take Hz as the unit and set the speed reference 13 under multi-speed reference selection for the digital signal | 0.0~60.0   | Hz | 0.0   | Be used only under multi-speed reference selection |
| P03.20 | speed reference14 | Take Hz as the unit and set the speed reference 14 under multi-speed reference selection for the digital signal | 0.0~60.0   | Hz | 0.0   | Be used only under multi-speed reference selection |
| P03.21 | speed reference15 | Take Hz as the unit and set the speed reference 15 under multi-speed reference selection for the digital signal | 0.0~60.0   | Hz | 0.0   | Be used only under multi-speed reference selection |

| P03.22        | Parking speed select                                | Parameter No. of parking creeping speed                                                                                                                                                                   | 0 or 3.07~3.21 |      | 0.00            | Be used only under multi-speed reference selection                                                                                                                                                                                                                     |
|---------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P03.23        | Parking speed                                       | Parameter No. of parking speed                                                                                                                                                                            | 0 or 3.07~3.21 |      | 0.00            | Be used only under multi-speed reference selection                                                                                                                                                                                                                     |
| P03.24        | Slow down time                                      | Parking process from crawling to stop the process of deceleration time                                                                                                                                    | 0~360.00       | s    | 5.00            |                                                                                                                                                                                                                                                                        |
| Function Code | Designation                                         | Content                                                                                                                                                                                                   | Setting Range  | Unit | Factory Setting | Remarks                                                                                                                                                                                                                                                                |
| F04           | Torque reference and torque compensation parameters |                                                                                                                                                                                                           |                |      |                 |                                                                                                                                                                                                                                                                        |
| P04.00        | Torque reference selection                          | Set torque reference selection:<br>0: set on the panel<br>1: AI0 analog torque reference<br>2: AI1 analog torque reference                                                                                | 0/1/2          | ×    | 0               | In the majority of circumstance, torque reference selection is not used and the parameter is set to be 0. If torque reference selection is adopted, then speed reference selection (P03.00 is set to be 0) will not be used.<br>Valid only when P00.02 is set to be 2. |
| P04.01        | Torque compensation reference method                | Torque compensation reference method:<br>0: no torque compensation<br>1: set the compensation on the basis of light/heavy load switch<br>2: AI0 analog torque reference<br>3: AI1 analog torque reference | 0/1/2/3        | ×    | 0               |                                                                                                                                                                                                                                                                        |
| P04.02        | Torque compensation direction                       | Set torque compensation direction:<br>0: positive<br>1: reverse                                                                                                                                           | 0/1            | ×    | 0               |                                                                                                                                                                                                                                                                        |
| P04.03        | Torque compensation gain                            | Set torque compensation gain                                                                                                                                                                              | 0.0~200.0      | %    | 100.0           | Valid only when P04.01 is set within 2~3.                                                                                                                                                                                                                              |
| P04.04        | Torque compensation offset                          | Set torque compensation offset                                                                                                                                                                            | 0.0~100.0      | %    | 0.0             | Valid only when P04.01 is set within 2~3                                                                                                                                                                                                                               |
| P04.05        | Light load switch compensation Quantity             | Set the compensation quantity of downward torque when the light load switch acts                                                                                                                          | 0.0~100.0      | %    | 0.0             | Valid only when P04.01 is set to be 1                                                                                                                                                                                                                                  |
| P04.06        | Heavy load switch compensation quantity             | Set the compensation quantity of upward torque when the heavy load switch acts                                                                                                                            | 0.0~100.0      | %    | 0.0             | Only valid when P04.01 is set to be 1                                                                                                                                                                                                                                  |
| P04.07        | Output torque limit                                 | Set the output torque limit, which is the percentage data                                                                                                                                                 | 0~200          | %    | 175             |                                                                                                                                                                                                                                                                        |

|               |                                            | of rated torque                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |       |                 |                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P04.08        | ARD running speed                          | Running speed in ARD mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0~655.35        | Hz    | 0               |                                                                                                                                                                                                                                                                                                                                                                    |
| P04.09        | ARD torque limit                           | Setting ARD torque limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0~200           | %     | 150             |                                                                                                                                                                                                                                                                                                                                                                    |
| Function Code | Designation                                | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Setting Range   | Unit  | Factory Setting | Remarks                                                                                                                                                                                                                                                                                                                                                            |
| <b>F05</b>    | <b>Digital input definition parameters</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |       |                 |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.00        | X0 terminal input function definition.     | Set the function code of the digital input port:<br>0: no function (invalid corresponding port);<br>3: speed reference 0 for digital multi-speed;<br>4: speed reference 1 for digital multi-speed;<br>5: speed reference 2 for digital multi-speed;<br>6: speed reference 3 for digital multi-speed;<br>7: up command;<br>8: down command;<br>13: external reset signal;<br>14: external fault signal;<br>15: external encoder phase angle rectification command;<br>16: emergency power source operation;<br>17: weighing compensation input (dedicated to special users);<br>18: base block signal;<br>19: light load compensation switch;<br>20: heavy load compensation switch<br>21: detection signal of output contactor<br>22: detection signal of brake contactor<br>23: detection signal of brake switch<br>34: jog input signal<br>Others:stand by | 0~20<br>103~120 | ×     | 0               | Factory Setting:<br>P05.02=3: X0 terminal input is multi-speed reference 0;<br>P05.03=4: X1 terminal input is multi-speed reference 1;<br>P05.04=5: X2 terminal input is multi-speed reference 2;<br>P05.06=7: X6 terminal input is up command signal;<br>P05.07=8: X7 terminal input is down command signal;<br>P05.05=18: X5 terminal input is base block signal |
| P05.01        | X1 terminal input function Definition.     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |       | 0               |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.02        | X2 terminal input function Definition.     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |       | 3               |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.03        | X3 terminal input function Definition.     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |       | 4               |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.04        | X4 terminal input function Definition.     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |       | 5               |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.05        | X5 terminal input function Definition.     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |       | 18              |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.06        | X6 terminal input function Definition.     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |       | 7               |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.07        | X7 terminal input function Definition.     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |       | 8               |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.08        | Digital input filtering times              | Digital input filtering times of digital input port                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1~99            | times | 5               |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.09        | Point operating frequency                  | If enter No. 34 jog function, and the operating frequency of the corresponding input point is valid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0~655.35        | Hz    | 0               |                                                                                                                                                                                                                                                                                                                                                                    |
| P05.10        | Accelated time 1                           | Accelated time when point running                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.1~360.00      | S     | 5.00            | Only can be used in point running                                                                                                                                                                                                                                                                                                                                  |
| P05.11        | Moderate time 2                            | Moderate time when point running                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0.1~360.00      | S     | 5.00            | Only can be used in point running                                                                                                                                                                                                                                                                                                                                  |

| Function Code | Designation                                 | Content                                                                                                                                                                                  | Setting Range   | Unit | Factory Setting | Remarks                                                                                                                                                                                      |
|---------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F06           | Digital output definition parameters        |                                                                                                                                                                                          |                 |      |                 |                                                                                                                                                                                              |
| P06.00        | K1 port (relay) output function definition. | Set the function of each digital output terminal:<br>0: means that the corresponding output port is not in use;<br>For the meaning of other values, please refer to the following note 1 | 0~15<br>101~115 | ×    | 0               | The three terminals of relay output corresponding to K1 are respectively 1A, 1B and 1C and the contact between 1A and 1B is normally closed, while that between 1B and 1C are normally open. |
| P06.01        | K2 port (relay) output function definition. |                                                                                                                                                                                          |                 |      | 0               | The three terminals of relay output corresponding to K2 are respectively 2A, 2B and 2C and the contact between 2A and 2B is normally closed, while that between 2B and 2C are normally open. |
| P06.02        | Y0 port (relay) output function definition. |                                                                                                                                                                                          |                 |      | 3               | Y0 terminal is defined as the inverter run signal                                                                                                                                            |
| P06.03        | Y1 port (relay) output function definition. |                                                                                                                                                                                          |                 |      | 2               | Y1 terminal is defined as the inverter fault signal                                                                                                                                          |
| P06.04        | Y2 port (relay) output function definition. |                                                                                                                                                                                          |                 |      | 0               |                                                                                                                                                                                              |
| P06.05        | Y3 port (relay) output function definition. |                                                                                                                                                                                          |                 |      | 0               |                                                                                                                                                                                              |
| P06.06        | K1 terminal output delay                    | Set the delay acting time of K1 terminal output condition after the actual signal is ON                                                                                                  | 0.0~60.0        | s    | 0               |                                                                                                                                                                                              |
| P06.07        | K1 terminal reset delay                     | Set the delay reset time of K1 terminal output condition after the actual signal is OFF                                                                                                  | 0.0~60.0        | s    | 0               |                                                                                                                                                                                              |
| P06.08        | K2 terminal output delay                    | Set the delay acting time of K2 terminal output condition after the actual signal is ON                                                                                                  | 0.0~60.0        | s    | 0               |                                                                                                                                                                                              |
| P06.09        | K2 terminal reset delay                     | Set the delay reset time of K2 terminal output condition after the actual signal is OFF                                                                                                  | 0.0~60.0        | s    | 0               |                                                                                                                                                                                              |
| P06.10        | Y0 terminal output delay                    | Set the delay acting time of Y0 terminal output condition after the actual signal is ON                                                                                                  | 0.0~60.0        | s    | 0               |                                                                                                                                                                                              |
| P06.11        | Y0 terminal reset delay                     | Set the delay reset time of Y02 terminal output condition after the actual                                                                                                               | 0.0~60.0        | s    | 0               |                                                                                                                                                                                              |



|               |                                                  | signal is OFF                                                                                                                                                                                                                                                                                                                                                |               |      |                 |                                                                                            |
|---------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|--------------------------------------------------------------------------------------------|
| P06.12        | Y1 terminal output delay                         | Set the delay acting time of Y1 terminal output condition after the actual signal is ON                                                                                                                                                                                                                                                                      | 0.0~60.0      | s    | 0               |                                                                                            |
| P06.13        | Y1 terminal reset delay                          | Set the delay reset time of Y1 terminal output condition after the actual signal is OFF                                                                                                                                                                                                                                                                      | 0.0~60.0      | s    | 0               |                                                                                            |
| P06.14        | Y2 terminal output delay                         | Set the delay acting time of Y2 terminal output condition after the actual signal is ON                                                                                                                                                                                                                                                                      | 0.0~60.0      | s    | 0               |                                                                                            |
| P06.15        | Y2 terminal reset delay                          | Set the delay reset time of Y2 terminal output condition after the actual signal is OFF                                                                                                                                                                                                                                                                      | 0.0~60.0      | s    | 0               |                                                                                            |
| P06.16        | Y3 terminal output delay                         | Set the delay acting time of Y3 terminal output condition after the actual signal is ON                                                                                                                                                                                                                                                                      | 0.0~60.0      | s    | 0               |                                                                                            |
| P06.17        | Y3 terminal reset delay                          | Set the delay reset time of Y3 terminal output condition after the actual signal is OFF                                                                                                                                                                                                                                                                      | 0.0~60.0      | s    | 0               |                                                                                            |
| P06.18        | Non-zero current detection threshold for parking | During setting non-zero current detection threshold for parking when there is the current higher than the setpoint flowing through the inverter at the time of parking, the non-zero current detection signal is valid. It is a data expressed in percentage, but the actual value equals to the data multiplying the rated current and then divided by 100. | 0.0~100.0     | %    | 2.0             |                                                                                            |
| P06.19        | Any frequency detection speed                    | It is a frequency detection benchmark data used together with P06.20.                                                                                                                                                                                                                                                                                        | 0.00~60.00    | Hz   | 1.00            |                                                                                            |
| P06.20        | Any frequency detection width                    | It is a frequency detection width data used together with P06.19.                                                                                                                                                                                                                                                                                            | 0.00~20.00    | Hz   | 0.20            |                                                                                            |
| Function Code | Designation                                      | Content                                                                                                                                                                                                                                                                                                                                                      | Setting Range | Unit | Factory Setting | Remarks                                                                                    |
| <b>F07</b>    | <b>Analog input port definition parameters</b>   |                                                                                                                                                                                                                                                                                                                                                              |               |      |                 |                                                                                            |
| P07.00        | AI0 analog input type definition                 | Set the type of AI0~AI1 analog input ports with the connotation of each parameter as follows:<br>0: 0~10V<br>1: -10V~10V                                                                                                                                                                                                                                     | 0 / 1         | ×    | 1               |                                                                                            |
| P07.06        | AI1 analog input type definition                 |                                                                                                                                                                                                                                                                                                                                                              |               |      | 1               |                                                                                            |
| P07.01        | AI0 analog input function definition             | Set the function of AI0~AI1 analog input ports with the connotation of each parameter as follows:                                                                                                                                                                                                                                                            | 0 / 2 / 3 / 4 | ×    | 2               | AI0 analog input port is set to be analog speed reference input when it leaves the factory |

| P07.07        | AI1 analog input<br>Function definition     | 0: invalid (not in use)<br>2: analog speed reference<br>3: analog torque reference<br>4: analog compensation torque reference                                                                                                                                                                       |               |      | 0               | AI1 analog input port is set to be analog torque compensation input when it leaves the factory. |
|---------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|-----------------|-------------------------------------------------------------------------------------------------|
| P07.02        | AI0 analog input<br>Offset                  | Set the offset voltage value of AI0 analog input                                                                                                                                                                                                                                                    | 0.000~20.000  | V    | 10.000          |                                                                                                 |
| P07.03        | AI0 analog input<br>Gain                    | Set the gain amount of AI0 analog input which is a data expressed in percentage                                                                                                                                                                                                                     | 0.0~100.0     | %    | 100.0           |                                                                                                 |
| P07.04        | AI0 analog input<br>Filtering time          | Set the filtering time of AI0 analog input signal                                                                                                                                                                                                                                                   | 0~30          | ms   | 10              |                                                                                                 |
| P07.05        | AI0 analog input<br>Voltage chopping        | Set the AI0 analog input voltage chopping value                                                                                                                                                                                                                                                     | 0.000~10.000  | V    | 10.000          |                                                                                                 |
| P07.08        | AI1 analog input<br>Offset                  | Set the offset voltage value of AI1 analog input                                                                                                                                                                                                                                                    | 0.000~20.000  | V    | 10.000          |                                                                                                 |
| P07.09        | AI1 analog input<br>Gain                    | Set the gain amount of AI1 analog input which is a data expressed in percentage                                                                                                                                                                                                                     | 0.0~100.0     | %    | 100.0           |                                                                                                 |
| P07.10        | AI1 analog input<br>Filtering time          | Set the filtering time of AI1 analog input signal                                                                                                                                                                                                                                                   | 0~30          | ms   | 10              |                                                                                                 |
| P07.11        | AI1 analog input<br>Voltage chopping        | Set the AI1 analog input voltage chopping value                                                                                                                                                                                                                                                     | 0.000~10.000  | V    | 10.000          |                                                                                                 |
| Function Code | Designation                                 | Content                                                                                                                                                                                                                                                                                             | Setting Range | Unit | Factory Setting | Remarks                                                                                         |
| F08           | Analog output port<br>Definition parameters |                                                                                                                                                                                                                                                                                                     |               |      |                 |                                                                                                 |
| P08.00        | M0 analog output port function definition   | Set the function of M0~M1 analog output ports with the each parameter defined as follows:<br>0: not defined                                                                                                                                                                                         | 0~44          | ×    | 1               |                                                                                                 |
| P08.01        | M1 analog output port function definition   | 1: U phase current<br>2: V phase current<br>3: W phase current<br>6: speed reference<br>7: feedback speed<br>13: speed regulator output<br>14: current regulator IQ reference<br>15: current regulator ID reference<br>30: current regulator IQ output<br>32: DC bus voltage<br>44: speed deviation |               |      | 2               |                                                                                                 |

(Continued)

| Function Code | Designation | Content | Setting Range | Unit | Factory Setting | Remarks |
|---------------|-------------|---------|---------------|------|-----------------|---------|
|---------------|-------------|---------|---------------|------|-----------------|---------|

|        |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |   |        |  |
|--------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|--------|--|
| P08.02 | M0 analog output Offset                  | Set the voltage offset of M0 analog output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.000~20.000 | V | 15.000 |  |
| P08.03 | M0 analog output Gain                    | Set the gain size of M0 analog output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.0~1000.0   | % | 100.0  |  |
| P08.04 | M1 analog output Offset                  | Set the voltage offset of M1 analog output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.000~20.000 | V | 15.000 |  |
| P08.05 | M1 analog output Gain                    | Set the gain size of M1 analog output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.0~1000.0   | % | 100.0  |  |
| P08.06 | Select U01 display data content of LCD   | <p>The operator has LCD and LED display screen. LED can only display one data, while LCD can set 8 display datum U01~U08. The set parameter specifies the signal contents of the display data. Their corresponding relationship is shown as follows:</p> <p>0: not defined<br/>1: feed(rpm)<br/>2: speed reference(Hz)<br/>3: feedback speed (Hz)<br/>4: output current (A)<br/>5: output voltage (V)<br/>6: output torque (%)<br/>7: bus voltage (V)<br/>8: analog input 1 signal(V)<br/>9: analog input2 signal(V)<br/>13: static self-study magnetic pole angle<br/>16: zero servo torque(%)<br/>17: disturbed times of encoder Z phase<br/>18: the disturbed times of encoder 's phase A and phase B<br/>23: load compensation torque(%)<br/>24: given rotary speed(rpm)<br/>25: speed deviation(rpm)</p> | 0~31         | × | 24     |  |
| P08.07 | Select U02 display data content of LCD   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |   | 1      |  |
| P08.08 | Select U03 display data content of LCD   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |   | 25     |  |
| P08.09 | Select U04 display data content of LCD   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |   | 4      |  |
| P08.10 | Select U05 display data content of LCD   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |   | 6      |  |
| P08.11 | Select U06 display data content of LCD   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |   | 16     |  |
| P08.12 | Select U07 display data content of LCD t |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |   | 7      |  |
| P08.13 | Select U08 display data content of LCD   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |   | 5      |  |
| P08.14 | Select the display data content of LED   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |   | 1      |  |

| Function code | Designation                     | Factory setting | Setting range | Unit | Remarks              |
|---------------|---------------------------------|-----------------|---------------|------|----------------------|
| P09.00        | Accumulated on time             |                 |               | h    | Read-only parameters |
| P09.01        | Accumulated run time            |                 |               | H    | Read-only parameters |
| P09.02        | Highest temperature of heatsink |                 |               | ℃    | Read-only parameters |

|        |                                                                 |        |            |           |                                                     |
|--------|-----------------------------------------------------------------|--------|------------|-----------|-----------------------------------------------------|
| P09.03 | Hardware version                                                |        |            | ×         | Read-only parameters                                |
| P09.04 | Software version of control panel                               |        |            | ×         | Read-only parameters                                |
| P09.05 | Rated power of inverter                                         |        |            | <b>KW</b> | Read-only parameters                                |
| P09.06 | Torque direction                                                | 1      |            |           |                                                     |
| P09.07 | Current loop Kp                                                 | 1.40   |            |           |                                                     |
| P09.08 | Current loop Ki                                                 | 1.00   |            |           |                                                     |
| P09.09 | Current loop Kd                                                 | 0.00   |            |           |                                                     |
| P09.10 | Current loop bandwidth                                          | 400.00 |            | Hz        |                                                     |
| P09.11 | Flux loop bandwidth                                             | 0.8    |            | Hz        |                                                     |
| P09.12 | Current loop selection                                          | 0      |            |           |                                                     |
| P09.13 | Stand by                                                        |        |            |           |                                                     |
| P09.14 | Filter time of the current loop(stand by in the latest program) |        |            | ms        | Generally do not modify                             |
| P09.15 | PWM modulation mode                                             | 1      | <b>0~2</b> |           |                                                     |
| P09.16 | Zero servo compensation                                         | 0      |            | %         |                                                     |
| P09.17 | Contactor opening delay                                         | 0.8    |            | S         |                                                     |
| P09.18 | Brake opening delay                                             | 0.4    |            | S         |                                                     |
| P09.19 | Contactor off delay                                             | 1.0    |            | S         |                                                     |
| P09.20 | Brake delay                                                     | 0.1    |            | S         |                                                     |
| P09.21 | Output turn-off delay                                           | 0.3    |            | S         |                                                     |
| P09.22 | Zero speed threshold                                            | 0.20   |            | Hz        |                                                     |
| P09.23 | Special funtion selection                                       | 0      |            |           |                                                     |
| P09.24 | Three-phase current balance coefficient                         | 1.043  |            |           |                                                     |
| P09.25 | Light troubleshooting                                           | 1      |            |           | 0:do not output fault relay<br>1:output fault relay |
| P09.26 | Automatic fault reset time                                      | 10.0   |            | S         |                                                     |
| P09.27 | Times of automatic fault reset                                  | 3      |            |           |                                                     |
| P09.28 | Time of Radiator overheating                                    | 0.50   |            | S         |                                                     |
| P09.29 | Overspeed protection                                            | 120.00 |            | %         |                                                     |

|        | coefficient                                                     |       |         |       |                                                                                                                                                                                                                                                     |
|--------|-----------------------------------------------------------------|-------|---------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P09.30 | Overspeed protection time                                       | 1.00  |         | S     |                                                                                                                                                                                                                                                     |
| P09.31 | Input phase voltage threshold                                   | 55    |         | V     |                                                                                                                                                                                                                                                     |
| P09.32 | Confirmation times for short circuit of braking resistor        | 10    |         | times |                                                                                                                                                                                                                                                     |
| P09.33 | Confirmation for Encoder disconnection                          | 2     |         | times |                                                                                                                                                                                                                                                     |
| P09.34 | Confirmation for inputting open phase                           | 2.000 |         | S     |                                                                                                                                                                                                                                                     |
| P09.35 | Voltage for relay failure                                       | 65    |         | V     |                                                                                                                                                                                                                                                     |
| P09.36 | Frequency dividing coefficient of encoder<br>(PG card required) | 0     | 0~7     |       | 0: ( no frequency dividing), 1:( 2 frequency dividing), 2: (4 frequency dividing),3:(8 frequency dividing), 4: (16 frequency dividing),5 (32 frequency dividing),6:(64 frequency dividing), 7: (128 frequency dividing)<br>Note: (PG card required) |
| P09.39 | Synchronous motor study angle or not when power on              | 1     | 0/1     |       | Determine whether synchronous motor conduct angle self-study or not when power on , 0 for no study, 1 for study                                                                                                                                     |
| P09.40 | Current gain when self-study                                    | 150   | 0~400   | %     | Current gain when synchronous motor conduct angle self-study                                                                                                                                                                                        |
| P09.41 | Confirmation threshold of Encoder phase CD failure              | 300   | 0~65535 |       | No 28 failure reported in case that the D-value of the absolute position and computing position of encoder exceed the setting value.                                                                                                                |
| P09.43 | Protection threshold of ABZ encoder disconnection               | 20    | 0~100   |       | Protection in case of speed feedback deviation of synchronous motor confirmed for more than the setting value                                                                                                                                       |
| P09.44 | Protection times of IGBT <sub>s</sub>                           | 2     | 1~65535 | times |                                                                                                                                                                                                                                                     |
| P09.45 | Protection option of I <sub>2t</sub>                            | 0     | 0~65535 |       |                                                                                                                                                                                                                                                     |
| P09.46 | ID_0                                                            |       |         |       | Internal parameters, do not modify                                                                                                                                                                                                                  |
| P09.47 | ID_1                                                            |       |         |       | Internal parameters, do not modify                                                                                                                                                                                                                  |
| P09.48 | ID_2                                                            |       |         |       | Internal parameters, do not modify                                                                                                                                                                                                                  |
| P09.49 | ID_3                                                            |       |         |       | Internal parameters, do not modify                                                                                                                                                                                                                  |

|        |      |  |  |  |                                    |
|--------|------|--|--|--|------------------------------------|
| P09.50 | ID_4 |  |  |  | Internal parameters, do not modify |
| P09.51 | ID_5 |  |  |  | Internal parameters, do not modify |
| P09.52 | ID_6 |  |  |  | Internal parameters, do not modify |

## B.2 Fault sheet

| Fault code | Failure display               | Possible causes                              | Countermeasures                                                                                                    |
|------------|-------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1          | Module overcurrent protection | Too high voltage at the DC end               | Check the grid power supply for fast shutdown due to no energy braking under high inertia load                     |
|            |                               | Short circuit on peripheral equipment        | Check the motor and the output wiring for short circuit and for short circuit to the ground.                       |
|            |                               | Phase absence from output                    | Check the motor and the output wiring for loose connection.                                                        |
|            |                               | Encoder phase fault                          | Check encoder phase                                                                                                |
|            |                               | Motor phase fault                            | Check motor phase                                                                                                  |
|            |                               | Phase angle self-study fault                 | Restudy phase angle                                                                                                |
|            |                               | Lack current when phase angle self-studying  | Increase current gain                                                                                              |
|            |                               | Encoder failure                              | Check the encoder for damage and improper wiring.                                                                  |
|            |                               | Hardware bad contact or damage               | Invite professional technician to maintain                                                                         |
|            |                               | Loose plug-in pieces inside the inverters    | Invite professional technician to maintain                                                                         |
|            |                               | Induction motor slip setting unreasonable    | Adjust the induction motor slip                                                                                    |
|            |                               | No-load current coefficient set unreasonable | Adjust the no-load current coefficient                                                                             |
| 2          | ADC failure                   | Damaged current sensor                       | Replace the current sensor                                                                                         |
|            |                               | Current sampling circuit in problem          | Replace the control panel                                                                                          |
| 3          | Heatsink overheat             | Too high ambient temperature                 | Reduce the ambient temperature, and enhance ventilation and heat dissipation                                       |
|            |                               | Air duct block                               | Remove the foreign material such as dust and cotton wool from the air duct.                                        |
|            |                               | Abnormal fan                                 | Check whether the power supply line to the fan is in good condition or replace the fan with that of the same type. |
|            |                               | Temperature detection circuit failure        | Invite professional technician to maintain                                                                         |

| Fault code | Failure display                             | Possible causes                                                            | Countermeasures                                                                      |
|------------|---------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 4          | Braking unit failure                        | Damaged braking unit                                                       | Replace corresponding drive module                                                   |
|            |                                             | Short circuit on the external braking resistor's line                      | Check the braking resistor's wiring                                                  |
| 5          | Fuse blown                                  | Too high current results in fuse wire blown                                | Check the fuse circuit for break and loose connections                               |
| 6          | Output over torque                          | Too low input power supply voltage                                         | Check the input power supply                                                         |
|            |                                             | Motor is blocked or experiences serious load mutation                      | Prevent the motor from being blocked to reduce load mutation.                        |
|            |                                             | Encoder failure                                                            | Check the encoder for damage or improper wiring                                      |
|            |                                             | Phase absence from output                                                  | Check the motor and the output wiring for any loose connection                       |
| 7          | Speed deviation                             | Too short acceleration time                                                | Extend the acceleration time                                                         |
|            |                                             | Too high load                                                              | Reduce the load                                                                      |
|            |                                             | Too low current limit                                                      | Boost the current limit within the permissible scope                                 |
| 8          | (accelerating) bus overvoltage protection   | Abnormal input power supply voltage                                        | Check the input power supply                                                         |
|            |                                             | The motor restart quickly during high speed rotation                       | Restart after the motor stops running                                                |
|            | (decelerating) bus overvoltage protection   | Too high load moment of inertia                                            | Use proper energy-consuming braking assembly                                         |
|            |                                             | Too short deceleration time                                                | Extend the deceleration time                                                         |
|            |                                             | Too high braking resistance or no resistor is connected                    | Connect proper braking resistors                                                     |
|            | (constant speed) bus overvoltage protection | Abnormal input power supply                                                | Check the input power supply                                                         |
|            |                                             | Too high moment of inertia                                                 | Use proper energy-consuming braking assembly                                         |
|            |                                             | Too high braking resistance or no resistor is connected                    | Connect proper braking resistors                                                     |
| 9          | Busbar undervoltage                         | The power supply voltage is lower than lowest working voltage of equipment | Check the input power supply                                                         |
|            |                                             | Transient outage occurs                                                    | Check the input power supply and restart after the input voltage resumes and resets. |
|            |                                             | Too high voltage fluctuation of input power supply                         |                                                                                      |
|            |                                             | Loose wiring terminal of the power supply                                  | check the input wiring                                                               |
|            |                                             | Abnormal power supply to the inside switches                               | Invite professional technician to maintain                                           |

| Fault code | Failure display                                          | Possible causes                                                                        | Countermeasures                                                                                                                                             |
|------------|----------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                                                          | There is load with high starting current in the same power supply system               | Change the power supply system to make it satisfy the specifications                                                                                        |
| 10         | Phase absence from output                                | Abnormal wiring at the output side of the inverter: missed connection or broken wires. | Check the wiring at the output side of the inverter according to the operation procedures to exclude the possibility of missed connection and broken wires. |
|            |                                                          | Loose output terminals                                                                 |                                                                                                                                                             |
|            |                                                          | Too small motor capacity below 1/20 of the maximum applicable capacity of the motor    | Adjust the capacity of the inverter or that of the motor                                                                                                    |
|            |                                                          | Imbalance among the three output phases                                                | Check the motor for its wiring                                                                                                                              |
| 11         | Motor low speed and overcurrent (accelerating)           | Low grid voltage                                                                       | Check the input power supply                                                                                                                                |
|            |                                                          | Abnormal motor parameter setting                                                       | Set the parameters correctly for the motor                                                                                                                  |
|            |                                                          | The running motor starts at a high speed directly                                      | Restart after the motor stops                                                                                                                               |
|            | Motor low speed and overcurrent (decelerating))          | Too low grid voltage                                                                   | check the input power supply                                                                                                                                |
|            |                                                          | Too high load moment of inertia                                                        | Use proper energy-consuming braking assembly                                                                                                                |
|            |                                                          | Abnormal motor parameter setting                                                       | wet the parameters correctly for the motor                                                                                                                  |
|            |                                                          | Too short deceleration time                                                            | extend the deceleration time                                                                                                                                |
|            | Motor low speed and overcurrent (running constant speed) | Load mutation during running                                                           | Reduce the frequency and amplitude of load mutation                                                                                                         |
|            |                                                          | Abnormal motor parameter setting                                                       | set the parameters correctly for the motor                                                                                                                  |
| 12         | Encoder failure                                          | Incorrect encoder connection                                                           | change the wiring for the encoders                                                                                                                          |
|            |                                                          | The encoder has no signal output                                                       | Check the encoder for proper operation and power supply conditions                                                                                          |
|            |                                                          | Broken encoder connecting wire                                                         | Repair the broken wires                                                                                                                                     |
|            |                                                          | Abnormal function code setting                                                         | Confirm the function codes related to the encoders of the inverter are properly set.                                                                        |
| 13         | Current is detected at the time of parking               | The current flow is not effectively cut off when the motor stops.                      | The synchronous motor slips.                                                                                                                                |
|            |                                                          |                                                                                        | invite professional technicians to maintain                                                                                                                 |
| 14         | Speed reverses                                           | The speed reverses during running                                                      | Check for external load mutation                                                                                                                            |



| Fault code | Failure display                                                        | Possible causes                                                         | Countermeasures                                                    |
|------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------|
|            | during running                                                         | The encoder and the motor have different phase sequence                 | Change the phase sequence of the motor or encoder                  |
|            |                                                                        | When started, the motor run backward and the current reaches the limit. | Too low current limit or mismatched motor                          |
| 15         | Speed is detected at the time of parking                               | The lock brake loosens and the elevator slips                           | Check the lock brake                                               |
|            |                                                                        | The encoder is disturbed or loosens                                     | Tighten the encoder and remove the disturbance                     |
| 16         | Wrong motor phase sequence                                             | Reverse wiring of the motor                                             | Reverse the wiring or adjust the parameters                        |
| 17         | Overspeed in the same direction (within the permissible maximum speed) | The synchronous motor de-magnetizes and races                           | Check the motor                                                    |
|            |                                                                        | The synchronous motor angle self-study is wrong                         | Self-study again                                                   |
|            |                                                                        | Set the encoder's parameters incorrectly or be subject to disturbance   | Check the circuit of the encoder                                   |
|            |                                                                        | Too high positive load or load mutation                                 | check the load mutation for external causes                        |
| 18         | Reverse overspeed (within the permissible maximum speed)               | The synchronous motor de-magnetizes and races                           | Check the motor                                                    |
|            |                                                                        | The synchronous motor angle self-study is wrong                         | Self-study again                                                   |
|            |                                                                        | Set the encoder's parameters incorrectly or be subject to disturbance   | Check the circuit of the encoder                                   |
|            |                                                                        | Too high positive load or load mutation                                 | check the load mutation for external causes                        |
| 19         | UVW encoder wrong phase sequence                                       | Wrong wiring of the encoder or incorrect parameter setting              | check the wiring or change the parameters                          |
| 20         | Encoder communication failure                                          | encoder failure                                                         | Check the wiring of the encoder and restart the encoder self study |
| 21         | Abc overcurrent (three-phase transient value)                          | The motor is short circuited to the ground at a single phase            | Check the motor and output line circuit                            |
|            |                                                                        | encoder failure                                                         | Check the encoder for damage or incorrect wiring                   |
|            |                                                                        | Encoder phase fault                                                     | Check encoder phase                                                |
|            |                                                                        | Motor phase fault                                                       | Check motor phase                                                  |
|            |                                                                        | Phase angle self-study fault                                            | Restudy phase angle                                                |

| Fault code | Failure display                       | Possible causes                                                                   | Countermeasures                                                                                                              |
|------------|---------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|            |                                       | Lack current when phase angle self-studying                                       | Increase P09.40 current gain                                                                                                 |
|            |                                       | Something wrong with the drive board's detection circuit                          | Replace the drive board                                                                                                      |
|            |                                       | Induction motor slip setting unreasonable                                         | Adjust the induction motor slip                                                                                              |
|            |                                       | No-load current coefficient set unreasonable                                      | Adjust the no-load current coefficient                                                                                       |
| 22         | Failure is detected on the lock brake | The output relay doesn't act                                                      | Check the control circuit of the relay                                                                                       |
|            |                                       | The relay action brake is not On.                                                 | Check the power line of the brake for loose and broken wires.                                                                |
|            |                                       | No signal is detected on the feedback elements                                    | Adjust the feedback units                                                                                                    |
| 23         | Input overvoltage                     | Too high incoming voltage                                                         | Check if the incoming voltage matches the inverter                                                                           |
|            |                                       | There are problems in the voltage detection circuit for the switch power supply   | invite professional technicians to maintain                                                                                  |
| 24         | UVW encoder wire break                | There are problems in the wiring circuit of the encoder                           | Loose wiring end or damage and breakage in the line                                                                          |
| 25         | Standby                               |                                                                                   |                                                                                                                              |
| 26         | Encoder fails self study              | The synchronous motor fails to study the angle of the encoder                     | Carry out encoder self study                                                                                                 |
| 27         | Output overcurrent (virtual value)    | Too long time exposure to overload , the higher the load and the shorter the time | Stop running for a period and if the same occur after it restarts, then check if the load falls within the permissible scope |
|            |                                       | The motor is blocked                                                              | Check the motor or the lock brake                                                                                            |
|            |                                       | Short circuit on the motor coil                                                   | Check the motor                                                                                                              |
|            |                                       | Encoder phase fault                                                               | Check encoder phase                                                                                                          |
|            |                                       | Motor phase fault                                                                 | Check motor phase                                                                                                            |
|            |                                       | Phase angle self-study fault                                                      | Restudy phase angle                                                                                                          |
|            |                                       | Lack current when phase angle self-studying                                       | Increase current gain                                                                                                        |
|            |                                       | Output short circuit                                                              | Check the wiring or the motor                                                                                                |

| Fault code | Failure display                                                  | Possible causes                                                                   | Countermeasures                                                                                                                           |
|------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 28         | Sincos encoder failure                                           | Damaged encoder or wrong circuit                                                  | Check the encoder and circuit                                                                                                             |
| 29         | Phase absence from input                                         | Abnormal voltage at the input side                                                | Check the grid voltage                                                                                                                    |
|            |                                                                  | Phase absence from the input voltage                                              |                                                                                                                                           |
|            |                                                                  | The wiring terminal loosens at the input side                                     | Check the input terminal wiring                                                                                                           |
| 30         | Overspeed protection (beyond the maximum speed protection limit) | Wrong parameter setting for the encoder or disturbance                            | Check the circuit of the encoder                                                                                                          |
|            |                                                                  | Load mutation                                                                     | check the load mutation for external causes                                                                                               |
|            |                                                                  | Incorrect overspeed protection parameter setting                                  | Check the parameters                                                                                                                      |
| 31         | Motor high speed overcurrent                                     | Low grid voltage                                                                  | check the input power supply                                                                                                              |
|            |                                                                  | Load mutation during running                                                      | Reduce the frequency and amplitude of the load mutation                                                                                   |
|            |                                                                  | Abnormal motor parameter setting                                                  | wet the parameters correctly for the motor                                                                                                |
|            |                                                                  | Incorrect encoder parameter setting or disturbance                                | Check the circuit of the encoder                                                                                                          |
| 32         | Grounding protection                                             | Wrong wiring                                                                      | Correct the wrong wiring in reference to the users' manual                                                                                |
|            |                                                                  | The motor is abnormal                                                             | To replace the motor, test the insulation to ground first                                                                                 |
|            |                                                                  | Too high current leakage to the ground at the output side of the inverter         | invite professional technicians to maintain                                                                                               |
| 33         | Capacitance aging                                                | Capacitance aging of the inverter                                                 | invite professional technicians to maintain                                                                                               |
| 34         | External failure                                                 | External input fault signal                                                       | Check the causes for the external failure                                                                                                 |
| 35         | Output imbalance                                                 | Abnormal wiring at the output side of the inverter : missed wiring or wire broken | Check the wiring at the output side of the inverter to exclude missed wiring and broken wires in accordance with the operation procedures |
|            |                                                                  | The three phases of the motor imbalance                                           | Check the motor                                                                                                                           |
| 36         | Wrong parameter setting                                          | Incorrect parameter setting                                                       | Modify the parameters of the inverter                                                                                                     |
| 37         | Current sensor failure                                           | Failure on the hardware of the drive board                                        | invite professional technicians to maintain                                                                                               |
| 38         | Braking resistance short circuit                                 | Short circuit on the external braking resistor                                    | Check the wiring of the braking resistor                                                                                                  |

| Fault code | Failure display                                        | Possible causes                                                                                 | Countermeasures                                                                 |
|------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 39         | Too high transient current                             | When Ia, Ib and Ic are idle, alarm occurs due to too high transient current on the three phases | invite professional technicians to maintain                                     |
| 40         | KMY detect failures                                    | The KMY detect contact signal inconsistent KMY control signal                                   | Check the KMY the control and detection contacts                                |
| 41         | Brake switch detects failure                           | Brake switch detects contact signal inconsistent control signal                                 | Check Brake switch                                                              |
| 42         | IGBT short-circuit protection                          | Periphery short circuit exists                                                                  | check the motor and output wiring for short circuit, or short circuit to ground |
| 44         | Voltage drop during operation                          | The power supply capacity is insufficient, leading to input voltage drop during operation       | Check the power supply capacity                                                 |
|            |                                                        | Inverter charging relay is damaged                                                              | Contat the manufacturer for repair.                                             |
| 45         | I <sub>2t</sub> instantaneous overcurrent protection   | Same as failure 21 、 27                                                                         | Same as failure 21 、 27                                                         |
| 46         | I <sub>2t</sub> effective value overcurrent protection |                                                                                                 |                                                                                 |

## Appendix C Inverter Complies With The Standards



### 1. European Low Voltage directive

AS320 Series converter products comply with EN61800-5-1 :2007 standards, which comply with the Low Voltage Directive (Low Voltage Directive 2006/95/EC)。

**The inverter also meets the following standards:**

**EN61800-5-1: 2007:Adjustable speed electrical powerdrive systems –Part 5-1:Safetyrequirements –Electrical, thermal and energy**

### 2. European EMC regulations

When you follow the recommendations in this manual for installation, the AS320 series inverter product meets the following EMC standards:

**EN12015.1998 Electromagnetic compatibility-Product family standard for lifts, escalators and passenger conveyors-Emission.**

**EN12016.2004 Electromagnetic compatibility-Product family standard for lifts, escalators and passenger conveyors-Immunity.**

**EN61800-3:2004: Power drive D of Part 3**

### 3. ISO9001 Quality Management System

Shanghai Sigriner STEP Electric Co., Ltd will manage its quality management system in accordance with the ISO9001 standards.

# A letter of Advice to Clients

Dear clients,

RoHS is the abbreviation for *The restriction of the use of certain hazardous substances in electrical and electronic equipment* which was implemented by EU on July 1<sup>st</sup>, 2006. It stipulates that in the newly launched electrical and electronic equipment, the following six hazardous substances are restricted: lead, mercury, cadmium, sexivalence chrome, PBB and PBDE.

In our country, *the Electronic Information Products Pollution Control Management Measures* was issued on February 28th, 2006 jointly by the Ministry of Information Industry, State Development and Reform Commission, Ministry of Commerce, General State Administration for Industry and Commerce, Administration of Customs of the P.R.C, General Administration of Quality Supervision, Inspection and Quarantine and State Bureau of Environmental Protection, becoming an RoHS direction of Chinese Version and enforced. On February 1<sup>st</sup>, 2008, *electronic waste environmental pollution prevention and control management measures* issued by the State Bureau of Environmental Protection of the P.R.C began to be executed, clearly specifying that the users of electronic and electrical products shall provide or entrust the electronic waste to the disassembling and disposing units (including small individual businesses) with corresponding business scope listed in directory (or temporary directory) to disassemble, make use of or dispose.

Our company follows the requirements in *the Electronic Information Products Pollution Control Management Measures* and RoHS directive in the aspects such as purchasing and selecting the types of electronic parts and components, PCB filter plates, wiring harness material and structural parts and strictly controls the above-mentioned six hazardous substances. Meanwhile in the production process, PCB parts and components are welded on XinChi lead free welding production line with a lead free welding technology.

Hazardous substances which may be contained in the following assemblies:

| Type of assembly              | Electronic components                                                              | Electronic printed circuit board | Sizing sheet pieces | Radiators | Plastic pieces | conductors |
|-------------------------------|------------------------------------------------------------------------------------|----------------------------------|---------------------|-----------|----------------|------------|
| Possible hazardous substances | six hazardous substances: lead, mercury, cadmium, sexivalence chrome, PBB and PBDE |                                  |                     |           |                |            |

1) analysis on environmental influence: Our electronic products will produce some heat in use, which may lead too the emission of individual hazardous substance but will not cause serious influence on the surrounding. Once an electronic product is discarded after the expiry of its life, the heavy metallic and chemical hazardous substance in it will severely pollute the soil and water resources.

2) The life cycle of electronic products and equipment. Any electronic product and equipment has a life cycle and can be damaged and discarded. Even if it can still be used, it will be replaced and washed out by new generations of electronic products. Our products and equipment normally have a life cycle not more than 20 years.

3) The treatment of discarded electronic products. If the discarded electronic products can not be treated properly, they will pollute the environment. Our company requires our clients establish a reclaiming system in accordance with related national regulation and not throw away them as ordinary domestic waste. The products shall be stored and used in environment-friendly ways or reclaimed by qualified units by strictly complying with the *electronic waste environmental pollution prevention and control management measures* issued by the State Bureau of Environmental Protection of the P.R.C. Any individual or unit having no such qualification is prohibited conducting the activity of disassembling, making use of and disposing electronic wastes.

Please don't throw away electronic waste together with ordinary domestic waste, but call the local waste disposing agencies or environment protection agencies for suggestion on how to deal with the electronic waste.

**Shanghai Sigriner STEP Electric Co., Lt**